

THE AUSTRALIAN ABORIGINAL

BY

HERBERT BASEDOW

M.A., M.D., PH.D., B.Sc., F.G.S., ETC.

SOMETIME CHIEF MEDICAL INSPECTOR AND CHIEF PROTECTOR OF ABORIGINES
IN THE NORTHERN TERRITORY ; SPECIAL ABORIGINES' COMMISSIONER
FOR THE FEDERAL AND STATE GOVERNMENTS ; LOCAL CORRESPONDENT ROYAL ANTHROPOLOGICAL INSTITUTE OF GREAT
BRITAIN AND IRELAND ; HONORARY FELLOW ANTHROPOLOGICAL SOCIETY OF GÖTTINGEN, ETC.

With 146 Illustrations

ADELAIDE :
F. W. PREECE AND SONS

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Title: The Australian aboriginal

Author: Herbert Basedow

Release date: October 23, 2023 [eBook #71940]

Language: English

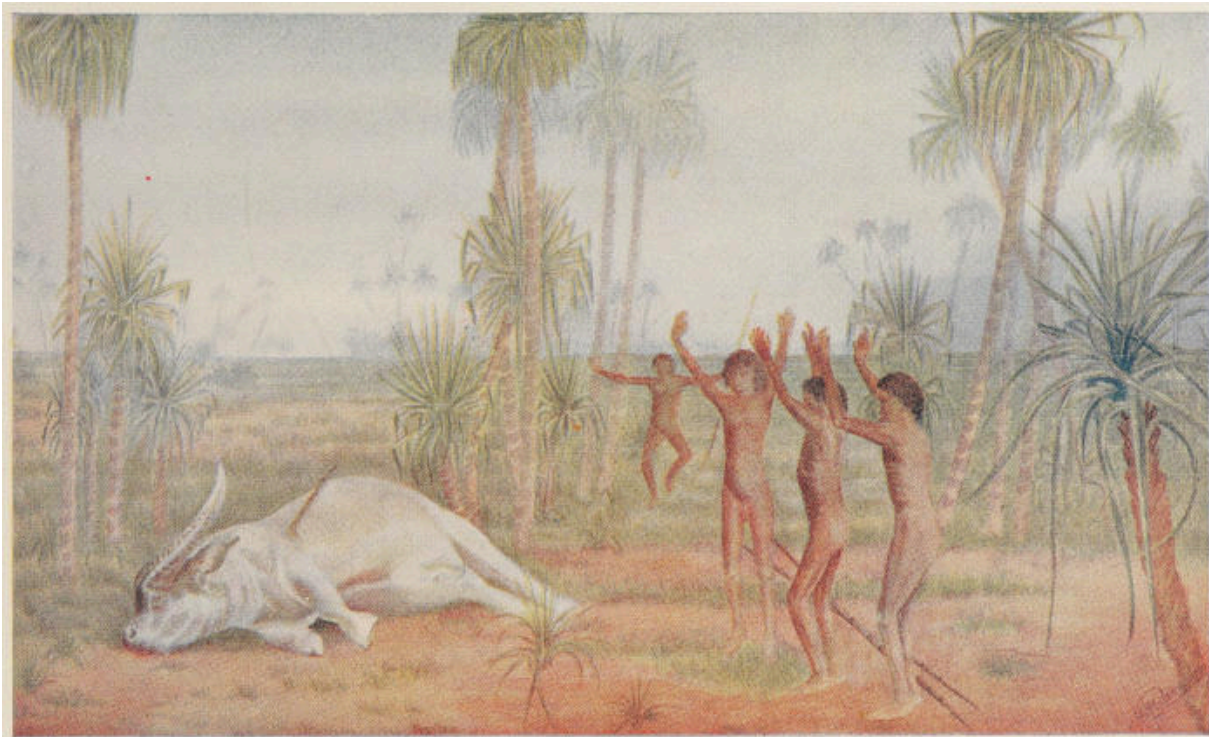
Original publication: Adelaide: F. W. Freece & Sons, 1925

Other information and formats: www.gutenberg.org/ebooks/71940

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AUSTRALIAN ABORIGINAL ***

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THE BUFFALO HUNTERS' TRIUMPH

H. BASEDOW, *pinx.*

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Registered by the Postmaster-General for
transmission through the post as a book.

Wholly set up and printed in Australia
at THE HASSELL PRESS, Adelaide.

TO THE MEMORY OF MY MOTHER
ANNA CLARA HELENE BASEDOW

PREFACE

For some years past it had been my intention to write a progressive series of treatises on the Australian aboriginal, embodying observations as they were being made. As time wore on, however, my procrastination amassed so great a fund and so great a variety of notes that my original plan became impossible. The delay was not altogether through any fault of mine, but through a run of fortunate happenings which allowed me to spend the better part of every year in the great unbounded spaces of central and northern Australia, to live among the uncontaminated tribes, and to study Australian anthropology at the fountain head. It was through my repeated and prolonged absences from the city (and civilization in general) that I could not attend to the publication of any lengthy scientific papers.

Apart from my private and professional journeys, the following are some of the better known expeditions I have accompanied or led:

The Government North-West Expedition,

Government Expedition of Geological Exploration in the Northern Territory,

Cruise of the s.s. "White Star" under His Excellency Sir George Le Hunte, Governor of South Australia,

Mineralogical Survey of the Flinders Ranges,

Sir Joseph Verco's Deep Sea Dredging Expeditions,

Exploration of Melville and Bathurst Islands,

Expedition in Search of Muniton Minerals in the Northern Kimberleys of Western Australia, officially subsidized by the Premier (Hon. John Scaddan, M.P.),

Expedition in Search of Muniton Minerals in Central Australia,

Three South Australian Medical Relief Expeditions among the Aborigines,

Expedition in Search of Reported Remains of Leichhardt, under
auspices of S.A. Geographical Society,
Commonwealth Medical Survey of Aborigines in the Northern
Territory,
Mararoa Geological Expedition from Darwin to Northern
Kimberleys,
Viceregal Expedition to Central Australia, under His Excellency, Sir
Tom Bridges, Governor of South Australia,
Commonwealth Railways Commissioner's Journey from Mildura to
Port Augusta, under Hon. P. G. Stewart and N. G. Bell, Esq.,
Expedition through the Interior of Australia, under His Excellency,
the Earl of Stradbroke, Governor of Victoria.

During the terms that I held official positions, firstly as a State Geologist, and secondly as Chief Medical Officer and Chief Protector of Aborigines for the Commonwealth Government in the Northern Territory, opportunities were afforded me of continuing my investigations among the indigenous population of Australia; especially fruitful were my researches when, as a Special Aborigines' Commissioner, I medically overhauled the tribes of South Australia and of the southern region of the Northern Territory.

In a more private capacity the southern districts and goldfields of Western Australia, the coastal and south-western districts of Queensland, and the north-western areas of New South Wales were traversed. Quite recently, too, a professional excursion to Java considerably enriched my knowledge of Melanesian ethnography and helped to explain the existence of several cults in the northern districts of Australia which border on the Indian Ocean.

On the other hand, not long before the war I continued study abroad and was privileged to be associated with the late Professor Hermann Klaatsch, under whose admirable guidance my researches were conducted in the Anatomical School of the University of Breslau. It will be apparent, therefore, why many of the results enumerated in the chapters dealing with the racial characteristics of the Australian aboriginal are based upon the doctrines of this eminent authority. In London Sir Arthur Keith courteously placed the whole of the valuable collections of Australian skulls and

skeletons in the Hunterian Museum of the Royal College of Surgeons at my disposal; while Captain T. A. Joyce, in an equally generous way, facilitated my investigations in the anthropological galleries of the British Museum; I am taking advantage of this opportunity of expressing my sincere gratitude to these two gentlemen. I also desire to thank the Directors of the National and University Museums which I visited in France, Germany, Switzerland, and Holland, who so willingly responded to my enquiries and brought me into personal contact with any matters I was especially interested in.

While engaged upon one of my later commissions in central Australia, my duties took me to the Hermannsburg Mission Station on the Finke River, where for a fortnight I enjoyed the companionship of the Rev. Carl Strehlow and received every assistance in my researches among the western groups of the Arundta tribe. Through the recent pathetic and heroic death of Strehlow, Science has lost an indefatigable and conscientious worker, and the aborigines a staunch and faithful friend.

In the present volume I have endeavoured to sift my subject matter in such a way as to keep the text in a suitable sequence and to make it of general interest. The principal difficulty has been to delete matter in order to keep down the bulk of the book. The latter remark applies equally well to the selection of illustrations; it was with a heavy heart that I found myself obliged to reduce the number of plates, all of which illustrated interesting points referred to in the text. Nevertheless, I feel that I am greatly indebted to the publisher, Mr. F. W. Preece, for allowing me to include so many more illustrations than he had originally counted upon.

Now that the manuscript is in the hands of the printer, I feel that there are several chapters I should like to have done more justice to; those, for instance, dealing with tribal organizations, initiation ceremonies, religious ideas, and art might have been considerably expanded if it had not been for the want of space. In those on religious ideas and art, I realize that I am launching in new directions so far as Australian anthropology is concerned, but I trust that the evidence which I have produced will be sufficient to prove that I did not arrive at the results too hastily, or, indeed, without deliberation and substantiation. Phallic worship had long been suspected in Australia, and some of our foremost authorities, such as the late R. Etheridge, Jun., were inclined to explain the occurrence of certain stones and other objects in the tribal areas of Australia (present or past) on that

hypothesis. I came upon the evidence accidentally in 1916 in the form of a stone phallus erected in the ground at Success Strait in the far north-western corner of Australia. The stone was surrounded by a cleared cirque where much blood had been spilt at a recent ceremony. A few weeks later I for the first time witnessed an actual performance on the shores of Cambridge Gulf at which wooden phallus were produced. Since then I have been able to trace the existence of phallic worship of some form or other in several districts of central and northern Australia, an outline of which appears in a subsequent chapter. I trust that the facts I have been able to collect may help to illumine the somewhat doubtful question of how an aboriginal looks upon the process of procreation. Phallicism is closely related to such forms of Nature worship as are practised in order to make any species of animal or plant proliferate, or, for that matter, to bring down a shower of rain in times of drought. The phallus might gradually merge into a tjuringa. The painted "*Ngadanji*" and "*Ilbarinam*" tjuringas of the Arunndta tribe are regarded as images of the reproductive organ of a spirit which can generate life; in that sense they might be classed as phallus. I find that ancestor worship is generally indulged in; it is difficult at times to distinguish between an original spirit ancestor and a deity, but a Supreme Spirit or Deity is believed to exist and to rule over all creation.

With regard to totemism, I have shown the beliefs in a manner slightly different from those hitherto recorded. The mystical relationship between individual and object is traced to a mythical semi-human forerunner which was common to both and is now a spirit. The Arunndta call the spirit "*Knaninja*." The animal or plant relative of an individual is what has been commonly called the "totem" in Australian ethnology, while the symbolic representation of this object, which becomes the crest of the individual, corresponds to the "*kobong*" first described by Sir George Grey. Emblematic representations of both the "totemic" object and of the Knaninja are embodied in the tjuringa of the individual.

The essay on art, it will be observed, has been written on evolutionary lines with respect to both technique and design. It will be understood that the material upon which the conclusions are based was collected in many parts of Australia and during many years of travel. The cults in question are in many cases distributed all over the continent, but occasionally are quite local. As an instance of the latter kind, I mention the famous drawings

recorded by Sir George Grey, some of which I was fortunate in being able to locate and study on the Glenelg River in the far north-west. I might mention that, with very few exceptions, the designs appearing in the book as text figures are actual tracings reduced by photography to a size in keeping with the dimensions of the page. There is no doubt that primitive art in Australia is a fascinating study which has not received the attention it merits; and unfortunately it is rather late in the day to think of making a start. The system of conventionalism derived from the numerous pictographs and carvings is the basis of all characters and messages one finds on sticks, stones, and persons; it is the only key to an understanding of sacred tjuringa symbols. We have for too long looked upon aboriginal designs as meaningless, and upon aboriginal art production as being idle concoctions out of nothing which were invented just to make a thing "look pretty." This is anything but the true position. An aboriginal artist knows no such thing as a design without motive or origin; to him the shortest line or the smallest circle conveys a thought. In the chapter dealing with stone implements, I have, among other things, described a new type of scraper which was used by the now extinct Adelaide tribe for trimming skins of animals.

The article on language is not intended to be at all comprehensive; my main object was to give a general idea of the construction, together with a few examples of the idiomatic uses, of the aboriginal tongue. I hope at a later date to be able to present a complete account of the Arunnda and Aluridja dialects, including the syntactical rules and grammatical forms.

I have to offer an apology to any authors who may claim priority to some of the facts which I mention in this book. I have written this account of the Australian aboriginal without attempting to consult previous literature, for the simple reason that, had I started looking up all necessary references, the volume might never have been completed. My time at headquarters has been so limited during the last fifteen years that, in the absence of a library near at hand, it was impossible for me to adopt any other method than to write up my observations at first-hand and run the risk of a certain amount of trespass. Nevertheless, I trust that the authors so affected will realize that there was no slight intended and will treat my transgression in the spirit of independent corroboration.

Our knowledge of Australian ethnology is so meagre that every man who has had first-hand experience among the tribes should consider it his bounden duty to place on record any facts he possesses, however trivial they may be. Every year the number of people who have seen the unsophisticated savage is dwindling. When I look back to the time of my first meeting with the tribes of central Australia, just twenty years ago, and compare the conditions of then and now, I shudder to think how quickly the romance of aboriginal affairs, together with all the scientific treasures it encompassed, has vanished, and is now irretrievably lost to the world. The rising generation will not have the advantage of men of even our time. Bones, stone artefacts, and wooden implements will remain in our museums for ever, but the habits, laws, beliefs, and legends are doomed to rapid extinction.

I do not claim to be an initiated member of any tribe. To be candid, I several times tried to qualify by impressive exhibitions of surgical skill and exaggerated munificence, but, although I gained the confidence and goodwill of the old men, I was informed that I could only be accepted provided I passed through the different grades of initiation and submitted to the attendant mutilations in the orthodox way. The medicine men, however, usually claimed me as a “*Kata*” or colleague, and allowed me to witness most of their rituals and sacred ceremonies, which they carefully explained to me. In this way I was able, also, to secure a very great series of photographs depicting intimate scenes from aboriginal life, many of which are unique. The only photographs illustrating this book which were not taken by me are those reproduced in Plates [XLVIII](#) and [XLIX](#); for these I am indebted to the late Mr. Nicholas Holtze.

I could not allow this opportunity to pass without making brief reference to the causes of the early extinction which is threatening these inoffensive, useful, and scientifically important people. We have only to cast our eyes in the direction of any wave of settlement to behold the disastrous effects our occupation of the land has had on the natives. Take, for instance, the Lake Eyre region, which embraces the Dieri, Yantowannta, Ngameni, and Yauroworka tribes. Official reports show that only forty years ago the population was so dense that the white settlers asked for greater police protection; the four tribes mentioned numbered many thousands. During a recent survey on behalf of the Government, I could barely muster three

hundred wretched and decrepit souls in this region, who, literally speaking, were waiting for a lingering death to relieve them of their misery. We content ourselves by saying that civilization is the cause of the increased mortality, no doubt a plausible but very vague explanation. As a result of my investigations, I venture the opinion that the factor which has wrought the greatest havoc among the tribes is disease. The principal scourges are syphilis, pulmonary tuberculosis, and trachoma. Unless we realize the obligations which rest on our shoulders and give our natives a medical protection similar to the successful measures adopted by the United States, Canada, and New Zealand, they will continue to vanish and soon be classed as an extinct race.

H. BASEDOW.

Kent Town, South Australia, 2nd November, 1924.

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CHAPTER I

INTRODUCTION TO AN AUSTRALIAN TRIBE

The lonely bush of Australia—The silence is broken—A mysterious call-note—A human figure in the distance—Coo-ee!—A voluntary but cautious escort—The official approach and salutation—Friendship established—Tribal introductions.

Let us imagine that we are travelling with a caravan over a lonely tract of ground, in a remote district of the Australian bush, which has not been contaminated by any disturbing influence of our civilization. In consequence of the serenity and the deserted aspect of the scene around us, we would be wondering whether the place holds any mortal creatures but our party. We might even have resigned ourselves to the inevitable desolation.

Suddenly the spell is broken by a faint sound falling upon our ears—a long-drawn, shrill, yet melodious note—then all is silent again.

What could it have been? We are in doubt as to whether this was the call of a bird or animal, or a phenomenon unknown to us; being so far away from any centre of industry, a locomotive or factory whistle is quite out of the question. Eagerly we wait for a possible repetition of the singular sound.

Soon, indeed, it comes again; and, with the attentiveness our expectation has solicited, we now recognize the human character of the note. Presently it is repeated, then again, and yet again. But where does it come from? So far we know not.

In the meantime we continue forging through the sand, and, being now on the alert, we in due course espy, in the direction whence the sound is coming, but a considerable distance off, a slim, dark figure gliding from the cover of a rock to that of a bush. Presently it again shows itself a short distance on; and our attention is further attracted by the appearance of one or two other dark bodies running from cover to cover in a line parallel to our course.

The calling is continued at frequent intervals; and, as near as we can represent it by our alphabet, it sounds like the word “Coy!” with the “y”

specially emphasized and spun out. One has no difficulty in recognizing in this call-note, which is met with all over Australia, the derivation of the familiar “Coo-ee!”

It is, of course, assumed that we in no way betray a feeling of uneasiness or give these dusky fellows the impression that we are preparing for hostilities. Having satisfied themselves in respect of this, and after manoeuvring for a considerable time in the manner described, our uninvited escort become more trusting, even daring. They run or walk for longer distances away from cover and gradually bring their line of travelling closer in to ours. When eventually they realize that we seem more like friend than foe, they drop behind our caravan, and, at a measured distance, in our trail. Whenever we move, they move, and when we stop, they stop. If any of our party go towards them, they turn to the side and cleverly disappear into the bush ([Plate I](#)).

When we have arrived at the end of our day's journey and camp, they pull up short and squat for a while; although it may not appear so to the inexperienced traveller, they are taking stock of all our doings.

Before very long, two men stand up and beckon to a small boy among their party, who immediately jumps to his feet and walks to a place in front of his elders. Each of the men now holds one of his hands upon the lad's shoulder, and, in that position, the little group moves towards our camp.

In the hands not so used, both men are now seen to be carrying small branches of *Cassia* bush, which they occasionally lift towards us. Thus they walk to within fifty or sixty paces of our encampment and again squat on the ground, arranging themselves in the same order as they walked in. It is apparent they want us to approach them; and we do so.

Statuary could not be more rigid than the persons we behold, upon arriving at the little group. They sit silently, with downcast eyes, and it is not until we address them that they rise to their feet. The old men, for such we now recognize them to be, start a meek conversation, nervously pushing the boy towards us. We are informed that this is an uninitiated boy, a child in fact, a piccaninny in the tribal relationship of son to the speakers. We are implored not to hurt this innocent being, and to extend that friendship to themselves and to all the members of the tribe. It is pointed out that they are

unarmed and that this is the guarantee of their goodly intentions towards us, the unknown wanderers.

The men now advance and pat us on the chest, and instinctively we return the compliment—for such it is intended to be—which is akin to any ordinary European method of salutation. The bonds of friendship have thus been sealed, and the men continue to jabber profusely on the more intricate tribal relationships existing between themselves and the rest of their party.

When the genealogical explanations have been concluded, the men turn towards their company, who are still squatting in the distance, and call aloud to them to come along: "*Pitchai, ngalla pitchai, waipella tami pu!*" which in the Wongapitcha dialect stands for "Come along, the white fellow is good." The invitation is quickly responded to, and ere many minutes have passed the whole group has arrived, which includes other men as well as women and children. The new arrivals, without hesitation, and with seeming confidence, join in the conversation.

So this is our introduction to the aboriginal, the primitive hunting man of Australia, and his family!

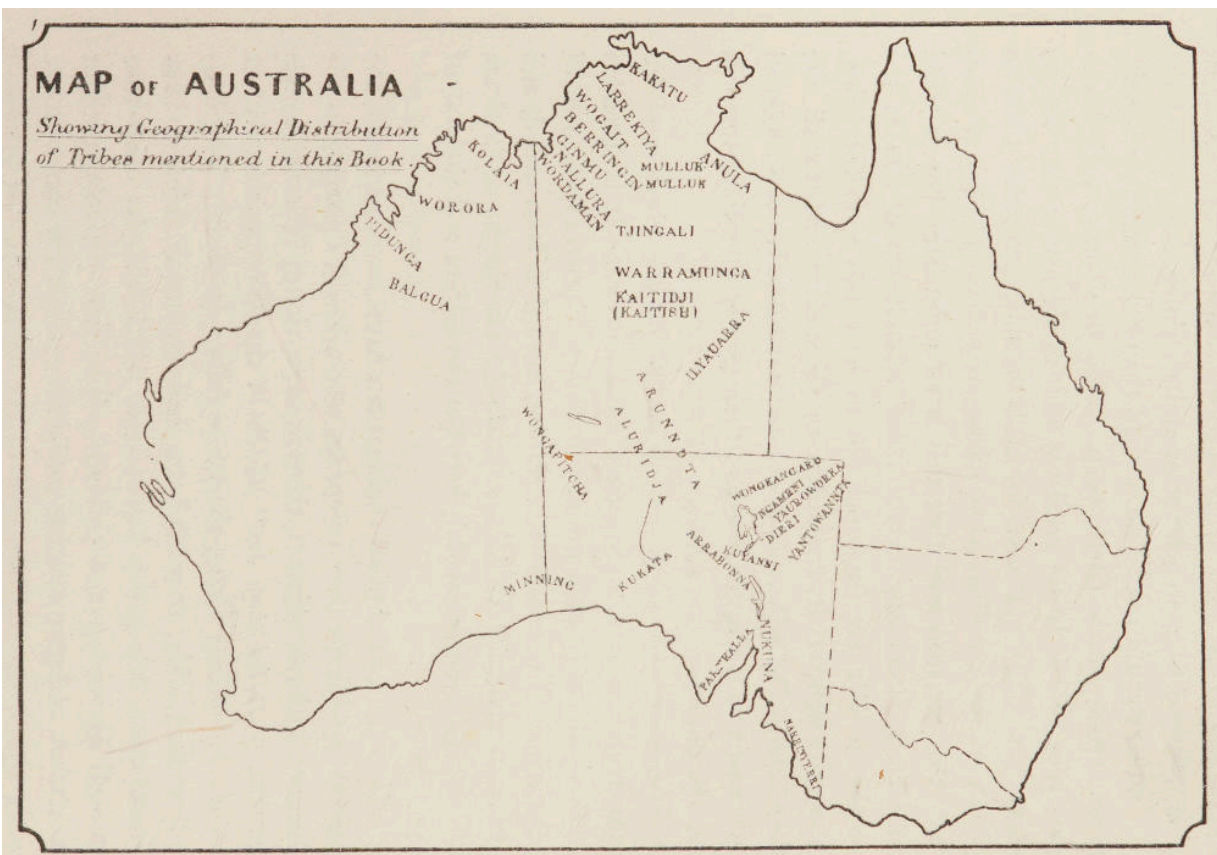


Fig. 1. Map of Australia showing geographical distribution of tribes.



Wordaman natives on the march.

"... they drop behind our caravan, and, at a measured distance, in our trail."

PLATE I

Wordaman natives on the march.

"... they drop behind our caravan, and, at a measured distance, in our trail."

CHAPTER II

RACIAL CHARACTERISTICS

Straightness of figure—Angular contours—Absence of fatty tissue—Nature's economy—Abnormal obesity—Straight spinal column—Flat and long dorsal curve—*Ensellure*—Strong cervical curve—Uniformity of sacro-lumbar curve—Flexibility of spine due to thick cartilaginous discs—Racial comparisons between length of vertebræ and that of intervertebral discs—Influence of upright attitude—Smallness of bones composing the spinal column and the explanation—Exceptions to rule—Narrow sacral bone—Peculiarities of fifth lumbar vertebra—Long extremities of equal length—Foot suggests tree-climbing—Evolution of foot—So-called "hand-footed" men—Feet used for lifting and carrying purposes—Function of peroneus muscle—Flatfoot rare—Correct placement of foot when walking—Tree-climbing and its effects—Peculiarity of Tasmanian's foot—The shape and skeleton of lower limbs—"Boomerang" legs—Shoulder and arm bones—Bodily height of male and female.

Let us study these interesting-looking people more closely and endeavour to find out their characteristic features, and in what respects they differ from ourselves. At the same time, let us in passing ascertain to what degree they resemble us and other peoples, past or present, and what peculiarities they might share with the man-apes or lower forms of the animal kingdom.

What immediately appeals to our critical eye is the strong contrast in the general outline of the figure when we compare it with our own. The round, full contours and shapely exterior of the European are replaced by an angularity and straightness in the aboriginal. The surface-padding or, more correctly, the subcutaneous deposition of fatty tissue, which makes the lines and curves of our bodies so uniform—and the female figure so beautiful—is, to a large extent, the result of long selective culture and of the comforts which civilized life has brought with it. In the case of the aboriginal, however, the forethought of Nature has not allowed the development of such paddings of fat to any considerable amount; they would only tend to impede the agility of the hardy desert roamer. He does not need a thick layer of fat beneath his skin. From an artistic point of view, the leanness of his body is quite becoming to himself. He lives in a country whose climate is hot, and his healthy hunting-life makes him immune from many of the ills to which the city dweller is heir. As a reserve storage of heat and nourishment, therefore, which might be called upon to aid his physiological constitution in times of need, the quantity of superfluous fat can safely be

reduced to a minimum. Nature has given enough, but not a measure in excess. Thus, without any indication of unhealthy emaciation, the integumentary accumulation of fat is so scant that parts of the internal anatomy of an aboriginal can readily be deciphered topographically. We can follow the shape of the superficial muscles and of the skeleton, and can palpate the outline of the abdominal organs with comparative ease. Although the muscles are small, they are, nevertheless, strong, firm, and wiry; this is particularly noticeable in the extremities.

It is a curious fact, however, that there is a natural predisposition in the aboriginal to produce fatty tissue once he gives up his active hunting career, like a sportsman out of training, and to develop a perceptible obesity when he lives under conditions which supply him daily with an abundance of nourishment. Under such circumstances, which are of course abnormal and only brought about by European influence, his skin is very apt to accumulate locally masses of fat known medically as *lipomas*. The Arundta natives call these tumours “*lurra*,” and connect their appearance upon their bodies with heavy weight-carrying. It is, indeed, a noteworthy fact that these fatty tumours frequently occur upon the shoulders of aboriginal wood-carriers, who are in the habit of collecting logs of timber for camp or station purposes.

Associated with the angularity and flatness of the bodily form, we notice, when looking upon the figure of an aboriginal in profile, and comparing it with that of an European in a similar position, a straightness of the spine. If, for instance, we were to make an accurate drawing of the spinal curvatures of the two subjects shown in [Plate II](#), we would find that the line representing the spinal column of the aboriginal would be very straight in the centre of the back, that is, in the dorsal segment; in fact it would be found that the dorsal curve is very slight. Careful comparisons have been made with frozen corpses of different races and the man-apes, bisected in the mesial plane, and it has been determined that this portion of the spine is flatter in the Australian aboriginal than in any other race of man; and indeed it is flatter than in the chimpanzee. Moreover, a larger number of vertebræ are involved in the dorsal curve of the Australian than there are in the other cases.

In the European subject of our illustration—a young Australian lady—the lumbo-sacral curve, known usually as the *ensellure*, is unusually prominent.

If now we examine the curvature of that portion of the backbone which constitutes the neck, we shall find it less pronounced in the European but strongly developed in the Australian aboriginal.

But perhaps the greatest difference between the two types is the manner in which the curvature breaks from the lumbar to the sacral portions of the spine. In the European this break is sharp and angular; in the Australian it is very gradual on account of a peculiar oblique position of the last lumbar vertebra. Should we, again, extend our observation to the chimpanzee, we would find that a number of the sacral vertebræ are included in the lumbar curve. Consequently the aboriginal's spine seems to occupy an interesting position in which the last lumbar vertebra stands almost as a connecting link between the lumbar curve above and the sacral curve immediately below it.

The lumbar curve is greater in the European than in the Australian, but it is decidedly greater in the chimpanzee. The difference in curvature is brought about mainly by the discs of cartilage which exist between the vertebræ, and that is why there is a considerable movement possible in the lumbar portion of the spine of the lower races of man; and it is quite possible that the lumbar curvature alters according to the position adopted by the individual, that is, according to whether he be in a standing or in his favourite squatting posture. In the European the corresponding portion of the spine is much more rigid.

The proportional lengths of cervical, dorsal, and lumbar spinal sections are much the same in both European and Australian, but there are considerable differences in the two spines so far as the proportions of the bony vertebræ and the intervertebral discs of cartilage are concerned, especially in the lumbar region. The lumbar vertebræ of the European are shorter than are those of the Australian, and the latter again are shorter than those of the man-apes. In other words, the lengths of the bones, which build up the lumbar portion of the spinal column, increase (*i.e.* in proportion to the size of the column as a whole) as one passes from the most highly cultured European through the primitive human stages, like the aboriginal of Australia, to the anthropoid apes, and finally to the lower types of monkeys. At the same time, as the length of the vertebræ increases, a reduction in the thickness of the cartilaginous discs takes place.

There is no doubt this phenomenon depends to some extent upon the acquisition of the upright attitude by man, since the cartilage between the bone acts as an effective shock-absorber—the percussion produced by the impact of the heel against the ground when walking being reduced before it reaches the brain. When the brain-box does not rest immediately above the point of percussion, as for instance in the semi-erect posture of the apes, the dangers of concussion are not so great.

While we are discussing the vertebral column of the Australian aboriginal, we might draw attention to the comparative smallness of the bones composing it. If we were, for instance, to compare the column of an Australian with that of a European of similar height, we would find that the vertebræ of the former are appreciably the smaller—their volumes being almost in the ratio of one to one-and-a-half. This is the more striking since we shall learn later that the Australian aboriginal often is quite as tall in stature as the European. The skeleton of the African negro, on the other hand, is decidedly more massive than that of the European.

The smallness of the bones composing the vertebral column undoubtedly favours the flexibility and agility which characterize the Australians as a hunting people. There are, it is true, certain variations in the structure of the spinal column of the Australian, which seem to contradict this general rule, as, for instance, a slightly stronger development of the vertebræ of the neck and a greater volume of the lumbar vertebral bones in the female. The former of these features is no doubt a primitive characteristic throwing back to the quadrupedal ancestry of the human species, the latter having to do with the processes of birth.

The sacral bone at the lower end of the vertebral column varies slightly in size, but is, generally speaking, much narrower than that of the European or of any other living race. It is principally on this account that the hip-bones of the Australians seem remarkably close together in both sexes.

The fifth lumbar vertebra of the Australian often exhibits certain sacral characters, which remind one of the orang outang; in that anthropoid the fifth lumbar bone is often fused to the *os sacrum* and in reality becomes the first sacral body. Occasionally this vertebra is asymmetrical, being normal, *i.e.* lumbar, on the one side and sacral on the other. Its posterior arch is at times wanting, the spines having failed to join, as ordinarily they do, in a

median line behind the main body of the bone. The last named feature is, however, not infrequently observed in the skeletons of other races as well.

Another very striking feature, connected with the anatomy of the Australian, is the great length of his arms and legs. This length of extremities is taken in a conjoint sense, and with regard to the height of the individual. The aboriginal is often said to have very much longer arms than legs. This is incorrect. In point of fact, no human type is known, living or fossil, with such a disproportion in the limbs. All types of mankind, individually considered, have arms and legs more or less equal in length; from this original condition the elongated arms, so typical of apes, have evolved, by secondary processes, in all probability through the acquisition of arboreal habits. There is, of course, no doubt that the length of the extremities, both upper and lower, so characteristic of the Australian, together with the relative slenderness of the vertebræ, points to an early evolutionary stage, which was common to the ancestral forms of both man and ape. The monkey has brought tree-climbing to such a degree of perfection that it practically lives in the branches. Primitive man, too, has not neglected the art, and, although the normal proportions of his extremities do not directly suggest tree-climbing, there is another development which does, especially in the Australian; and that is his foot.

When we consider the likely transformations which the human foot has undergone from an original hand-like form, resembling that of certain monkeys and lower primates, to its present condition, we shall find that two processes have been at work in the modelling of this important part of man's anatomy. Firstly, the big toe (originally a thumb) has taken up a position adjacent to that of the next digit (originally an index finger), and, by lying in the same plane with it, has forfeited its power of opposition. Secondly, the big toe has grown appreciably stronger, while the other digits have become smaller and weaker. That the big toe, in its ancient evolution, once stood in the same relation to the other toes as the thumb does to the fingers of the hand, is evident from the arrangement of the blood-vessels and nerves in this part of the foot, corresponding exactly to that of the hand, even though the gap originally existing between the first and second digits has been filled by fleshy tissue.

It is of considerable scientific interest to note that cases are occasionally observed among the Australian tribes in which indications of this ancestral

condition are retained. In the [Fig. 2](#) we see the feet of an aboriginal of the Berringin tribe in the north of Australia, whose big toes are remarkable for their shortness when compared with the second.



Fig. 2. Peculiar “hand-like” feet of the Berringin tribesmen. Tracing from a photograph.

It is, of course, a well-known fact that the newly-born European baby possesses a wonderful mobility in its feet; and such might also be acquired by people who have lost their arms; but the wearing of boots usually deprives modern nations of this freedom of movement. The aborigines of Australia make frequent use of their toes. A considerable lateral flexibility of the end phalanges enables them to lift small objects off the ground between the big and second toes. Spears are carried by warriors, between the toes of either foot, to conceal the weapons in the grass; and so the enemy is led to believe that the men are unarmed.

When collecting firewood, the gins never stoop to pick up the pieces, but lift them with their toes to the level of their hands. The hands then pile the fuel upon the head and hold it there until sufficient has been collected to carry back to camp.

The power of being able to use the toes in the manner described depends upon the development of a muscle, which arising from the outer side of the fibula and terminating in a long tendon, passes obliquely across the sole of

the foot, to insert itself into the metatarsal bone of the great toe. This is the long *peroneus* muscle, the function of which, in the monkeys at any rate, is to keep the big toe in opposition. In man, moreover, this muscle helps considerably to maintain the arch of the foot. Flatfoot is eminently rare among the aborigines; only one or two cases have come under observation.

When walking, the aboriginal carries his foot so that it points directly ahead of him, and not, as has been written, “with his toes well turned out.” If anything, the sole of the foot is slightly tilted so that the outer border touches the ground a little in advance of the ball.

There is no doubt the outer surface and the ball of the foot play an important role in the art of tree-climbing, as it is practised by the Australians and other primitive peoples. Several methods are in vogue; they will be described later. Suffice it, for the time being, to refer to one: In pursuit of small marsupials, young birds, honey, nuts, fruits, or any other things good to eat, the aboriginal often has occasion to ascend the tall smooth trunks of trees, which harbour such articles in abundance. This is done, often without the aid of any implement, in the following way: The hunter faces the tree and applies the palms of his hands to the opposite side of the butt. As he tightens his hold with his fingers, he springs from the ground and clutches the butt between the soles of his feet. The arches adjust themselves to the convexity of the trunk, whilst the pressure of the outer edges and balls of the feet prevent the limbs from slipping. In this posture, the hunter is virtually hanging by his arms, which are hooked by the hands, and is sitting upon his heels, which are fixed firmly against the tree, as described. Holding his head well back between the shoulder-blades, he suddenly lifts his body upwards with his thighs, while his hands, momentarily relaxing their hold, are pushed upwards also. Now the fingers again tightly clasp the trunk, and the feet are quickly lifted and tucked under the buttocks, to again support the weight of the body as before. The same actions are repeated, time after time, and it is not long before the climber reaches the nearest branches, when progress is simplified. Vide [Plate XIX](#), 2.

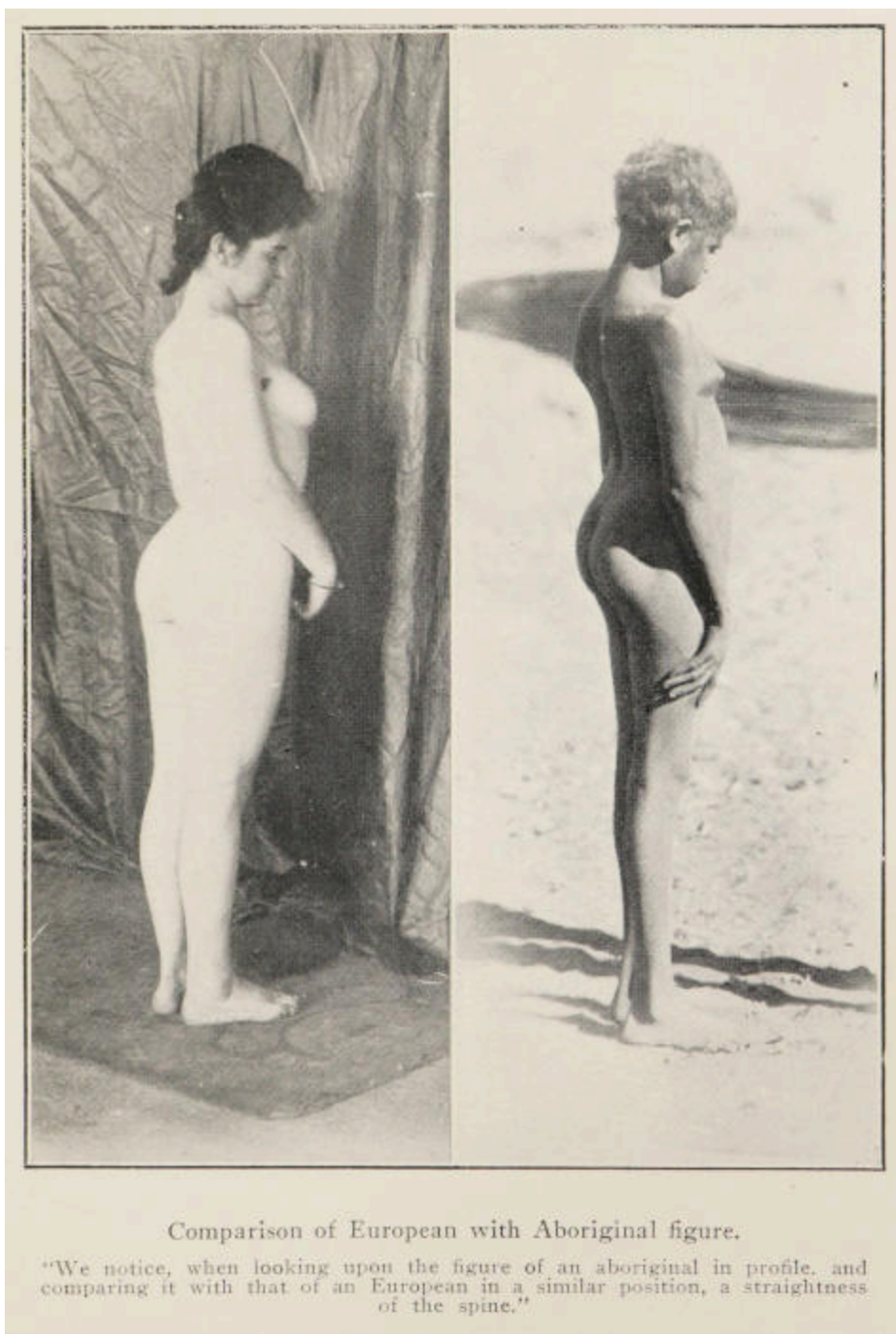


PLATE II

Comparison of European with Aboriginal figure.

"We notice, when looking upon the figure of an aboriginal in profile, and comparing it with that of an European in a similar position, a straightness of the spine."

This ancient custom of tree-climbing is not peculiar to the Australians, but is adopted by most primitive races. It is very probable, too, that the prehistoric races were to a large extent arboreal, and made use of similar methods of tree-climbing. When considering the evolution of the human foot, therefore, we will have to remember that it has been to some extent influenced by the tree-climbing factor, which, indeed, must be considered in the light of a forerunning stage in the acquisition of the upright attitude by man.

In this primitive method of ascending trees, by which the head is thrown so far behind, we see also a likely explanation of the greater cervical curvature we have noticed in the aboriginal's spine, when one compares it with the European's. We might even venture to say that these processes originally brought about the lumbar curvature, and thereby laid the foundation to the acquisition of the erect posture, by means of which man learned to balance his head upon the vertical spinal column. Then the foot, which had been to a great extent modelled through his arboreal activities, stood man in good stead, and he began to walk erect between the trees.

The foot skeleton of the Tasmanian shows a peculiarity, in which it differs from that of the Australian on the mainland. Under normal conditions, the heel-bone of the Australian, and of the European as well, has a small elongation or process on the anterior side which separates the two adjoining small bones, the *cuboid* and the *talus*, from each other. But in the case of the Tasmanian the two small bones named lie in juxtaposition. This phenomenon is only occasionally noticed in Australian skeletons, and is quite exceptional in European; it is abnormal even in the anthropoids.

The Australian's legs are often the subject of comment, if not ridicule; they are so thin and lanky. Even when the proportions of the chest and trunk as a whole are good, the legs usually remain unshapely. Even under the best of conditions, there is a paucity of flesh both in thigh and calf; the lower portions of the limbs are in the true sense of the word spindle-shanks.

Even the gluteal musculature is only moderately developed. Sedentary life and cosmetic culture seem to have been the principal factors at work in shaping this region in the modern European. Monkeys, on the other hand, show no considerable gluteal development at all. It would appear, therefore, that tree-climbing has not played a great role in developing these muscles,

but seems rather to have influenced the growth of the deltoid muscle, which extends from the upper arm to the shoulder-blade and collar-bone, and of the big pectoral muscle.

The thigh-bone, although it is slender, like the rest of the long bones of the Australian, is abruptly dilated at its epiphyses, and, in that respect, differs considerably from the European femur, which widens gradually towards the extremities, in trumpet fashion. The Australian's thigh-bone is more like the Neanderthal type, but the smallness of its head at once distinguishes it from the fossil. The slenderness of the shaft, together with the relative smallness of the condyles, brings the Australian femur nearer to the *Pithecanthropus*. Generally speaking, this bone is stronger in the Tasmanian than in the Australian.

One occasionally finds a strongly developed ridge or process in the upper portion of the Australian femur, which has been styled the third trochanter. At the lower extremity, the smooth depression on the anterior surface of the bone, between its condyles, is deep in the Australian and Tasmanian, and in that respect resembles the Neanderthal femur. The superior margin of the hyaline cartilage covering this depression is variable, and occasionally far exceeds the average European limit.

Among certain tribes of central and southern Australia, the tibia is often peculiarly flattened laterally, like a sword, whilst the anterior edge of the bone is remarkably prominent. This condition is known as *platycnemia* and has also been observed, quite frequently, in the skeletons of the extinct men of Europe and Egypt, and in the Negroid and Polynesian races.

Occasionally this platycnemic condition is associated with an exaggerated curvature of the anterior edge of the bone, a phenomenon which Dr. E. C. Stirling has described as *camptocnemia*. The popular name for it among bushmen is "Boomerang-Leg"; in some cases the tibia certainly has quite as large a curvature as some of the least bent of the familiar throwing sticks have.

In attempting to offer an explanation for this remarkable phenomenon, it is at the outset difficult to say to what extent it might be pathological, that is, the direct result of some constitutional disorder, like rickets, from which the individual, in whose shin-bone the curvature appears, might be suffering.

A theory has, however, been advanced to the effect that, since the anterior ridge of the bone represents part of the surface from which the *tibialis posticus* muscle arises, and since this muscle effects the adduction of the foot, when a person is walking, it is feasible that the altered shape and the increased bulk of the tibia may be due to that factor. The Australians, like other primitive hunters, are possessed of an astounding endurance when running down wounded game.

Dr. Ramsay Smith points out that there may be a connection between a platycnemic condition and the peculiar method the Australians have of lifting things from the ground with their toes, by which the *tibialis posticus* muscle is specially involved.

The fibula of the Australian is straight, and, especially in the case of the female Tasmanian, often has the end adjacent to the knee, which is known as its head or capitulum, prolonged in an extraordinary manner. This feature is of morphological interest because it harks back to a primitive condition in the evolution of the knee, in which the long bone of the lower extremity played a more important part in the action of the joint than it nowadays does in the human species.

In male Tasmanians the shoulder-blade is of considerable length, and its apparent narrowness is primarily due to the elongation of the infra-spinous fossa.

The humerus of the Australian shows a very small torsion, the angle being less than in any other human type. A foramen is not infrequently observed between the condyles of this bone. The Tasmanian humerus possesses a peculiar, laterally convex curve; and its internal condyle is often much enlarged.

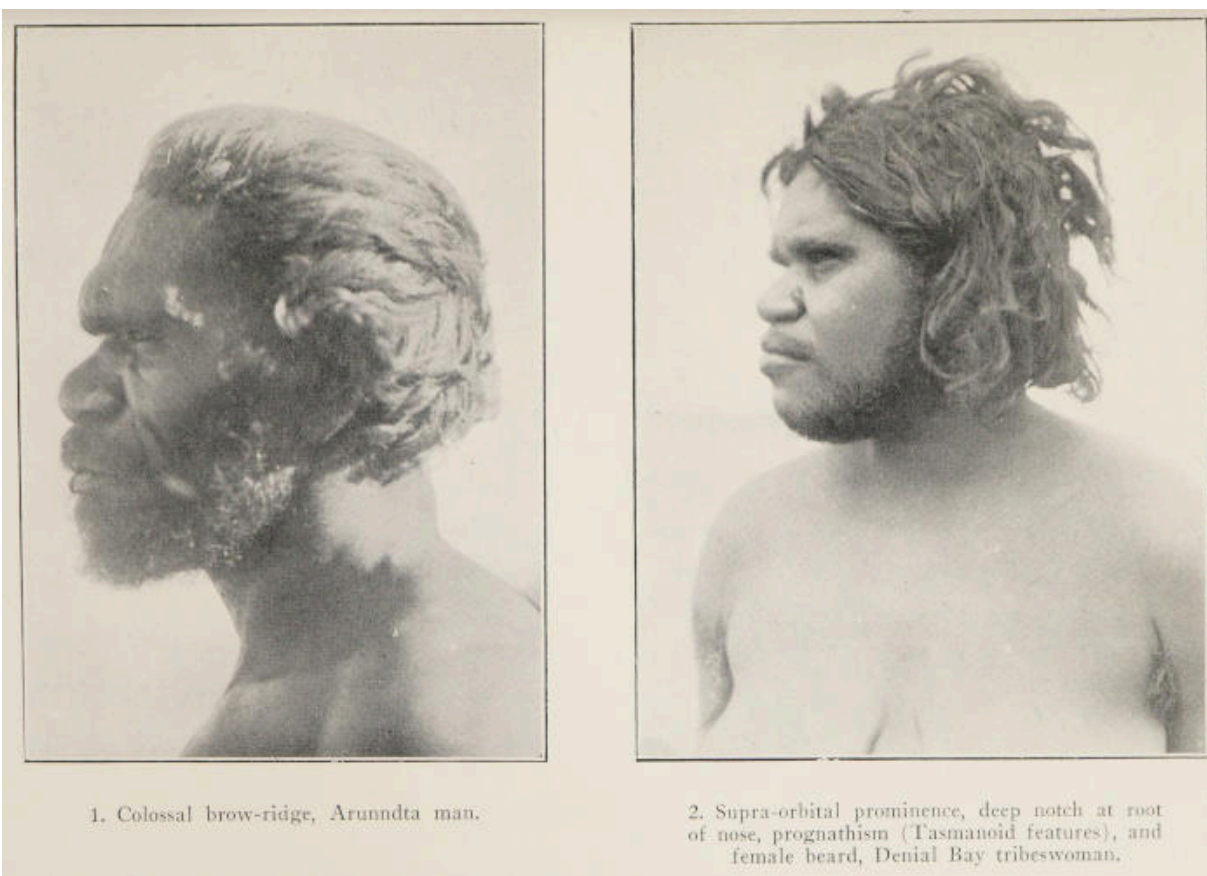


PLATE III

1. Colossal brow-ridge, Arunndta man.
2. Supra-orbital prominence, deep notch at root of nose, prognathism (Tasmanoid features), and female beard, Denial Bay tribeswoman.

As with ourselves, the bodily height of the Australian varies considerably, even within one and the same tribe. No great racial importance can on that account be placed upon statistical data in respect of height. The tallest individual I know of was a man of Yarrabah, near Cairns, in Queensland, who stood seven feet four inches high.^[1] Some of the smallest men I have ever seen lived in the Tomkinson Ranges in Central Australia, who barely measured four feet six inches in height; yet among the same tribe were many men who stood over six feet. The smallest gin, the mother of two children, who has come under my observation, measured four feet five-and-a-half inches. She lived in the Katherine River district. We might claim from five feet four-and-a-half inches to five feet six inches as a reasonable average height for the male, and about five feet for the female.

[\[1\]](#) Measured by the Rev. E. R. Gribble.

CHAPTER III

THE BREAST AND ABDOMEN

The female breast—Aboriginal ideas of shapeliness—Traditional cultivation by ceremonial—Prevalent shapes—Artificially induced lactation—The abdominal region—"Pot-belly"—The sign of surfeiting as well as of malnutrition—Living skeletons—Starvation a justifiable cause of cannibalism.

The female breast varies much in size and consistency, according to the age and physical (and physiological) condition of the individual. As in most matters that concern the aboriginal, his utilitarian inclination outweighs his æsthetic instincts, even to the extent that he regards the breast of his gin simply as that part of her "flesh," which at the required period contains or produces the nutrient "water," necessary for the rearing of his progeny. No breast, no matter how firm and classically hemispherical it might be in our estimation, would appeal to the aboriginal on account of its shapeliness. To him the voluminous, pendant, udder-like form, which can comfortably be handed over the gin's shoulder, or under her arm, to the babe riding upon her back, would seem the orthodox and perfect creation. Indeed, among most of the tribes the husbands endeavour to attain that type both by magical incantation and by actual manipulation.

When the hour arrives that signs of adolescence first manifest themselves in a girl, her future husband, to whom she has long been assigned (perhaps even entirely speculatively, on the chance of the sex, before ever she was born), sets about to conjure up her feminine qualities. He may be, and usually is, joined by other men, to whom tribal law has by descent given claim to periodic domestic privileges approaching those of the marital relationships which are to exist between the individual husband and his gin.

Without advertisement, the tender novice is quietly coaxed away from camp by the men, who, by talking kindly to her, have no need to apply coercion. At no great distance they halt, and the future husband anoints the areas surrounding both nipples, which are likely to bulge forth as the future breasts, with grease; the anointed areas are then covered with a layer of red ochre. Whilst this is taking place, all present sing to the budding milk-

gland, first softly, then vehemently, and with ceremony. During the performance on the North Coast, the female dugong, whose motherly devotion to her young is a recognised virtue, is frequently alluded to.

The painted areolas are frequently charmed by touching them with a magic stone, and at intervals the enchanters bring an anointed circle into contact with their lips, as if endeavouring to draw the nipple forwards, that it might grow.

Ultimately the girl is told to return to the women, who take her on a food-collecting expedition; during her absence from camp, it is quite possible that the signs of approaching maturity may become more definite.

The aboriginal breast begins to grow at an earlier age than the European, on an average at about the tenth year. Neglecting for the present the different phases in its development, from the puerile papilla through various shapes, depending upon the growth of the milk-gland and the deposition of fat about it, one type of breast is typical of early adult life, that is the pear-shaped form. In this type, especially in its earliest stages, the secondary bulge beneath the nipple often fuses imperceptibly with the basal hemisphere, so that a conoid shape results. In older women, the breasts at times assume extraordinarily large dimensions, especially when the individual is inclined to be on the well-nourished side. In very old gins, who have born and reared a number of children, the shape disappears entirely, the breasts becoming mere flaps of skin. A full dry breast is the exception rather than the rule; only in youthful cases, who have not become mothers, do we meet with firm and standing breasts. In later life, the inevitable sagging and attenuation are materially increased by a child in arms which, as the mother plies and looks for food, secures itself by firmly clasping one or both of its parent's breasts.

The breast is situated a little more laterally in the aboriginal than in the European; and in the former case it is also lower and more nearly mid-way between shoulder level and umbilicus. One often finds the breasts of one and the same individual unsymmetrically developed ([Plate IV](#), 1).

In connection with the female breast, I have a somewhat remarkable case of artificially induced lactation to record from the Alligator Rivers district. The mother of an infant of tender years having died, a younger sister of the deceased, who had no children, volunteered to adopt the helpless mite. The

foster-mother diligently treated her breasts with a pulp she made by mashing *Eugenia* leaves with ashes and sufficient water to make a paste; and heated stones were placed over the breasts at frequent intervals. The mammary glands and their surrounding tissues were at every opportunity plied with the fingers, and the babe's lips were as often put to the nipples. Within a short time, fluid formed in the breasts; and the child was suckled. The fluid was said to have been more watery than milk, but, nevertheless, made good nourishment for the child. This case is by no means unique. A number of records are available from different parts of the world, the most classical among which is perhaps that mentioned by Alexander von Humboldt of a South American man who sustained a child on his breast for five months during the illness of his wife.

In the Australian, the belly is flatter, the pudendum if anything, slightly more anterior, and the inguinal folds decidedly steeper than in the European.

One frequently has an opportunity, however, of observing a youngster with a remarkably big abdomen, a condition known in the bush by the name of "pot-belly." Such a picture might point to either plenty or to want. In good seasons, when animal and vegetable diet is to be had in abundance, the younger children soon acquire a "pot-belly" in consequence of ample feeding and gorging. But, on the other hand, a distended abdomen is more often found in consequence of malnutrition and starvation, which the children have to suffer during bad seasons of drought. The distention in this case is due to the swelling of some of the large abdominal glands.

The same sufferings manifest themselves similarly in the adults, and particularly in the aged. Among the latter one only too often finds hungering creatures whose flesh has wasted away to a mere parchment wrapped around the bones, living skeletons in fact. In these cases, too, enlarged glands give rise to an unhealthy nodular protuberance in the epigastric region.

Can one wonder if, under such conditions, a kindly club, wielded by a more robust relative, puts the sufferer out of his misery? It is during these trying times, too, that parents are obliged to resort to extreme measures, so that they might sustain the lives of their children. Driven to the verge of despair, and visibly moved at the thought of it, a father must occasionally

make the pathetic and gruesome decision to slay one child in order that another may be saved.

On account of his acting thus, when dire need compels him, people, who should know better, often call the Australian aboriginal a cannibal! Is this cannibalism? Have not shipwrecked people of our own colour, when in a similar plight, often been compelled to kill and eat one of their friends to save themselves from starvation?

CHAPTER IV

THE FACE AND ITS SKELETON

Fierceness of expression—European-like features—The eyes—Colossal brow-ridges—The iris—An unusual colour—The eye-sockets—The nose and its aperture—Natural and artificial flatness—“Primitive snout”—Prognathism—“Negative chins”—The ear.

When we look at an aboriginal in the face, the first impression it gives us is that of wildness and fierceness amounting often to repulsiveness. There are, of course, appreciable differences between different individuals, and often during a day's journey one encounters features which might be classed as decidedly pleasing and almost European. We might even go so far as to say that in some faces, especially those of the old women, we might feel inclined to establish an analogy with classical or historical types of our own colour. Such descriptions, however, convey no more to the person who has never seen an aboriginal than a statement like one, often heard, to the effect that the features are of the “usual Australian type.”

In order to arrive at a little more exact description of the facial appearance, let us assume that the individual we are considering possesses all of the “Australian” features, and let us analyze each of them separately. At the same time, we must remember that probably no single individual exists in whom all of these characteristics are present.

To help us better understand the various points we are about to introduce, it will be advisable, whenever possible, to draw into the discussion the morphological peculiarities of the bony skeleton and skull, which immediately underlie the fleshy parts of the face and head.

There is no doubt the eyes of an aboriginal largely account for the wild appearance of his countenance, already alluded to. These organs are deeply sunken in their sockets, which lie beneath a projecting bony roof and bushy eyebrows. Nature in this way protects the eyes against the scorching rays of the southern sun by an effective screen, which lies above and before them like the peak of a cap. For that matter, we ourselves often instinctively endeavour to obtain a similar protection, when we stare into a glaring light, by holding a hand against the brows. Within the scope of this protection

comes also, so far as the aboriginal is concerned, prevention, to a certain degree, of such accidental harm as might be caused by stakes or brushwood, during an exciting hunt through bush or forest land. The strong colour-contrast of the sclera against the swarthy skin, and the active, searching movements of the eyes, considerably help to intensify the sullen look.

In the fossil men of Gibraltar and Neanderthal, too, the eyes were overlain by very massive, bony ridges, but in those people the eyes stood further forward.

The supraorbital region is unquestionably one of the most prominent, and at the same time most striking, features of the Australian aboriginal's face. The high degree of development of the bony prominence, combined with a sloping forehead, are primitive characteristics which he shares with no other living race. Yet it is possible for a cultured people, like for instance the Europeans, with high forehead, at times to show a strong superciliary development. As opposed to this, one not infrequently discovers an Australian with strong brow-ridges combined with a comparatively full and steep forehead. But usually the area occupied by these bony prominences can be differentiated, from the forehead above it, by a dividing line or zone of crowded foramina—the outlets of small blood-vessels. In the aboriginal, the effect is intensified, not alone on account of the usually receding forehead, but also because the root of his nose is appreciably depressed between the eyes; and consequently the glabella appears to project extraordinarily far outwards ([Plate III](#)).

It was Thomas Huxley who first drew attention to the analogy existing between the skull of the fossil Neanderthal man and that of the Australian aboriginal. To quote the words of that famous anthropologist, “a small additional amount of flattening and lengthening, with a corresponding increase in the superciliary ridge, would convert the Australian brain case into a form identical with that of the aberrant fossil.” Since those words were written, a number of other fossil skeletons of men have been found, the examination of which has confirmed Huxley's observation upon the first Neanderthal skull.

The substance of the supraorbital prominence consists of massive bone. As in the fossil skulls, the sinuses lie behind this mass, not far from the inner surface. It appears that this colossal growth of bone takes place in

early adult life; so far, no superciliary ridges have been observed in children's skulls.

Originally this great thickness of bone in the supraorbital region of the Neanderthal calotte was regarded as a proliferation of bony tissue caused by disease. Then it was proclaimed to be a characteristic by which one might distinguish the skulls of fossil from those of all living races. The subsequent recognition of true supraorbital ridges or *tori* in the Australian completely disproved both these hypotheses.

The feature is, so far as our present knowledge goes, essentially masculine, and, as such, suggests a secondary origin comparable to the tremendous supraorbital developments of bone in the skull of the male gorilla. Professor D. J. Cunningham, who investigated this subject thoroughly from a comparative anatomical point of view, found "superciliary and supraorbital elements" even more or less developed in the lower types of apes. A supraorbital prominence is rarely observed in the female ([Plate III](#), 2).

To return to the eye: the colour of the iris, in its normal condition, is practically without exception dark brown. The only exception to this rule, that has come under my personal observation, was a young, full-blooded gin of the Mulluk Mulluk tribe in the Daly River district, north Australia, whose iris was a deep bluish grey.

There is great variety in the shape of the cavities which hold the eye-balls of the Australian. The orbits are large, and their outer margins, as in the Neanderthal type, very nearly form a circle. According to Professor Klaatsch's measurements, the ratio of the maximum vertical to the maximum horizontal diameter in the male Australian skull is as 39 millimeters is to 40. As a rule, the eye-cavities in children are slightly depressed horizontally, and occasionally this is also the case in the skulls of females.

The orbit's upper edge is very strong; and what is known as the *internal angular process* of the frontal bone, in the inner wall of the cavity, is very prominent in the Australian. The last named characteristic is, however, also observed in the skulls of Veddahs and other primitive people, as well as in those of the anthropoids. But we must not forget, when dealing critically

with a skull, that an internal angular process may not be so typical in the male as it is in the skulls of women and children.

The *external angular processes* are often well developed, the malar boundary being strong and broad, without the sharp edge usually noticed in European skulls.

The *ethmoid* bone in the inner wall of the orbit is, as in most of the primitive skulls, noticeably small.

The groove of the *lachrymal* bone, which *intra vitam* carries the tear duct connecting eye with nose, is usually very pronounced in the Australian.

In children's skulls a supraorbital notch divides the upper margin of the orbital cavity into two almost equal parts, the outer of which has a well-defined edge.

In the nose and its aperture, we again recognize primitive characteristics of considerable importance in the Australian. We have already had occasion to notice how deeply the root of the external organ seems to be retracted under the great glabellar prominence of the forehead. A bridge in the true sense of the word seems wanting, the nose consisting of a flabby body at its point, above which lies a saddle-shaped depression sloping imperceptibly into the retracted root beneath the forehead. Not uncommonly one finds a number among the males of all tribes whose noses are curved and give one the impression of Jewish features; the type is rare among women ([Plate VI, 2](#)).

The breadth of the nose is very great, and consequently the nasal aperture in Australian skulls is of corresponding dimensions. The width of the aperture often actually exceeds thirty millimeters.

It must not be supposed that these features are quite peculiar to the Australian; they are also present in the fossil skulls of Europe. If then we regard the latter as the ancestral stock, from whom modern peoples have sprung, and, at the same time, recollect that diverse admixtures of strain might have taken place periodically, it would not be amiss to expect indications of such primitive nose developments in the higher types of man. That such do occur, we can every day verify for ourselves.

Apart from being a racial characteristic, the flat broad nose is cultivated by many of the tribes. Mothers artificially flatten the noses of their children

when quite young by pressing upon them with their fingers, and often repeating the process. It is, therefore, often difficult to say whether a specially flat nose is natural or is the result of cosmetic culture. The wearing of a bone or stick through the septum would also tend to flatten and widen the organ to an abnormal extent ([Plate VI](#)).

The nasal aperture of a modern European skull almost invariably has sharp lateral margins, which unite at the base behind a prominent bony spine; for reasons which will become apparent below, we shall call this the inner boundary. In an anthropoid, like the gorilla, however, the cavity is bounded on its lower side by two ridges, which enclose a groove in front of the large aperture. Converging upwards, these ridges, on either side, unite to form the lateral margins below the nasal bones. In the monkeys there is no indication of a well-defined boundary at all, the lower surface of the cavity appearing more or less smooth, whilst the spine, so prominent in man, is barely recognizable.

In the Australian skull we often find an interesting transition stage connecting these extremes, the inner margin being present but associated with a pre-nasal groove at the base of the aperture. Indeed, the sharp lateral margin is often found to pass into a pre-nasal ridge which forms the anterior margin of the groove. Such a condition is of considerable interest, since it recalls a stage in our evolution, when the nose was closely connected with the mouth part; that is to say, that a portion now absorbed into the modern skull was originally the floor of the nose, and helped to build up the alveolar process of the upper jaw.

In fact, we are reminded of this condition when we look upon the living aboriginal; for his nose seems to ride upon the upper portion of his mouth, to which it seems rigidly attached, after the pattern of an animal's snout. We see the same sort of thing in the European embryo during the first few months of gestation.

This "primitive snout" is made the more conspicuous in the Australian on account of the strong *naso-labial folds* in the skin, one of which, on either side, encloses the angle of the mouth in a semi-circular fashion. With us Europeans, the elevating processes which our nose has undergone have tended to reduce the depth of these folds, in the upper portions at any rate.

This elevating process, by the way, has largely been in consequence of the recession our mandibular skeleton has suffered ([Plate V](#)).

The jaws of the Australian are, like those of most of the fossil skulls and of the Negroids, protubefent—a condition known as *prognathism*. In the Tasmanians, too, the strong development of the jaws, and of the teeth, has resulted in a general fullness of the same region ([Plate III](#), 2).

In aboriginal infants, one often finds the bony process, upon which the teeth subsequently grow, to be directed forwards, almost in a straight line with the floor of the nose. This hereditary predisposition towards a horizontal development of the alveolar region reminds one forcibly of features belonging to the anthropoid apes.

Yet, generally speaking, it must be admitted, there is a great variation in prognathism among the Australians.

In order to compare the degree of facial prognathism of the skulls of different men, a method was devised by Fraipont: The glabella is connected with the lambda by an imaginary plane, and another plane erected at right angles to this at the glabella. The latter plane usually cuts the alveolar plane at about the first or second premolar, occasionally at the first molar. Still another plane is imagined, extending vertically from the most anterior point of the alveolar to the glabella-lambda plane. Then the rectangular distance between the two upright planes represents, after Fraipont's method, the prognathism. The maximum prognathism of the Australian, determined by this means, is, according to Professor Klaatsch, twenty-five millimeters, and the minimum eight.

Let us now enquire into the possible origin of prognathism among the primitive races of mankind. We shall have to take note, in the first place, of the large occipital development of both the brain and the brain-box in the lower types of the human species. In order to antagonize the downward pull of this weight, the mandibular region has expanded and provides the balancing moment about a fulcrum on the spine. With this explanation fresh in our minds, we understand how the development of the frontal region of modern peoples would tend to modify the lower region of the face and establish the condition known as *orthognathism*.

Prognathism is usually associated with a receding chin. By elaborating the Fraipont method, Professor Klaatsch has added another vertical plane at the most anterior point of the cutting surface of the teeth. In primitive folks, like the Australians and the fossil Neanderthals, the chin lies behind this plane and is called a “negative chin”; in the Mongoloids (Malays, etc.) the chin practically lies against the plane and is called a “neutral chin”; and lastly, in the modern Europeans, the chin lies before the plane and is known as a “positive chin.”

In the Tasmanians, the chin was bluntly rounded, without much of the prominence so highly perfected in the modern peoples.

The mental foramen is usually situated at a point below and between the second bicuspid and first molar.

The Australian’s ear is large and longish, much the same in general appearance as the European’s; the Negroid’s ear is decidedly rounder. There is, however, no great racial variety in the human ear; man has, like the rodents, retained the primitive shape, whilst the monkeys have acquired more specialized forms.

The small process, known as the tragus, which partially covers the ear-hole, is mostly covered with bristly hairs in adult men.

The dependent portion, or lobulus, which carries the earrings of our European ladies and is often mutilated by the lower races, is not as a rule interfered with by the Australians. The custom of piercing the lobulus appears to be confined to the Cape York Peninsula in Queensland. The hole is pierced with a small pointed bone, after which a short cylindrical wooden rod or bone is inserted, frequently removed and replaced again, until the edge of the hole has healed. In the course of time, a series of rods, of gradually increasing diameter, are forced into the perforation, until a large pendant loop is formed. Upon special occasions, short painted rods of wood, two inches or more in diameter, are inserted into the loop.

The cheek bone, or zygomatic arch, is usually horizontal; but it may curve upwards from the squamous bone, thence downwards anteriorly. The *mastoid* process is comparatively small, but it is often associated with an unusual thickness of the wall of the ear-passage.

CHAPTER V

THE MOUTH

The lips—The vault and hard palate—The teeth—Excessive wear of grinding surfaces—Tooth-picks—Fourth molars—Dental rudiments—The canines.

The Australian's mouth is decidedly large, and his lips full. The latter, especially of the children, are as often as not becomingly arched and furnished with a shapely philtrum.

If we were to look into the mouths of a number of aborigines we would find considerable individual differences in the configuration of the vault. In some instances the roof would appear high and arched, in others low and flat. If, further, we extended our observations in the direction of any differences which might exist in the individual faculty of articulated speech, relative to the variations in height already noted, our efforts would be fruitless. It is very doubtful whether any such connection between the height of the mouth and freedom of tongue or speech does exist in the aboriginal's case. But it has been rightly pointed out that the hard palates of fossil skulls are flatter than they are in those of modern races.

Perhaps the finest natural gift of the Australian (and the same was true of the Tasmanian) is his strong set of ivory white teeth. In the primitive tribes, living apart from civilization, dental disease or caries is practically unknown. A common feature, however, is that the teeth are ground down on a level, to varying extents, depending upon the age of the individual examined. In many cases, the biting and grinding surfaces of the teeth have been worn to almost the alveolar or gum-level of the jaws, leaving only the roots with short truncated stumps to do the mastication.

This excessive wear of healthy teeth is mainly attributable to the large quantities of sand contained in the everyday diet. The aboriginal cooks nearly all his meals in hot ashes and sand; it is unavoidable, therefore, to include an appreciable quantity of gritty material in the articles which are consumed. The aboriginal, furthermore, during the course of a meal, might repeatedly call upon the strength of his teeth, as an easy way of crunching

bones of animals, and shell of molluscs and crabs, and many other things. Casually one might take notice of the fact that the teeth of the fossil of Gibraltar are worn in the same remarkable way.

An aboriginal does not take any particular care of his teeth, with the exception that after every meal, some considerable time is devoted to the removal of any remnants of meat which may have been retained. For this purpose, the dry seed-stalks of grass and small twigs are generally used. The old Kukata men were observed to possess permanent tooth-picks, consisting of short pieces of wood sharpened at one end. For convenience sake, they carried these, planted in their shaggy beards.

Should there be an aching tooth to cure, the native does it by heating the point of a small stick in a fire and inserting it into the cavity which is causing the trouble.

A most interesting circumstance in connection with the dentition of the Australian is the comparatively frequent occurrence of a fourth molar in the jaws. We know that in European subjects the third molar or wisdom tooth is smaller, and takes longer in coming to the surface than the other molars; its development is certainly on the down-grade with our kind; but the third molar of the aboriginal is strong and lasting.

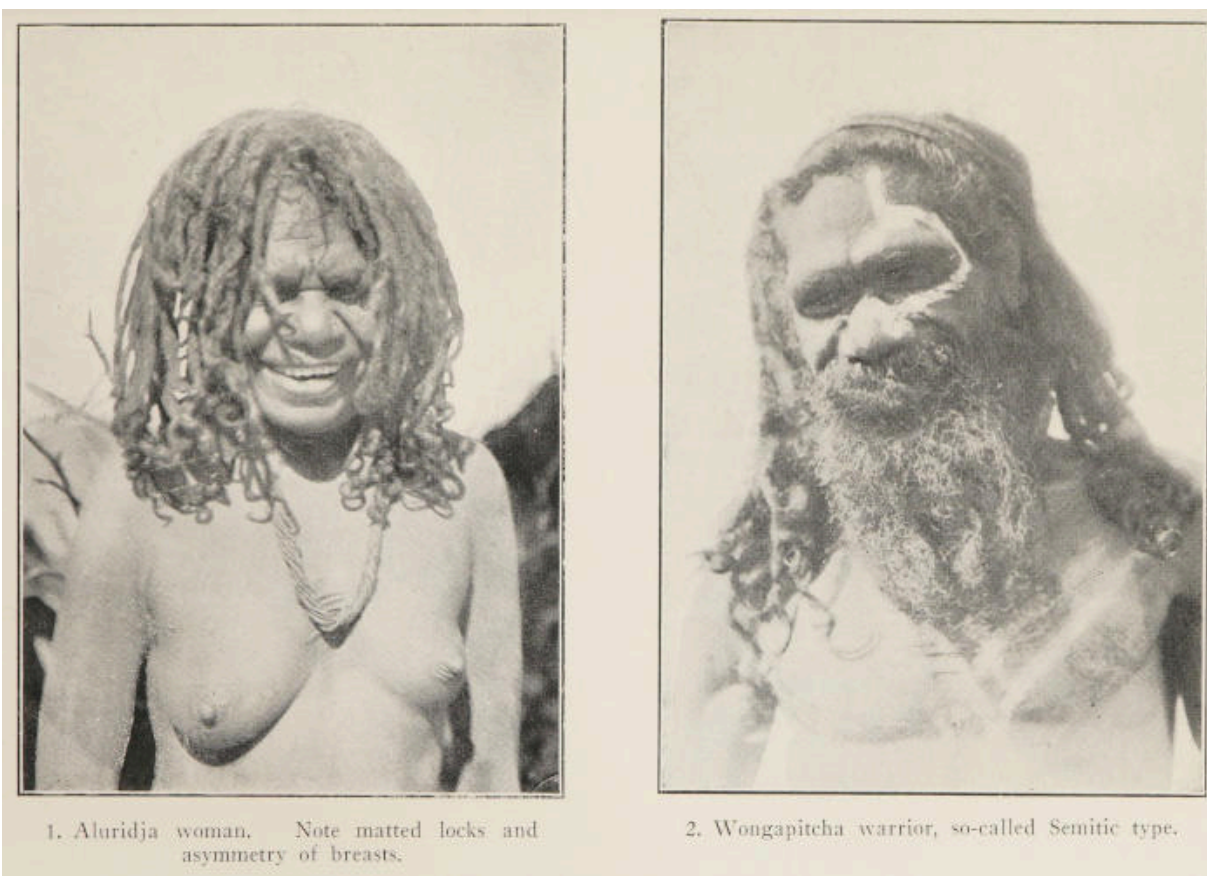


PLATE IV

1. Aluridja woman. Note matted locks and asymmetry of breasts.

2. Wongapitcha warrior, so-called Semitic type.

Even when a fourth molar cannot be found in toto, there is often present, behind the third molar, a peculiar prolongation of the alveolar groove, which seems to be indicative of a former existence, in the earlier evolutionary history of the Australian, of such a tooth. Indeed, the occurrence of a fourth molar in the human species, which in the aboriginal is certainly not sporadic, must be looked upon as a character originally common to the ancestral forms of both man and anthropoid. For this reason, we must not be surprised to hear that a fourth molar might occasionally be found in any race of man.

Professor W. L. H. Duckworth has described some small dental rudiments on the alveolar surface of the upper jaw, which might even suggest remnants of third premolars. Such rudiments usually occur between the second bicuspid and the first molar, and consist of dentine. If it can be

proved that we have before us true evidence of immature tooth-development, the phenomenon suggests a dental formula similar to that of some of the simians possessing three premolars. On the contrary, the formations may be the remnant masses of temporary milk teeth.

Supernumerary bicuspidis are, it appears, not very often observed in the Australian.

It is still questionable whether, as Charles Darwin suggested, the ancestor of the human species has ever possessed extra large eye-teeth or canines in any way resembling those of an anthropoid. In the Talgai skull, referred to later, the canines certainly seem abnormally large, but one could not be expected to draw definite conclusions from a single specimen, especially when it is known that, even among ourselves, we here and there see persons whose canines are quite the same size as those of the Talgai fossil.

CHAPTER VI

THE SKULL AND BRAIN

Scaphocephaly popularly misinterpreted—Sutures and *wormian bones* Dolichocephaly—Tasmanian skulls more globose—Forehead occasionally well developed—Absence of *tubera frontalia*—Fronto-squamous articulation—Occipital peculiarities—Massive skull-walls—Cranial capacity—The brain—Generally well developed—Important primitive characteristics—Rhinal fissure—Insertions of neck-muscles in occiput—Atlanto-occipital articulation.

Opinion is often expressed that the Australian has not sufficient brains to completely fill his brain-box! Such a statement is, of course, not in accordance with fact. A condition known as *scaphocephaly* is, however, not infrequently observed in the Australian skull, which gives one the impression of insufficient inflation of the cranial vault as a whole. A sagittal ridge is present which, with a little exaggeration, might be likened to an inverted boat ([Plate V](#)). A similar frontal ridge is also occasionally observed in the Negroid skull and some of the fossil calvaria, but rarely in the modern European and Asiatic races.

This median frontal ridge is not the result of any abnormal thickening of bone locally, but anthropologists believe it may be connected with the early fusion of the frontal suture. As a matter of fact, the frontal suture is extremely rare in adult primitive peoples' skulls, and so far only three have been recorded in the Australian by Dr. Ramsay Smith in specimens from the Northern Territory; one case has been found in the Tasmanian; and one or two in the Torres Strait Islanders. All the other sutures in the calvarium seem to be less complicated in the Australian than in the European and other more highly developed types of man; the most complex is the lambdoid suture; and it often has one or more small *Wormian* bones interposed within its course, which are, in all probability, connected with the growth of the brain, and with the consequent enlarging processes, which the enclosing bones have suffered. An *os inca* is occasionally seen.

Although scaphocephalic skulls are not rare among the Australians, we must not overlook the fact that occasionally we find specimens, dead or alive, exhibiting sagittal curves in no wise behind those of modern peoples.

And this seems to have been even more conspicuous in the case of the lost Tasmanians.

The Australian skull is remarkably narrow and long, in most instances *dolichocephalic*. In the Tasmanians, the parietal portion is considerably wider in proportion; the whole of that region seems to be inflated, when compared with the frontal portion. Whereas in the Australian we noticed a sagittal ridge, a peculiar median sulcus is usually present in male skulls, running along the line of the sagittal suture.

There is a remarkable uniformity in the contours of the male Tasmanian skulls. They resemble the female Australian much more than they do the male Australian, principally because the adult male Australian skulls vary so.

We have already referred to the sloping forehead and frontal region of the Australian skull; such is observed also in the Tasmanian and, for that matter, in all primitive men whether they be recent or fossil. But at the same time, one frequently observes crania of these primitive people in which the contours of the frontals are as steep as in any average European. The last remark applies especially to the female skull, which even might occasionally show a combination of a prominent forehead with a primitive superciliary ridge. The two conspicuous eminences of the forehead of the European skull, known as the *tubera frontalia*, one of which is situated over either eye-socket, are not developed in the Australian or Tasmanian.

The frontal bone of the Australian skull often lies in direct contact with the *squamous* portion; this is, however, the case in other primitive races also. The articulation between the bones mentioned is effected by means of a process which the squamous bone sends towards the frontal; the actual line of contact measuring several millimetres. In other instances the connecting process is replaced by a small epipteris bone. The articulation may occur on one or both sides of the skull.

The bony tuberosity of the occipital bone, being part of a muscle-attachment, varies appreciably in its position; its central point is anthropologically styled the *external inion*. A similar protuberance on the inner surface of the bone is the divisional line between the great and small hemispheres of the brain; it is called the *internal inion*. The internal inion is,

as a rule, situated lower than the external in the adult Australian; but in female and immature skulls the two points are at about the same level.

The angle which is contained by the occipital bone at the inion, in a sagittal plane, is less in the Australian than in the modern European skull. Many of the Australian skulls one examines, therefore, seem to be unusually flattened at the base, below the inion.

The cranial walls of no other race are so massive as those of the Australian. It is particularly in the supraorbital and the occipital regions that the bone is so thick; the thinnest portions lie in the temporal and lower parietals; these remarks apply principally to the adult male skulls. The consequent strength of the aboriginal's skull has almost become proverbial. Many are the club-duels which tribal law and honour demand to be fought. Upon these occasions the head is the mark. But also in the settlement of his domestic affairs, when a serious offence calls for punishment, the husband selects the head of his gin for beating with the weighty nulla-nulla. Is it a wonder, then, that one often finds the skulls of aborigines covered with dents, which have resulted from such a battering? Occasionally death might follow such treatment; and a few cases stand on record of blindness following the destruction of the centre of vision by a blow from the club. So severe is the impact that often, in the stillness of night, I have heard the falling of the blows upon a disobedient gin's head, although our camp might have been some considerable distance off.

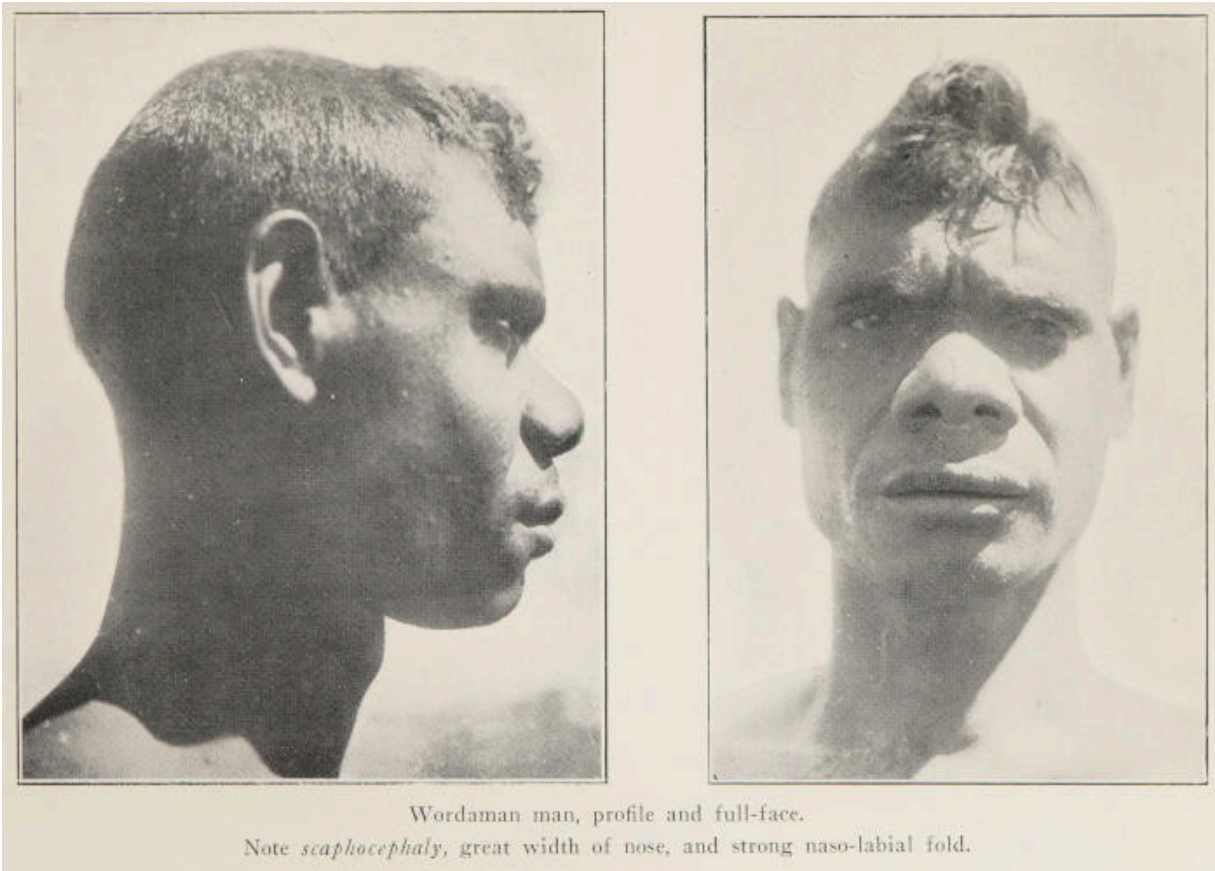


PLATE V

Wordaman man, profile and full-face.

Note *scaphocephaly*, great width of nose, and strong naso-labial fold.

This wonderful strength is largely due to the better development of the compact tissue of the skull-bone, when compared with that of the European. The external, as well as the internal, laminae of the cranial wall are thicker than ours, while, on the other hand, the intermediate layer, known as the *diploë*, is thinner in the aboriginal's skull. This condition serves the double purpose of protecting the brain against the mechanical injury referred to, as well as against the powerful rays of a fierce southern sun.

In regard to the cubic capacity of the Tasmanian and Australian skulls, we might accept as averages for the adult male and female Australian skulls 1,290 and 1,845 cubic centimeters respectively, and as similar averages for the Tasmanian, 1,315 and 1,155 respectively. The individual variations in the adult male Australian skulls range from 1,630 to 1,040, and in those of the opposite sex from 1,280 to 1,010 cubic centimeters. The corresponding

variations in male and female Tasmanian skulls are from 1,465 to 1,140, and from 1,225 to 1,060 respectively.

There are, however, considerable variations in the capacity of Australian skulls; many instances may be selected in which the capacity is quite as good as that of an ordinary European brain-box. On the other hand, cases have been recorded of capacities not greater than 940 cubic centimeters in adult female Australian skulls.

The brain of the Australian has not been studied to any considerable extent. The first impression a layman receives, upon beholding the brain of an aboriginal, is, perhaps, a little disappointing. Assuming him to be a man of low intellect, he expects to find a brain of inferior development. But such is not the case. In fact, to any but the specialist, there seems nothing to indicate a lowly intellectual capacity. The number of convolutions is about the same as one finds in Europeans of average intelligence; but the structure, as a whole, is, perhaps, a little less complicated and less tortuous than we are accustomed to see in our own sort. The large cerebral hemispheres completely cover the cerebellum. Certain features, like the operculum, are more strongly developed on the left side than the right.

In some respects the Australian brain preserves important characteristics, which indicate the genesis of the more modified or more specialized conditions seen in the brain of modern man. The length of the hemispheres and the small occipital development are unquestionably extremely primitive characters, which, among others, remind us of the common ancestry of man and ape. In the brains of the more cultured peoples, processes are at work, which are completely remodelling portions of the important organ, and thereby making it more and more unlike the anthropoid or simian brain. By such modifications in the occipital region, the human brain is gradually ridding itself of a feature strongly developed in the monkey's brain, which has been named the *sulcus lunatus*. German anthropologists call this sulcus "*Affenspalte*," which means "*Monkey-Cleft*," i.e., a cleft or sulcus in the posterior portion of the brain of primates, which is strongly developed in the monkeys, but disappearing in the brain of man. In the Australian's brain, the *sulcus lunatus* can often be more or less distinctly discerned, and its presence there affords us valuable evidence when tracing the remnants of the sulcus in the brains of other races, including those of the modern Europeans.

The posterior lip of the *sulcus lunatus* is occasionally operculated in the Australian's brain. In the parieto-occipital region, the outer convolutions are depressed and covered by an operculum-like flap; but this condition is also occasionally observed in European examples.

Another simian feature, rarely seen in European brains, is rather frequently found in Australian, in the shape of a *rhinal fissure*. It should be observed, however, that the European embryo clearly shows this fissure in the brain as it is developing.

The occipital bone varies in its appearance. The impressions made upon the surface, where, during the life-time of an individual, the strong muscles of the neck were attached, are, as a rule, well developed. The minor *posterior-rectus* and *complexus* muscles of the neck often leave deep hollows in the occipital bone at the points of their insertions.

A bony process is often noticed in front of the big foramen, which joins the occipital condyles; this is an atavistic condition, by means of which an extra articulation is occasionally effected between the occiput and vertebral column. The condyles vary considerably in their elevation above the occipital bone. The large foramen is mostly oval in shape, but often has a little median notch in its posterior margin.

CHAPTER VII

COLOUR OF ABORIGINAL'S SKIN

Unsuitable nomenclature—Aboriginal of Australia not a “Nigger”—Colour normally chocolate-brown—Lighter in infancy—Variations of shade due to several causes—Colour-classification schemes obsolete—Pigmentation very superficial in aboriginal's skin—“White blackfellows”—Pigment destroyed by disease and lesion—Actual colour—Its intensity and distribution—Effect of environment on aboriginal's skin colour—Climatic influence.

The Australian aboriginal is popularly spoken of as a blackfellow; at times one even hears him referred to as a nigger! Strictly speaking, the former appellation is not in accordance with obvious fact, and the latter in addition is scientifically grossly incorrect. The aboriginal is no more black than the average modern European is white, and, apart from his darker colour, he certainly has not many negroid features which we do not also possess, at any rate more or less sporadically. Under normal conditions, the colour of the Australian is a velvety chocolate-brown, somewhat lighter or more coppery in the female than in the male. The skin of a newly-born piccaninny is very much paler, with a distinct tint of fleshy red about it, which many people maintain reminds one of the skin of young murines, as it appears before it developes fur. For this reason, too, the inexperienced observer often accuses an aboriginal mother of infidelity; the colour of the infant's skin, when compared with that of its parent, indeed suggests a mixing of her blood with that of another lighter coloured race. The child's skin, however, soon darkens in colour; and, within a few weeks, attains a shade not appreciably different from that of the adults of its tribe.

Apparent gradations in colour are occasionally observed among different members of one and the same tribe. As with ourselves, circulatory disorders are not absent among the aborigines, and such materially affect the quality of colour in the aboriginal's skin. Simple anaemia, or even a temporary blanching of the tissues, through nervousness or fright during the time of an examination, will affect the appearance of the skin. In the same way, full-bloodedness, or a passing flush, will deepen the shade, the injection of blood into the underlying tissues being clearly noticeable through the epidermis. Pathological conditions like jaundice are also frequently

developed in the aborigines, and impart to their skin a sickly ashen hue; in this case the yellow colour of the conjunctiva usually indicates the disorder. The likelihood of any such conditions being present should be carefully investigated before applying the standard colour tables of modern anthropologists.

The oldest systems of colour-classification divided the races of man into five groups—the white, the yellow, the red, the brown, and the black. But nowadays, even the layman knows that such hard and fast divisions are impossible. We find that among individuals of one particular race, whatever its so-called, and somewhat arbitrary, colour might be, there exist noticeable variations in shade. Red Indians have yellow or brown skins almost as frequently as a genuine red; the “white” races of Europe often have so dark a “complexion” that they are in reality brown; and the skin of a negro at times has a distinctly reddish or brownish hue.

Early anthropologists thought that the “dust or tawny” colour was due to the accumulation of carbon in the external layers of the integument. But since the introduction of the microscope, which made the study of thin sections of human skin under great magnification possible, it was found that the colour is due to living cells, which carry pigment in their protoplasm, and are more or less migratory.

In the Australian aboriginal, these pigment-cells lie quite superficially in the skin. Some years ago Professor Klaatsch, of Heidelberg University, when in Australia, managed to obtain the corpse of an aboriginal, which he consigned to a large tank holding an ordinary preserving fluid. Hermetically sealing the lid of the tank, the Professor shipped the specimen to Europe, where it was to be dissected. Some months later, I joined him at Breslau University, and together we opened the tank. Imagine our surprise when we beheld what one might describe as an anthropological contradiction—a “white blackfellow!” It took us some time to recognize in the form in front of us that of the aboriginal we had seen in Australia. What had happened was that, during the continued movement of the preserving solution during the transport, the superficial layers of the skin had been removed, and, with them, the colour too. In other parts, the skin had blistered and become detached, leaving more or less adherent strips of epidermis in which the colouring matter could be recognized.

I have seen a similar condition of things in corpses of aborigines, in the remoter districts of the Australian bush, where the dead are placed to rest on artificial platforms in the branches of trees. When, during the processes of decomposition, the skin peels off, and is washed away by the rain, the corpse assumes a pinkish white colour, resembling the body of a white man, some time dead. No doubt it was on this account that, in the early days of European settlement, it was a general belief among the aborigines that the white man was one of their own dead warriors returned to life in a different colour. We have a classical example in the experience of the escaped convict, William Buckley, who lived for thirty-two years among the natives of Victoria, the latter regarding him as their dead chief returned to life transformed. It is quite possible that this belief, which is so common among the tribes, originated from the fact that the natives themselves had observed, as Professor Klaatsch and I did, that the decomposing bodies of their dead might, under certain conditions, become very much lighter in colour.

Throughout the Northern Kimberley district the natives maintain that a dead tribesman will “jump up all-the-same whitefellow” in colour.

A singular case, illustrative of the shallowness of pigmentation in the epithelium of an aboriginal's skin, was reported from Canowie Sheep Station by the late Rev. Tenison Woods. A native, suffering from an obstinate skin disease, was “dipped,” like a sheep, in a solution containing soft soap, tobacco, and arsenic, the last-named in the proportion of one ounce to the gallon of water. The native became very ill, lost his hair, and his finger and toe-nails. Eventually he became better, but his skin peeled off. He was then described as “presenting the appearance of a magpie during the time the process of decortication was going on.” Finally his skin became “smooth and as glossy as marble.”

In pemphigoid skin-eruptions, when blister-like bullae develop over different parts of the body, the lesions left in the skin for a while are pinkish and unpigmented. Scars resulting from a cut or burn remain red for a considerable time, but eventually turn the same uniform colour as the rest of the skin.

Under normal conditions, one may often find patches of pigment on mucous surfaces of the inner lips and mouth. The pathological condition known as leucoderma is, on the other hand, rather frequently observed

among the different tribes of Australia. I have seen natives, both in the north and south of this continent, whose skin over certain areas was devoid of pigment; the hands and feet seem particularly prone to be thus affected.

So much for the seat of the pigmentation. If we now enquire into the actual complexion, or colour-tint, of the aboriginal's skin, the question is not so easily satisfied as one might have thought. The colour is, of course, brown—a soft, velvety brown, like chocolate. Scientists tell us it is about the same as tint No. 3 of the colour scheme on Plate III, Notes and Queries on Anthropology, London. But the matter is not so simple as one might be led to believe. To begin with, we have not a homogeneous colouration before us. If, for instance, we wanted to paint a picture of an aboriginal, we would mix a fundamental chocolate-brown to deck the surface with; to obtain the shade of the back we should have to mix a blue or green with the brown; whereas the cheeks and chin would require a yellow or red. The intensity of pigmentation varies to a noticeable degree; it is deepest on the back and neck, and along the folds of the skin. The soles and palms are always very much lighter in colour than the rest of the body.

Environment plays an important role, because a native's skin has the remarkable power of what might be termed complimentary colour-adaptation, as a result of which a hard and fast definition, or fixing, of the shade is practically impossible. For this same reason it is conceivable why the sombre hue of the hunter's skin becomes neutralized by the sallowness of the arid Central Australian scrub, as well as by the deeply-shaded verdure of the tropical jungle. Upon a clear day, with an open, blue sky, an aboriginal always appears dark or dingy, while on a dull and cloudy day, his skin is more of a chocolate-brown; when he is swimming in the open sea, his colour may even become coppery and seem not much darker than that of a Javanese.

The adaptable tone-characteristics of their skin are well-known to the natives themselves, not only while in the hunting field and on the warpath, but also on the playground. At Opparinna, in the Musgrave Ranges, children were seen indulging in a game resembling "hide and seek," and often, in an endeavour to avoid the keen eye of the "seeker," one would duck in amongst the boulders of granite and imitate the rust-coloured contours of a boulder to perfection.

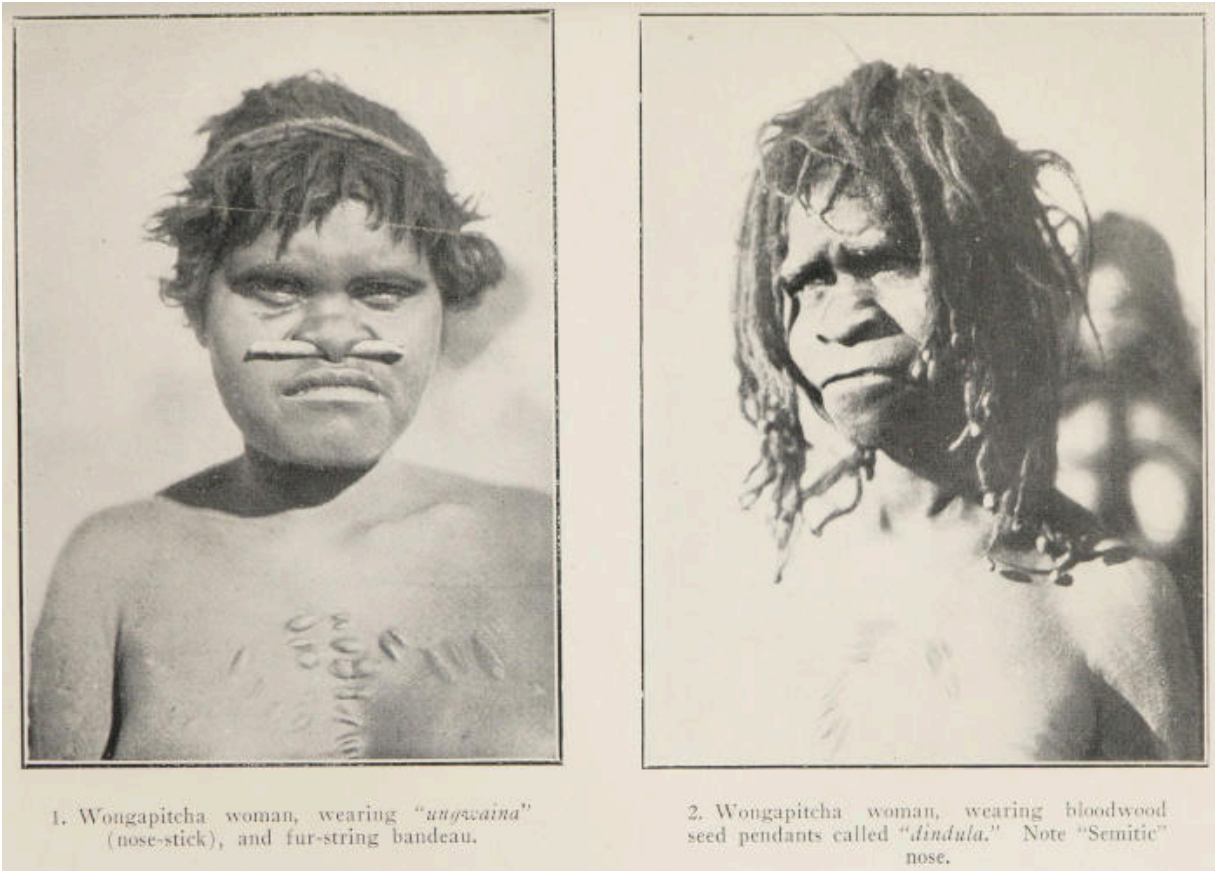


PLATE VI

1. Wongapitcha woman, wearing "ungwaina" (nose-stick), and fur-string bandeau.
2. Wongapitcha woman, wearing bloodwood seed pendants called "dindula." Note "Semitic" nose.

We know that the climate has an important bearing upon the subject of pigmentation in all races. Most of us have had our faces, hands, and other exposed parts of our body bronzed when holiday-making at the seaside. If we come straight from indoor life in the city, we might, in addition, find our skin develop a badly blistered condition known as sunburn. Yet the bushman, the coach-driver, or the sailor, who, by long exposure, has acquired a more or less permanent tan, can bask with impunity in the severest rays of the sun.

Exactly the same processes take place in the coloured man's skin. When cruising about the north coast, I frequently had occasion to observe that my Malay crew were more bronzed above than below the belt. This was obviously caused through continued exposure to the tropical sun, since

during the heat of day they would work, or lie about on deck, wearing nothing else than a serong hanging from their waist.

When aborigines are taken from their wild outdoor life, and kept under European conditions, more or less confined, their skin becomes unquestionably lighter; this is particularly noticed in their faces.

These phenomena indicate to us the method Nature adopts in protecting our skin, and with it our system, against scorching rays of the sun; and we also realize why it is that the coloured man can endure the disadvantages of a tropical climate so much better than we.

The same phenomena might also be made responsible for the wonderful absence of pigment in the skin of modern white peoples of European origin. There is no doubt, the great Ice Age and the living in caves and shelters (and huts) were the essential factors which ultimately established the “white” skin in man. In this hypothesis, we naturally assume that our Diluvial or earlier Tertiary ancestor had a moderately dark-coloured skin, which protected him against the tropical sun, which Geology has taught us, shone over Europe at the beginning or middle of that great period.

CHAPTER VIII

THE HAIR

The lanugo—Hairiness of body—Female beards—The hair of the scalp and how it is worn—Its colour—Aboriginal blondes—Albinism—Erythrism—Fair hair a likely “throw-back” to prototype—Influences of climate and geological antiquity—Other instances of fair-haired aborigines—Grey hair—Baldness—The beard and methods of dressing it.

Let us proceed with a discussion of the aboriginal’s hair. As in the youthful individuals of most races of man, including the European, the Australian is born with a rudimentary, short body-hair, known as the *lanugo*. This growth covers practically all surfaces of the child’s figure, but is thickest on the back.

The colour of this infantile coating of hair is not, as one might have expected, black, but fair, and casts a pretty golden sheen over the sombre skin. In later adult life this growth of hair becomes stronger, and darkens to complete blackness. In ripe old age, the hairs turn grey.

Many of the old men have a remarkable hairiness of the body, amounting almost to a *hypertrichosis*. In these cases the hairs are up to an inch long, and cover especially thickly the back, the chest, the thighs, and buttocks ([Plate VII](#), 1). Amongst the women quite pretentious beards are of rather frequent occurrence ([Plate III](#), 2).

With regard to the hair which covers the scalp, we find that, in the majority of cases, it is wavy. It is by no means uncommon, however, to find the Australian’s hair distinctly frizzy; straight hair is least frequently observed. The male aboriginal generally wears his hair in long loose curls; often these are matted together artificially with grease and red ochre into long, pendant, sausage-like masses. In the central region of Australia, the men pull out the hair growing on the upper portion of the forehead, each hair being removed separately. A hair-string band is worn over the cleared portion of the forehead, and this, at the same time, keeps back the locks of hair. Very often, in the central as well as in the northern districts, the hair thus tied back is worked up with a pad of emu feathers into a chignon, which is tied round and round with human hair-string ([Plate IX](#)). The

women are frequently asked to cut their hair short, and to deliver the clippings to their husbands, who work them up into coils of string, out of which they subsequently fashion hair belts and a variety of other articles in daily use. Occasionally one sees an aboriginal whose hair stands about his head after the type of a Struwwelpeter, or it may hang from the scalp like a mop. The last-named types were more plentiful on the north coast than in central Australia, but a number of cases were recorded among the Aluridja and Wongapitcha.

The men of the same west-central tribes decorate their hair with wooden pins whose surfaces they cut longitudinally over certain sections, so that the shavings, which form, curl outwards but still remain attached to the rod. The ornaments which go by the name of “*elenba*” remind one of trimmed skewers occasionally seen in butcher-shops. One or two of them are worn long at the back of the head ([Plate IX](#), 2), or a number of them short, as a fringe above the forehead. The women of the same districts try to make their hair look attractive by attaching to the tips of the matted locks numerous seed-capsules of the Bloodwood eucalyptus. The hairs are rammed into the open ends of the seeds and kept there by small plugs of wood or blebs of resin. The little ornaments dangle about the forehead and shoulders and are known as “*dindula*” The seeds of the Currajong are similarly used. Vide [Plate VI](#), 2.

The colour of the adult Australian’s hair is almost invariably black, but often of different intensities over different parts of the scalp. When a hair is pulled and examined under magnification, the part which had been embedded in the follicle is not infrequently discovered to be colourless or brownish.

By far the most remarkable, and genetically perhaps the most important, subject in connection with the colour of the Australian aboriginal’s hair is the juvenile *blonde*. In 1903, I first came across a number of children of the Ulparidja group of the Wongapitcha, in the Tomkinson Ranges, who had heads of hair of a flaxen colour. Since then, I have seen many similar cases among the Aluridja, in the Lilla Creek district, several Wongapitcha from south of the Musgrave Ranges, at Ooldea, and a limited number among the western Arunndta, on the Finke River, south of the MacDonnell Ranges. The colour of these children’s hair varied from a straw-yellow, through light brown, to dark brown ([Plate X](#), 1).

It appeared that, towards the age of puberty, the shade became visibly deeper; after which it gradually changed to dark brown or black. This change of colour I have recently been able to watch very closely in the hair of two young gins who have been under daily observation for over four years.

Among the adults, such blondes are unknown; two or three cases of brown hair in grown-up persons, however, came under our observation among the Aluridja, west of the Finke River.

True albinism has not been authoritatively established, as existing among the Australian aboriginals. The case of a young woman was reported from the Dépôt on the Victoria River, who is said to have been quite “white,” and to have had “red” eyes. This albino, I was informed, was photographed by the local constable shortly before she died in 1921.

Erythrism, too, is practically unknown; the only cases of the latter on record are four from the Tully and Bloomfield Rivers, discovered by Dr. Walter Roth. Red hair is, however, not appreciated by the aborigines, who incline to hold it to ridicule.

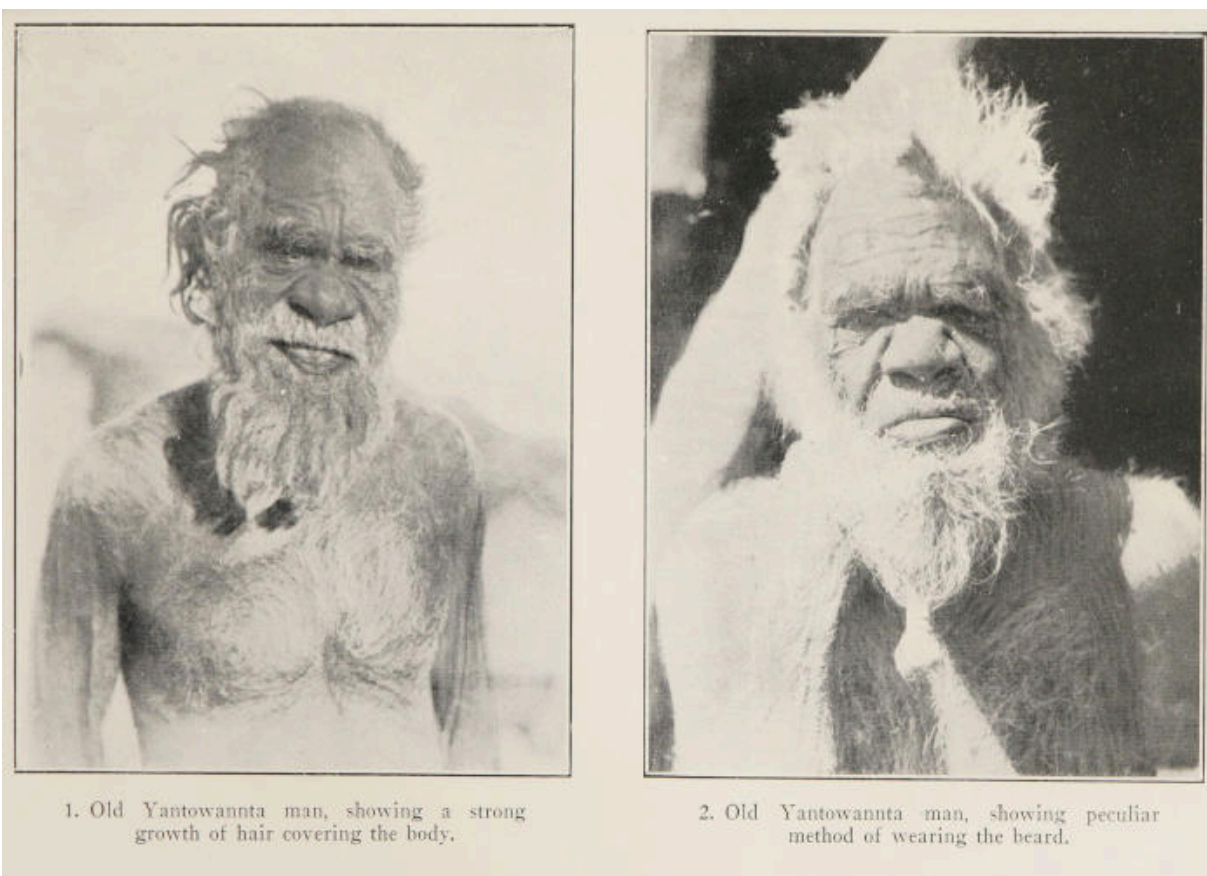


PLATE VII

1. Old Yantowannta man, showing a strong growth of hair covering the body.
2. Old Yantowannta man, showing peculiar method of wearing the beard.

When, therefore, we consider the likely significance of fair-haired aborigines, from an anthropological standpoint, we cannot ignore the claims of atavism. Noticing the phenomenon so abundantly developed, as it has been our good fortune recently to record, one is inclined to behold in it a primitive feature, which was originally typical of the ancestral hordes from whom the aboriginal Australian has sprung. This assumption is strengthened by the light colour of the lanugo regularly observed in the children.

The question arises whether the dark colour of the Australian's skin (and hair) is entirely a secondary development due to climatic influences. The superficial nature of the pigmentation in the aboriginal's skin is in support of such reasoning. It is known that the hair of some Arctic explorers, after a protracted sojourn in the frigid zones, has turned from dark to fair; and the same has been reported of alpine guides. We shall see presently that there is

evidence of great antiquity of man in Australia; his occupation of the land dating back in all probability to the early Tertiary period. Geology teaches us that the climate has fluctuated considerably since and before that time. Consequently, it is quite within reason to assume that, in the earlier days of his racial existence, there may have been no need for any considerable accumulation of pigment cells within his skin, as a means of safeguarding his system against a sun, anything like so severe as is nowadays reigning over Australia. From later Tertiary times onwards, however, the climate of central and northern Australia has been continuously hot or tropical.

We are further strengthened in our theory by the fact that the hair of the Tasmanians is known to have been generally lighter in colour than that of the Australians. Sydney Hickson even described the Tasmanian hair as light golden brown in colour. Tasmania has, we know, since the later Tertiary at all events, enjoyed a decidedly colder climate than Australia proper.

One point remains unexplained; namely, why the occurrence of light-coloured hair among Australian children should be geographically restricted. Apart from the tribal groups in central Australia, which I have mentioned, I know of no other record except one by Professor Klaatsch from a coastal district in Queensland.

The hair of an aboriginal turns grey at a riper old age than is the case of the European's. It seems, moreover, that the hair of the women retains its colour longer than that of the men.

Baldness is comparatively rare among the aborigines; only a limited number of cases have come before the writer's personal notice.

The old Arunndta men are very particular about their appearance. When one is stricken with baldness, he constructs a pad, resembling a skull cap, out of emu feathers, which he ties on top of his head with human hair-string and wears regularly to hide the bareness of his scalp. He refers to this feather-wig as "*memba*." Aluridja men adopt a similar fashion, but call the article "*lorngai*" ([Plate VIII](#)).

The men all over Australia, as a rule, can produce quite comely beards, but the methods they adopt of dressing them vary according to locality. In the River Murray and other southern districts, long square full-beards were the vogue. The Yantowannta and other tribesmen of the Cooper's Creek and

Lake Eyre region turn the point of the beard back upon itself into a loop, and, by winding fur-string around it, keep it fixed in that fashion ([Plate VII, 2](#)). North of the MacDonnell Ranges, and on some of the islands off Arnhem Land, the older men keep the upper lip clean by pulling out the hairs one by one. Along the north coast, from the Gulf of Carpentaria to the Buccaneer Archipelago, the men over a certain age are allowed to singe off, or shave with a sharp chip of stone or shell, the entire beard including the upper lip. The women of the King Sound tribes are required to help the men remove the hairs; a man will lie for hours, with his head upon his lubra's lap, whilst she busies herself pulling the hairs from her husband's chin. The old men of the Cambridge Gulf tribe twist each end of the moustache and surround it with a cylindrical layer of beeswax, from which the tips project on either side like the hairs of a paint brush. The beard is divided into two equal bundles of hairs, the ends of both of which are treated in the same way as the moustache. On some of the islands of the Buccaneer Archipelago, the men shave the upper portion of the moustache below the nose, leaving only a narrow fringe of hairs, immediately above the margin of the upper lip.

CHAPTER IX

LIKELY ORIGIN OF THE AUSTRALIAN ABORIGINAL

Deductions theoretical—Pre-historic men of Australia—Tennant's Creek calvarium—Talgai skull—Other finds—Alterations in world's ancient geography—Former land-bridges—Probable home of man in region now occupied by Indian Ocean—Early migrations—Three principal strains—Negroid—Mongoloid—Australoid—Tremendous upheavals at close of Triassic Period—Australia isolated—Early inhabitants unmolested—Primitive Australian anthropologically related to cultured Caucasian—Survival of the Stone Age in Australia—Living fossils—Short resumé—The mixing of European with aboriginal blood—What is a half-caste?

Having satisfied ourselves in regard to some of the principal somatic characteristics of the Australian aboriginal, we shall proceed to discuss briefly his likely origin. In the present absence of more material facts relating to his ancestry, and of a more thorough comparative knowledge of races in general, we are lamentably handicapped in this direction, and many of our conclusions are necessarily theoretical.

So far as possible, we shall take into consideration his present relationships to other living races and peoples, as well as his affinities with the ancient hunting peoples, who inhabited various parts of the world in bygone eras, and are now only known in a fossilized condition.

This introduces the geological element of time—hundreds of thousands, yea, millions perhaps, of years have passed since man left records of his being; definite traces have been found, embedded in the same deposits as contain the mammoth, on the one hand, and the *Diprotodon*, on the other.

The evidences of pre-historic and fossil men in the Old World are too numerous and well-known to need elucidation here; we shall confine our attention to Australian records.

Some years ago a specimen was submitted to me for identification which had been found in Pleistocene (or Pliocene?) gravels S.S.E. of the Tennant's Creek district. It was so completely petrified and "stony" looking that the organic origin was doubted, but a thin section viewed under the microscope revealed the true structure of bone. After cleaning the fragment thoroughly, I recognized it as portion of a human skull, viz. the posterior half of the left

parietal. The anterior fracture is vertical and at about the centre of the parietal eminence; the thin squamous edge is also broken away. The lambdoidal border is still quite characteristic and shows the complex nature of the parieto-occipital suture. Both the external and internal surfaces are rough and pitted through exposure, age, and mineral precipitations, but the temporal ridge is still discernible and can be traced posteriorly right up to the parieto-occipital suture. There is no indication of a parietal foramen. The bone is thick about the posterior inferior angle, but the groove of the lateral sinus has broken away. The specimen, when struck, has a clear metallic ring, like that of earthenware or porcelain. When treated with acid, the surfaces as well as the "bone-substance" effervesced briskly, proving that a thorough intermolecular substitution of organic matter by mineral was in progress. This calvarium, fragmentary though it is, is of considerable importance from a prehistory point of view, since it gives us another definite link in the somewhat meagre chain of evidence which has been established in connection with the geological antiquity of man in Australia.

The most important find of an extinct Australian type was made at Talgai, in south-eastern Queensland, as far back as 1884, in the shape of a fairly well preserved skull; but it was not until a few years ago that a description of it was published by Dr. S. A. Smith. Although no other bones were discovered in association with the skull, numerous remains of extinct creatures like the Diprotodon, the Nototherium, and horny reptiles have been unearthed not many miles remote from the site of the interesting discovery. Dr. Smith sums up his observations as follows: "This fossil human skull of a not yet adult Proto-Australian presents the general picture of a cranium similar in all respects to the cranium of the Australian of to-day, combined with a facial skeleton of undoubtedly Australian type, in the palate and teeth of which there are to be found, in conjunction with the most primitive characters found in modern skulls, certain characters more ape-like than have been observed in any living or extinct race, except that of *Eoanthropus*."

Other less convincing discoveries have been recorded in the shape of human and dingo bones from the Wellington Caves, human remains and artefacts from beneath the basalts of Victoria, and the fossil footprints of an aboriginal in the upper Tertiary beds of Warrnambool.

It would seem, therefore, that sufficient facts have been forthcoming to prove that man was in existence at any rate in late Tertiary times; and since he was then perfectly developed, it would not seem unreasonable to assign to him a very much greater antiquity.

During these long ages, tectonic forces, and the ever active denuding agents of the atmosphere, in all their phases, have wrought considerable transfigurations in the surface of the globe. Some portions of the earth's crust have been swallowed by the ocean, whilst others have been wrenched from the depths by upheaving processes. Thus the geography of our present world would be a *terra incognita* to the earliest progenitors of the human kind, who lived in the dim dawn of man's ascending tendencies, while, on the other hand, we would require a new army of intrepid explorers to pave the way for civilization if we were suddenly placed back into the world as it stood in the beginning of primeval days.

Old land connections then existed between entities which now are parted by abysmal depths. Such evidence of once-existing continental links is afforded by what has been termed a "biological consanguinity" between organic creations on both sides of gaps now occupied by ocean water.

There is no novelty about all this. Our best scientists have long recognized that such connections have existed beyond all doubt. They become evident when one enquires into the present geographical distribution of botanical and zoological species, and when one correlates geological strata in different parts of the world, on the basis of palæontological evidence contained in them.

The same principles apply when we consider the probable original home of man, and the subsequent migrations and racial evolutions of the pristine hordes, which followed.

That once a chain of land linked together the shores of Australia, South Africa, and India seems certain. The continental masses, which in past eras supplied this link, zoologists have christened *Lemuria*, while geologists refer to the lost land as *Gondwana*. It is somewhere within the area once occupied by this submerged continent, perhaps not far remote from Australia, that we must look for the cradle of the species *Homo*. Although most of the evidence has been irretrievably lost to scientific investigation, much might yet be expected from any of the contiguous continents or

islands in this region, upon which occur Tertiary or later sedimentary formations. The discovery of the oldest fossil, which appears to be human—the *Pithecanthropus erectus*—in Java, was by no means accidental. Professor Dubois, before leaving for that island to undertake a fossil-hunting expedition there, declared that in all probability he would discover the remains of a primitive creature related to man.

From some point, then, upon this ancient, vanished continent, perhaps no great distance north of our present Australia, we believe migrations of the earliest representatives of the human species took place. The directions in which these migrations took place would be governed according to the lie of the land as it was then determined by the impassable waters of the ocean. In all probability, the families or groups wandered in various directions, at first keeping more or less in contact and on friendly terms with each other, but as time, and eventually ages, wore on, these migratory groups, by selective culture, environment, climate, and, maybe, sundry other causes, became differentiated into peculiarly distinct strains, all of which we are nowadays able to reduce to three fundamental races.

One of these migrations was along a western course, which led the wandering groups into the region now represented by the continent of Africa. This established the Negroid element.

Another strain moved northwards and spread itself, like the rays of a rocket, across the land now known as Asia. Some of these “rays” reached what is now Lapland, while others found their way, via the region of modern Esquimaux Land, across to what we now call North America. This march evolved the Mongoloids.

Yet another body of primitive hunters, who interest us most, worked their way north-westwards, on a course between the former two, and took possession of any portions of the dry land of the globe, the present relics of which are India, south-western Asia, and Europe.

Then came the catastrophe! The exact period is not determined. It must have happened since the advent of the “human” type, but there the evidence fails. Upheavals or subsidences of land usually take an age to make themselves noticeable. It is scientifically established that the close of the Triassic period was characterized throughout the world by great tectonic changes. Beds of rock were faulted to lofty heights on one side, and to

dizzy depths on the other. The height of the Blue Mountains plateau of New South Wales is evidence of such upheaval, whilst the broken coastline, with its “drowned” rivers and myriads of islands along the north-west of Australia, together with the coastal fringe of coral reefs along the north, are all evidences of comparatively recent subsidence *en bloc*.

By these processes Australia was gradually isolated from its former land-connections, but, being near to the original home of man, it is only natural to suppose that the land was peopled.

From that time on Australia remained, whether as an island continent or a group of associated islands does not concern us here, isolated from the rest of the world. The original inhabitants whiled away their time in comparative ease. They had nothing to fear. Their former companions who had, through their nomadic migrations, been so far removed from them, would, no doubt, have now posed as formidable rivals, if the barriers had not come between. Until the recent arrival of the European explorers and settlers, and the periodic visitations to the north coast by Malay *bêche-de-mer* fishers, this great Southern Land had remained the undisputed property of the comparatively sparse progeny of the first primitive possessors.

There were no ferocious animals to molest these early prehistoric Australians. Apart from a few dangerous, but usually non-aggressive, reptiles, the large animals were almost without exception of the ancient marsupial order, and, although perfectly harmless, offered excellent opportunity for the chase.

Thus it happened that the primitive hordes could roam at large in a congenial climate, and under peculiar conditions, which were everywhere much the same; and, in their subsequent wanderings, they met only with people of their own descent and inclinations. In consequence, they were spared many of the bloody brawls and conflicts, which the competitive waves of culture continually showered upon the other hordes that were struggling northwards under decidedly more adverse conditions of climate.

The great struggle for existence did not make itself felt so keenly to the ancient Australians because they were strictly insulated, and thus kept outside the sphere of exotic influence and interference; their only troubles amounted to an individual club-duel, or occasionally an inter-tribal warfare,

which evoked more irate words than actual blood drawn by their sharply-pointed spears.

So the Australian has remained just what he was ages ago. And on that account the evolution of his pristine contemporaries, who were seized by the flood wave of culture, becomes the more comprehensible, when we measure the differences, but recognize the affinities, existing between the extremes. A line drawn across the map of the world indicating, so far as it is at this stage possible, the areas whose populations show, or before their extinction showed, the strongest affinities with him will represent roughly the direction of migration and incidentally of evolution of the Australoid strain.

This line of anthropological relationship connects the Australian (including the Proto-Australian) with the Veddahs and Dravidians of India, and with the fossil men of Europe, from whom the Caucasian element has sprung. In other words, the Australian aboriginal stands somewhere near the bottom rung of the great evolutionary ladder we have ascended—he the bud, we the glorified flower of human culture.

In the living Australian then, we see the prototype of man as he appeared in Europe in the Stone Age. Australia has upon other occasions proved to be extraordinary in a scientific sense. The kangaroo is known only in the petrified condition in the Tertiary deposits in other parts of the world. The *Zamia*, which is still found living in Australia, is a conspicuous plant of the coal-measures in every other country. The ornamental mollusc, known as *Trigonia*, had been regarded as extinct until it was re-discovered in Australia. Most of the great river systems of central Australia have had their day; they have flourished in the past; yet, occasionally, after a prolific downpour, their dry courses swell temporarily to majestic streams. And, lastly, we see in the aboriginal yet another palæontological overlap—a living fossil man—the image of ourselves, as we appeared many ages before we learned to record the history of our progress, and of the world in general.

When one wades more deeply into the subject, only skimmed above, the following points suggest themselves to one: Our line of racial development was very early dissociated from the Mongoloid and Negroid lines; and geographically it ran between the latter two. There are considerable racial

differences between the other races and the Australoids, the most highly specialized and cultured division of which is now represented by the modern Caucasian. The last-named deductions are entirely supported by the shallowness of the pigmentation in the aboriginal's skin, and by the fair hair of children found among certain tribes of central Australia. In fact, the colour question, so far as the Australian aboriginal is concerned, is a relative conception, the difference in the amounts of pigment in his skin and in the "white" man's being in all probability due to climatic influences extending over long periods of time. It is doubtful whether the primitive Australoid or the Proto-Australian possessed a skin so dark as that of the present-day Australian. We may now understand why it is that the quarter-blooded progeny derived from the union of a half-blooded aboriginal woman with a European father is always lighter in colour than its mother, and the octroon lighter still. Unions further on the European side produce children practically white; and no case is on record where the colour in a later generation reverted to the darker again. The latter, we know, happens only too often when there is a taint of Negroid blood running in a family, even though the mixing of race took place generations back.

Apart from its great scientific significance, this matter is of considerable social and national interest to citizens of Australia, and we might well ask ourselves: "Are we justified in referring to the half-blooded aboriginal, with European parentage on one side, as a half-caste, or in even stigmatizing him as a bastard?"

CHAPTER X

AN ABORIGINAL'S BIRTH

Recognition of pending maternity—Peculiar beliefs in connection with the cause of pregnancy—Larrekiya legend and maternal dietary—Maiyarra's accouchement—Birth—Twin births—After-treatment—Artificial termination of pregnancy—Preparing the new-born—Children's lot decided by peculiar group-relationships—Parents' affection—Children unclothed—How they are kept warm and reared—Different methods of carrying and nursing children.

It had been talked among the old men for some time past that the lubra Maiyarra was giving cause for suspicion. Her husband Pitjala agreed; to his knowledge there had been no occasion for her to leave his camp for some moons past. His mother, old Indarrakutta, had told him that when she and Maiyarra were gathering roots down by the Womma waterhole, many of the gum trees were covered with manna and they partook freely of the sweet meal, which, as he knew, does not often come to their district. The old woman had cautioned the girl and growled at her when she did not obey, because she knew Maiyarra was of the Yalliadni clan and should not be allowed to eat the manna. This disobedient gin had, however, not eaten much before she became sick and was obliged to lie in the hot sand of the creek where the bullrushes stand. Indarrakutta had stood aghast, Pitjala explained to the old men, when unexpectedly disturbing a snake from the bullrushes, she observed that the creature, in gliding over the ground, touched the body of Maiyarra with its tail and, in its great haste to disappear, had left portion of its glossy slough beside her. "*Yakai*," gasped the men, as if from a single mouth, "then it is clear the ever wakeful spirit of Womma has caught the neglectful Maiyarra sleeping and it is certain she is with child."

Such was the history of the case as narrated to us. It corroborated previous observations from central and northern tribes. The recognition of maternity is not connected primarily with any conjugal liberties a husband or number of tribal husbands may be privileged to enjoy, but more with the recollection of any accidental contact with an object by which it is supposed a spirit child can enter the body of a woman. The spiritual ingress may take place in a variety of ways, but as often as not it is believed to be by means

of a hollow object of some description. In the present instance it was a snakeskin. On the Victoria River the gins have a dread of the whirlwind, thinking that if such should pass over one of them, a spirit child would immediately enter the woman. In the Cambridge Gulf country, young women very reluctantly go into a water hole in which lilies are growing, fearing that as they step over the leaves, which are hollow, a similar fate may overtake them.

In the ancestral days of the Larrekiya in the Port Darwin district, for instance, it is believed that a baby boy was once seen to spring from the burrow of a rabbit bandicoot; whence he had come no one knew. He was invited to come to the Larrekiya camp and live with them, but he refused. Some time after, when the boy had become a man, he was again met by the tribe, who once more invited him to their camp; but he declined as before. Thereupon the men became angry and dragged him to a waterhole, and threw him in. The stranger immediately sank, and five bubbles of air rose to the surface as he disappeared. The men sat down and watched the water, when suddenly the man's face reappeared. The Larrekiya hurled a spear at him, and he was killed because they knew he had no father and no mother and was the accomplice of the evil spirit, who, it is asserted by the Wogait, makes a big fire, from the smoke of which he takes an infant and places it, at night, into the womb of a lubra; and she must then give birth to the child.

In the same district, when it becomes known that a happy event is pending, the husband goes out with his lubra and kills a certain animal or collects certain vegetable products, which he hands to the woman to eat, believing that these articles when swallowed will ensure a successful birth.

To return to our story: Maiyarra was groaning with pains in the abdomen. She was alone with the old woman Indarrakutta, who was her mother-in-law, well beyond hearing distance from the main camp. A small fire was burning sluggishly by their side and throwing a thin column of bluish white smoke into the air. Maiyarra was sitting upon a small patch of ground cleared of the burrs, with her legs stretched before her. She was propping her writhing body, sloping slightly backwards, with her arms against the ground. The old woman sat closely behind, with her arms thrown around Maiyarra's waist, and with her lower limbs, bent in the knee, enclosing and pressing against the younger woman's buttocks on either side. Occasionally the old woman would relinquish her hold and make for the fire, over which

she warmed her hands to subsequently massage the patient's abdomen. Now and then she might even rub warm ashes over it. Then the two sat in patient expectation, and, whenever there came a pain, the old woman would tighten her grip, while she spoke encouragingly to the parturient Maiyarra. This method is very generally employed, except that when the final stage has arrived, the Arundta and other neighbouring tribes in central Australia request the gin to squat on her toes, with her buttocks resting over her heels.

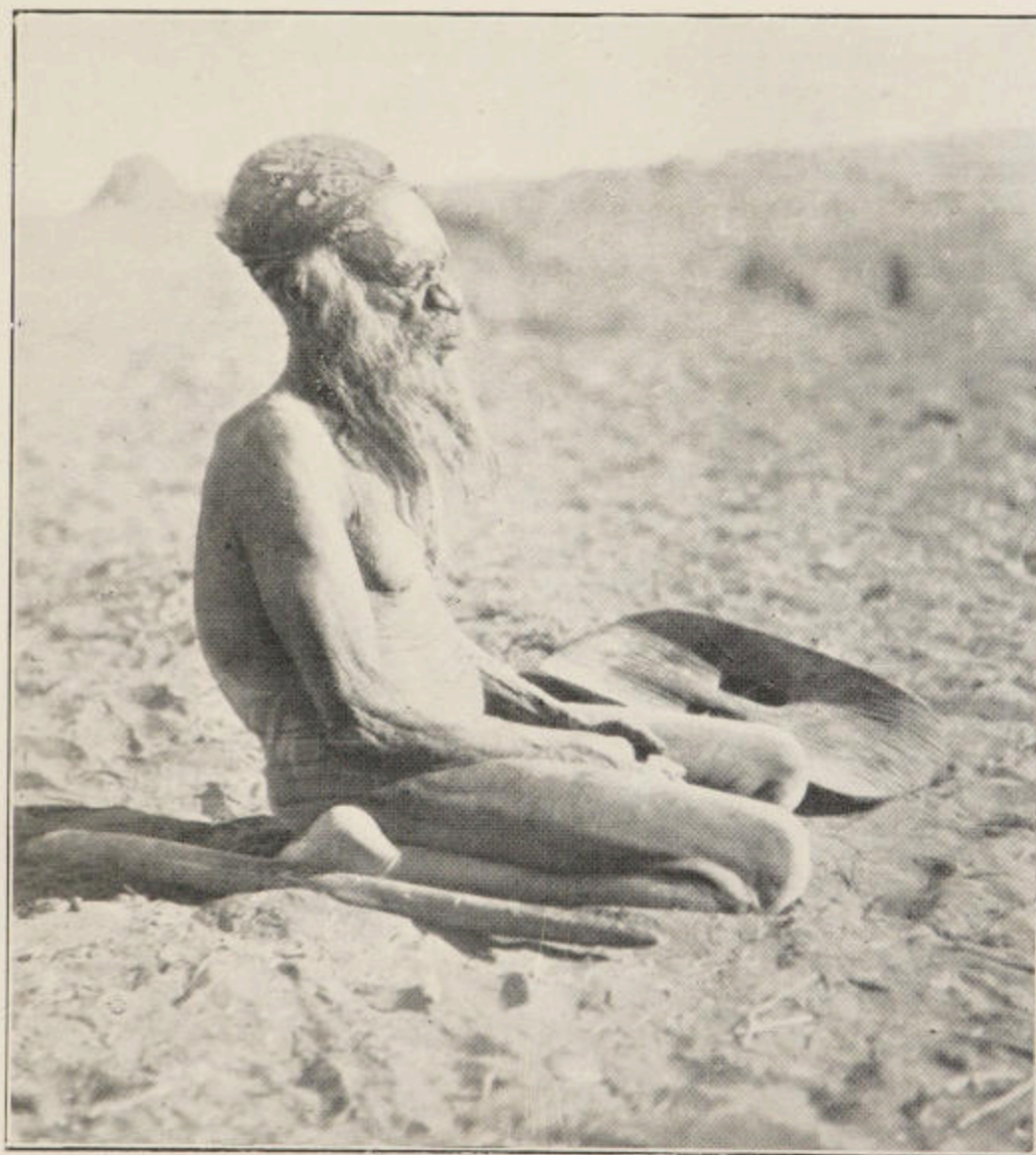
The event is almost invariably spontaneous. In my experience I have very rarely seen complications, and then usually when the lubra has been living under civilized conditions.

Twins are very exceptionally seen; we do not mean to imply, however, that multiple births do not occur more often than one sees or hears of. No authentic observations are available to satisfy our curiosity in regard to this point. We have been repeatedly assured that when twins are born, one has arrived as the result of the evil spirit's witchcraft. The child, one is informed, will do no good for itself, and, on account of the evil within it, it will contaminate others with whom it comes into contact, and, if it were allowed to grow up, it would be in league with the evil spirit, whom it would look upon as a brother, and to whom it would betray all the tribal secrets. The evil spirit would carry this information to the enemy and their tribe would surely be wiped out of existence. In consequence of all this, the suspected one of the two infants is destroyed, usually by one of the old women in attendance, who places a red-hot coal in its mouth or smothers it with sand.

The placenta is waited for, and then the umbilical cord is severed two or three inches from the child's abdomen in one of the following ways: It may be twisted off, cut with a sharp fragment of shell or splinter of rock, or pinched off with the finger-nails, or even bitten off with the teeth. Another method is to batter it through with a stone, after which the small remaining portion is packed with warm ashes. When it falls off, it is tied around the child's neck with a piece of fur-string, where it is worn for a while as an amulet. The placenta is either burned or buried.

Intentional interferences with pregnancy are rare among the unsophisticated tribes, but rather frequent when the natives are living under more civilized conditions. At Fowler's Bay a gin, who wishes to rid herself

of prospective motherhood, collects a number of black beetles, known as “yarralyi,” which she roasts and reduces to powder. Of the powder she rubs some into her armpits, and some over her breasts and pubes.



Old Kai-Kai, the leading medicine man of the western Arunndta.

“The medicine man is not so much an individual who has the knowledge of medicinal values of herbs and surgical practices as one who is the recognized sorcerer. . . .”

(Note also the emu-feather skull cap, light-wood shield, and “Kutturu.”)

PLATE VIII

Old Kai-Kai, the leading medicine man of the western Arunndta.

“The medicine man is not so much an individual who has the knowledge of medicinal values of herbs and surgical practices as one who is the recognized sorcerer....”

(Note also the emu-feather skull cap, light-wood shield, and “*Kutturu*.”)

The newly-born infant, as it lies upon the sand, is rubbed all over and dried with ashes; then it is usually transferred to a sheet of bark or a trough-shaped bark food-carrier, in which it is carried about during the first few months of its existence, the mother, at feeding time and other odd moments, taking it up into her arms. On Sunday Island the bark food-carrier, there known as “*oladda*,” is used as a cradle; one often might see a busy mother, attending to duties which occupy her hands, putting her child to sleep by simultaneously rocking the receptacle containing it with her foot ([Plate XI](#)).

The Aluridja smear ochre, ashes, and fat over the body to protect it against the hot wind and the flies. Some of the south-eastern tribes, now practically extinct, did likewise.

Among the Kolaia near Cambridge Gulf the common practice is to apply mother’s milk to the infant’s body and sprinkle it with charcoal. In their endeavour to make a young mother’s breast as productive as possible, the Aluridja and Arundta burn sticks of the mulga and stroke the breast with the charred ends.

The Arundta singe the infant’s hair with a fire-stick and rub the skin over with charcoal to bring about a darkening of the colour as speedily as possible.

In the same way as girl-piccaninnies are assigned to their tribal husbands before even they are born, according to certain group-relationships, so are the boys of the Port George IV district apportioned by the same law to the old men, whom they must obey, when called upon, throughout the term of the elders’ lives.

An aboriginal gin is often charged with callousness towards her offspring. Such an accusation, apart from proving the informant’s ignorance, amounts to a slanderous injustice. The aboriginal mother is as fondly attached to her babe as most white women are to theirs, and the way she can endear herself to it is pathetic. The men, too, exercise a chivalrous and honourable guardianship over the innocents of their tribe as well as over the children of any white settlers, who happen to reside in their district. Those who have lived among the Australian natives, like the northern

squatters, know only too well that under ordinary circumstances their children could not be in safer custody than when entrusted to the care of the aborigines.

An infant is never clothed. On Sunday Island a single strand of human hair-string is tied around its hips and pubes. Such is, of course, in the first place to decorate the body, and secondly to charm away the evil-bringing spirits which may surround it.

To bring warmth to an infant during the night, it is cuddled by its mother or other near relative; during the day, when the mother's hands are otherwise occupied, a piccaninny is often kept snug in its bark-cradle by bedding it upon, and sprinkling it with, warm ashes.

A child is not weaned until it is at least three or four years old; at times it is kept at the breast for even a year or more longer. Nevertheless, a mixed diet is offered the suckling very early in life; one often sees a baby, but a month or two old, vigorously sucking the smooth head-end of a big bone and apparently thoroughly enjoying the treat.

Different methods have been devised to assist the gins in carrying their infants with as little inconvenience as possible when on the march.

When the babe is very young, the bark-carrier is indispensable; it is either carried under the arm or cleverly balanced upon the head. In the latter case a circular cushion or ring-pad is first placed on the head to steady the weight.

One precaution is constantly preached to young mothers, namely, not to allow the child's legs to hang over the edge of the wooden carrier lest they grow crooked.

The tribes north of the Great Australian Bight swing the infants in skins or plaited vegetable fibre mats over their backs, the corners of the receptacles being tied in front of the mother's neck. The nearly extinct tribes of the lower River Murray and surrounding districts, as far as western Victoria, used to adopt the same method.

When the child attains a riper age it sits in the bag-shaped receptacle, its head being the only exposed part of its body which is visible. The natives maintain that they originally learned this dodge from the kangaroo, which carries its young in a pouch.

When the child is a little older and has arrived at the toddling stage, it is allowed to ride pick-a-back style upon its mother's back, where it secures its position by catching hold of the gin's shoulders, neck, or breasts. Another favourite method is for the gin to straddle the child upon one of her hips and hold it there with her arm.

Occasionally the child sits upon either parent's shoulders and hangs on to the elder's head or hair. More for the excitement created than as a recognized way of transport, the parent, usually the father, may seat the child upon his head and hold both his arms up for the rider to clasp. After a short run with its father in this position, the child usually asks to be let down again.

The most peculiar custom is that in vogue among the Wongapitcha of the Tomkinson and other associated ranges in central Australia. The child is laid across the small of the mother's back, face forwards, and is kept in a horizontal position by partly lying upon the gin's buttocks; it is supported by the mother's arms, one of which is held beneath its neck, the other beneath its knees. By adopting this method of carrying, the gin has both her hands free. The same method is adopted during the transport of a favourite dog, the women maintaining that it is a very comfortable occupation in the cold weather because the animals help to keep them warm ([Plate XVI](#), 1).

When off duty, that is when not on the tramp, hunting, or wood-collecting, a gin will carry, rock, and caress her offspring much like a European mother does, by tenderly clutching it in both her arms.

If work permits, the mother often sits on the ground and lays her offspring across her lap; by lifting her thighs towards her body, she forms a trough, in which the babe lies most comfortably.

On the north coast one might occasionally see a gin swinging her babe upon an aerial root or branch of a tree, or upon the flexible stalk of a tropical climbing plant.



1. Men of Kolaia tribe, Cambridge Gulf, wearing the hair tied at the back around a pad of emu feathers.



2. Wongapitcha men wearing ornamental wooden hair-pins known as "*elenba*." Note charcoal rubbed over the foreheads.

PLATE IX

1. Men of Kolaia tribe, Cambridge Gulf, wearing the hair tied at the back around a pad of emu feathers.
2. Wongapitcha men wearing ornamental wooden hair-pins known as "*elenba*." Note charcoal rubbed over the foreheads.

CHAPTER XI

CHILDHOOD

Much freedom given to children—Entertained and amused by parents—Taught songs and dances—Drawing tracks in the sand—Importance of learning to track—Playing with sand, mud, and water—Sliding and mud-balling—Tobogganing—Tree-climbing practice—Chasing wind-driven objects—Spearing moving targets—“Hand-ball”—“Catch-ball”—“Tip-cat”—Throwing contests—“Hide and Seek”—Toys—Playing at “Father and Mother”—“Dolls”—Fireless cooking—Toy throwing-sticks—Sham-fights and hunts—Emu game—Toy boomerang—Toy raft—The “*Kukerra*”—Spinning tops—“Cratch-cradle”—Children rarely attend ceremonies—Discipline and obedience—Girls trained by mothers—Boys taught how to make and use weapons—Girls’ stick practices—Spartan principles—Animal and bird pets.

So soon as the child is able to walk and run, independently of its mother, it is allowed every freedom, but never far away from the watchful eye of its parent; quite occasionally, however, one might meet with a toddler roaming about the bush all alone, and miles away from the main camp. Recently we saw a little chap near Running Waters on the Finke River, who would wander away from camp and spend days alone in the sandhills. The only nourishment he could find during his absence was a handful of small bulbs, which grew along the sandy banks of the Finke. It must be mentioned that this little fellow was an orphan, and nobody seemed to take much notice of his absence for the first day or so, after which a near relative would set out, pick up the wanderer’s track, and bring him back to camp.

Parents devote much of their time to the entertainment and amusement of their children; but the economical side of play is never forgotten. If during a game, a practical wrinkle can be taught, which will prove useful when the playful moments are left behind and the more serious stage of life is entered, the opportunity is never missed.

Much time is spent in the evenings teaching the younger generation songs and dances, which allude to ancestral traits, to the tricks of the chase, and to the damage the evil spirits can do. The notes and calls of the different wild animals and birds, with which the tribe has daily to do, are cleverly imitated and explained, disregardless of the numerous repetitions, which are begged, to satisfy the childish curiosity. For instance, the plover is by the Western Arundta called “*kurreke tata*,” which is softly and musically

rendered in imitation of the bird's familiar cry. The plover is described as a rain-maker, which is able to bring the water from a cloud whenever it desires. Even the European settler often refers to this bird as a "rainpiper"; the connection between the species and rain no doubt having arisen from the fact that plover usually follow up showers and remain in the vicinity of any pools which collect upon the ground. During any rain-making ceremonies the plover is frequently mimicked. Another of their favourite items is the imitation of a whining and howling dingo, which they accomplish with wonderful accuracy.

The dances, too, are largely imitative. One of the most popular of the Arundta repertoire is the frog-dance. The child adopts a sitting attitude and passes its arms from the outside, behind the knees, and forwards to the ground. In this position, it moves about on "all fours," with a peculiar hopping motion, adding greatly to the hilarity of the meeting.

Great pleasure is evinced by the beaming young faces when an adult prepares to draw pictures in the sand. A small circular patch of ground is cleared by the entertainer, and the children seat themselves around it. Having smoothed the surface with the palm of his hand, he proceeds to "draw" by scratching the design into the sand with a small pointed stick. Although the pictures are crude, and often nothing short of puzzles to the European, the artist talks all the while to the children in such a convincing way that, even assuming their eye incapable of comprehension, their interest is excited or persuaded to such an extent as to almost render the few lines in the sand a living reality. "Here is the man," explains the artist, as he draws a vertical line, "walking about" (a number of small holes are tapped into the sand), "he sees a lizard" (a longer line on a slope crossed by two shorter bars at right angles), "away it runs" (pairs of taps slantingly opposite to each other), "the man after it" (single taps between the former pairs), "he throws a boomerang" (the familiar shape of the weapon is outlined), "the lizard goes down a hole" (a hole is scratched into the ground), "the Kurdaitcha take it, it is gone!" (he slaps the spot with the flat of his hand). "*Yerrai!* What is that? A snake!" (emerging from the hole he draws a curved line), "the man has lost his boomerang but he hits the snake with his waddy" (the curved line is smacked several times with the small drawing stick the artist holds in his hand). "*I, i, i!* he has finished (*i.e.* killed) it." And so the narration might go on for a considerable time.

Commendable pains are taken by the adults in imitating the tracks of all the animals of chase, and the children are invited to compete in reproducing them. For instance, an “emu track” is obtained by pressing the inner surfaces of the index finger and thumb, held at an angle of about forty-five degrees, into a smooth patch of sand; then, without lifting the index finger, the thumb is moved to the opposite side and there pressed into the sand, at about the same angle as before. Often the impression of the “pad” of the bird’s foot is indicated by dabbing the round point of the thumb into the sand immediately behind the intersection of the three “toes.”

A kangaroo track is simple, and is made by imprinting a finger or big toe twice in the sand, an inch or two apart, so that the resulting marks are two parallel grooves supposed to represent the impressions of the long central toes of the marsupial. A shorter mark is made at the centre of either of these, at an angle of about forty-five degrees, to indicate the lateral toes, when the track is complete. At times a small scratch or hole is made at the end of each of the “toes,” to suggest the claw-marks.

A dog track is made with the fingers alone. The tip of the thumb makes an imprint, which is to represent the pad, whilst the finger-tips supply those of the four toes, ranged in a semi-circle about the former. The claw-marks are added in the same way as described of the kangaroo track.

A human track is imitated by imprinting the outer edge of a half-closed hand, the left hand being used for the left foot and the right for the right. This impression will give the ball, the outer surface, and the heel of the required track; the toe-marks are dabbed in with the finger-tips.

Where the camel is known, its track is reproduced. A piccaninny is momentarily sat upon a smooth patch of sand and lifted away again; the imprint of its stern supplies the outline for the required track. The lower half of ridge left in the sand by the cleft between the child’s buttocks is obliterated, when the “track” is ready for the never-failing applause. Occasionally the upper angles, representing the camel’s toes, are improved by making them more acute and deepening them to show where the claws are supposed to have cut into the ground.

The study of animal-spoors in all their specific and various intricacies, and especially the art of individualizing the human foot-print, rank among the most important and earliest occupations of the aboriginal child’s mind.

Parents are required by law to see that the children receive constant instruction and exercise in this department. It is a common thing for a mother to purposely slip away from her child and not to respond to the imploring wail, which follows when her absence is discovered. The only sympathy some relatives or friends might proffer is to direct the child's notice to its mother's tracks and at the same time urging it to follow them up.

Whereas the average European can distinguish between the tracks of a dog and a cat, it is a decidedly more difficult matter for him to discriminate between those of a mastiff and a wolf on the one hand and, say, those of a sheep-dog and a fox or jackal on the other. The aboriginal, however, learns to recognize not only the class, or species, or variety, as they are known to us, by the spoors, but can particularize each single individual. By looking at a track, for instance, which we can only describe as a "dog track," an aboriginal can immediately tell us whether it is that of a "wild-dog" or of a "whitefellow-dog," whether the animal is young or old, male or female, and whether it passed over the ground sometime to-day, yesterday, the day before, or a week ago; finally, he will tell us whether he has ever seen the dog, and, if so, he will probably know who the owner is. And all this information comes at a glance!

The children love basking in the sand; and spend hours playing with the glistening white and red drifts so plentiful in central and northern Australia. Little mounds are built, upon which they sit to defy the others to topple them; and oblong holes are scratched, into which some of the younger are laid and almost completely covered with sand. When the buried ones presently throw the sand from their bodies, the rest of the players scamper off with a hullabaloo and tell their elders the Kurdaitcha (devil-devil) is coming.

Along the north coast of Australia, the mud-banks of the tidal rivers are a great source of amusement. When the water recedes during the heat of day, the young folk make for the blue mud-banks to indulge in sport and play. Mud-sliding is usually the first item of the programme. Taking a long run over the firmer ground, the performers reach the mud-bank with considerable velocity, and in consequence of this, they slide over the slimy surface in much the same style as our children in the Old Country do over the ice. They maintain their balance with their arms. The sliding is effected

either singly or in long chains they form by joining hands. They endeavour to make their slide reach the water, which, if they are successful, they enter with no end of splashing and shrieking “Ai! Ai! Ai!”

Another method often tried is to run and then take the “slide” lying upon chest and belly. As one is sliding this way, he turns his body round its long axis and eventually disappears into the water. To the observer a child sliding in this way looks deceptively like a stranded dugong or other big fish endeavouring to make back to water. Occasionally one of the sporters breaks the monotony of the game by negotiating the “slide” with his stern. Or he may bring along a small oblong sheet of bark, upon which he sits or kneels and propels himself along with his foot. In modern language this toy might be called a “mud-scooter.”

On Cambridge Gulf the girls have evolved yet another variety of this sport. One lies flat on the mud, face downward, whilst another stands behind her. The one lying now places her arms forwards and holds the palms of her hands together; at the same time she bends her legs to a rectangle in the knees, and keeping them together she holds them rigid in that position. Now the girl standing behind seizes the legs of the one on the mud at about the ankles and pushes the human sleigh along the “slide.” As a special favour a piccaninny might be allowed to take a seat upon the sliding girl’s back.

In the same district the boys delight in carrying each other pick-a-back to the brink of the softest mud, to precipitate the rider backwards into the slush or into the water.

To break the monotony, the children place themselves in a row, each with a number of flattish pebbles previously collected on the dry land, and take it turn about to pelt the surface of the water at a very sharp angle in order to make the stone ricochet as many times as possible.

A new game is begun by the girls stamping the water with their feet, as at a corroboree; the boys are preparing for a sham-fight. They cover the whole of their body, including the hair, with thick mud, which they maintain is the same as the ochre the warriors apply to their bodies when on the war-path. They pick sides and stand face to face on a bank, about half a chain apart. Upon a given signal they commence bombarding each other with mud-balls! One is reminded of the snowballing feats of European school

children. The lads endeavour to dodge the mud-balls thrown by their adversaries with as little movement as possible, by just contorting the figure or lifting a limb to allow a missile to fly harmlessly by. Much gesticulation takes place during the friendly combat, and often does one hear an excited “Ai! Ai!” announcing the fact that a player has effected a narrow escape. The climax, however, is not reached until a hit is recorded, and a ball, too carefully aimed, spatters the body of a neglectful opponent. When the game is over, all participants rest for a while upon the surface of the muddy bank, then dive into the water and wash the slush from their heated bodies.

At Kurrekapinnya Soakage in the Ayers Ranges the bare, inclined surface of a granitic outcrop is utilized by the children for tobogganing. The same track has been in use for so long that the “slide” has become remarkably smooth from the constant wear. The tobogganer gathers a bundle of rushes at the soakage and makes for the top of the outcrop. There he places the bundle upon the polished “slide,” sits upon it, and starts himself moving down the slope. Considerable speed is attained by the time he reaches the bottom of the rock, whence he shoots into the sand adjacent to it. The performance is repeated over and over again.

The Victoria River tribes arrange competitions among the boys in tree-climbing, the lads being required to clamber up a number of selected trees, and down again, in the quickest time possible.

On Bathurst Island a favourite amusement of the younger folk on a breezy day is to collect the light globular seed-heads of the “spring rolling grass” (*Spinifex hirsutis*), that grow on every sandhill near the coast, and take them to the beach to release them on the hardened sand. Driven along by the wind, these seeds travel over the surface at no mean pace. Allowing them to gain a fair start, the children bolt after them, endeavouring to overtake them and pick them up from the ground while dashing past at full speed in “cow-boy” fashion.

In the same locality the children assemble on the beach and compete in running and long jumping.

The Arunndta and Dieri children collect the dry tussocks of the “roly poly” (*Salsola kali*) upon a windy day and take them to a big clay pan. There they liberate them, and, as the wind whips them over the level ground, the youthful gang makes after them with toy spear or boomerang,

each endeavouring to either stake a tussock with the first-named weapon or shatter it with the latter.

In the Fowler's Bay district the tussock is replaced by an artificial target, such as a ball of fur-string, which is rolled over the surface by an elder.

The Arunndta boys on the Finke River cut discs out of the bark of eucalyptus trees, which they roll over the hard ground and chase with toy spears. In the same district I have seen the bark disc replaced by an iron ring the boys had been given by a teamster; this was carefully kept in one of the huts and only produced when the lads were at liberty and felt inclined "to tilt at the ring."

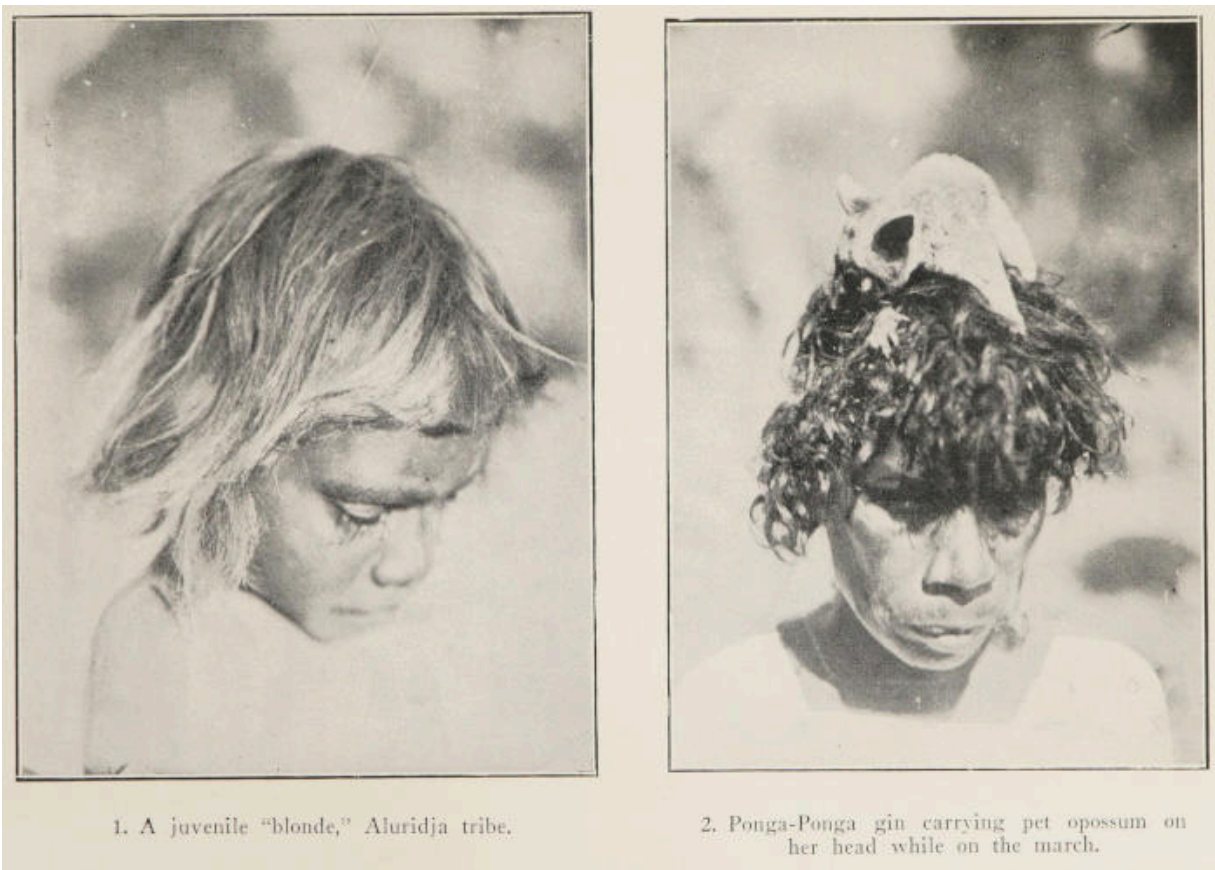


PLATE X

1. A juvenile "blonde," Aluridja tribe.

2. Ponga-Ponga gin carrying pet opossum on her head while on the march.

A modification of the game was observed in the Humbert River district, Northern Territory. Two parties, of about half-a-dozen each in number, take

up positions opposite each other and about a chain or chain and a half apart. A circular piece of thick green bark is thrown overarm by a member of one party swiftly towards the other, so that it strikes the ground a little distance in front of the latter and rolls along the ground past them. The waiting party stand in a row, with their spears poised, and each in succession rushes forward to pierce the rolling disc by hurling his missile at it. If one is successful the fact is immediately announced by loud cheering. Then one of the party, who have thrown, returns the disc in a similar way to the opposite side whilst several of his fellows collect the spears for the next turn. Thus the players are alternately “active” and “passive.” Special short wooden spears are used, about five feet long, pointed at one end. The disc is shaped out of an irregular piece of bark by biting off the angular points until a more or less circular piece is obtained. Men and youths play for hours at a time at this game, which they call “*gorri*”; and even children are tolerated by them, although the last-named are often growled at and told to keep out of the way ([Plate XIII](#), 1).

A kind of hand-ball is practised on Bathurst Island. The seeds of the *Zamia* (*Cycas media*) take the place of a small ball. Two lads stand facing each other and hit the seed to and fro with the palms of their hands, after the style of a modern game of tennis. On the Victoria River, the children made similar use of the green seed capsules of the cotton-tree.

In the Meda district of north-western Australia, players at the same game employed flat pieces of wood resembling cricket bats, the balls being fashioned out of the woody fruits of the *Pandanus*.

Catch-ball is played by the children of all Australian tribes. The “balls” might consist of anything; the Arunndta of the Finke River country use the seeds of the *Macrozamia Macdonnelli*, the natives of Melville and Bathurst Island *Zamia* and *Pandanus* seeds, the Larrekiya of Port Darwin small bags stuffed with fur, the scrotum of a kangaroo being often used for the purpose. It is surprising, however, that despite the quickness of their eyes and the keenness of their sight, the natives, as a rule, are very backward at catching with their hands any object which is thrown at them.

The boys of the Arunndta and Aluridja tribes construct a small cylindrical stick sharpened at both ends, which they lay on the ground; then, with a longer stick held in the right hand, they strike one end of it, to make it

bounce into the air, and, as it rises, hit it with considerable force. Competitions are held to see who can, by this method, drive the small object farthest. The game is much the same as our familiar “tip-cat.”

From Sunday Island I have already recorded a peculiar type of throwing contest, which the girls were indulging in at the time of our visit. Two rows of seven or eight each were standing a couple of chains apart, and, whilst facing one another, were bombarding their opponents with consolidated cakes of cow-dung. The missiles were thrown with considerable skill, even by the smallest girls, due allowance being made for the curvature in flight. It was part of the game to dodge the flying bodies, and, whenever a hit was recorded, a triumphant cheer would ring from the opposite side. In rushing wildly about the space, the light calico skirts of the playing damsels would fly high in the air, exposing their slender limbs beneath. In their eagerness to hit, and avoid being hit, they repeatedly exclaimed: “*Arre minya, arre minya.*” Some of the more experienced throwers showed their proficiency by using flat slabs of stone in lieu of the cakes of dung.

The games we generally refer to as “hide-and-seek” are known to the tribes of Australia. In one game a number of persons hide behind bushes and boulders and are sought by one or more children; in another an article is hidden by one and looked for by the rest of the party. The players endeavour to mislead the “seeker” by obliterating their tracks and substituting them by all sorts of “back-to-front,” “devil-devil,” and other deceptive tracks. When a “find” is made there is a loud, jubilant cry; and the “hider,” unless he can escape in time, is pommelled by the “seeker”; the players do not, however, run back to a crease or “home,” as we do in the European game.

Once one considers toys supplied by adults for the special benefit of children, the question resolves itself into an analysis of endless possibilities of creation achieved by the happy combination of inventive mimicry and lucid interpretation. It would be as futile to attempt an exhaustive discussion upon so big a subject as it would be to even try to describe all the artificial objects one classes as toys, however crude they might appear, which are manufactured for the purpose of entertaining and instructing the child. And, indeed, the young folks themselves are neither idle nor behindhand in augmenting the collection of playthings supplied by others.

Most of these articles are, however, made on the spur of an impulse and serve their purpose just for the time being.

One of the favourite occupations of the children of all Australian tribes is to build small brushwood shelters and wurlies for themselves to sit and talk in like grown-ups. Occasionally they occupy these miniature domiciles as “father and mother,” but more often as “father” or “mother,” with a number of “children” to “look after.”

When the “children” are not available as living playmates, inanimate objects, such as stone, bits of wood, leaves and flowers, are selected to take the “children’s” place. These receive names and are placed in a row before the “foster parent,” who talks to them and frequently changes them about. But the best part of the game is when one of the “children” is supposed to misbehave itself, and in consequence must receive a good flogging with a stick, kept handy for that purpose. At other times one of the objects is taken up, nursed, and spoken to most affectionately.

Here then we have the idea of the doll, simple though it be. But after all, simplicity in method, so far as the training of children is concerned, is perhaps the readiest means of stimulating the imitative, and with it the creative, force which Nature has endowed them with.

The doll is usually just a plain stick or stone, with perhaps some distinguishing feature upon it, like a knob at one end which represents the head. Occasionally it is painted with red ochre. Dr. W. E. Roth found that on the Tully River in Queensland a forked stick is chosen so as to permit the child fixing it on its neck like a mother carrying her baby, with its lower limbs dangling over the shoulders.

Imaginary fireless cooking is also a pastime the little girls never tire of. A shallow hole is scooped, into which a few handfuls of cold ashes are thrown; this represents the fireplace. Upon the ashes is laid a pebble, a leaf, or any other article which they make up their minds to “cook.” Having covered it with sand in the orthodox way, the girls sit and talk, whilst they make themselves believe the dish is in course of preparation. They invite each other to the prospective feast, each explaining what she is cooking; one might have a wallaby, another a lizard, and still another a yam.

Quite apart from accompanying their mothers on the regular hunting expeditions, the little boys often go out alone. They carry toy weapons, with which they say they are going to slay a kangaroo or anything else happening to come their way. In the Fitzroy River district the young hunters collect, or cut out of a gum-tree butt, several pieces of bark, dry or fresh, and shy these into the crown of a boabab (*Brachychiton Gregorii*), hoping to fell a nut or two. If they are successful, they proudly return to camp with their spoil and obtain permission to roast it at the fireside. The small bark missiles are looked upon by the boys as quite equivalent to the "*kaili*" (boomerangs) of their fathers; and there is no doubt they can throw them with greater skill. I have seen the little fellows stalk a flock of foraging cockatoo and, when within range, fling several of the toy weapons into the birds as they are rising; invariably one or two birds are brought to fall.



Rocking a child to sleep.

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The trimmed stalks of bullrushes and reeds make excellent toy spears, which are thrown with the heavier end pointing forwards and the thinner end poised against the index finger of the right hand. With these “weapons” the lads have both mock fights and mock hunts. In the latter case, one or two of their number act the part of either a hopping kangaroo or a strutting emu and, by clever movements of the body, endeavour to evade the weapons of the hunting gang.

One of the favourite pastimes of young and old among the Kukata is to play at “emu.” The players take a stick, about three feet long, and tie a bundle of grass, brushwood, or feathers to one end of it, to represent the neck and the head. The performer clasps this stick with both hands and holds it erect in front of him; then he bends his body forwards from the hips, whilst other persons cover him with skins, and tie a tussock of grass over his stern to indicate the tail. The actor next begins to walk around, as truly as possible imitating the actions of the bird he is representing. As he walks, he nods his “head,” while some of the children scamper round him in great excitement, others flee from him shrieking with terror. After a while he stops short, turns his head and shakes his body, finally running away in a zig-zag course. As he runs, he frequently imitates the peculiar deep note of the old emu and occasionally the shrill whistling cry of the young. A most amusing little incident, which I saw in connection with this game, happened near Mount Eba. A man, fully rigged as an emu, was entertaining the camp with some very clever, bird-like antics, when suddenly a vicious mongrel darted at the performer from a wurlie he was passing. The “emu,” without any deliberation at all, dropped its head, scampered across the ground, hotly pursued by the dog, and, in its terror, climbed the nearest tree.

The King Sound men construct miniature “*kaili*,” barely an inch in length, and practically straight, which they project, before the admiring eyes of their juvenile audience, by using their fingers only. The little toy is held between the second and third phalanges of the left index finger, so that a good half of its length projects above the hand. The inner tip of the right index finger is pressed strongly against the outer surface of the left thumb

and suddenly allowed to slip over the top edge and strike the projecting part of the toy. The little slab of wood is jerked into the air, whirls through space in a parabolic curve, and, when well managed, returns to the hand of the projector. The children often try this feat, but, with the exception of a rare fluke, never succeed; in fact, it is not every adult man who can do it.

On Sunday Island, small models of the raft ("*kaloa*"), locally used, are made for the children to play with. These toys are exact replicas of the craft described in a subsequent chapter and are neatly constructed in every detail.

A kind of dart is made by the children in the Northern Kimberleys of Western Australia out of the root ends of grass seed-stalks, six to eight inches long. These are held, one at a time, between the palms of the hands parallel to, and between, the middle fingers, beyond which they project but a fraction of an inch. In this position the hands are turned so that the fingers point towards the body. Then taking careful aim at an object, the child throws its hands vigorously forwards, at the same instant opening them and shooting the dart in the desired direction.

Among the Dieri, Yantowannta, and Ngameni, principally, and to a less extent among the Arunndta, Aluridja, and Kukata, a playing stick is found which is commonly known by the name of "*kukerra*." Although a toy, the men only were observed to use it, not only to amuse the children, but for the benefit of the whole camp. The *kukerra* is a slender, club-shaped stick made out of the Mulga. Its length is about three feet six inches, of which the thickened end occupies something like nine inches; the head, *i.e.* the swollen portion, is up to an inch or slightly more in diameter, whilst the "stick" is not thicker than an ordinary lead pencil; each end terminates in a blunt point. The Dieri *kukerras* are lighter and more slender than the Arunndta or Aluridja. The playing stick is seized at its thin end and, swinging it with a straight arm, it is made to strike a bush or tussock in front of the thrower; whence it bounds through the air in an inclined position, and, after striking the ground, glides along the surface in a snake-like manner.

Natives are fond of spinning any suitable objects which fall into their hands; small pebbles, gall-nuts, and the larger varieties of conical and bell-shaped eucalyptus fruits are all made to spin upon a level surface just to amuse the children. The Yantowannta, Wongkanguru, and other tribes of the

Cooper Creek region are very clever at moulding tops out of clay, with real pegs, upon which the toys revolve. These tops are undoubtedly an indigenous invention. The spinning is usually accomplished by rubbing the toy between the palms of both hands.

Skipping is indulged in by little boys and girls alike. A long vine is used by the Wogaits on the Daly River, which is swung to and fro like a pendulum by two of the players, whilst others jump over the line as it passes beneath them. The “rope” is not swung overhead.

At Engoordina, Arunndta children and women were noticed to entertain the tribe by artful tricks with an endless piece of string. By an intricate method of inter-looping and threading, a long cord, tied together at its ends, so as to form a complete ring, is transformed into different patterns of squares, triangles, and circles, the composite groups of which are intended to represent different natural objects. The string is held at different points, according to the complexity of the design, by the fingers, toes, and mouth of the performer; occasionally, indeed, the services of an assistant are required to support the pattern whilst it is being constructed. The more elaborate articles when completed resemble a loosely netted or knitted fabric, the plainer are more after the style of a few loops or meshes lightly held together. Some of the designs bear a recognizable resemblance to the objects they are intended to represent (*e.g.* birds, animals, men, etc.), others appear to be (to the European at any rate) rather far-fetched. Very often a complicated-looking design can be instantly reduced to the original piece of string by simply pulling one of its component loops or ends.

Dr. W. E. Roth has very ably described a series of such figures, made with one or two endless strings, which he found to be commonly constructed throughout north Queensland, where the tribes play a game resembling the European “cratch-cradle.”

Children are not allowed to attend many tribal ceremonies; consequently one does not often see them with their bodies decorated or ornamented. There are occasions, however, when their presence is tolerated, such as, for instance, at receptions to relatives who are returning from a fight or long hunting expedition. White earth or kaolin is invariably used for beautifying the appearance, it being maintained that it is unwise to apply much red, the token of blood, to the body of one who has not sacrificed some of his blood

during the course of such ceremonies as will elevate him to the status of the tribe's manhood. The kaolin is applied in the form of a thick paste in a series of thin lines. On Sunday Island these lines pass from the centre of the shoulder, on either side, diagonally to and along the breast-bone; and horizontally across the thighs and forearms. The face has a smear of white straight down the nose, and two semi-circular lines, which enclose the mouth and converge to a point opposite the ear on either side. The child in addition wears a belt of twisted human hair-string, from which pends a pearl shell ornament; it might also carry a plume of white cockatoo feathers in its hair. There are slight variations in the patterns and designs chosen for child-decoration both in the same tribal group and among different tribes, but no matter what part of Australia is considered, the effect completed is decidedly less elaborate than the complex and ornate colour schemes seen in the ceremonial displays of performing men. The children take no active part in such proceedings, but usually walk or stand about at points where they do not interfere with the proceedings in general. There is no objection to the child carrying one of the boomerangs of its father, but under no conditions is it permitted to handle a spear-thrower.

Children are early accustomed to discipline and obedience. They are not required to obey any but their individual fathers, tribal fathers, and tribal uncles. They need not pay heed to the orders of their contemporaries, but only to the word of such as they consider "grew them," that is, men of a previous generation to themselves. Women need not be obeyed by law, but, with the knowledge and sanction of a father, a mother can chastise and punish a child as much as she pleases. The father reserves the right to interfere at any moment.

The little girls accompany their mothers whenever collecting rambles are undertaken. They receive instructions in the methods of locating and gathering grubs, lizards, seeds, and roots; and during this time they are required to daily handle the yam-stick in the correct manner as shown them by their seniors. Subsequently they are taught how to clean, cook, and prepare the meals to be placed before the men.

The boys are early in life schooled in the practices of carpentry, so far as they are applied to the making and shaping of domestic utensils and weapons with the few crude implements at their disposal.

Further, they are instructed in the knacks and arts of handling and throwing weapons of chase, attack, and defence. The lads take to this instruction enthusiastically. For instance, whilst being taught the art of boomerang-throwing, one might daily see a youngster, even in the absence of his master, posing in the attitude demonstrated to him, without actually letting the piece of wood, which answers the purpose of a weapon, go out of his hand ([Plate XIV](#), 1).

Boys are not allowed to handle real boomerangs, spears, or shields before they have undergone the first initiation ceremony. If they did so, the offence would be looked upon as an insult to the dignity of the men who have qualified and are thus entitled to the privilege of carrying such weapons on parade. The offence is in fact, on a point of decorum, similar to the case of a fellow in the “rank and file” wearing the sword or insignia of his superior officer.

When, at a later stage, the elder boys of the Northern Kimberleys of Western Australia become well-skilled in throwing, sham-fights are arranged. Pieces of bark are broken from the mangroves, out of which the combatants make missiles resembling straight boomerangs. Sides are picked under the supervision of the men and the signal given to start. In a moment the air becomes alive with the whirr and buzz of the flying pieces of wood, which the youths throw straight at one another. Often severe gashes and wounds are inflicted upon the bodies of the “fighters,” but such are taken in good faith and looked upon as being part of the game.

As a means of self-defence and protection against such throwing-sticks and the small toy-spears previously mentioned, the Arunndta construct for their boys light bark shields. A piece of green bark is cut out of the butt of a eucalyptus, oblong-oval in shape and about two feet long and six inches wide. Two holes are cut in the central line of this piece, about six inches from either end, and through them two or three fairly stout, green twigs are stuck, from the under, concave surface, to form a handle. The points of these twigs stick out from the top surface, some two inches, but they are left to prevent the ends of the handle from slipping out. The bark is then bent in the required shield-shape and dried over a slow fire or in hot ashes ([Plate XIV](#), 1).

The girls, too, are encouraged to indulge in stick-practice to prepare them for the “*kutturu*” duels they will have to take part in, in later years. The principal mark is the foot, which each alternately tries to strike, while the other is “on guard” with her stick. At other times they stand face to face, with the palms of their hands pressed tightly together. Presently one voluntarily bows her head, when the other immediately gives her a severe crack over the scalp with the small finger side of her hands. Then the other has a turn; and the process may be repeated. The object of this strange procedure is to “harden” the head in anticipation of the real blows it will receive in time to come. To make the performance appear genuine, the girl, whose turn it is to strike, may be heard to feign a curse: “*Atutnia, arrelinjerrai!*”

The child’s mind is early imbued with the importance of hardening the body and nerve against pain, and thereby making the system less susceptible to the hardships of life, which they know to be inevitable.

The system of personal mutilation, described in a subsequent chapter, has to a certain extent been evolved for a similar reason.

Camp life brings many little accidents with it, but the Spartan principles which are cultivated lead to an almost complete ignorance of the existence of pain as might be brought about by small cuts or burns. Just for the sake of competitive amusement, the boys of the Kukata tribe take a live coal from the fire and lay it upon the naked skin of their forearm. A red-hot coal, about the size of a pea, is usually selected for the purpose and momentarily “cooled” or “blackened” by covering it with a handful of sand. The black coal is then placed upon the forearm at any suitable spot and touched with the red-hot point of a firestick. The coal on the arm immediately turns red again and in that condition is allowed to remain there until it falls to white ash. The first effect is naturally to raise a blister, but this is soon burnt through and the raw skin is exposed, upon which the coal gently fizzles. Whilst this is going on, the boy is seen to bite his lips together and to clinch the fist of the suffering arm, as if to suppress the pain. The lesion will, of course, leave a permanent scar. Some of the lads have many of such marks upon both arms, and they seem quite proud of them.

In the camps of any of the tribal groups throughout Australia, who are still enjoying an uncontaminated life, one might see captive birds and

animals temporarily tethered or kept for the amusement of children. Such are usually brought home by the men returning from their hunting expeditions. A young wallaby, for instance, is let go on an open flat and all the children set after it. They are not allowed to hurl stones or sticks after the fleeing game, but must retake it alive by the use of their hands only. Although it is against the rules to harm a captured animal, it is a curious circumstance that a native, even if grown up, invariably forgets to feed it, although entertaining the idea of keeping it alive. Occasionally, however, it happens that a creature survives and looks after its own needs; in this case the animal or bird becomes a real pet and is not made the object of children's coursing matches any more.

The King Sound natives catch the small ring-tailed opossums, which live in the mangroves, and hand them to their children. The Ponga-Ponga gins become very attached to these marsupial pets, which they carry about with them on their days' outings planted in the locks of their hair. The opossums seem quite contented to abide there whilst their mistresses are on the march and hang on by means of their claws and tail ([Plate X](#), 2). Occasionally one might even see an affectionate gin suckling her pet at her breast.

On Sunday Island several cockatoos were kept by a fishing party in their camp not far from shore. The birds had their flight feathers pulled and were allowed to roam about the country in search of food. The cockatoos seemed to regard the huts as their home, to which they invariably returned; they had, moreover, picked up many phrases of the aboriginal tongue.

Although the native animals and birds of Australia have always been, and still are, the daily object of aboriginal chase, it is a remarkable fact that great friendships are made between the hunter and his would-be prey when the latter is in captivity. Indeed, the instinctive fear of an animal or bird is ever so much greater when a white man approaches than when a native does. I have seen cases where semi-wild cockatoos, magpies, and other birds have allowed themselves to be handled by natives without much concern, but the moment a European attempted to do likewise, the bird would become unmanageable, terrified, and vicious. Partly domesticated birds seem to have a predilection for perching themselves upon the legs of their native masters when the latter are sitting or lying on the ground. Talking of instinct reminds me of the occasions I took aborigines to the Zoological Gardens to see the favourite attraction, viz. the "monkey-

wurlie.” Whilst European visitors were coming and going, the apes would appear unconcerned, phlegmatic, and blasé; but the moment the dark-skinned people arrived, the animals would instantly become electrified and bounce towards the iron bars, which they seized and shook frantically. The natives, on the other hand, would evince no fear, but endeavoured to edge as close as possible to the monkeys, although they had never seen one before. The monkeys, in appreciation of the coloured visitors’ benignity, would grin, wink their eyes, and make guttural noises.

At the time of my visit to the Forrest River Mission Station, a tethered monkey was kept on the premises. The natives regarded it as a real “little man,” and many of them, especially the children, were on excellent terms with it, but this could not be said of some of the old men. The monkey and the children were inseparable, and usually at play; but occasionally disputes arose which always ended in an open tussle, during which hands, teeth, and finger-nails were used. Although young humanity did not always fare best, the monkey would never take a mean advantage nor resort to extreme tactics. When it found itself victorious, it would jump triumphantly on to its perch and cry “*Arre Arre.*”

CHAPTER XII

THE DAY'S MARCH

Orders of the day—Selection of camp site—Feminine water carriers—Great variety of bark vessels—Skin water-bag—Bailers and drinking cups—Natural water supplies—Water-bearing trees—Modes of drinking.

“When another sun will come, and when he is still a piccaninny, Punya umberri (everybody) will walk to the big stone (hill), lying in the gum-trees, where Kuddoguddogu (a landmark) holds up the clouds of the Pindanol's country. Narrawiddi and Wetninnya will carry my angamma (bark-wrap with small personal belongings), and all other women will take many naramarragam (bark food-carriers) and fill them with yams on the way. Plenty water sits upon the ground. The men will run the kangaroo's track with me.”

Upon an order like this from one of the old men, the following day's itinerary is cast. Brief though it seems, it is sufficient because, although the chances of the coming expedition might widely separate the members of the group, they keep in constant touch with each other by signs and signals best known to themselves.

The site for a camping ground is thus always selected by one of the old men in authority. Preference is given, other things being equal, to a spot near to a natural water supply. There are, of course, numerous occasions when there is no water available. When, for instance, the natives are hunting in the sandhills during a good season, they either carry water with them for miles, or rely on the succulent parakylia and other water-holding plants.

It falls to the lot of the women to carry water upon such occasions. The fluid is contained in bark carriers of different designs, which they either skilfully balance upon their heads or carry under their arms. The water is kept from splashing over the sides, in the first place by the naturally graceful gait of the women; but, at the same time, an intentional addition of twigs and branchlets further checks any undue movement of the fluid which might be produced in the vessel during the march.

The Dieri, Yantowannta, Ngameni, Arunndta, Aluridja, Wongapitcha, and other central Australian tribes use shield or trough-shaped carriers cut out of the bark of the eucalyptus, shaped and hardened over the fire. The shield type is flat, with more or less open ends; the trough type has higher sides and ends, and is therefore more capacious. There is, however, no hard and fast division between the two. The surfaces of these are either smooth or longitudinally grooved with a stone scraper. The largest were observed on Cooper's Creek, measuring three feet in length, one foot in width, and five inches in depth, while those of the Arunndta and Aluridja are not quite so long and wide, but they may be deeper. The utensils go by different names, according to tribe and locality; three of the most commonly heard are "*mika*," "*pitchi*," and "*cooleman*." In addition to taking the place of water-holders, they are also used as food-carriers.

North of the MacDonnell Ranges, similar articles are cut out of solid wood, usually the Northern Territory Beantree (*Erythrina vespertilio*).

The Warramunga and Kaitish (or Kaitidji) tribes in addition make large canoe-shaped carriers out of similar material. Two varieties are met with. The first is more or less flat-bottomed with steeply inclined sides coming to a sharp edge at each end; the second is uniformly curved, shield-like, with all its sides standing at about the same level at the open end. The former is grooved longitudinally on the outside surface only, the inside being left in the rough; the latter is finely grooved on the inner, as well as the outer surfaces. Both types are generally painted over with red ochre. It is a decidedly laborious job to remove the wood, which originally fills the inside of this carrier, a fact which will be realized when one considers that it has all to be done by burning with live coals, and gouging and scraping with stone implements.

The Sunday Islanders take a rectangular sheet of bark of the woolly-butt eucalyptus, fold both ends for a distance of three or four inches, into pleats (like a concertina), and stitch them together with split cane. The utensil is used throughout the north-west coast as far as Cambridge Gulf.

On Bathurst and Melville Islands similar structures are made out of the bark of the paper-bark tree (*Melaleuca*). An oblong piece is bent upon itself lengthwise, both its ends folded, as in the previous case, and kept together by binding with cane or by spiking with short wooden pegs.

The same pattern, slightly modified here and there, is found along the shores and islands of the Gulf of Carpentaria and the Cape York Peninsula. We might say, therefore, that it occurs throughout the entire length of the north coast of Australia.

Another type, perhaps more food than water-carrier, is common on Melville and Bathurst Islands; it is made of a single piece of the “stringy-bark” eucalypt. An oblong sheet, say a good yard long and nearly half as wide, is freshly cut and folded transversely at its centre. The edges of both sides are pared down, laid flat, one over the other, and sewn or laced together with plain or “run-on” stitches. A row of slanting and overlapping stitches is often inserted along the open edge a short distance down; and occasionally part of the same edge may be cross-hemstitched and plastered with beeswax; the object of these stitches is to prevent the bark tearing along the fibres. The mouth of the carrier is nearly circular, or at any rate oval. Ordinarily the bark is left in its raw condition, but upon special occasions elaborate designs, consisting of circles, and other figures, with cross-hatched line-patterns, are drawn on the outer surfaces in red, yellow, white, and black.

An article is in use locally among the Worora at Port George IV, which perhaps interests us most on account of its similarity to the orthodox water-carrier employed by ourselves, viz. the bucket. What makes the fact more interesting still is that this unique type of water-vessel is found in a locality, than which even at the present time none other is further remote from civilization. The bark-bucket of the Worora, known vernacularly as “*wirrauwa*,” is beyond doubt an indigenous evolution. It is much like a bushman’s billycan in shape—a cylindrical vessel closed at one end and with a handle at the other, measuring from four to twelve inches in height, and from six to nine inches in width. A circular piece of woolly-butt bark is cut for the base, and this is surrounded by another sheet which forms the cylinder. The joints are carefully stitched together with threads of split cane, using a bone-awl to prick the holes; then melted resin from the eucalyptus tree is applied over the seam to render it water-tight. The edge of the open mouth may be strengthened by cross-stitching and applying resin. The handle is made of human hair-string, several pieces of which are threaded diametrically across the open end of the bucket, through holes previously made with a bone-awl, and tied. The outer surfaces of the vessel are often

painted. The usual device consists of alternate bars of red and white or red and black, joined at the top and bottom by horizontal lines of red; occasionally the whole surface may be splashed or daubed with white, or the above designs may be embellished with regularly spaced dots and “emu tracks.”

Lastly we shall briefly refer to the skin water-bag which is used (or has been used) by the desert tribes of central Australia, from central Western Australia to Western Queensland. A kangaroo, wallaby, euro, or dingo is killed and the animal's skin removed almost in toto by making a circular cut around its neck, and, whilst one or two men hold on to the head, others detach the skin from the carcass and pull it off inside-out. The neck-hole forms the mouth of the bag, but all the other openings are tied, stitched, or pinned together. The limbs are cut off near the paws, the tail near its root, and the resulting holes securely tied with string. The limb-pieces are tied together and act as straps to assist the native carrying the bag when filled with water.

To fill these vessels with water, bailers are available either in the form of specially constructed or of naturally occurring objects; no matter which they are, they usually also answer the purpose of drinking cups. Along the north coast of Australia the large melon shell is perhaps the handiest; it is either used as it is found or its inner whorls and columella are broken away, leaving just the spacious outer shell to hold the water like a bowl. The same remarks apply to the large *Fusus pricei*, and other molluscs.

The Narrinyerri and other tribes south of Adelaide used human calvaria as drinking vessels. The facial skeleton of a complete skull was broken away so as only to leave the brain-box; and this held the water.

The broken shells of the large boabab nuts are similarly used in the Northern Kimberleys of Western Australia, and now and then the broken shell of the emu egg also makes a very serviceable cup.

A miniature bark-cooleman is constructed by the Wongapitcha, Aluridja, and Arunndta tribes, like that described on [page 92](#), about eight inches long and half as wide, which serves the purpose of a bailer, drinking-vessel, fire-shovel, and special food-carrier. It is strongly convex lengthwise, and therefore comparatively deep.

The Bathurst Islanders tear or cut a piece of bark from a tree, usually the ti-tree or “paper-bark,” out of which they fashion a cup. The piece of bark measures about twelve inches in length, and eight in width. It is first folded longitudinally at about its middle, and then both ends of the doubled piece are folded transversely at about one-quarter the whole length. The overturned parts of the inner sheet of the first fold are clasped between the fingers on the inside and the thumb on the outside, when the cup is ready for use.

The natural water supplies available over so vast an expanse of territory as is embraced by the continent of Australia and its subjacent islands, occur, as one might have expected, in great variety. There is no need for us to consider such familiar supplies as rivers, creeks, lakes, billabongs, waterholes, and springs; we shall just briefly consider a few of the more uncommon cases, which are of special interest. The native has a wonderful instinct for locating hidden supplies of water; and many a European wanderer has perished in the Australian bush, within a stone’s throw of the life-saving fluid, all for the want of that gift, which to the primitive inhabitant of the desert central regions means his very existence.

Along the superficially dry, sandy beds of “rivers” in arid Australia, he is able to pick sites, at which, by shallow digging with his hands and yam-stick, he can in quick time produce a “native well,” sufficient to supply the needs of all the camp. The water is often exposed within a foot or two of the surface, but at times he has to dig to a depth of from five to six feet, which so far as my experience goes seems to be the limit. When not in use, or when the camp moves on, the natives always take care to cover the mouth of the well in order that wild animals cannot reach the water and pollute it. When the well is deep, its sides are made secure with pieces of timber and brushwood, and cross-pieces are left to serve as a ladder whereby the native can attain the water. Similar wells are constructed in the catchment basins adjacent to the hills.



PLATE XII

Juvenile Types.

1. Full-face, female, Wongkanguru tribe.
2. Profile, female, Aluridja tribe.

Rock-holes in granite (Musgrave Ranges), quartzite (Krichauff Ranges), or limestone (Nullarbor Plains) are favoured on account of the cool, clear water which they generally contain. Where such are of a cavernous nature, and opening from a bare inclined surface, the natives often build a small bank of clay across the slope to direct the flow of water, resulting from a downpour, towards the hole. A unique variety of this type was discovered by us at Ullbönnaenna, east of the Musgrave Ranges. Through a hole in the barren slope of gneiss, a communication has been established by atmospheric denudation with a small reservoir below. To obtain the water contained in it, the natives keep a broom-shaped piston handy, with which they pump the fluid to the surface, as required. The piston is merely a rod, about five feet long, round one end of which a bundle of brushwood is

securely tied with string. The size of the brushwood bundle is such that it exactly fits the hole in the rock (about six inches). The implement is inserted, brushwood foremost, and slowly pushed down into the water, and, after a short interval, quickly withdrawn again. The water, which had collected behind this “piston-head,” is thereby forcibly ejected, and is collected inside a small enclosure of clay built around the hole.

The aborigines are most particular about preserving their water supplies against pollution, especially where such is brought about by excremental and decaying animal matter. In the Musgrave Ranges, the natives did not in the slightest object to our camels being watered at the supplies they were dependent upon, but when the animals dirtied the rocks above the hole and there was a chance of the discharge running into the water, they immediately set to and built a barrier of earth to intercept the flow before it reached the hole.

In the Northern Kimberleys of Western Australia valuable pools of water collect upon the boabab trees. The branches of this species surround the “gouty” stem in a circle at the top, like the heads of a hydra, and by this means form a concavity between them, which is capable of storing a considerable volume of cool, clear rain-water. To reach this water, the natives construct ladders by simply driving a series of pointed pegs into the soft bark of the tree one above the other.

Certain desert trees like the Currajong have the property of retaining considerable quantities of water in their tissues, even under the worst conditions of drought, for periods of many weeks or months. This water the native obtains by felling the tree and setting fire to the crown; the water oozes out from the cut trunk and is collected in bark carriers. The “Bloodwood” (*Eucalyptus corymbosa*) has similar properties.

In the Denial Bay district a remarkable mallee (*Eucalyptus dumosa*) grows, whose roots supply the natives with water. This mallee is a rather big tree, which lives in association with other smaller species of the same genus. It appears, also, that not every specimen of the particular species referred to contains water; it requires the experience of an aboriginal to predict which of the trees is likely to carry such. Having selected his tree, the native proceeds to expose one of the lateral roots, which grow in the sand at no great depth from the surface. The root is then cut in two places,

three or four feet apart, and lifted from the ground in a horizontal position; finally it is turned on end over a bark cooleman, when water, clear as crystal, begins to drip from the lower end into the vessel. Sufficient water can thus be collected to sustain the camp, if need be, for even a longish period.

Other trees in central Australia are known to possess similar properties though to a lesser extent, as for instance the Needle-Bush (*Hakea lorea*, var. *suberea*).

When, after a good day's march, the natives have the luck to strike a big waterhole, each of the party immediately quenches his thirst by literally "filling up" on the spot. Different methods are adopted to accomplish this. Some prefer to remain prone at the water's edge, whilst others wade into the deeper parts of the hole, and, placing their hands upon their knees, stoop and drink off the surface.

The Cambridge Gulf tribes pull a long-stalked leaf of the water-lily, which, after they have cut it at top and bottom, acts like a tube and permits them to suck the cooler fluid from a depth.

After quenching his thirst, a native will on a hot day often cool his system by pouring water on to his head. The women-folk and children are very fond of splashing the head with cold water, which they might do repeatedly during the day, provided the opportunity is afforded. Bathing during the heat of the day is also commonly enjoyed by all the northern tribes, especially those resident in the tropics where water is abundant.

CHAPTER XIII

CAMP LIFE

Preparation of camping ground—The bed and its coverings—Sleeping order—Brushwood shelters—Various habitations—Vermin-proof platforms—Common position during sleep—Friendly meetings and salutations—Sitting postures—Sense of Modesty—Bird-like attitude—Gins procure firewood—The campfire—Methods of cooking—Fire-shovels—Fire-stick—Fire-whisk—Fire-saw—Women the recognized transport agents—Care of weapons—Sundry occupations while in camp—Absence of wearing apparel—Pubic coverings—Cosmetics—Hair-belts—Pristine philosophy—Removing thorns—The aboriginal loves his dog—The dingo.

Having arrived at the chosen camp-site, each family group at once busies itself clearing a patch of ground of any obstacles, like stones and lumps of earth, the biggest of which are picked up by hand or crushed by foot, the smaller brushed aside with the sides of the feet. In addition, the women may be told to sweep the ground with branches to clear it of grass-seeds and burrs. Should it be that the spot is only reached after dusk, the natives set fire to one or two dry bushes, the glare of which supplies them with the necessary light.

Each adult scoops a “bed” for himself on the sand, and lights a small fire on one or both sides of it.

The northern coastal tribes very often spread sheets of paper-bark over the sand, and they might also cover their bodies with similar material. Should the mosquitoes become a great nuisance, one often sees them completely covered with sand.

The south-eastern tribes of Australia, including those of the River Murray and Adelaide Plains, used skins and rugs made of kangaroo and opossum skins, neatly sewn together, to lie upon and under.



1. The game of "gorri," Humbert River, Northern Territory.



2. A "Kutturu" duel, Aluridja tribe.

PLATE XIII

1. The game of "gorri," Humbert River, Northern Territory.

2. A "Kutturu" duel, Aluridja tribe.

The children sleep with or close to their parents. When an aboriginal has more wives than one, his camp is subdivided according to their number, and

he sleeps with his favourite.

The strangest conditions reign on Groote Island in the Gulf of Carpentaria, where the women practically live apart from the men during the whole of the day, and only come into camp after sundown to deliver the food supplies they have collected over day. When on the march, every adult female carries two big sheets of paper-bark with her, which she holds with her hands, one in front and another behind her person. Whenever a stranger approaches, they duck behind these sheets of bark, as into a box, for cover.

No matter when or where an aboriginal camps, he constructs a brushwood shelter or windbreak at the head-end of his resting place. This consists of a few branches or tussocks stuck in the ground or piled against any bush, which might be growing upon the patch of ground selected. Under ordinary circumstances, this is the only shelter erected.

Even under the best of conditions, the night's rest of an aboriginal is hard, and at times very cold and wet. It is not an uncommon experience for a person to sit up part of the night, hugging a fire, and when the sun is up to lie in its warmth to make good the sleep lost.

During a run of wet weather or when the camp is to be of a more permanent nature, different kinds of structures are erected, or already existing habitations selected, which will afford a better shelter than the crude structures referred to.

Natural caves or shelters beneath a sloping wall of rock, although frequently chosen as a mid-day camp, are not favoured on account of the superstitious dread of the evil spirit, whose haunts are supposed to be in the rocks. Caves are in any case only occupied during the heat of summer, the rocks being considered too cold to sit and lie upon in winter. If possible, a tribe will always make for the sandhill country in the winter, the sand making a very much softer and warmer bed. Such caves as are regularly occupied almost invariably have the walls and ceiling decorated with ochre drawings.

The opportunity of camping under large hollow tree-trunks, when available, is never neglected in wet weather. In the southern districts, as for instance the Adelaide Plains and along the River Murray, the large red-gums, especially such as have been partly destroyed by a passing bush-fire,

supply the best covers of this description, whilst on the north coast of Australia the boabab occasionally becomes hollow in a like way, and makes a very snug and roomy camp.

Huts are constructed after different patterns according to the materials available. In the Musgrave Ranges, as in most parts of central Australia, the usual plan is to ram an uprooted dry trunk of mulga into the sand in an inverted position, so that the horizontal root system rests at the top, generally about five feet from the ground. Making this the central supporting column, branches of mulga and other bushes are placed in a slanting position against it, so that they rest between the roots at the top and form a more or less complete circle at the base, measuring some eight or nine feet in diameter. An opening is left, away from the weather-side, large enough to permit of free access. The spaces and gaps between the branches are filled with small bushes, tussocks, and grass, and on top of it all sand is thrown.

Very often the branches are placed around a standing tree for a central support, and now and again they are simply made to rest against one another in the required conical fashion.

Roof-like shelters are made by piling branches and brushwood either upon the overhanging branches of a tree or across two bushes which happen to be standing close together.

On Cooper's Creek, in the extreme south-western districts of Queensland, these huts are more carefully constructed. A solid, almost hemispherical framework is erected consisting of stout curved posts, with a prong at one end, so placed that the prongs interlock on top and the opposite ends stand embedded in the sand in a circle. Vide [Plate XV](#), 1. The structure is covered with the long reeds that abound along the banks of the large waterholes of the Cooper. Some of the huts are indeed so neatly thatched that they have quite a presentable appearance.

The eastern Arundta groups, in the Arltunga district, cover a light framework of mulga stakes, erected after the general central Australian pattern, entirely with porcupine grass ([Plate XV](#), 2).

At Crown Point, on the Finke River, other groups of the same tribe cover their huts with branches and leaves of the Red Gum.

When camped on the great stony plains or “gibbers” of central Australia, it is often very difficult to find a suitable covering for the huts, the vegetation being either unsuitable or too scanty. On that account the Yauroworka in the extreme north-east of South Australia utilize the flat slabs and stones which abound in that locality to deck their more permanent domiciles with. The supporting structure must, of course, be made particularly strong to carry the weight of the stones. The crevices between the stones are filled with clay to render them water-tight, and earth is banked up against the base of the walls both inside and outside.

Along the north coast of Australia, from the Victoria River to Cape York, the prevalent type of hut is a half-dome structure, whose frame consists of a series of parallel hoops, stuck into the ground and held in position by a number of flexible sticks tied at right angles to the former with shreds of Hybiscus bark. The ends of the cross-pieces are poked into the ground on that side of the framework which will be opposite the entrance of the hut when completed. The hoops are made of slightly decreasing size from the entrance towards the back, and so correspond in height with the upward curve of the cross-pieces. This skeleton-frame is covered with sheets of “paper-bark” (*Melaleuca*) and grass; and the floor of the interior is carpeted with similar material; a small space is however left uncovered to hold the fire. Such a hut measures about five feet by five feet at the base, and is four feet high. In Queensland palm leaves may take the place of the paper-bark sheets.

In districts where the mosquitoes are very troublesome, the dome is completed by erecting hoops on the open side as well, and making the cross-switches long enough to be lashed to them all and to be stuck into the sand at both ends. The whole structure is covered with bark, but three or four small holes are left along the base for the people to slip in by; and a ventilation-hole is left at the top of the dome to allow the smoke to escape from the fire, which is burned inside to keep out the insects.

Provisional rain-shelters are made by cutting a big sheet of bark from one of the eucalypts, usually the “stringy bark.” To do this the bark is chopped through circumferentially on the butt in two places, about seven or eight feet apart, slit vertically between the two incisions, and removed by levering it off with two chisel-pointed rods. The sheet is folded transversely at its centre and stood upon the sand like a tent.

At times the sheet of bark is simply laid length-wise against two or three sticks previously stuck into the ground. In the more durable structures of this type, two forked poles are rammed into the ground so that they can carry a horizontal piece after the fashion of a ridge-pole of a tent. Against the latter then are stood several sheets of bark at an angle of about forty-five degrees.

Occasionally a scaffold is erected with four poles and cross-pieces at the corners of an oblong space, and sheets of bark are then laid across the top. The sheets are of sufficient length to hang over the sides of the frame so that their weight bends them into an arch along the centre.

Commander Lort Stokes found similar structures near Roebuck Bay in the north of Western Australia, but in place of the bark they there had a slight, rudely-thatched covering.

In districts where thieving dogs, ants, or other vermin become troublesome, the men construct platform-ladders, upon which any reserve supplies of meat are laid. Hawks are the greatest nuisance in camp. It is astonishing with what fearlessness such birds fly right into camp and swoop the meat from the natives. As a protection against theft of this description, the campers cover their stores with branches.

The position favoured during sleep is to lie upon one side, with the legs drawn up towards the stomach and bent in the knees. The head rests upon one or both hands; should one hand not be so occupied, it is usually placed between the closed thighs. At Delamere in the Victoria River country, the natives were observed to have convex pieces of bark in their possession which were used as head-rests during the night.

It goes without saying, of course, that the aboriginal might at any time change his position during sleep to one of the many commonly adopted by European or other people.

During the warm summer months, the campers are up with the first glimpse of dawn, but when the nights become cold, they often remain huddled by the fires until the sun is high up in the sky.

When a messenger or visitor approaches a camp at night, he will not do so without announcing his arrival in advance by loudly calling from afar to the groups at the fireside. Should a person be discovered prowling the

surroundings of a camp, without having heralded his coming, he runs grave risk of being speared, on the chance that he be on no good business.

In the way of salutations, hand-shaking and kissing are unknown, but when two friends meet it is quite the usual thing for them to walk together for a while, hand-in-hand. When a person, who has been long absent, returns to camp, everybody is so overcome with joy that he starts crying aloud as if his joy were grief.

On the occasion of friendly Arundta groups visiting, all members of both parties, male and female, approach each other with their spears, boomerangs, shields, and fighting sticks. The visitors first sit down in a body while the others walk around them, in a widening course, flourishing their weapons high in the air and shrieking with joy; later they return the civilities by acting similarly.

When seating himself, an aboriginal always prefers the natural surface of the ground to any artificial or natural object, which might serve him in a manner suggestive of a chair. Rocks and fallen tree-trunks might occasionally be used, and children are sometimes seen sitting upon the lowest big branches of trees, especially if they slope downwards to the ground. But even in these cases they rarely allow the legs to dangle, preferring to draw them, bent in the knee, close against the body, and usually with the arms thrown around the legs or resting upon the knees.

The men use the same method, when squatting at ease upon the ground, keeping their thighs apart, heels touching and close against the buttocks, with their elbows resting upon their knees and their hands usually joined in front. When the hands are to be used, the sitter acquires greater stability by placing the feet further apart and swinging the arms over the knees.

Another common posture is to double the shins under the thighs and rest them half-laterally upon the ground. This method is frequently combined with the previously mentioned by holding one leg one way and the other the other.

From either of these positions, the sitter may change by tucking the shins well under the thighs and rolling on to the side of one of his thighs.

Again, he may change by simply stretching his legs forward full length.

These methods are made use of by men, women, and children alike. Unless it be that the person prefers his legs to remain in close apposition, whilst squatting in any of the positions indicated, he will endeavour to hide his shame behind one of his feet. This is particularly characteristic of the women, and their natural sense of decency is prettily described in the narration of the voyage in search of La Perouse as follows: "Though for the most part they are entirely naked, it appears to be a point of decorum with these ladies, as they sit with their knees asunder, to cover with one foot what modesty bids them conceal in that situation."

By their method of standing at ease on one leg, the natives of Australia have evolved a remarkable posture which reminds one of birds. In this position, a man rests the sole of his unoccupied foot against the knee of the standing leg, and usually props his body with a spear-thrower ([Plate XVI, 2](#)).

Strictly speaking, it is the duty of the women to gather firewood, although very often, when there is a supply close at hand, the men will also drag a few logs to the family camp. The women, on the other hand, are required to collect sufficient to keep the fire going, during the day for cooking purposes and during the night for warming and lighting purposes. At times this entails weary searching and long-distance marching. We have already referred to the way they lift the pieces of wood from the ground, between the toes of one foot, to the hand on the opposite side, by passing the piece behind the body; the same hand next stacks the wood upon the head, where the other holds and steadies it. By this method, the gin has no need to stoop, and can in consequence build up astonishingly high piles of wood upon her head ([Plate XVII](#)). A small pad is usually first laid upon the head to prevent the scalp from chafing and the wood from slipping. Arrived at the camp, a gin throws her load to the ground and breaks the longer pieces across her head with her hands.

A native's idea of a good fire is to keep it as small as possible, but, at the same time, to derive a cheerful glow from it. His opinion of the European traveller's camp-fire is that it is so ridiculously big that one cannot lie near to it, without being scorched. In the winter the native often selects a large dry log if available and keeps this aglow at one end throughout the night; in the absence of such a log, he will at frequent intervals find it necessary to attend to his fire during the cold hours of the night. So diligently, indeed,

does he nurse his fire that his eyes often become inflamed in consequence of the continued irritation by smoke when he fans a smouldering flame with his breath.

A small fire like this, especially when it has burned for some time, is quite sufficient to cook all the smaller articles, which constitute the daily bill of fare, as for instance roots, tree-grubs, and lizards.

When, however, big game like a kangaroo is to be prepared, larger fires are essential and special culinary rules observed. A method, which has been in use practically everywhere in Australia, and is still found in use among the uncontaminated tribes, is to burn a big fire for a while upon a sandy patch, and then to lay a number of flat stones upon the red-hot coals and cover everything with sand. After a while the sand is scraped aside and the oven is ready for use. In the Northern Territory the stones are substituted by brick-like lumps broken off one of the tall termite-hills, which abound in that country.

In south-eastern Australia and along the River Murray the stones selected are usually composed of travertine or limestone.

In the Musgrave Ranges oven-stones are not in use, the game being simply laid upon, and covered with, hot ashes and sand.

“Big” cooking is done by the men, whilst the women are required to attend to the preparation of all smaller articles like yams, grubs, and seeds.

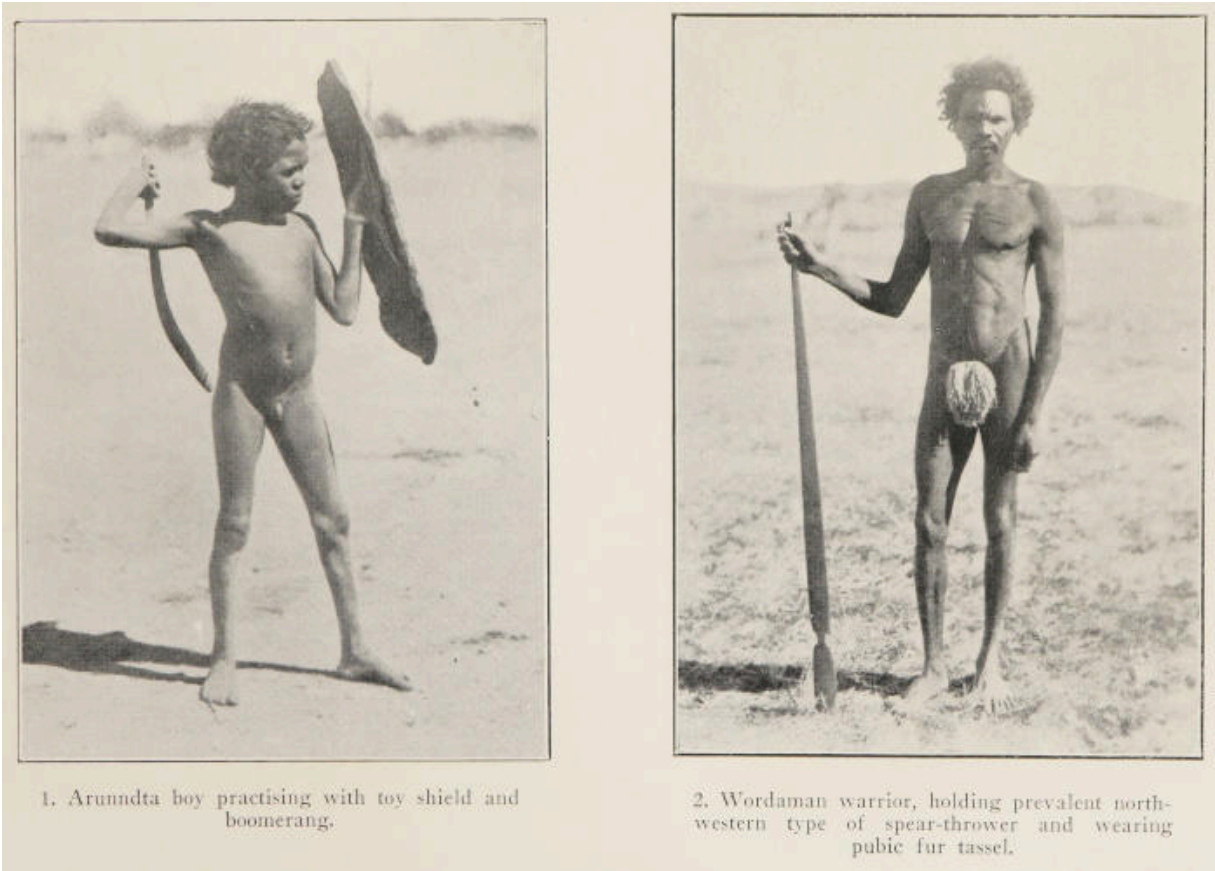


PLATE XIV

1. Arunnda boy practising with toy shield and boomerang.
2. Wordaman warrior, holding prevalent north-western type of spear-thrower and wearing pubic fur tassel.

Slight variations are noticed in the method of cooking a kangaroo according to the locality. In the Musgrave Ranges, the animal is prepared whole. The skin is not detached, whilst the bony paws may, or may not, be removed beforehand in order to secure the sinews, which are used all over Australia for tying purposes in the manufacture of their implements and weapons. The carcase is laid upon its back and completely covered with hot ashes and sand, and thus permitted to cook. When sufficiently, and that according to our ideas often means only partly, cooked, the skin can easily be removed. The belly of the baked carcase is cut open and the gut laid aside. What remains is then pulled to pieces by hand and the portions distributed among all members having a right to such. The meat is tender and juicy when cooked this way. Even the intestines, after their contents have been squeezed out by the aid of two fingers, are eaten by the less

privileged members of the tribe. The Arunndta call the last-named dish "*uttna kalkal*." Most of the bones, if not crushed between the jaws, are shattered between two stones and the marrow eaten.

The Wogait and other tribes on the north coast break the legs of the animal and tie them together in pairs with shreds of Hybiscus bark. The carcase is opened at one side to remove the entrails, and an incision is also made into the anus to clear it. In the case of a kangaroo, the tail is cut off and cooked separately in ashes. The skin is not removed. When thus prepared, the animal is transferred to an oven as described above and first covered with a piece or two of "paper-bark," then with hot sand and ashes.

In order that they may readily scoop out a fireplace, scrape the sand to and from the roast, and handle the meat, vegetable, seed-cake, or whatever the article in the oven might happen to be, the Northern Kimberley tribes have invented a long wooden shovel. This is a slightly hollowed blade, about three feet in length, four inches wide at the lower end, and decreasing in width at the hand end. The implement is mostly cut out of a sheet of eucalyptus bark.

The central tribes generally make use of a discarded or defective boomerang, which seems to answer the purpose very well.

Perhaps the most important article a native possesses is the fire-stick. No matter where he might be, on the march or in camp, it is his constant companion. Important as it is, the fire-stick is only a short length of dry branch or bark, smouldering at one end. It is carried in the hand with a waving motion, from one side to another. When walking in the dark, this motion is brisker in order to keep alive sufficient flame for lighting the way. A body of natives walking in this way at night, in the customary Indian file, is indeed an imposing sight. Directly a halt is made, a fire is lit, to cook the meals at day and to supply warmth during sleep at night. When camp is left, a fresh stick is taken from the fire and carried on to the next stopping place.

In consequence of carrying the fire-stick too close to the body during cold weather, most of the natives have peculiar, irregular scars upon abdomen and chest which have been caused by burns. The Wongapitcha call these marks "*pika wairu*."

If by accident the fire should become extinguished, a fresh flame is kindled by one of the methods depending upon the friction and heat which are produced by rubbing two pieces of wood together. Two methods are in use, all over Australia and the associated islands to the north; the one is by means of the “fire-whisk,” the other by the “fire-saw.”

In the first-mentioned case, two pieces of wood are used, usually a flat basal piece, with a small circular hollow in its centre, and a long cylindrical stick, rounded at one end. The native assumes a sitting position with his legs slightly bent in the knees. He places the flat piece of wood upon the ground and holds it securely beneath his heels. The rounded point is now inserted into the small hollow, and, holding the stick vertically between the flat palms of his hands, the native briskly twirls it like a whisk ([Plate XXII](#)). The twirling action is, however, not backwards and forwards, but in one direction only. After a while, the wood dust that accumulates by the abrasion begins to smoke, then smoulder. Suddenly the native throws his stick aside, and quickly stooping over the smoking powder, gently blows upon it whilst he adds a few blades of dry straw or other easily inflammable material. When the smouldering dust has been coaxed into flame, more straw and twigs are added, then larger pieces of wood, until eventually a blazing fire results.

Often a small notch is cut at the side of the central hollow in order that the smouldering powder might find its way down to a piece of bark placed beneath the basal stick, and there, by the aid of gentle blowing, ignite the dry grass, which was previously laid upon the bark for that purpose.

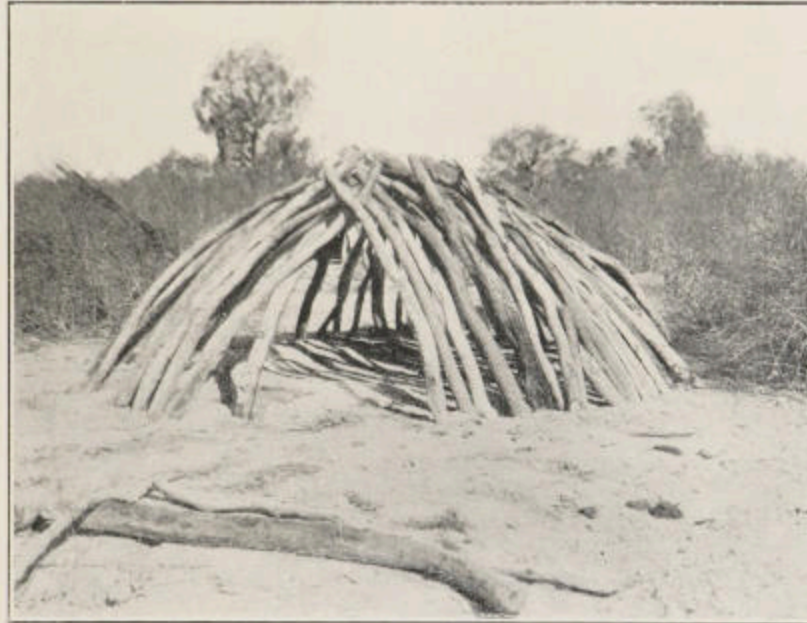
Usually, during the process of twirling, a little fine sand is placed upon the hollow to increase the friction. A curious practice was observed among the Larrekiya at Port Darwin, which seems to be opposed to the friction principle. When the fire-maker has, by careful twirling, adjusted the point of the upright stick, so that it fits nicely into the hole in the basal piece, he squeezes a quantity of grease from the sebaceous glands of his nose, which he scrapes together with his finger-nails and transfers in a lump to the ankle of his left foot. Then he resumes the twirling, and, so soon as the stick begins to smoke, he applies its hot end to the grease, which spreads itself over the point. The stick having been thus lubricated, the process is continued as before.

The central tribes, like the Dieri, Wongapitcha, and Aluridja, usually make the basal piece short and flat, and wider than the twirling stick. The Dieri select needlebush for the twirling stick, and Hack's Pea (*Crotalaria*) for the basal piece. The other tribes mentioned combine the needlebush wood with that of a mulga root.

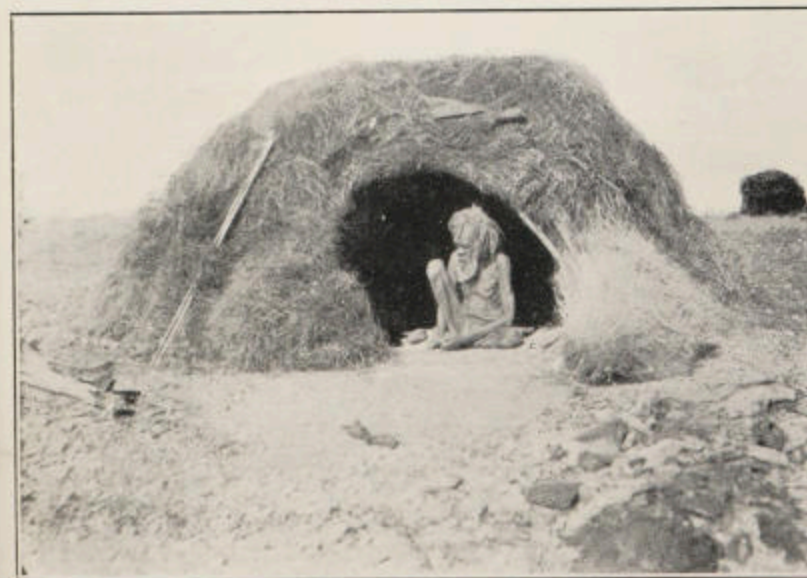
The northern tribes almost invariably employ two long sticks, one of which has a rounded point at one end, the other a series of shallow circular pits, into which the point of the twirling piece just described fits when the implement is in use.

The Mulluk-Mulluk, Ponga-Ponga, and other tribes of the Daly River district carry a number of these sticks about with them, especially in the rainy season, when there is always a chance of the fire-stick being extinguished by an unexpected tropical deluge. The fire-making apparatus is carried in a receptacle, which consists of a single segment of a bamboo, with a septum at the bottom. The sticks are stuck into this cylindrical holder, which keeps them perfectly dry even during a prolonged season of rain, after the fashion of arrows in a quiver.

Fire-making implements are carried by the men, whilst the fire-stick is, as often as not, carried by the women also. As we have already seen in connection with the carriage of water, it is the concern of the women to undertake the transport of the camp-belongings from one site to another. When moving they pack themselves with the domestic implements, collecting-vessels, personal paraphernalia, and their infants, whilst their husbands burden themselves only to the extent of a few spears, a spear-thrower, and the fire-producing sticks just mentioned. The men declare that it would be most unwise to be burdened with any impediment themselves, while on the march, because at any moment, and when least expected, they might be pounced upon by an enemy, who would make good use of their unpreparedness. So also, should game of any kind suddenly come into view, the men, who are the recognized hunters, must always be ready for quick action, or in a position to take up the chase immediately. Hence it comes about that in the best interests of the tribe the women are required to undertake the transport.



1. Framework of hut in course of construction, Cooper's Creek, S.W. Queensland.



2. Hut decked with porcupine grass, Arltunga district.

PLATE XV

1. Framework of hut in course of construction, Cooper's Creek, S.W. Queensland.

2. Hut decked with porcupine grass, Arltunga district.

The men spend hours at a time in camp making or sharpening spears. When, moreover, the weapons are not in use, the hunters are most

punctilious in preserving them from harm. The rule of the camp is never to lay a spear upon the ground for any length of time for two reasons; firstly, to prevent it from warping, and secondly, to eliminate the risk of breakage by somebody carelessly walking on to it. For these reasons the men, when camped, always take the precaution to stand their spears in a more or less upright position against the entrance of their huts, or against any bush or tree which happens to be growing close at hand.

Whenever possible, the opportunity of a sojourn in camp is seized for conducting a festive dance and song. For this purpose head-gears and other decorations have to be manufactured, plumes and permanent ornaments renovated, and pigments prepared to adorn the persons taking part in the performance. Upon such occasions a native is never seen idle.

Quite apart from preparing himself in anticipation of an extraordinary event, however, a native might take advantage of a delay in camp to manufacture an article with which he can barter with an adjoining tribe. The most common article thus prepared is ochre. Many tribes do not possess a deposit of this natural pigment and they are most anxious to do business with their neighbours. In exchange for the ochre, they offer such things as weapons, pitjuri-leaf, fish, or yams. The ochre is carried to its destination either in lumps or prepared as a fine powder. In the former case it is packed in small fibre or fur-string bags; in the latter the powder is wrapped in thin sheets of bark and tied together with string into neat parcels. The preparation of the ochre-powder entails much grinding between the stone surfaces of a hand-mill. The ochre is used for decorating the body, as well as implements and weapons.

If now we enquire into the method of dress adopted by the aborigines of Australia, we find that most of the tribes originally walked about in the nude, and, apart from a few small personal decorations, possessed nothing in the shape of a covering which might be described as a dress. The advent of civilization has largely interfered with this ancient practice.

The now practically extinct south-eastern tribes, including those along the River Murray, used more of a body-covering than any others. Opossum, wallaby, and any suitable marsupial skins were collected and carefully sewn together, and with these rugs the natives could, if need be, cover the greater part of their bodies.

By far the most common mode is to tie a string around the waist, from which is suspended a tassel to cover the pubes. These tassels vary considerably in size according to the tribe which wears them. The smallest are found among the Wongapitcha in the Mann and Tomkinson Ranges; the appendage is there only worn by the men and is barely large enough to cover the part. It is made of human hair strands fastened at the knot of the tassel directly to the pubes; the covering is known as the "*moiranje*." As a general rule, it might be said that the northern tribes have larger coverings than the central, although the Yantowannta and other Barcoo River tribes wear them as large as any. Even the Arunndta and Aluridja at special functions suspend large pubic tassels of fur-string from the waist-band.

The northern type of tassel consists of a great number of strings, usually of opossum fur, bound to a central piece, which is attached to the waist-band, either by two separate terminal strings, or by means of one single tie from the top of the tassel. The completed covering hangs from the waist like an apron. This type of pubic tassel is known to all north-central and northern tribes, and might be worn by either adult male or married female.

In place of the tassel a small sheet of the paper-bark is popular amongst the north coastal tribes. A narrow strip, from eight to twelve inches long, is folded transversely at its middle and hung over the waist-band, from which it pends like an apron as described of the tassel. Captain Matthew Flinders is perhaps the first European to have observed this custom, as far back as 1803, at Caledon Bay, where he observed a girl wearing "a small piece of bark, in guise of a fig leaf, which was the sole approximation to clothing seen."

The women of Bathurst Island carry folded sheets of the paper-bark or large food-carriers about with them, which, upon the approach of strangers, they hold in front of their person.

Along the whole length of coast line of north Australia, the large shell of the pearl-oyster is made use of as a pubic covering. Even among such tribes as live remote from the sea, one may occasionally find the shell so used, in which case, of course, it has been acquired from a coastal tribe by barter. In order to hang the shell, two holes are drilled through it near the hinge line, at the top, and a string passed through them, with which it is tied to a belt. The rough exterior surface of the shell is ground smooth; and it is this side

which lies against the body. The nacreous inner surface is frequently decorated with either painted designs or carvings subsequently tinted with ochre. The Sunday Island natives are especially adept at this type of decorative art, which will be referred to later.

Although the aboriginal does not wear much clothing, he is very particular about regularly anointing his supple skin. This precaution no doubt gives him greater protection against the changes of weather than all the modern ideas of clothing could do. What he principally applies is fat of emu and goanna, and on the north coast that of some of the larger fish as well. The emu in particular, and especially during a good season, accumulates masses of fat under its skin, which are readily removed, when slain by the hunter. This grease the native rubs over the whole surface of his body to shield the skin from the painful sting of the broiling sun and of the arid wind. In addition he covers certain parts of his body and face with red ochre and charcoal, both for cosmetic and protective purposes. The application of coloured pigments for purely decorative and ceremonial purposes will be discussed later.

Hair-belts are worn by young and old, male and female. Children have only one or two twisted fur-strings tied around the waist. Among the coastal tribes of the Northern Territory, men wear belts made of twisted human hair. A skein of about thirty strings is tied at two points diametrically opposite, and, making these the ends, the sixty strings are loosely twisted into a hank about two feet in length. The belt thus completed is tied around the waist with a piece of human hair-string. The article is of practical use since it permits of carrying various implements and weapons, which a man sticks between the belt and his body. A Wogait warrior was seen with a tomahawk thus placed at the back of his body; to stay the swinging of the handle he held it securely in the cleft between his buttocks. The same type of belt is used by the tribes of the Northern Kimberleys, and there they are always chosen when a man is wearing the pearl-shell appendage.

The Worora construct more elaborate articles by winding much human hair-string circumferentially (i.e. spirally) round a thick inner skein like the one described above. The finished belt looks like a cylindrical ring about an inch in thickness.

Other kinds of belts are made, but they are more for gala occasions, as when ceremonies are performed and tribal dances arranged.

When his affairs are working harmoniously, game secured, and water available, the aboriginal makes his life as easy as possible; and he might to the outsider even appear lazy. Blessed with a fair share of pristine philosophy by heredity, his motto might be interpreted in words to the effect that while there is plenty for to-day never care about to-morrow. On this account an aboriginal is inclined to make one feast of his supplies, in preference to a modest meal now and another by and by. The result is that, when a beast has been roasted, the whole of it is eaten, even though the participant family or group be small in comparison with the bulk of the spread. In consequence of this custom, the surfeiters find it necessary in times of plenty to frequently lie in camp, in undisguised idleness, until such time has lapsed as Nature must demand of their systems to overcome the discomforts which the reckless gorging had brought about. During this period of digestive recovery, an aboriginal endeavours to spend most of his time in sleeping off the objectionable after-effects of his temporary indiscretion. As an apology, however, one must admit that only too often the same individual is compelled to go for many days without even a mouthful to eat, and possibly, at the same time existing on a minimum of water, under the most trying conditions imaginable—conditions whose origin must be traced to the cycles of drought the great southern continent is heir to, and which have become more drastic in their effects, since the coming of the white man, through the extermination of many indigenous animals and plants the original owners of the land used to depend upon for their existence.

It is during the leisure hours of any stay in camp that attention is paid to such operations as hair-cutting and beard-removing previously referred to. When the natives have been on the march for a time, and especially after they have been out hunting or collecting, numbers of thorns, prickles, and splinters are picked up by the soles of their feet, in spite of the thick horny nature of the skin. Many of these break off short and in due course set up irritation, necessitating their removal. Firstly the sufferer tries to remove the foreign body from the skin with his finger-nails; failing to succeed by this method, he cuts a small piece of wood the shape of an awl, and with its sharp point removes the obstacle. The prickle is often completely buried

and quite invisible to the eye, yet it has to be removed. In order to locate it under those conditions, the aboriginal resorts to the method, not infrequently applied by the modern surgeon, of gently pressing the skin at different places with the tip of the instrument until the seat of pain has been located. At that spot he cuts away the skin to a depth sufficient to expose the hidden body, which he can then in most cases express with his fingers. During these operations the natives repeatedly give vent to a sharp, yet subdued "*irr*," combining the expression of pain with that of disgust or temper.

The method adopted by the natives of walking one behind the other, where possible, is partly to minimize the risk of picking up prickles with their feet, and partly to obliterate the individual tracks of the party.

The dog is the aboriginal's constant companion. In the original tribal areas the dingo, Australia's wild dog, is captured and tamed young; in the more civilized districts the European dog has been acquired and bred by the natives in alarmingly large numbers. The animals are kept by both man and woman—in a single wurley one might count as many as fifteen dogs living with the human occupants. The yelping hordes are useless, except perhaps that they raise the alarm when strangers approach the camp. Some of them are indeed dangerously vicious. The natives have the dogs about them merely for the love they bear towards them; it is on account of the unreasonable amount of petting and pampering, received at the hands of their masters, that the dogs become so thoroughly useless. A native just holds the unruly mob about him for company sake; he prefers to rely upon his own skill and instinct when hunting, and rarely allows his dogs to go with him; in fact, there seems little inclination on the part of the dogs to accompany the chase with their master. They are so well looked after, and regularly steal so much from the general supplies of the camp, that they grow fat and lazy. When a dog seems to be off colour, or has been accidentally hurt, it is nursed like a sick child; it is placed by the fireside, upon the best rug available, and covered with other rags, the natives themselves going without any covering. One might occasionally find a gin going so far as to even suckle a pup at her breast.

Interesting discussions have taken place as to whether the dingo is indigenous to Australia or whether it has come hither from some other land, possibly with man. The wild dog found in the mountains of Java certainly

resembles the dingo very closely. Whether or no, the dingo has existed in Australia quite as long as the primitive tribes. Osseous remains of the wild dog have been found contemporaneous with the extinct Diprotodon and other pre-historic monsters. Indeed in the "mammalian drifts" filling the ancient valleys of the ranges in the Noarlunga district, south of Adelaide, bones of such animals have been found showing distinctly the teeth-marks of the dingo upon them. In Victoria, and New South Wales also, dog remains have been found in old cave deposits in company with fossil-mammals and struthious birds, often buried beneath the basaltic flows and ashes of Mount Gambier and other volcanoes, which have long since become dormant. It seems most probable, therefore, that the dingo existed in Australia in the Pliocene period, or at any rate in that immediately following it. It is a strange circumstance that the dingo has never been traced to Tasmania, although, immediately opposite that island on the mainland, the dog was most plentiful in by-gone times. The surmise is that the animal had in its migration not reached so far south before Tasmania was severed from the Australian continent by the breaking through of Bass Strait. It is reasonable to assume then that the dingo came to the south of Australia subsequent to the aboriginals who inhabited Tasmania.

CHAPTER XIV

HUNTING

True sportsman's instinct—Comprehensive list of game—Land-snails—Fresh-water mussels—Marine molluscs—Caterpillars—Grubs—Tree-climbing—Trees felled by burning—Witchedy hook—Eggs of birds and reptiles—Snakes and lizards—Fishing methods described—A turtle hunt—Crocodile—Dugong—Hawk traps—Wild geese and other birds—The emu—Big game hunted by men—Opossums—Burrowing marsupials—Wallaby—Kangarooing expeditions—The buffalo—Wild bee honey—The honey ant.

Nothing surpasses the pleasure of real pristine chase. The aboriginal's ideal of life is attained when he finds himself in hot pursuit of the game, which shares with him the wilds of his ancient haunts. He lives at an accelerated pace; his pulse quickens, and in his excitement he completely dissociates his mind from everything but the spoor of his prospective prey. His vision is focussed rigidly upon the fleeing animal—he is blind so far as any other objects are concerned—and, behind it, he beholds just the one picture of his ambition realized, *viz.* the “kill.”

The love of the sport, the keenness of the senses, and the astounding powers of endurance are natural attributes, which the aboriginal alone knows how to use to their fullest. These are the hereditary gifts of man which characterize the primitive hunter; and these are the instincts which modern representatives of the human species have deplorably neglected.

The object of the chase is, of course, in its original phase, to find the means wherewith to sustain the hunter's existence. Although he loves the sport so well, a native will never kill wantonly; whatever is slain is eaten; to kill just for the pleasure of the thing is beyond his comprehension and clashes seriously with his profound notions of justice and fair-play being meted out to all his fellow-creatures.

Apart from some of the flabby marine creations, there are few things in the animal world which the aboriginal does not eat, either raw or cooked in ashes. Generally speaking, the male only hunts the larger mammals and such things as require expert knowledge to locate, or the taking of which is

associated with adventure and skill. We shall consider a few items separately.

The larger land-snails are collected by the women in their food-carriers. After a good downpour of rain such come out from their hiding places in great numbers and can be collected in large numbers, but even in midsummer, in the ranges of central Australia, a meal of snails can at any time be secured by searching under tussocks and beneath stones. The principal species eaten by the Aluridja, Wongapitcha, and to a less extent by the Arundta tribes, is the *Helix perinflata*. When sufficient have been collected, they are merely thrown upon hot ashes to roast and then picked out of the shell with a small pointed stick.

Fresh water mussels are gathered from the mud, roasted and consumed. These molluscs, known as the *Unio*, are very plentiful in some of the permanent water-holes, such as exist along the Cooper and Strzelecki Creeks in the Yantowannta, Wongkanguru, and Dieri country. Along the banks of the River Murray great heaps of the shells of such mussels are encountered by travellers even nowadays, indicating to what an enormous extent the molluscs were eaten by the extinct river tribes.

Countless marine species, both heliiform gastropods like the periwinkle and cockle-like bivalves, contribute towards the daily meals of the coastal tribes. All along the south-eastern shores of South Australia, that is, upon the cliffs occurring between Kingston and the Glenelg River and in Victoria, many kitchen middens are still to be found containing heaps upon heaps of the large *Turbo undulata*—the remains of what the local tribes have feasted upon. Associated with these heaps are the stones of their ovens and other camp refuse.

The cliffs and reefs, which fringe the north coast of Australia, are profitable hunting grounds where all kinds of shell-fish, especially oysters, abound. The last-named are gathered and thrown upon hot ashes; when the valves open, the “fish” are detached without any difficulty and swallowed.

The Wongapitcha and other desert tribes do not hesitate to consume quantities of green caterpillars, but such usually only at the beginning of a good season, when fresh herbs are available, and the morsel is in consequence claimed to have acquired a sweetish flavour. The only treatment the caterpillars receive is to be thrown upon hot ashes until they

expand and straighten with the heat. The small hairs covering them are thus singed off, but the caterpillars are far from being cooked when eaten.

The most popular and at the same time most widely distributed article of diet in the insect line is the larva of the big *Cossus* moth, commonly known as the witchedy grub ([Plate XVIII](#)). The two varieties which make themselves most conspicuous are, firstly, one living in the roots of such shrubs as the Cassia and certain species of Acacia, and, secondly, one which bores into the butt of the eucalyptus. The first cannot usually be located by the eye, but its presence is determined by ramming the yam-stick into the ground under the root of the shrub and testing its resistance to leverage—if the grub is present, the root will readily snap, whereupon the native soon unearths it by digging with the stick and his hands. This variety is smaller than the tree-grub and is mostly of a yellow colour.

The abode of the tree-grub is detected by the native's keen eye in the small holes the young larva bores into the bark and lives in until it attains the mature moth stage. The larva lives in the butt or in any of the larger limbs of the tree; consequently it may at times be concealed in the bark high above the ground. In the latter case the native will have to climb the butt and effect an ascent, no matter what the shape of the tree happens to be. Various methods are made use of, one of which we have already considered on [page 13](#).

One of the commonest methods is to cut shallow notches in the bark of the butt, one above the other, and so placed that the toes of either foot of the climber can be placed into them, alternately right and left. The climber, as he ascends, cuts fresh notches into the bark with his tomahawk; and if the butt is so big that he cannot hold on to it, he is obliged to cling to a notch with the fingers of one hand whilst the other hand is used to cut the new notch above it. In this manner he works his way upwards to the nearest lateral branch, whence the ascent is in most cases easier. It is wonderful with what assurance and ease the native accomplishes this dangerous task; and one marvels how it is possible for him to retain his balance against the vertical trunk of the tree. The notches, too, which he cuts into the bark, are so shallow that only the very toe-tips can be inserted. One uncertain movement, or one slip of either toe or finger, and he would fall to the ground; and considering the height to which he occasionally climbs, this

might mean certain death. All grub-holes are examined on the way up and the occupants extracted. Vide [Plate XIX](#), 1.

When the diameter of the butt is not too large to prevent the native from holding his hands around the distant side, the notches are dispensed with, and the climber simply “walks” up the tree, keeping his arms extended and allowing his hands to slide along the surface as he ascends. The method is in vogue principally in the tropical portions of Australia, where small-trunked trees and palms are plentiful in the jungles surrounding permanent water-holes and rivers.

In the same region, strong vines and creepers, which are interlaced with the branches of a tree to be explored, are climbed, hand over hand, by the hunter, who gets his hold by gripping the stem between the big and second toes.

In north-eastern Queensland, tree-climbing is accomplished by the aid of a long and strong piece of the lawyer-cane, which is passed around that side of the tree-trunk lying away from the climber. The native seizes both ends of this loop, one in either hand, and keeping it taut by throwing his body backwards into a sloping position, he places his feet against the tree. Momentarily relaxing the strain upon the cane, by a jerk from his arms, he thrusts the loop a short distance up the trunk, at the same time lifting one foot above the other as though he were walking up a ladder. The same operation is repeated, when the other foot is lifted; and thus he makes quick progress towards the first lateral branches of the tree. Should at any time the climber want to use one of his hands, he passes one end of the cane under the knee of the same side as the hand to be liberated and holds the end between the big toe and that next to it.

Often the natives do not climb the tree, but prefer to light a fire at the base of it and wait until the flame has eaten its way through sufficiently far to fell it. After a giant of the woods has crashed to the ground, it is comparatively easy for the hunters to obtain any spoil concealed in its bark, hollows, or branches.

Although we have considered the methods of tree-climbing in connection with the witchedy grub, it must be understood that the same methods are employed in hunting small animals, in bird-nesting, in honey-collecting, and so forth.

The witchedy grub is extracted from its hiding place by means of a light hooked stick. This implement is from four to six inches long and is usually cut from a small pronged twig, one arm of which is left the required length, the other cut short and sharpened to form the hook. The stick is inserted into the hole occupied by the witchedy grub, hook foremost, and pushed in until the grub is penetrated; then it is withdrawn, the hook bringing the grub with it. As the hole is usually small at its entrance, the bark is first cut away to a small depth with a tomahawk in order to avoid the constriction when the grub is being withdrawn. The witchedy-hook is known throughout central and southern Australia; the Arunndta word for it is "*ullyinga*."

The witchedy grub is prepared like most things already described, namely, by throwing it upon hot ashes for a few moments until it straightens and expands, but does not burst. Although we Europeans have become adverse to eating anything in the grub line, there are many bush people who regularly partake of the witchedy; indeed, by many the grub is regarded as a very tasty dish. The flavour of the cooked witchedy is like that of scrambled egg, slightly sweetened.

The eggs and fledglings of all birds yield abundant food supplies during favourable seasons. In central Australia such seasons are dependent entirely upon the rains. Birds breed usually after the setting in of rain, which might be once or twice a year, but in the driest regions, like the Victoria Desert, perhaps only once every few years. There is no doubt that emu, black swan, and native goose are amongst the biggest suppliers of eggs. Of the two last-named birds, in particular, enormous harvests of eggs are occasionally wrested during exceptional seasons. At these times the tribes who have been so bounteously favoured carry on a regular trade with neighbouring tribes, who have perhaps not had the same opportunity or good fortune.

The eggs of the larger birds mentioned are laid upon, or into, hot sand and frequently turned to ensure them cooking on all sides. The desert tribes of the Kimberley district have a knack of snatching the egg, as it lies upon the hot ashes, spinning it in the air, catching it again, and replacing it on to the ashes. The process might be repeated two or three times. The idea is to stir up the contents of the egg, in order that they may cook uniformly, much after the style of an omelette or scrambled egg.

The eggs of lizards, crocodiles, turtles, and other reptiles are also feasted upon. Of turtles in particular great numbers of eggs are collected along the north coast of Australia. The female turtle comes out of the ocean and lays many eggs in the sand, a short distance above high water mark; between fifty and sixty eggs are commonly found in a single nest. The turtle lays the eggs into a hole it previously scoops out, and covers them with sand it piles up with its paddles. The aboriginal locates the nest by tracking the characteristic spoor across the sand. When the nest has been discovered, the hunter probes the pile with a pointed stick or spear to ascertain whether the eggs are still available. This he can presage by looking at the point of the stick when he withdraws it: if the eggs are freshly laid, the point will be covered with yellow yolk, if partly hatched blood will show itself. The eggs are eaten in either condition. The lucky hunter, immediately he finds a nest, digs out the eggs with his hands and yam stick, and carries them in a food-vessel back to camp. Their preparation is much the same as that of birds' eggs, but, in the case of the turtle's, the white of the egg does not coagulate.

Snakes and lizards, especially the larger species, contribute towards the daily meals. They are tracked to their holes and hiding places and dug out. Great catches are made in the northern coastal districts by setting fire to areas covered with long, dry grass. The hunting party surrounds the burning patch and kills the reptiles, as they are driven out of their hiding places by the heat. Many creatures are overtaken by the flames and partially roasted before they can escape. These are collected as soon as the ground permits of walking over it, but very often such morsels fall into the claws of the birds of prey, which hover over the place directly the fires are started. In order to ensure a rapid spread of the flames, the natives make use of a stick, about a yard long, with a hook at one end. With this stick in their hand, they pick up some of the blazing grass at the hook-end, and run with it along and through the grass, setting fire to as much as they can, and in as short a time as possible.

Of the lizards, the most favoured are the species of *Varanus*, popularly known as the printhy and the goanna. These species live in central and northern Australia and attain a considerable size, the printhy in particular, living in the MacDonnell and Musgrave Ranges, attain a size of over six feet. These lizards in a good season are considered a delicacy, and the fat is prized as nutriment, as medicine, or as cosmetic ointment alike. The lizards

are slain either in the caves they inhabit or as they are running from the hunter. In sandy stretches of country, the smaller goannas are often dug out of the holes they dwell in, and are killed on the spot.

Fishing is indulged in wherever the conditions permit of it. Opportunities are naturally rare in central Australia, and are restricted to only a few permanent water-holes along the courses of river systems like the Cooper, Diamantina, and Finke. In the northern rivers, which are nearly all permanently flowing, as for example the Fitzroy, Prince Regent, Victoria, Daly, Alligators, Roper, and Leichhardt, there is always an abundant supply of fish available. A common method practised both in central and northern Australia is to form a fishing party of men and boys, who enter a water-hole at one end and drive the fish before them, by making as much noise and splash as possible, at the same time gradually working their way towards the shallow water. Great care is taken not to allow any of the prey to make its escape by darting back through the line of the party into deep water again. Suddenly a final, united drive is made, through which most of the terrified fish find themselves in disastrously shallow water. In their frantic endeavour to escape, they entangle themselves in the mud, and can easily be grabbed by the members of the party. As fast as the fish are taken by the men, they are thrown on to the dry bank, where some gins are in readiness to seize them and dispatch them by crushing their heads between their teeth.

Many of the larger fish living in the water-holes have the habit of throwing themselves out of the water when the mud is stirred up in it; lying high and dry upon the muddy banks of the hole, they are easily seized by the natives and killed.

In the western rivers district of the Northern Territory, after the water has dried up in the creeks, leaving only shallow pools behind, the imprisoned fish are driven towards small inlets by means of long bundles of grass and twigs held horizontally in the water. In this way they are pushed forward on to the bank, and the water filters back to the hole. The fish are retained by the brushwood, from which they are easily extricated by the hunters.

The Carpentaria tribes build ingenious races at suitable sites, as for instance at a point where a river drains a billabong. By means of a series of embankments, the water is conducted along a number of constricted channels to long hollow logs, through which it flows, and, subsequently,

empties itself over stacks of brushwood. The fish are entangled in the brushwood, whence they are removed by the hunters to a place of safety.

Crawfish are traced in shallow pools by keeping a constant look-out for the ends of their antennæ, which stick out above the surface of the turbid water; the native seizes these, and, with a jerk, hurls the crustaceans on to the bank.

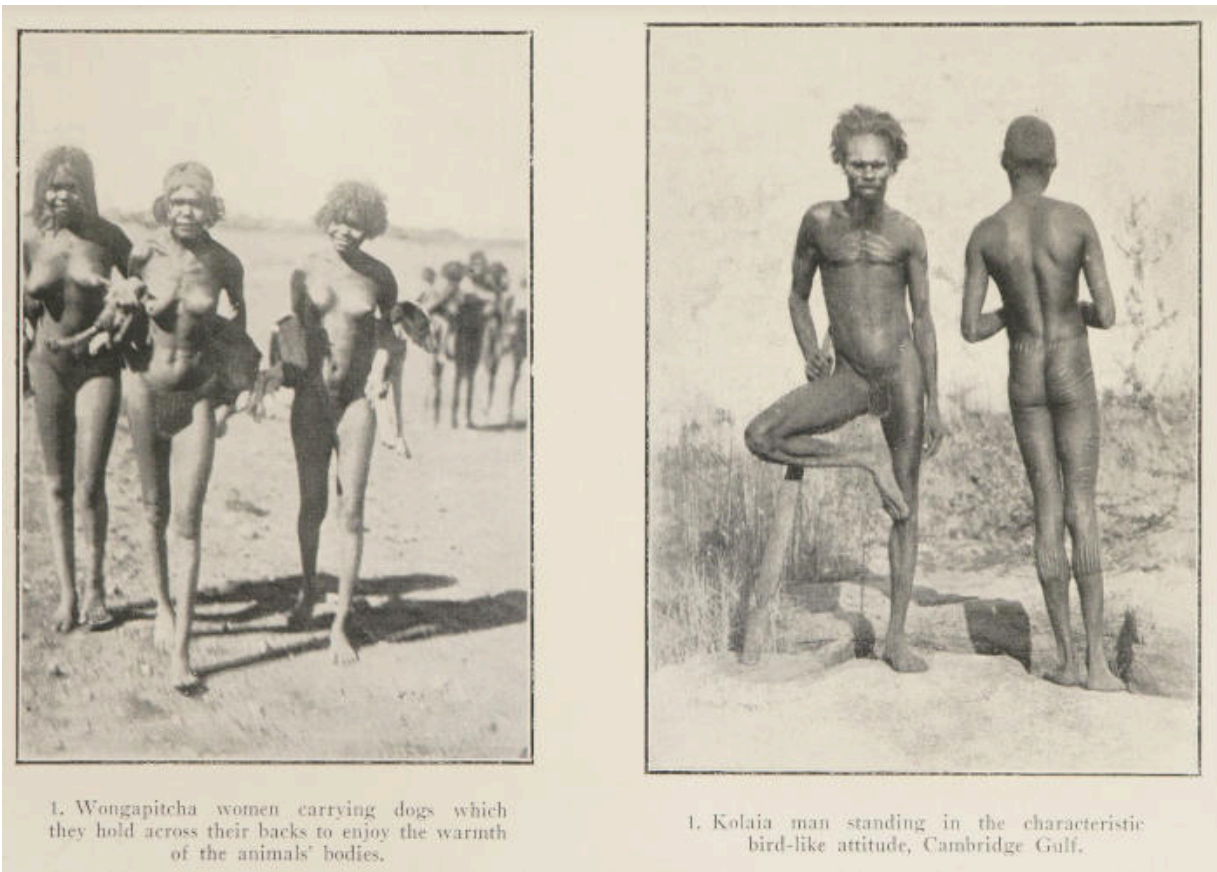


PLATE XVI

1. Wongapitcha women carrying dogs which they hold across their backs to enjoy the warmth of the animals' bodies.

2. Kolaia man standing in the characteristic bird-like attitude, Cambridge Gulf.

In the Victoria River district of the Northern Territory, where there are suitable constrictions in shallow tidal-inlets and creeks, the natives sit in a line across the opening at ebb-tide after the following fashion: Each person squats with his legs doubled in the knees, the thighs resting well apart and in contact, one on either side, with those of his neighbours, who are seated just as he. When the last of the water ebbs out, the fish endeavour to make

through the line of the fishers, but the moment one invades the angle contained by an aboriginal's thighs, it is immediately pounced upon, caught, and thrown across to the women waiting on the shore.

In place of the human line, very often a net is spanned across an ebbing inlet, being kept in a more or less vertical position by a number of stakes, which are driven into the sand. These nets are made of vegetable fibre twine, strung or knitted together, after a loop-within-loop pattern, into pieces many yards in length.

Barriers are also built across shallow inlets, which upon the recession of the tide may occasionally retain large numbers of fish. Such structures are plentiful in the Berringin territory along the north coast, and in the Carpentaria Gulf country.

More temporary structures are made of branches and strips of paper-bark, stayed by vertical piles, driven into the sand at short distances apart. This type of barrier was seen mostly across the beds of creeks such as the McKinlay, Cullen, and Lennard.

A kind of noose is made by the Daly River tribes consisting of a long piece of big meshwork, which is loosely suspended across a narrow arm of water, or a creek, known to be frequented by larger varieties of fish. Whilst endeavouring to swim through the meshes of this contrivance, which at first offers no material resistance to the attempt, the fish, in carrying the noose onwards, forces its body partly through one of the meshes. Some of the prey might succeed in slipping through, others will pull the noose over their heads and fix the mesh in such a position that it will move neither forwards nor backwards. In this case, the fish will be obliged to linger until the natives come along to ascertain what luck they have had, when it is removed, together with any others which have met with a similar fate.

Practically all coastal tribes of Australia have made use of fishing nets at some time or other. Nowadays only the far northern coastal tribes still practise netting. The Daly River tribes, the Wogait, Sherait, Larrekiya, Berringin, and others construct nets after one and the same principle. Two hoops are made of the long shoots of *Spinifex*, growing upon the sandhills on the shore, usually by twisting two pairs of such pieces together, respectively, and tying their ends so as to form a complete ring, measuring about five feet in diameter. Round the inside of this ring is tied a circular

net made out of fibre twine or of Hybiscus bark. The net is made by hand, after the loop-within-loop method previously referred to. To use the net, two natives, usually females, seize it with one hand placed at either side of the rim. As they wade out to a suitable depth, they hold the net between them, partly submerged and slightly inclined, so that the lower edge is in advance of the upper. In their free hands the gins carry branches, with which they frequently beat the water on either side, so as to drive any fish, within the beating radius, towards the centre of the net. So soon as a fish is noticed to go inside, the net is quickly turned up into a horizontal position and the captive bagged ([Fig. 3](#)).



Fig. 3. Berringin women netting fish.

A constant watch is kept for large fish, which may be swimming close in to shore, in order that they might be dispatched with a stone or throwing stick. Feeding stingrays are often captured this way. When saw-fish come into shallow water, the natives wade in, seize the fish by their tails, and throw them up on to dry land before the dangerous “saws” can do any harm.

The spear is a favourite weapon with which to obtain a fish. A special type is used by the Larrekiya, Wogait, the Alligator River, and other coastal tribes in the north. The spear is about eleven feet long, has a shaft of reed or bamboo, and three barbed prongs of ironwood grouped in a circle around the head of the shaft. These spikes, measuring from twelve to fifteen inches, are attached with resin and string or paperbark. The barbs are short and directed backwards, each prong containing from eight to eleven, gradually increasing in size from the point towards the shaft. This trident-spear is used mostly for salt-water fish, and is thrown with or without a spear-

thrower. The idea of the three spikes is to jam the fish so that it is held by the retrorse barbs. The spearing is done either off reefs and rocks, or simply by wading out into shallow water and securing the prey as it emerges from sea-weed or swims near the sandy bottom; many of the coastal and river tribes do much of their spearing from the bow of a canoe. Some tribes make similar fish-spears, but with two prongs only.

The Kimberley tribes of Western Australia, the island tribes off the north coast and the north Queensland coastal tribes use straight, pointed spears of mangrove wood, those of the Crocker Island being exceptionally well finished and of harder wood.

The tribes living more inland in the river districts of the north also use a singly pointed, straight fish-spear. A native does most of his fishing by perching himself upon a high bank, rock, or fallen log, at a place where he knows fish are being attracted to the surface of the water by flies or other causes. He stands rigidly for hours, with his spear poised, patiently waiting for the opportunity, which will allow him to thrust the sharp point into the body of the unsuspecting prey. The spear is thrown with such force that it either sticks in the mud below or disappears under water for a while. In either case, the native has to wade or swim out to the missile, and when he retakes it, the fish, if he has been successful, is found sticking to the end of the spear.

The hook and line are used by the Daly and Alligator River tribes of the Northern Territory. Two types of hook are employed. One is cut out of a pelvic bone or shoulder blade of the kangaroo, or out of the pearl shell. It is of a strongly bent sickle-shape, pointed at one end and containing a notch at the other, to which a line can be securely tied.

The other hook consists of two pieces of bone of unequal length. The shorter and thinner piece, about two inches in length, is pointed at one end and tied at an angle to the bigger piece, which is about four inches long. The joint at the angle is strengthened with wild bees' wax.

A fishing line, made of vegetable fibre twine, is looped once around the hook, and made secure by tying with another piece of string, the union being covered with bees' wax. The line is held either in the hand or is attached to a long rod. The hook is baited with grubs. It is by this means

that the Barramundi is caught in large numbers in most of the northern rivers.



PLATE XVII

Female wood-carriers, Aluridja tribe.

“The women ... are required to collect sufficient to keep the fire going, during the day for cooking purposes and during the night for warming and lighting purposes.”

A turtle hunt is conducted in King Sound after the following style: The game is sighted, floating upon the surface of the sea, either from the mainland or from a craft. Instantly a gesticulative appeal is made to all to keep quiet. “*Hai! Kurdemilla!*” (Look out, a turtle!) “*Sh! Sh! Sh!*” comes the voice of the person who has made the discovery, and others repeat it in an undertone. Everybody who has noticed the turtle bends his body to escape detection, and beckons to all others to do likewise. A number of the men then creep to the water’s edge and cautiously board a craft (i.e. if the observation was not made from a craft), in which they carefully paddle towards the prey. As the boat draws near, the keen eyes of the hunters endeavour to satisfy their curiosity upon the point as to whether the object ahead is just one turtle resting upon the surface in the warmth of a tropical sun, or whether, indeed, there might be a pair, coupled in the water. The female turtle is much preferred by the natives on account of the possible nutritious contents of the ovaries, whose appearance is much like that of a cluster of yellowish dates. When within reasonable distance of the prey, two men glide inaudibly over the side of the craft and disappear from view. Upon a signal, two or three others jump into the water, apparently disregardless of splash and noise, and swim towards the now startled game, which lifts its long neck and looks in the direction of the disturbance. Having perceived the imminent danger, the turtle at once endeavours to escape from the peril by ducking under water. But, at the same moment, it finds its head clasped by one of the men below to be thrust back above the surface of the water. One of the swimmers has now reached the turtle, whose shell he promptly seizes by the collar-like rim at the back of the neck, and pulls it backwards at the same time as the head is being thrust upwards from beneath the water. The turtle works its paddles in a frantic endeavour to escape, but, being held in the sloping position mentioned, the more it exerts itself, the more persistently it remains upon the surface, because its paddles are driving it there. If it be a large turtle, the man who is pulling from behind lifts his body on to the shell, upon which he sits astride. His weight tends to further ensure the inclined position of the turtle, which involuntarily acts as his carrier. The prey is then speedily dispatched by a blow over the head with a tomahawk or waddy. There is no doubt the much-

criticized De Rougemont must have witnessed such a scene as here described, and embodied the inspiration in his narration.

During the season that turtles come ashore, moonlight parties are arranged which overtake the amphibians as they are clumsily working their way over the sandy beach. The creatures are overturned and either slain immediately or are left in their helpless position over night to be killed in the morning.

Crocodiles (*C. porosus et Johnstoni*) and dugongs (*Halicore Australis*) are located below the surface of water by watching for bubbles of air, which might rise, or for any little swirls and disturbances in the water due to the movements of the creatures below. In the case of the dugong, also known as the sea-cow, its presence might be betrayed by small nibbled pieces of seaweed, which come to the surface when it is feeding. Both crocodile and dugong are speared or harpooned from the bow of a raft or canoe. A native often ventures into a water-lily pool or billabong, known to contain crocodile, prodding the mud with his spear, as he advances, to feel for the horny skin of the hidden prey. When one has been located, a large hunting party wade in a line through the water splashing it vigorously and shouting loudly to drive the terrified crocodile from the pool. Once it appears on dry ground, it is chased and pelted with heavy spears. Natives do not appear to be the least concerned about their safety when they wade into water containing crocodiles; in fact, there are few cases known of natives having been attacked by the reptiles, although occasionally one finds an individual partially incapacitated or scarred in consequence of an encounter with a crocodile. The scaly monsters seem loth to attack a coloured man, but for a white man to take the risk a native does would mean courting certain death.

Young crocodiles are caught by hand from the bow of a canoe whilst cautiously drifting upon them as they are floating upon or near the surface of the water. The natives explain that the teeth of the young reptiles have not hardened sufficiently to do any harm.

Along the north coast, and on the adjacent islands, a regular watch is kept for the dugong. This peculiar marine mammal lives in the deep sea, but comes near to the shore to feed. It might often be observed, especially at high tide, in small, calm bays and inlets, frolicking at the surface. Full-grown animals attain a length of twelve feet or more, and weigh up to 1,500

pounds; they are entirely vegetarian, browsing for the most part on salt-water algae. At intervals it is necessary for the animal to rise to the surface to breathe. The drawing in of the air can be heard a great distance off, and is generally, though erroneously, referred to as the "blowing." The man on the look-out watches for the dugong to show itself upon the surface, and his keen eye can usually detect it, even upon a slightly rippled water, at some considerable distance out. The most favourable times are at dusk and on moonlight nights.

When the signal has been given that a dugong is in sight, the hunters set out in a canoe and cautiously paddle towards the place at which it was last seen. It may be that long before the crew reach the particular spot indicated the animal has moved away, but the trained eye of the observer, who now stands at the bow of the canoe, is able to follow the movements of the dugong, even though it be some depth below the surface. At night the prey is observed on the reflected beam of the moon. When the game comes up again to take breath, it may be the canoe is still too far off for action, but the strictest caution is observed not in any way to cause a disturbance. All men in the boat remain rigid, and the paddles are held stationary; the man at the bow, holding his harpoon, poised in readiness above his head, stands like a statue. The moment the dugong goes below again, he indicates to the oarsmen with his hand how to steer. Thus the unsuspecting game is followed around from station to station until it comes to the surface within throwing distance. When this occurs, and it may be before the animal actually reaches the surface, the man in front sends the harpoon forward like lightning, with almost infallible precision, to penetrate the body of the rising dugong and firmly embed itself in it. The terrified animal plunges forward with a tremendous splash, tearing the line attached to the harpoon along with it to its full length. The canoe is pulled along with some velocity through the water, but the wounded animal soon weakens through loss of blood and the want of air. It is compelled to rise to the surface to fill its lungs, but no sooner does it come near than another missile flies from the hand of the hunter to also stick in the back of the exasperated animal. Again it shoots away, with a renewed effort to escape from the cruel harpoon, but in vain. Before long the want of air again necessitates a return to the surface, only to be met with a similar treatment as before. By this time the unfortunate dugong is so much weakened that it cannot travel far without coming to the surface frequently; and every time it does another spear is

planted into its body. Before long it caves in; a final swish with its powerful tail, a quiver throughout the body, and its helpless carcase is in the hands of the elated hunters. The victorious shouting or singing of the men in the canoe is echoed by those watching the hunt on shore, the latter, moreover, in their excitement usually starting a wild sort of dance on the sand. A noose is placed round the dugong's tail, by which it is towed towards the land. If any resistance is proffered, the rope is gathered in, and the creature drowned by keeping its head under water, or one of the hunters suffocates it by thrusting two of his fingers into its nostrils. This accomplished, the jaws are tied together to prevent the carcase from water-logging and sinking.

Often the hunters submerge their canoe, and, by swimming alongside, pull it under the floating carcase. The water is then bailed out of the craft and in this way the dugong is lifted. The hunters then row their prize ashore and drag it on to the beach, where it is quartered and cooked. Practically the whole of it is eaten.

The rich flesh of the dugong is relished by all northern coastal tribes from the east of Queensland to the west of Western Australia as far south as the 24th degree of latitude.

The hunters recognize an "old man" dugong by its tusks; and they assert that often a female is seen carrying its young upon a fin, suckling it. According to aboriginal information, a young dugong might also be observed riding on the back of its mother. A pregnant dugong is considered to make the richest dish of all.

Occasionally a dugong or a big fish, like a whale, is stranded during a gale, and this is indeed a red-letter day for the fortunate tribe upon whose territory the find was made.

Birds of every description are hunted with stone, stick, or spear. It is astounding how adroitly an aboriginal can project the light reed spears; to fell a dove at a distance of from forty to fifty paces is child's play for an experienced thrower.

There are, however, a number of species which are hunted in a peculiar way; and these will now be described.

In the northern coastal districts, where hawks are very plentiful, the natives build small stone covers resembling a surveyor's cairn. A hunter

conceals himself in one of these and holds a dove or other small bird, which he has speared or captured, in his hand above a small loop-hole left at the top of the structure. He moves it about to attract the attention of the birds of prey soaring on high. Presently one of the hawks swoops down upon the dove and grabs it with its claws. But at the same moment, the hunter drops the dove and with lightning quickness seizes the legs of the hawk and pulls the bird under, to quickly kill it. This method is largely practised in the Victoria River district, there being an abundance of stones available for the construction of the covers. The occurrence of many cairns of the type here described has hitherto puzzled travellers who observed them.

The northern desert tribes of Western Australia adopt a similar principle, but in place of the stones they use the tussocks of porcupine grass to cover themselves. In order to attract the hawks they are hunting, they set fire to one or two plants of porcupine grass growing close by.

In the same district, and more especially on the Daly and other rivers in the Northern Territory, wild geese are captured much after the same principle. Large flocks of these semi-palmated geese (*Anseranus melanoleuca*) are in the habit of visiting one and the same place year after year. The natives know these places well, and during the absence of the geese make an excavation in the ground, which they cover with twigs, pieces of paper-bark, grass, and soil, leaving only one or two look-out holes. When the birds have returned, a couple of natives sit in the excavation and watch for the geese to draw near. As soon as a head or a leg of a goose comes near to an observation hole, one of the natives seizes the bird, draws it below, and wrings its neck. In this way many birds may be bagged without disturbing the flock.

On other occasions the natives climb trees, in which they build platforms to seat themselves upon and await the arrival of geese at dusk. The birds come in such large numbers that dozens are caught at a time; they are simply seized by hand and killed on the spot. By cleverly imitating the call of the birds: “*nga ngang, ngang-ngang-ngang,*” the hunters entice as many birds as they like to the platform. But even at daytime, a native often hides in a tussock of grass and imitates the cry of the bird, which, when it unsuspectingly draws near, is either grasped with the hand or hit on the head with a stick.

At times the hunter plucks a large water-lily leaf, into which he cuts two holes for his eyes to look through. Holding this leaf over his face, he swims out to some geese he has observed on a lagoon, and, when within grasping length of the prey, he simply pulls a bird under by its legs and strangles it.

The note of the whistling duck (*Dendrocygna eytoni*) is also accurately reproduced, by which flocks of them are attracted and killed with a throwing stick while hovering around the spot which conceals the native. Cockatoos, plovers, and many other birds are secured in a similar manner.

The flesh of an emu is valued, if for no other reason than for the size of the carcase and the large amount of grease which lies beneath the skin. The northern tribes of Western Australia have discovered a simple means of capturing the big struthious bird in that they poison a water known to be frequented by the game. When the bird has quenched its thirst, it is stupefied to such a degree that it is an easy matter for the natives, lying in ambush, to overtake it and crack it on the head. The poison used is supplied by the leaf of *Tephrosia purpurea*, which the natives call “*moru*”; the active principle is a saponine. In central Australia the pitjuri leaf is largely used for the same purpose.

The natives also take advantage of the inquisitive nature of the bird by enticing it into a cul-de-sac or other trap by waving a conspicuous object, as for instance a corroboree plume, from behind a boulder or bush. When the bird is near enough, it is either rushed with waddies or speared by a number of chosen, astute men.

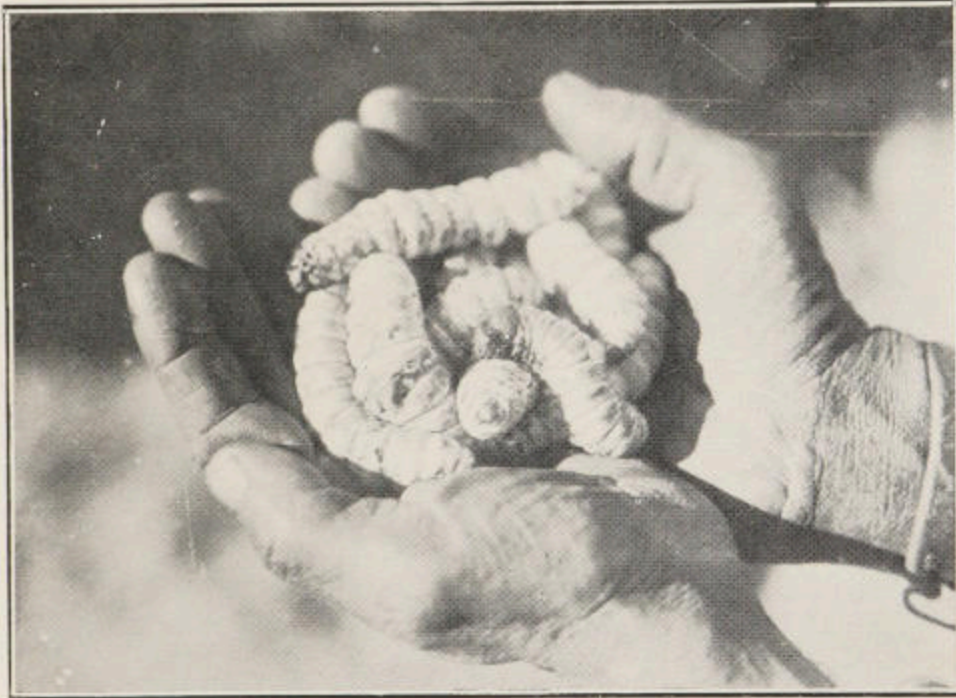
The Larrekiya and Wogait tribes conceal themselves in the branches of a tree, the seeds of which are known to attract the emu. The hunters ascend the tree in the early hours of the morning and remain there perfectly quiet until the prey arrives. At an opportune moment, the bird is speared with a specially heavy spear known as “*nimmerima*.”

The south-eastern tribes used to select one or more men, who would be “dressed up” as emu after the style of the Kukata men playing emu described on [page 81](#). In the case of the hunters, however, a real emu skin is usually employed, with the head attached and held erect by means of a stick, which passes through the neck. Very cleverly imitating the strut of the emu, the men carefully approach their prey, drawing their spears, which they firmly hold between two toes, along with them through the grass.

Carefully and very slowly encircling the birds, the hunters gradually work towards the birds, when presently one or two of them are espied. The moment this happens, the curious emu rush towards the strangers ruffling their feathers and emitting peculiar guttural sounds. Now the critical moment has arrived because the hunters know that, when their faked plumage is recognized, the birds will decamp. They stand and lift their spears with their feet. The birds are now in all probability within throwing distance and very likely on the point of turning. That is the time selected for throwing the missile. Having previously selected their mark, the hunters, with a mighty flourish, let the weapons fly through the air with almost infallible accuracy. Then sounds the triumphant whoop; the men, discarding their disguise, rush towards the wounded victim and promptly put an end to it. In place of assuming the guise of an emu, the south-eastern tribes, when in grass-tree (*Xanthorrhœa*) country, cut the crown from a spreading tree and carry this as a cover.

The real chase, that is the hunting of larger animals, reptiles, and birds, is strictly the business of the men, although the children and women often employ themselves at digging out lizards, snakes, and the smaller marsupials.

Opossums are driven from their hiding places in the hollows of trees by smoking them out. A fire is lit at the bottom of a tree which is known to be hollow, to burn through the enclosing wall at one side. Then green twigs are thrown upon the flame to make as much smoke as possible, which works its way upwards through all the hollows and emerges wherever there is an opening. The half-stifled animals make for the openings and usually drop to the ground; if not they are brought to fall by spear or throwing stick. Often the greater half of the butt is thus burned through and the tree crashes to earth. In this case a diligent search of all the hollows and nooks is made in order that all things to eat, quite apart from the opossums, may be bagged.



Two handfuls of witchedy grubs.

"The most popular and at the same time most widely distributed article of diet in the insect line is the larva of the big Cossus moth, commonly known as the witchedy grub."

PLATE XVIII

Two handfuls of witchedy grubs.

"The most popular and at the same time most widely distributed article of diet in the insect line is the larva of the big Cossus moth, commonly known as the witchedy grub."

Often, too, notches are cut into the butt of the tree in step-like manner to allow the hunters to ascend for the purpose of chopping out their prey from the hollows. Whilst some are thus busying themselves aloft, others are waiting below in readiness to secure any which might attempt to escape.

Most of the burrowing marsupials, as well as the dingo and the imported rabbit, are dug out of the ground. The largest among these is the wombat, which is nocturnal in habit. The native knows, moreover, that when the weather is excessively hot, the animal often comes to the surface and sleeps in front of its burrow. He therefore stealthily surveys the recognized haunts of the wombat at such times, and, should he be successful in locating one, he spears it on the spot.

North of the Great Australian Bight the small wallaby is captured as follows: The hunter ties a bundle of feathers to the top of a long pole, up to twenty feet in length, and this he whirls around his head, high in the air, as he walks across the tussocky plains known to harbour the game they call "*wilpa*." The wallabies, apparently taking the whizzing feathers to be an eagle hawk, squat in fear, and, for the moment, do not attempt to escape from the native. Before the animal recognizes the fraud, the treacherous spear of the hunter has pinned it to earth.

The larger species of these marsupials are hunted differently; they are mostly stalked and killed with the spear. It seems almost incredible that a native can approach a grazing kangaroo on a more or less open plain to within spear-throwing distance without being detected; but such is actually the case. He has so perfected his stealthiness that he utilizes every momentary opportunity, at which the animal's attention is directed away from him; and slowly he approaches step by step. His swarthy colour in itself gives him a natural protection; but more, he has learned the value of artificially colouring himself with the earth or mud of the terrain he is about to scout. Thus upon a "blue mud flat" his body is painted a slaty blue, whilst on a lateritic soil he applies red ochre or clay. His work is considerably simplified when the ground contains such features as ant-hills or dark-coloured boulders, which he can simulate. A native, when stalking a kangaroo in this way, will always have his spear poised in readiness to be thrown instantaneously if need be. Vide [Plate XX](#).

In central as well as northern Australia, hunting parties are arranged as follows: Several men hide themselves at different points of a known pad, along which kangaroos are in the habit of travelling to water or cover. A large party of "beaters," consisting of men, women, and children, disperse in the direction of where kangaroos have been reported to be feeding. On drawing near to the animals, all members of the beating party begin to sing and shout. In the Larrekiya tongue this sounds like "*Ye-we o-ho, ye-we o-ho*"; in the Arundta more like "*Yerrewai, yerrewai*." They scamper through the bush until a kangaroo is actually sighted, when it is pursued amidst the cries of "*Yackäü, yackäü*" in the former, and "*Yackai*" in the latter tongue. The frightened animal usually makes straight for the beaten pad, along which it tears at a terrific rate. Upon hearing the cry of the battue, the men in hiding along the pad place themselves in readiness; and when the animal

leaps by, the nearest hunter quickly rises and discharges his spear. If he is successful in felling the animal, he raises a loud, triumphant shout of “*Käu*,” as a signal to the driving party, who as quickly as possible assemble at the spot. Should the spear of the first thrower miss the kangaroo, the chances are that the next man, further along the pad, will have a chance of trying his skill.

A native considers one of the big hind legs to be the most effective part to wound a kangaroo in, especially if the leg can be broken. If the animal is hit in any other part of its body, it will in all probability make off and that will necessitate perhaps a whole day’s tracking, before it can be overtaken and killed. If only lightly wounded, the hunters will experience considerable difficulty in bringing the game to bay; and the shrewdest strategy might be needed to outwit the watchful animal. In the latter case the pursuers often split their company, and, whilst some are attracting the kangaroo’s attention in the distance, the others endeavour to crawl towards it under cover, until they are near enough to impart the death blow.

Whilst pursuing a wounded animal, a native simply flies over the ground. He cares for no obstacles and seems instinctively to presage the stability of doubtfully inclined and pivotted rocks, lying upon hilly slopes or partially concealed among tall grass. Thus he, with great confidence, jumps from point to point, with the agility of an antelope, and makes rapid headway, whereas a white man would hesitate and come to grief.

In the Musgrave, Mann, and Tomkinson Ranges in central Australia, long brushwood fences are constructed of a more or less zig-zag shape, the angles of which lie upon beaten pads, which are known to be used by kangaroos and wallabies living in the particular area. At each “angle,” the natives dig a large, deep hole, the mouth of which they cover with thin sticks, pieces of bark, and subsequently the whole with sand and grass to give the trap as natural an appearance as possible. So much completed, a log of timber is placed across the pad, at that side of the hole, from which the fleeing game is expected to come. The idea of the log is to make the animal jump over it and land upon the flimsy cover of the hole on the other side. Quite frequently an aboriginal places himself in hiding behind the fence at one of the “angles” and spears the game as it emerges; in this case the hole is dispensed with.

A kangaroo-hunting expedition often takes a tribe far away from a main camp, and the party may be absent for two or three weeks at a time. A native knows that kangaroo follow the new grass, which appears upon patches recently visited by a thunder-cloud or, as is the case upon the north coast, by a bush fire. When either of such phenomena has been chronicled, and after a short time has lapsed, a party of experienced men leave the main camp and prospect the ground for game. When they return, they report the results of their mission to the old men, and, if favourable, arrangements are immediately made for the expedition. The best time for the hunt is considered to be the later afternoon; in the morning and during the heat of the day, the animals are resting, and the hunter knows that under those conditions his chances are not nearly so good as towards evening, when they leave their haunts to feed.

When in 1828 the military settlement at Fort Dundas disbanded, Sir Gordon Bremer let a number of Timor buffaloes, which had been used by the residents, roam at large. Since that time, the animals multiplied to such an extent that large herds were found by later settlers both on Melville Island and the mainland opposite. Thousands have been shot by European hunters, principally for their hides. The natives, too, have learned to recognize these beasts as an important asset to the objects of their chase, although, it must be admitted, the flesh is not relished to anything like the extent of that of the indigenous game; often, in fact, a buffalo is slain merely for a slice or two of the flesh, usually the tongue. The cattle of the European, on the other hand, is eaten with distinct pleasure. Buffalo are hunted by stalking with the spear. This is not a task which demands much skill or laborious strategy. The buffalo spends most of its time out on the plains, more or less under cover of the tall, rank grass, which grows up after the "wet season." In consequence of this, it is a simple matter for the native to avail himself of the same cover when approaching his prey. In nearly every case, the wounded animal makes off, and the excited hunting gang follow it until it collapses through loss of blood. It might even be necessary to throw another spear or two during the chase to finally bring the beast to fall. The jubilation which takes place during the final stage of a buffalo-hunt is depicted in the *Frontispiece* of this book, an actual scene from life witnessed on Melville Island.

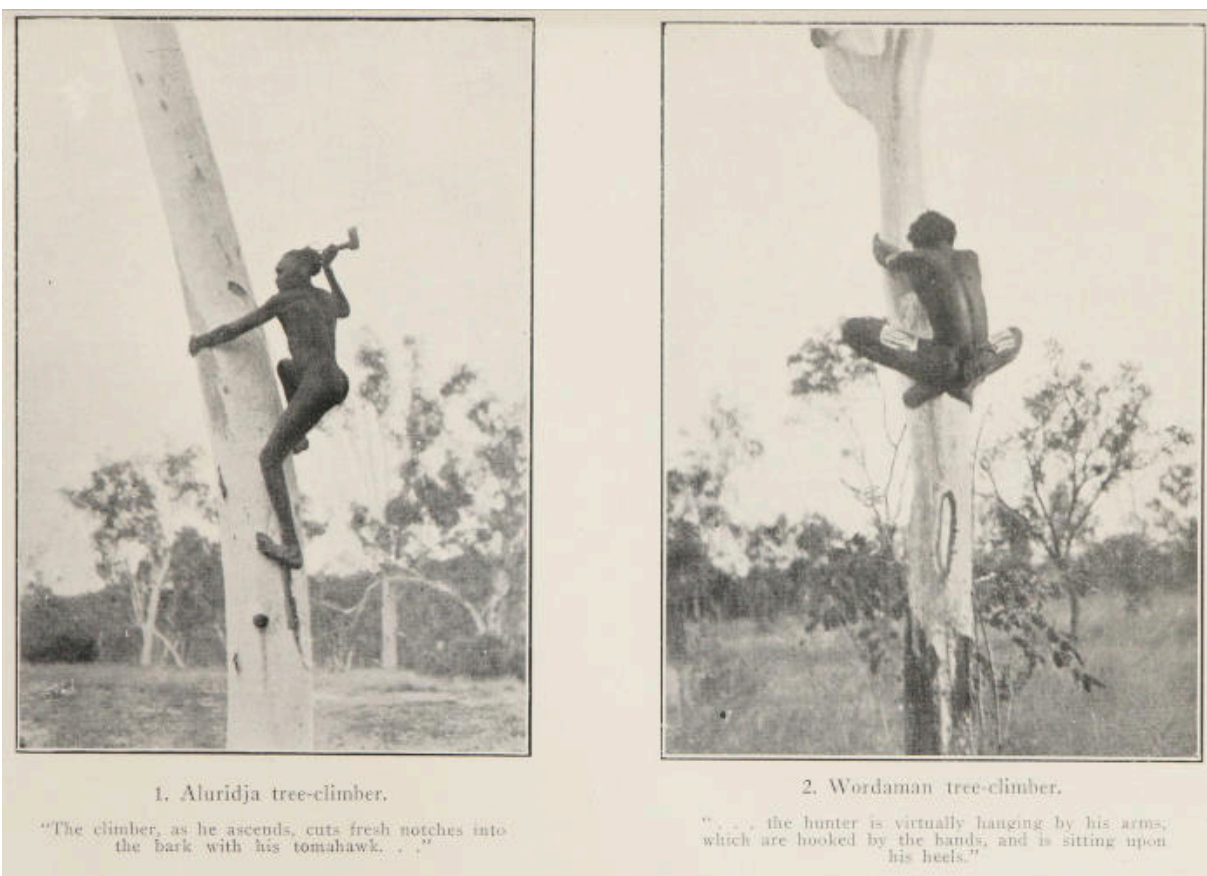


PLATE XIX

1. Aluridja tree-climber.

"The climber, as he ascends, cuts fresh notches into the bark with his tomahawk...."

2. Wordaman tree-climber.

"... the hunter is virtually hanging by his arms, which are hooked by the hands, and is sitting upon his heels."

All along the north coast, a welcome addition to the daily fare is wild bees' honey, or as it is now generally called by the semi-civilized tribes "sugar-bag." The wild bee establishes its hive either in a hollow tree or in a crevice in the ground, and the hunting native—man, woman, or child—is ever on the look-out for it. When the exit of a hive has been discovered in the ground, from which numerous bees are flying, the lucky finder immediately begins to carefully dig down along the narrow channel until he reaches the honeycomb. If the supply is limited, it is usually removed in toto by hand and lifted to his mouth without further ado. If, on the other

hand, there is a goodly amount available, the whole of the comb is collected and placed in a cooleman or other food-carrier and taken to camp.

When a hive is located in a hollow tree, the native places his ear against the butt and listens; by frequently altering the position of his ear like one undertaking a medical auscultation, he can gauge the exact position of the hive by the murmur and buzz beneath the bark. It is then a simple matter for him to cut into the cover and collect the honeycomb. Some of the experienced hunters can “smell” their way for a considerable distance to a wild-bee hive.

The Victoria River tribes have invented an ingenious device, by means of which they can secure honey from otherwise inaccessible fissures in rocks or hollows in stout-buttressed trees. A long stick is selected, to one end of which is tied a bundle of vegetable fibre or pounded bark. With the bundle forward, the stick is poked into the cleft leading to the hive, and, when the honey-comb is reached, it is turned around and allowed to absorb some of the honey. Then the stick is quickly removed and the absorbed honey squeezed from the fibres into a receptacle. The process is repeated, time after time, until the greater part of the honey has been obtained.

Wild bee honey is very liquid, but, nevertheless, quite as sweet and tasty as that of the Ligurian bee. The wild bee, moreover, possesses no sting, and so offers no serious resistance to the enthusiasm of the collector. The bee itself is comparatively small, about the size of an ordinary house fly.

There are no wild bees in central Australia, but in their stead appears the honey ant (*Melophorus inflatus*). These remarkable insects live underground, usually in the red sandy loams carrying forests of mulga. Throughout the MacDonnell Ranges, and the country north and south-west of them, and in the Musgrave Ranges district, they are eagerly looked for by the local tribes. When the entrance to a nest has been discovered, a gin at once sets to by inserting a thin stick as a guide and digging down the course of the hole. This is a somewhat tedious undertaking, and not infrequently she has to dig to so great a depth as to completely bury herself. On several occasions I have unexpectedly come across a woman thus engaged, and neither was she aware of my coming, nor I of her presence, until right opposite her. The “honey-ant” itself is a modified worker of the colony, which is so overfed by the ordinary workers that its abdomen swells to the

size of a marble, about three-eighths of an inch in diameter, in consequence of the liquid honey stored within. With the exception of the few transverse plates, the abdominal walls are reduced to an extremely fine membrane, through which the honey can be clearly seen from outside. The insect's viscera are compressed into a small space near the vent. The ant, in this condition, is naturally unable to move from the spot. It appears that the inflated ants in this phenomenal way provide for the needs of the colony during the barren season of the year, acting in the capacity of living tanks or barrels, which can be tapped as required.

The gin collects numbers of these ants, as she burrows her way downwards, and lays them in her cooleman; when the nest has been ransacked she returns with her prize to camp.

When a native wishes to partake of the honey, he grips one of the ants by the head, and, placing the swollen abdomen between his lips, he squeezes the contents into his mouth and swallows them.

In regard to the taste, the first sensation the palate receives is a distinct prick of formic acid, which is no doubt due to a secretion produced by the ant in self-defence. But this is both slight and momentary; and the instant the membrane bursts, it is followed by a delicious and rich flavour of pure honey.

The Aluridja and Wongapitcha call this wonderful ant “*winudtharra*,” whilst to the Arunndta it is known as “*yerumba*.”

In many parts of central Australia the leaves of the red gums (*Eucalyptus rostrata*), growing along the river-beds, are covered with *lerp manna*—white, conical structures, about the size of a small lentil, which are secreted by the larvæ of an insect known as *Psylla*. On account of their sweetish taste, large quantities of the cones are collected and eaten. The Arunndta refer to manna as “*prelja*.”

CHAPTER XV

VEGETABLE DIET

Women collect vegetable products—"Yelka"—"Munyeroo"—"Nardoo"—Water lily tubers—Native truffle—The "Kaula" or Native Pear—Gall-nuts of the mulga and bloodwood.

Whatever contributes towards the vegetable diet of a tribe is procured essentially by the women. Daily excursions are made by the women, young and old, collectively or in small groups, to lay in a stock for the family meal, which is prepared when the sire returns to camp. The articles which are collected are almost unlimited in variety, the time of the year usually determining which kind in particular is made the object of the day's outing.

The children accompany their mother, and although they help in the general collecting, as decided by the mother, they find many little "luxuries," like the seeds of the mistletoe and the nectar contained in the calyx of a flower, which they partake of as opportunity affords.

Throughout the central and west-central regions, one of the commonest vegetables, which is eaten in very large quantities, both raw and roasted, is the tunicated corm of the *Cyperus rotundus*, which grows in the sandy banks of practically all the river-courses in the area mentioned. These little bulbs grow not very deep below the surface, and, being covered by a comparatively loose sand or sandy loam, are easily obtained. The gins use "wanna" or yam-sticks which they mostly hold in the fist of one hand and apply the chisel-point to the ground like a pick, whilst the other hand scoops or scratches the sand out of the hole. As the bulbs come up, they are placed into the bark carrier ([Plate XXI](#)). The best time to dig for the bulbs is when the grass-like blades of the plant have dried off. When a tribe has been camped for a while near a favourable collecting ground, many acres of soil are turned over, giving one quite the impression of a cultivated field. The Arundta call this bulb "yelka" or "irriakutta," the Aluridja "dunnmördta." The bulb is about the size of a field-pea. To eat it, all that is required to be done is to rub it between the palms of the two hands and then blow away the light shell, which peels off during the process. The natives usually take up five or six at a time and treat them thus, when they are to be

eaten raw; children are especially fond of them raw. The bulbs have a sweetish, nutty flavour.

When “*yelka*” is to be roasted, the bulbs are laid for a short time on hot ashes, then taken up, rubbed between the hands, as described above, and eaten.

Of equal importance in this region is a little, fleshy-leaved plant, resembling a portulaca, which is known generally as “*munyeroo*,” and bears the botanic name of *Claytonia*. It springs up all over the sandhills of central Australia after a good rainfall. When on the march, the natives eat the leaves of this plant raw, both as nourishment and as a thirst-quencher; at times they are thrown on to hot ashes, and, after baking for a short time only, eaten hot. But by far the most important product of the munyeroo is its seed, which occurs as tiny jet-black grains enclosed within a capsule. Though to us it would seem most tedious work, it does not take the gins so very long to collect large quantities of the seed in their bark carriers or coolemans. Seen in bulk like this, the seed reminds one forcibly of gunpowder. With sufficient collected for a meal, they return to camp and clean the seed by “pouring” it from one carrier to another in the wind, when the dry shells will be blown away. If there is no wind available, the gin takes up a handful of the seed, and holding it over a cooleman, blows the husks away with her mouth. When tolerably clean, the seed is placed, little by little, upon a flat grinding stone and reduced to a mealy consistency with a pebble, which is worked by hand. Every now and again, a little water may be added to the mass, which, after a while, is scraped into a cooleman with the side of the hand. The paste may be eaten raw, but more frequently it is mixed with more meal to make a dough, and then baked in hot ashes. The Arundta call this seed “*ingwitega*,” the Aluridja “*waketo*.”

On the eastern side of the great central region, especially along the Cooper Creek, the small plant commonly known as “*nardoo*” is economically the most important to the local tribes. Nardoo grows only on clayey flats, on which water has been stagnant for a while. It is not altogether unlike a small-leaved shamrock; and its scientific name is *Marsilea quadrifolia*. Forming a ring around the stem, just above the surface of the ground, each plant develops a bunch of spore cases, which, when matured, are gathered in great numbers by the tribes. Those of the readers who are familiar with the history of Australian exploration will

recollect that the members of the ill-fated Burke and Wills expedition were for an appreciable time sustained by nardoo cakes, which were given them by the Yantowannta tribe on the Cooper Creek. The spore cases are flattish-oblong in shape, about the size of a small lentil, and extremely hard. The natives collect, grind, and prepare nardoo in precisely the same way as the other tribes mentioned treat the munyeroo. On account of the extensive use which is made of the stone hand-mill, in the Cooper Creek district, consisting of a large flat slab of stone, upon which a pebble is worked to and fro, the implement has earned for itself the name of nardoo-stone. The Arunndta refer to nardoo as "*parapara*," the Dieri as "*kalumba*."

By the same process the seed of many acacias, which is collected in considerable quantities, is made into cakes. A peculiar ingredient is added to the flour by the Victoria Desert tribes in the form of white ants, which they knead into the dough and bake with it.

On the north coast, from King Sound east to the Gulf of Carpentaria, quite a variety of grass seeds, also those of the lotus lily, are ground and baked.

One of the regular articles of vegetable diet in the tropics is the tuber of the water-lily, which is gathered by the score and roasted in ashes; it tastes almost like a potato, with perhaps a distinctive flavour about it resembling that of the Jerusalem artichoke. In the Northern Kimberleys, the species most frequently eaten is the beautiful *Nymphaea stellata*, which is variously known as "*kapa*," "*kadje*," or "*toki*." But perhaps the most popular dish in the tuber line in the Northern Territories is the yam. A great variety of these is available, but it often requires the genius of an aboriginal to locate them, especially when there are no leaves showing on the surface. It is not every yam, however, which is esculent; and some are so "hot" that even the smallest portion of one applied to the mouth will severely blister the mucous membranes. Here again the expert knowledge of the native is of inestimable service to the inexperienced, for it is he who can at a glance tell which is fit for consumption and which is not; and it is he who can treat some of the peppery varieties in such a way as to eliminate the objectionable taste. Some kinds he discards entirely because he knows that, if he ate them, they would cause a painful "fire in the anus."

Within a somewhat restricted area, extending from north of the Musgrave Ranges eastwards to the Dépôt Sandhills, a fungus exists, which is known to the Aluridja and Wongapitcha as “*widida*,” and to the Arunndta as “*oridja*.” In general appearance it is much like the European truffle, and, like it, grows mostly below the surface of the ground; indeed it is difficult for the untrained eye to detect a *widida*, except under the direction of an aboriginal. At times one finds old sweetish juice. The inner layer of the shell is white, soft, and specimens showing above the ground, which have been exposed by wind or rain, but when this is the case, the fungus is not really fit for consumption, because its richness will have attracted many blowflies, and it will, in consequence, be teeming with maggots. A few specimens of this interesting fungus were collected by me and submitted to Mr. A. Grant, of Sydney, who determined it to be a species of *Scleroderma*. The *widida* may be eaten raw, but more commonly they are cooked in hot ashes.

In the mulga country of the Flinders Ranges, and all over central Australia, a species known as *Marsdenia Leichhardtii* is rather common. It is a creeper with slender stalk and smallish, elongate leaves, and bears a pear-shaped fruit, consisting of a thin green skin, which encloses a mass of silky seeds. When broken the plant exudes a thick milky sap. The fruit, stalks, and leaves of the plant are eaten; they have quite an agreeable, sweetish taste. The Wongapitcha call the plant “*päuya*,” the Arunndta “*langu*,” whilst in the Flinders Ranges the recognized name for it is “*kaula*.” On account of the shape of its fruit, this plant is referred to by the settlers as “native pear.”

Gall nuts and excrescences, when obtainable, are also on the daily menu. The most popular is one which is found on the twigs of the mulga. It is usually referred to by the settlers as “mulga apple,” and grows up to the size of a walnut. The whole of the growth, with the exception of a small kernel-like structure, containing the insect, is edible. The taste, though slightly eucalyptine, might be compared with that of a “tasteless apple.” The Wongapitcha call the mulga apple “*jarrulge*,” and the Arunndta “*takul*.”

Another variety of gall nut is found on the smaller branches of the Bloodwood (*Eucalyptus corymbosa*), and is, in consequence, spoken of as the “bloodwood apple.” It is a nodular, warty, and woody growth, about the size of a billiard ball, the inside of which is hollow and contains, besides the

parasite, a sweetish juice. The inner layer of the shell is white, soft, and edible; the whole reminding one of a miniature cocoanut.

CHAPTER XVI

BEVERAGES

Honey solution—Pandanus cider—Human blood.

Although, naturally, the principal and practically only drink of the various tribes is water, there are one or two special beverages deserving of notice. In central Australia, the Aluridja, Arunndta, and Wongapitcha collect many handfuls of *Eremophila* flowers, commonly called honeysuckle by the local white settlers, in their bark food-carriers, on to which they pour a quantity of water. The flowers are stirred around for a while with a stick and then skimmed off with a piece of bark or by hand. The drink is ready for consumption immediately after; it has a slightly sweetish taste, and is relished by young and old. Another source of nectar is the beautiful red flower of *Brachysema Chambersii*, which grows in abundance in the sandhills both north and south of the MacDonnell Ranges and is known by the Arunndta as “*aumba*.”

On the north coast of Australia, the wild-bee honey is upon occasions dissolved in water and drunk. This is nearly always done when the comb, obtained in the first place, is mixed with sand and grit, or when the honey is absorbed in the fibres of the collecting implement described above ([page 146](#)).

The Roper River tribes pick the large fruits of the corkscrew palm or pandanus, which are not unlike pineapples in appearance, but very hard and stringy, and, after bashing them between heavy stones, they keep them immersed in water for some time before they drink the solution. The water absorbs the sweetly stringent juice and produces a refreshing toddy. It being necessary to keep the fruit in water for some time to extract as much of the palatable ingredient as possible (it may be, for that matter, that the natives leave the solution behind in a cooleman, while they go on a hunting tour, returning for it in the course of some days' time), opportunity is given for the solution to start fermenting; a mild pandanus-cider is the result. It actually happens that upon great festive occasions, when large quantities of this beverage have been made some time beforehand, the natives imbibe

more than ordinarily, and thereby bring themselves into a condition of indubitable merriment. The Katherine and Victoria River tribes make a similar beverage, but do not store it for any length of time. This is the only instance I am aware of where Australian natives, intentionally or unintentionally, make an intoxicating drink.

When men are on a long-distance stage, as, for instance, during a drought, when water is scarce and the sun is relentlessly fierce, they are occasionally obliged to resort to the old tribal custom of drinking each other's blood to escape perishing of thirst. They open a vein in the arm and collect the blood in a cooleman, or they allow one or more of their companions to drink straight from the wound. In certain cases of sickness blood is also given to the patient to drink.

CHAPTER XVII

PITJURI

Distribution of native tobacco—Collecting grounds—Native names—Pitjuri habit—Preparation of leaves—Stimulating and comforting properties—Ash added to liberate the alkaloid—Properties of piturine.

Most of the central Australian tribes have learned to recognize the narcotic properties of the tobacco-like plant commonly called pitjuri. All tribes, from the Wongapitcha eastwards to beyond the borders of Queensland and New South Wales, know the value of the plant, and even if it does not actually grow in the particular tribal area, its leaves are obtained from adjoining tribes by barter. The Dieri, Yantowannta, Wongkanguru, and Ngameni are all required to procure their supplies from further north, because the plant does not grow in the Cooper Creek district. The Arunndta, and latterly the Aluridja as well, regularly collect as much pitjuri as they want in the valley of the Finke and other gorges of the MacDonnell Ranges, whilst the Wongapitcha have their resources in the Musgrave and Everard Ranges.

The collecting grounds are as a rule owned by a circle of old men, each of whom clearly defines his boundaries by placing a number of stones upon the ground. A proprietor may give another person the necessary permission to gather leaves on his plot according to certain terms agreed upon. The owner usually takes a share of the leaves, and, in addition, levies other articles in exchange for what the collector has removed.

The plant is known by different names among different tribes; the Arunndta call it "*engulba*," the Wongapitcha "*peturr*," and the Aluridja either "*mingul*" or "*warrakinna*." Scientifically it goes by the name of *Duboisia Hopwoodi*.

The leaves and stalks of pitjuri are chewed by both men and women, and in many cases by children also. It cannot be denied, once a person starts chewing pitjuri, he soon develops a craving for it, like a habitual smoker does for tobacco. The usual plan is to partially dry the leaves in the sun, or over warm ashes, on the spot, and subsequently pack them into bundles to

take home, with the intention of storing them for future use. But once camp is reached, the future aspect becomes entirely inconsequential, because so long as pitjuri is known to be available, the supplies are drawn upon; the result is that the larder soon becomes depleted.

The men have a way of their own when preparing the pitjuri. Some of the dried leaves are ground between two stones and the powder brushed on to a small piece of bark. Then a few twigs of acacia or eucalypt bark are burned to white ash, which is mixed with the powder, the whole being subsequently worked into a softish mass with saliva. Of the final mixture a quantity is taken and rolled between another dry leaf of the pitjuri, cigar-fashion; and it is ready for mastication. A plug of pitjuri does not always remain the property of one individual, especially when the supplies are running short, but often passes from one mouth to another, until it has done the necessary round. When not in use, the plug is secured behind the owner's ear, after the style an office clerk carries a pencil.

The natives admit the stimulating benefits they derive from chewing, or, as they say, "eating," of pitjuri, both when they feel off colour or fagged after a strenuous day's outing. On the other hand, they look upon pitjuri-chewing in company as a social comforter, which fosters mirthfulness and friendly fellow-feeling. When natives meet, even though they be comparative strangers, an exchange, loan, or presentation of pitjuri takes place, as a token of friendship. In the same spirit, a native considers the gift of a stick of tobacco from a European stranger, who, according to tribal ideas, unlawfully passes over the hereditary boundary, as a mere formal obligation, which expresses the intruder's peaceable intentions.

The burnt acacia ash, which is added to the powdered leaf of the pitjuri plant, has a somewhat important function to perform; and one marvels at Nature having given the unsophisticated aboriginal the hint to add it. One of the favourite species, which is burnt for the purpose, is *Acacia salicina*. A. J. Higgin has determined by analysis that the ash of this plant contains the astounding amount of 51.15 per cent. of calcium sulphate, mixed with a little carbonate of lime. It is the alkali in this ash which liberates an alkaloid, known as piturine, from the crushed pitjuri leaves when the two substances meet in the presence of moisture supplied by the spittle; and this piturine is much the same in its action as nicotine. An alkaloid is nowadays manufactured from the leaves of the Australian plant which is used in

medicine as a powerful sedative and hypnotic. It is not difficult, therefore, to understand why an emu, drinking from a water poisoned with the leaf of pitjuri, should become stupefied. Vide [page 139](#).

CHAPTER XVIII

NAVIGATION

Floating log—Log rafts—Paddles—Outfit carried on board—Bark canoes of different patterns—Used in southern and northern Australia—"Housing" of canoes—"Dug-outs"—With or without outriggers—Sails.

We have on several occasions alluded to the fact that the natives make use of some kind of craft while hunting and fishing. A few remarks, therefore, upon aboriginal navigation in general may be appropriate at this juncture.

The simplest type of float is no doubt the log of light timber used along the north and north-east coast. The straight trunk of a mangrove is selected, and from it a log is cut, about five or six feet long, which is stripped of its branches. Where a river or an estuary has to be crossed, such a log is slipped into the water and the native lays his body over it, lengthwise, with his legs straddling it. With his head and shoulders well above the surface of the water, the swimmer propels himself along by means of his legs; occasionally he also uses his arms, but then primarily for steadying his body above the log. The natives maintain that this method gives them a certain amount of protection against the attacks of crocodiles, since, when viewed from below, the man and the log together resemble one of the reptiles in form. For the same reason the lower thin end is often left tapering to a point, to simulate the tail of a crocodile.

When two or three, or more, of these light logs of mangrove are lashed together, a simple raft results—a type in frequent use along the eastern shores and rivers of north Australia. The craft is propelled by either a pole or a paddle, the man standing in the former case and sitting in the latter.

The same contrivance is used when a man wishes to cross a river or a bay, and carry his children or belongings across, without swamping them. In this case, he usually swims alongside the raft and propels it by powerful leg-strokes.

In the north-western corner of the Australian continent (i.e. the King Sound—Glenelg River districts), navigation is undertaken in large rafts. These are constructed as follows: From six to ten poles are cut out of the trunks of a tall, straight-growing mangrove, resembling a pine in shape. The poles are cut into twelve-foot lengths, and are then trimmed longitudinally, so that they taper from about one-quarter their length downwards, like an elongated club; the two ends are pointed off. In their thickest part, the poles measure about six inches in diameter. Two of these pieces are now laid upon a level patch of ground, side by side, with the thick ends all pointing in the same direction, and “nailed” together with stakes of hard wood, at various distances along the entire length of the poles. The remaining poles are linked to the original two in a similar way; and so a strong platform results, in which the poles converge in the direction of the thin ends like the arms of a fan. Another platform is constructed exactly similar to the one just described. The only tools used in the making of these structures are tomahawks and large stone and shell scrapers.

All completed, one of the platforms is dragged down the beach and floated; then the second is taken to the water and lifted so that it rides upon the former with the converging ends reversed. The raft is now ready for use ([Plate XXII](#), 2).

Crudely fashioned paddles are used, about six feet long, and similar to those of the Melville and Bathurst Islanders. The local name for these is “*kanbanna*.”

One or two natives usually go out with a raft like this, and it is astounding with what skill and celerity the clumsy-looking structure can be handled and paddled along.

The local name for the raft is “*kaloa*.” The principal use to which it is put is fishing and turtle-hunting; the mainland tribes moreover use such rafts for general ferrying, when they make their periodic visits to the islands included within their tribal possessions.

One or two cushions of grass or reeds are laid upon the platform before leaving, to afford dry seating accommodation; and the hunters never go without taking a fairly solid fire-stick, which is stuck in an upright position between two poles of the raft. A few spears and a long harpoon (about ten feet long), with a barb at the pointed end, are carried, the latter being

secured to the raft by means of a good length of rope. A heavy boomerang is also added to the outfit, with which the hunters might kill the spoil when they haul it on deck.

Similar log-rafts are in use on some of the islands in the Gulf of Carpentaria, but one platform only is constructed, and the logs are simply lashed together with vines.

Any observant visitor to the River Murray will not fail even nowadays, when much of the original timber has disappeared, to observe the numerous trees, growing at or near the banks, from which large sheets of bark have been removed years ago by the local natives. The bark was used for making canoes. Sheets were cut from the eucalyptus trees, measuring from twelve to twenty feet in length by from three to four feet in width. These were laid horizontally upon the ground and moulded into shape while hot ashes were applied to them, the edges being propped up all round while the bottom was kept more or less flat. Several stakes were placed crosswise to keep the sides in position, both at the ends and at the centre. One end was usually more pointed than the other and slightly more elevated; this acted as the bow of the canoe. When thoroughly dry, the craft was launched and carried up to six or seven passengers. In addition, a small bed of clay was built upon the bottom, which carried a fire. The canoe was propelled by a man, who stood near the stern and either poled or paddled it along with a long oar.



Kangaroo hunters, Aluridja tribe.

"It seems almost incredible that a native can approach a grazing kangaroo on a more or less open plain to within spear-throwing distance . . ."

PLATE XX

Kangaroo hunters, Aluridja tribe.

"It seems almost incredible that a native can approach a grazing kangaroo on a more or less open plain to within spear-throwing distance...."

A number of different types of canoes are in use on the north coast, constructed out of one or more pieces of bark. In the Gulf country, a piece of bark is freshly detached from a tree, folded along its length, and laid upon the ground in a horizontal position. The ends are then heated, to render them pliable, and securely clamped between two upright stakes, and tied closely together above and below the folded sheet. Stakes of a length equal to that of the required width of the craft are next propped from side to side, to give the canoe its shape, and the ends trimmed on either side with a sharp stone-knife or fragment of shell. The bottom corners are usually bevelled or rounded off. The edges are finally held together by sewing them with strips of cane. Long, thin saplings, stitched along the inner top edges

of both sides, act as gunwales and considerably strengthen the structure. One or two ties of lawyer cane are stretched from side to side to prevent the bark from bulging in the centre. When afloat, a native squats low in the canoe near the stern and makes good headway by paddling with a small, oblong piece of bark, first on one side and then on the other.

In some cases, the bark sides are stiffened by poking flexible U-shaped hoops under the saplings which form the gunwales; and in others the sides are kept in position by a number of such hoops, together with stretchers and ties, without any special gunwale at all.

The Melville and Bathurst Islanders use large bark canoes up to nearly twenty feet long, which they construct after the following principle: A single sheet of bark is cut from either the woollybutt (*Eucalyptus miniata*) or the stringybark (*E. tetradonta*) by chopping through it circumferentially at two heights from the ground, the distance between which represents the required length of the canoe that is to be. Slitting this piece once vertically for the whole length, it is removed by forcing the edge of a chisel-pointed stake under the bark and levering it off. The outer surface of this piece of bark is rough and becomes the inside of the canoe. Transverse cuts are made about two feet from each end, and half the thickness of the bark removed with a sharp bivalve shell (*Cyrena*). The ends, which have by this treatment become pliable, are further softened by holding them over a fire. The sheet is folded lengthwise along its middle and clamped at its ends with stakes rammed vertically into the ground. The bottom corner of the fold is bevelled off by one or two sloping cuts, along which the two pieces are sewn together with close, overcast stitches; then the pieces are stitched together horizontally at the top corner, for a distance of three or four inches. Thus secured, an angular or curved piece is cut away from the bark, lying between the two sewn corners, in imitation of a fish-tail, and neatly laced together with strips of the lawyer vine. Holes are previously drilled through the bark with an awl made out of the leg-bone of a wallaby. The joints are made secure by plastering them with wild bees' wax, and the corners are caulked with plastic clay and fibre or resin. Along the top, inner edges of the canoe, on both sides, thin, straight poles are lashed with "run on" stitches. These, however, do not extend the whole length of the canoe, and, being straight, do not enclose the stern and bow of the craft. In other

respects the structure is much the same as that in vogue in the Gulf of Carpentaria country.

When not in use, the canoes are “housed” on a level piece of ground under the overhanging branches of a banyan or other shady tree. They are laid in a normal, upright position (not inverted), and are kept so by short pieces of timber, which are propped against the sides. The bark thus dries in the required shape and does not become lopsided. Each canoe has its recognized place. When a dense growth of mangroves skirts the foreshore, a regular approach to the water is kept clear by cutting away the trees as they grow up. The paddles are laid within the canoes.

When the occasion demands it, quite a large number of natives may be carried in a canoe, but usually, when on a simple turtle or dugong hunting expedition, two persons only man the craft. The boatmen, while propelling the canoe, squat with their buttocks resting upon the heels, and with their knees pressed against the vessel’s sides. The weight of the bodies being thus well within, the stability of the canoe is considerably increased.

Although these canoes are mostly used for navigating the various rivers and estuaries of Melville and Bathurst Islands, and especially Apsley Straits, occasionally, when wind and weather are favourable, the natives venture far out to sea, and not infrequently do they make the journey across to the mainland, some forty or fifty miles away, where in former days they carried on a bitter warfare with the Larrekiya and other tribes.

It is at times imperative that a canoe be attached to a hunting or warring party, which is travelling overland and later might want to drift down, or paddle up, a river or inlet to reach its destination. Under such circumstances, six or eight men carry the craft upon their shoulders as they walk alternately left and right of it.

Paddles are made of hard wood, having a single, well-shaped blade and a rounded handle. The edges of the blade are parallel, or taper slightly towards the end, which is either square or rounded. They are from three to five feet in length. When rowing, the natives clasp the handle with both hands and dip the blade on one side or the other, just as the steering requires it.

Certain north-eastern tribes of Queensland used to make their canoes of two or three sheets of bark. In the first instance the sheets would be stitched along the keel, and in the second a lenticular or oval piece was inserted, which acted as a flat bottom.

Dug-outs are found all along the north coast, but it is very probable that they are of foreign origin, presumably Melanesian or Polynesian. A suitable tree having been felled, its ends are shaped and the inside chopped, gouged, and burned out, so that only the outer walls remain. Some very big boats of this description were seen in use among the Larrekiya, and their seaworthiness was proved time after time.

Some of the Queensland tribes attach one or two outriggers to their canoes, which, of course, give them additional safety when by chance they might be overtaken by a rough or choppy sea.

The Groote Islanders in the Gulf of Carpentaria carry a mast in the centre of their dug-outs, to which they lash two long horizontal bamboo-booms and spread a sail between them. This circumstance is remarkable, since the Groote Islanders are among the least known of the Australian tribes and have come less into contact with Europeans than other tribes who might have learned the use of sails.



1. Arunndta girl digging "Yelka."

"The gins use "wanna" or yam-sticks, which they mostly hold in the fist of one hand . . ."



2. Arunndta gin cleaning "Yelka" in bark pitchi.

" . . . all that is required to be done is to rub it between the palms of the two hands . . ."

PLATE XXI

1. Arunndta girl digging "Yelka."

"The gins use "wanna" or yam-sticks, which they mostly hold in the fist of one hand...."

2. Arunndta gin cleaning "Yelka" in bark pitchi.

"... all that is required to be done is to rub it between the palms of the two hands...."

CHAPTER XIX

DUELS

Bragging preferred to fighting—Duels frequent among the women—Petty provocations—The “*Kutturu*”—Men use similar sticks and boomerangs for striking purposes—Waddies and clubs described—The “*Damatba*”—Wooden swords—Duels with reed spears—Stone dagger duels—Heavy spear duels—Chivalrous methods—“Bone-pointing” and other methods of suggesting death to an enemy—“Pointing” sticks—How the “boned” person is affected—Counter-charm the only cure—The medicine man or “*Nangarri*”—His witchcraft—The recovery.

Although under ordinary circumstances the aboriginal of Australia is a peaceable, placid individual, who prefers to talk of what he could do to his enemy rather than look for trouble in a hostile camp, yet, being human, there are naturally extenuating circumstances, which might thrust the obligation upon his shoulders to pick up arms and fight for the sake of his individual honour or of his tribe’s safety. In the former case a duel is arranged, in the latter a regular warfare is waged, which might last a day or continue, off and on, for years at a time.

Duels are perhaps more frequently fought among the women than the men, the cause in most cases being trivial. A common disturbance of the peace is brought on by petty theft. One woman might, intentionally or otherwise, appropriate a small article belonging to another. When the article is missed by the owner, an argument ensues, which soon warms up to a strained pitch of excitement. Abusive epitaphs become prolific, which repeatedly embody references to excrement and other filth. Eventually the irate hags can constrain themselves no longer and each produces her fighting stick, known throughout central and northern Australia as “*kutturu*.” Walking towards each other, and all the time striking the ground in front of them, from left to right, and from right to left, the women continue their vilification. In the Arunndta tongue this is something after the following style: “*Utnnarranduddi, uttnatikkia, atutnia, arrelinjerrai!*”

The ground is struck with the heavy sticks immediately in front of the opponent’s feet, so vigorously that dust and dirt fly into the air. It is not long before the foot of one of the gins is struck; and then the fight begins. The gin that was hit immediately lifts her *kutturu* and aims a blow at the head of

the offender. But the latter in all probability will have been prepared to ward off the blow.

The kutturu consists of a heavy “ironwood” stick, on an average about three feet long and of circular section; it is bluntly pointed at each end and usually has a carved decoration upon its surface. The parrying party holds the stick with its pointed ends between her palms, and, by moving or swaying it from side to side in an inclined position, diverts the force of the impact from her head. The duellists take it turn about to strike and parry. The head is the principal mark, but it is not against the rules to aim at the fingers. When the latter are struck, it not infrequently happens that one or two of them are broken.

If the antagonist is too clever at warding off a blow, a gin might occasionally alter her tactics and try to stab the head opposite her with the point of the kutturu. If the attempt proves successful, a very deep gash often results, followed by a prolific flow of blood. The damaged gin wails aloud and drops her kutturu whilst she catches the blood, which is pouring from her wound, in the hollow of her hand and throws it in the other’s face.

The triumphant assailant does not take a mean advantage of her “score,” but replies to the blood-slinging by rushing to the nearest fire, from which she scoops a double handful of hot ashes to throw at the lamenting one. This is by no means the end of the trouble, but really incenses the combatants to more desperate action.

So soon as the wounded gin has overcome the shock, she plucks fresh courage and again takes up the argument with her kutturu. The fight continues until one of the gins receives a blow on the head severe enough to disable her, or until both have kept the strife going to a stage of complete exhaustion.

When a gin has been disabled, and lies more or less in an unconscious condition upon the ground, the victor stands over her, triumphantly swinging her kutturu, whilst her tongue dispels the hatred by talking incessantly without opposition.

During the whole time of this heated altercation, the camp has been generally disorganized. Other women are vociferating wildly, children are screaming, a few score dogs yelping, and the men are sitting around quietly

and gloomily, with their eyes turned from the scene of the duel, and only occasionally exchanging a few words in a subdued whisper.

The method the men adopt for settling their disputes among themselves is not unlike that of the women, but more systematic. They place themselves face to face upon a clear piece of ground, with their kutturus in their hands, and about half a chain apart. After the customary abusive preliminaries, the psychological moment arrives, when one man rushes at the other carrying his weapon in both hands behind his back, and, as he runs, preparing for a monstrous blow. When he reaches to within striking distance of his opponent, he pulls up short, and, with the momentum created by his run behind him, deals an awful whack. But the other man has placed himself in the defensive attitude, and, as the weapon falls, he springs forwards and upwards to parry the blow destined to crash upon his skull. The striker now retraces his steps and prepares to receive the onslaught from the opposite side. This procedure of alternate attack and defence continues until one of the men falls or both combatants are thoroughly exhausted. All through the fight, however, there is a wonderful display of power, agility, and chivalry, the figures of both the striker and receiver being conspicuously graceful in their movements ([Plate XIII](#), 2).

Where the boomerang is known it, too, is extensively used, in conjunction with the shield, by duellists to settle minor altercations. The offended party throws one of his missiles into the camp of his rival as a summons to the fight, whereupon the latter immediately responds by throwing another back, and walks out into the open, carrying with him a single boomerang and a shield. Both men now start a war-dance, during which they gradually approach each other, lifting their legs high in the knees, brandishing their boomerangs in the air, and holding their shields in front of their bodies. After a while, they close in; and the real fight begins. Whenever an uncovered spot presents itself on either man, the opponent, with the quickness of lightning, attempts to strike it with his weapon. The hands in particular are selected as the best marks to quickly put the rival out of action; and this opportunity is never missed when it presents itself to the quick eye of the native.

Waddies and clubs of various forms are used all over Australia, both with and without shields, to decide the rights or wrongs of individual grievances.

There is no hard and fast line of demarcation between a waddy or fighting-stick and a club. The original conception of either is a short stick or truncheon, which is used both for beating and throwing.

The commonest form is a cylindrical rod of hard wood with a smooth or vertically grooved exterior and rounded ends. It is either straight or curved.

The Bathurst Islanders have a type similar to the above, but with a slightly swollen distal end. The stick often carries a sharp spike, which projects from the same end.

The largest fighting-sticks are to be seen in the Forrest River district in the far north-western district of Western Australia, measuring up to four and a half feet in length. The stick tapers from the top towards the handle end, and has a flat face at either extremity. A gripping surface is made by roughly incising the thinner end all round for a distance of four or five inches.

A peculiar combination of implement with weapon was used by the women of the lower River Murray tribes. It consisted of a stick with a blade at one end and a knob at the other, the one moiety serving as a digging stick, the other as a club.

Along the Cooper Creek, a large, stout baton of mulga, with a globate knob at the handle end, was used by the Wongkanguru, Yantowannta, and other tribes.

Some very shapely clubs belonged to the Narrinyerri. They were made of casuarina wood, and had a heavy, inflated head, which was usually pointed off at the top. The handle was moderately thin and had a number of circular grooves cut near its end to prevent the hand from slipping when the wood was wielded. Further west, on the Nullarbor Plains, the thick end was not pointed off, but, on the contrary, was perceptibly flattened. In both types mentioned, the surface was well smoothed and polished, although the clubs of the coastal tribes along the Great Australian Bight were generally longitudinally grooved.

A rather fanciful form of club, reminding one of the medieval spiked clubs, was found in the possession of the fast disappearing Yantowannta tribe at Innamincka. A stick, nearly two feet six inches long, and circular in section, had an enlargement near the head-end, which was deeply grooved

vertically and, in the upper portion, circumferentially also, the intersection of the grooves producing a number of pointed prominences.

Used in conjunction with a heavy three-sided shield, the south-eastern tribes of South Australia fought most of their duels with a dangerous type of waddy, some two feet or more long, which had an attenuated knob at the handle end and a flat, angular projection at the opposite end; the latter was sharp and pointed. The weapon was known as "*lionila*," and, from our point of view, might be classed as a battle axe.

In the Roebuck Bay district, a flat, hard-wood club is found, the sides of which are straight and slightly tapering towards the handle end; the edges being rounded off. One of the flat sides is usually ornamented with an engraved geometrical pattern.

The Larrekiya and Wogait construct a flat throwing weapon, not unlike a small cricket bat in shape, from six to twelve inches long, which they call "*damatba*." It has a short handle and very sharp edges, and, being hurled at an enemy edgewise, it flies through the air with a revolving motion. If any part of the native's naked body is struck with this weapon, a very deep wound is always inflicted, from the effects of which the unfortunate victim might easily bleed to death.

Under the category of fighting-sticks we might also mention the wooden swords, of which some form or other is known all over the continent.

In its normal form, the sword is a long and narrow, lanceolate blade of hard and heavy wood, up to five feet in length. The sides are convex, and the edges fairly sharp. The surfaces are either smooth or longitudinally grooved, and in addition decorated with finely incised patterns of different descriptions. These swords are much used by the Minning tribe of the Eucla district.

The Cooper Creek tribes make the weapon with a slight curve in it, like a boomerang, the length being about the same as in the previous type. Among the Dieri it goes by the name of "*marriwirri*." The Arunndta and Aluridja forms are not made so long, but slightly wider, and of very heavy mulga wood.

In the northern coastal districts, the type is different in so far as it tapers towards that end, which is to serve as the handle, and terminates there in a

flat or slightly concave base. The haft is not infrequently bound round with vegetable fibre, and subsequently covered with beeswax, to prevent the hands from slipping.

These weapons are all used with two hands for striking and parrying blows during a duel. When about to “receive” a blow, the native takes the precaution to keep his elbows close against his body to avoid the risk of having his arm broken by the ricochet of the heavy weapon. A favourite mark is the opponent’s hands; and the fighter has to carefully guard them by adroitly and instantly shifting the sword sideways the moment he perceives that the blow has been aimed at them.

The northern and north-western tribes use light reed spears when settling quarrels between two individuals. The spear is about five feet long and has a tapering head of mangrove wood, which is inserted into the top end of the reed shaft. All coastal tribes, from the Adelaide River westwards to the Ord River, use this spear, but it is not known in central Australia. The opponents, standing about forty or fifty feet apart, throw the weapons at each other with wonderful precision, but before the dangerous little missile can reach its mark, the would-be recipient dodges it with equal skill. A number of such spears is carried by each combatant. Whilst the duel is proceeding, the two natives dance in defiance to taunt the rival, grotesquely jumping from one foot to the other, holding the arms semi-erect and bent at the elbows and wrists, whilst the body is thrown forward and the head kept back. When a spear passes very near to or grazes its mark, the native greets his escape with a short but strangely articulated exclamation sounding like “*irr.*” The throwers are constantly on the move, and, whenever possible, one stoops to pick up a new spear from a bundle of them lying at his side. The little missiles are so light that, when they hit the ground, they skip along the surface and can be recovered uninjured. Although these spears seem more like toys than weapons, they are nevertheless most formidable on account of their sharp point and the velocity with which they travel.

A method which is in vogue among the central Australian tribes, like the Arunndta and Dieri, is the dagger duel. The dagger employed is a long stone-knife with a grip or haft of porcupine resin; the Arunndta name for it is “*putta ildurra.*” The combatants hold one of such daggers in one hand and a light shield of kurrajong wood in the other, and thus equipped they approach each other. After some preliminaries in the way of dancing and an

accompaniment of excited shrieking, the men close in upon one another. With the shield they not only catch the well-directed stabs before any bodily injury is done, but they also thrust the opponent back to keep him at arm's length. It is not an uncommon event for such a duel to be fought for a considerable time without any bloodshed, the skilful parrying checking many a fatal blow until eventually utter exhaustion appeases the thirst for revenge and soothes the hatred, which was only too evident at the beginning of the duel. At other times severe gashes are inflicted, which occasionally terminate fatally. Vide [Fig. 4](#).



Fig. 4. Two Arundta carvings of scenes in a dagger-duel ($\times 1/3$). Tracing.

The most serious of all duels is undoubtedly that in which the heavy spear is used. The method is similar to that of the light spear fights, with the distinction that those engaged in the strife stand further apart, and do not run the same amount of risk by exposing themselves to the same extent as in the former case. It is recognized that a "hit" by one of the large heavy-bladed spears will result in a dangerous wound, and, therefore, the men face the ordeal in all conscientiousness, knowing that if they can prove themselves equal to their opponent's dexterity for a reasonable length of time, the moment will arrive when the strife might be terminated by arbitration. One occasionally meets with a person who has a broken portion of a spear-head deeply embedded in the musculature of his thigh or other region of his body, where a violent inflammation around the lesion has caused the sufferer much pain because he has not been able to extract the offending piece of wood or stone from the tissue.



1. Sunday Islander making fire by the twirling process during a ceremonial.



2. "Kaloa" or mangrove raft, Worora tribe, Glenelg River district.

PLATE XXII

1. Sunday Islander making fire by the twirling process during a ceremonial.

2. "Kaloa" or mangrove raft, Worora tribe, Glenelg River district.

In the Balmaningarra district of the northern Kimberleys of Western Australia, the warriors hold the shield (which they call "*karwinnunga*") in

their left hand, but, if necessity arises, they quickly change it to the right. As a spear whizzes by harmlessly, or is skilfully warded off, the defendant sounds a short triumphant “*p-r-r-r*” with protruding and vibrating lips. When about to “receive” a missile, the native stands in much the same attitude as that adopted by a modern rapier-duellist. The moment, however, he realizes that his antagonist’s aim has been inaccurate, or he has been successful in dodging or warding off the death-bringing stick, he balances himself on one leg, cocks the other under his buttocks, and defiantly throws his arms above his head, jeering at his enemy and inviting him to throw once more, crying: “Look! I am defenceless, and I am exposing my body to you! I have no fear of you; your spears can only scratch the ground as the yam-stick of a woman digging for grubs!”

Barely has he finished his taunting sentence, and before the other man has had time to prepare for another throw, he bounds forward and projects his weapon. During the fight, the antics of the men, especially of the one on the defensive, are remarkable, jumping from leg to leg, at one time holding the shield straight in front of the body whilst covering the head, at another, standing erect with the shield at about the level of his chest.

During their duels, an admirable spirit of chivalry is displayed by the combatants. If, for instance, a shield breaks, the man who has the advantage refuses to fight until the broken weapon has been replaced. The same regulation applies to a broken spear, but not to a broken spear-head; if the latter breaks, the circumstance is looked upon as the result of bad workmanship, and the fight continues. Only a certain number of spears is carried by the parties; and any damaged weapon might be replaced from a bundle held in reserve by the gins, who stand in readiness. When the missiles have been spent on both sides, the parties change ends to collect their spears, after which the hostilities are resumed.

Irrespective of any of the methods here described, whereby the natives actually fight with weapons for the sake of honour or revenge, it is quite as customary among all Australian tribes to bring about the downfall of a rival or enemy by the magic influence of suggestion. This is the wonderfully potent method of “pointing” death at a man, who may or may not be present or visible. The process is usually referred to as “pointing the bone,” or simply “boning.”

In the Alligator district of the Northern Territory, the excrement of a man, who is to be sent to his death, is collected and roasted over a fire; after which a little of it is taken and mixed with the resin of porcupine grass. A ceremonial dance follows during the night, which is of a very secret character, men as well as women participating. The chants which are sung implore the birds of night—the owl, the plover, and the curlew—not to betray the men who are seeking revenge. At the far end of the ground, cleared for the occasion, a hole is dug, in which a fire is burned while the ceremony is in progress. Enchanting songs are now rendered, which are to entice the spirit-father of the doomed man to attend. A little later one of the principal performers marches forwards, carrying a small ball of resin mixed with the burned excrement, together with a short stick representing the spirit-father. At a given moment, he begins to dance and, with his free hand, catches hold of his scrotum. When he arrives at the hole, in which the fire is burning, other men snatch the glowing embers and clear the ashes out of the way. The dancer throws the resin-ball containing the excrement into the hole and covers it with hot sand. The moment it strikes the hot ground, the resin fizzles and crackles; and those sounds are taken to be the voice of the spirit calling the victim from the earth.

Shortly after the ceremony these facts are made known to the unfortunate who has been selected to die; they are usually conveyed to him by one or two eye-witnesses of the ceremony. Overcome with consternation and terror, the fellow immediately begins to fret; and death will inevitably be the outcome, unless the counter-influence of a medicine man or other tribal power can make itself felt beforehand.

Upon other occasions in the same district, the footprint of a man, who has been decreed to die, might be found upon a clay-flat or a river bank. The track must be intact; if it be in the least degree imperfect, it is considered useless for the purpose. Taking for granted, then, that it is clear and well-defined, the mould is cut out of the clay in toto and buried in an anthill. There it is secreted until such time as the spirit of the doomed man's father is supposed to be in attendance at a tribal ceremonial, when it is fetched and broken over a blazing fire. This act answers the same purpose as the burning of the resin and excrement in the previous case.

An old Arundta custom was to “cut the shadow” of a man, who was to die, with a sharpened mussel-shell knife called “*langa langa*.”

When a man of the Worora tribe dies, his relatives resolve to avenge his death, which they suppose was the work of an enemy, whom they name. During the obsequies, a bone is taken from the arm of the deceased, usually the humerus, but occasionally the radius, and small portions cut or scraped off it, to be handed to the nearest of kin, who officiate as the avenging party. The little group sit facing the direction in which the supposed murderer is seen, or at any rate is known to be residing. Together they place some of the pieces of their relative's bone, which they call "*gibba*," into their mouths, and, after chewing them for a while, they spit the pulp towards their victim, at the same moment naming him as the perpetrator of the deed—the cause of their bereavement. The Sunday Islanders adopt a similar method, but refer to the bone as "*käu-käu*".

Most of the tribes are in possession of differently shaped sticks and bones, with which the death-pointing is done. These are usually about three or four inches long, pointed at one or both ends, and containing a small bleb of resin at one end, to which a piece of human hair-string is attached. When the instrument is of bone, it is usually a piece of the dead man's skeleton. The Aluridja take the fibula of the man whose death is to be avenged, and construct a flattish "bone," pointed off at both ends. The Arunndta select the same bone, or the ulna, which they scrape down to a long tapering point at one end; to the opposite extremity they attach a little porcupine resin. Occasionally one finds these objects carried in hollow bone or bark-receptacles. Another common form of the Arunndta is a short stick, at either end of which a blunt arrow head knob is carved, round one of which a long piece of human hair-string is tied. A simple punctate design is at times burned into the stick. Some of the southern tribes of the Northern Territory have pointing stones, which are shaped much like a stone-knife, hafted with porcupine resin and suspended by a long piece of human hair-string.

A number of such pointing instruments are constantly carried about by certain men of the tribe, whilst others are kept buried in places only known to a few. Often a stick is constructed just when the occasion requires one, and when there happens to be none available.



Aluridja men "pointing" the bone.

"One man cowers upon the ground . . . whilst the second, kneeling at the side of him, holds his pointing stick at arm's length . . ."

PLATE XXIII

Aluridja men "pointing" the bone.

"One man cowers upon the ground ... whilst the second, kneeling at the side of him, holds his pointing stick at arm's length...."

Any fully initiated men may make use of a pointing-bone or stick, but when the grievance concerns the tribe in general, the operation is performed by the magician or medicine man. Women do not generally carry these sticks, but the Aluridja, and no doubt others also, allow their gins to charm their yam-sticks, with which they then "kill" their antagonists. A charmed yam-stick is believed to paralyse the arms of any person, whom it touches, when appealed to by the owner; consequently one of this kind is chosen for duelling whenever possible.

When a man has been condemned to death, the person or persons, who are to administer the fatal charm, are nominated. The "pointing" apparatus

are produced, and with them the men take up a kneeling position a little distance away from the camp. Facing the doomed man's habitation, they lift the bone, or stick, to shoulder height and point it at the victim. The long piece of hair-string, which is attached to the instrument, is tightly tied around the charmer's arm, above the elbow. This is done to endow his system with the magic influence of the pointing-stick he is holding; and that magic, he believes, passes into the destructive words, which he is uttering: "May your skeleton become saturated with the foulness of my stick, so that your flesh will rot and its stench attract the grubs, which live in the ground, to come and devour it. May your bones turn to water and soak into the sand, so that your spirit may never know your whereabouts. May the wind shrivel your skin like a leaf before a fire, and your blood dry up like the mud in a clay-pan."

There is a great number of different methods employed in administering the fatal charm of the pointing-stick, all of which, however, are after much the same principle. A common practice amongst the Aluridja is for the man, about to use the stick, to leave the camp and seclude himself behind a tree or other obstacle. He squats upon the heel of one foot which he has tucked under his body. He points the bone or stick straight at the man who is to die, or, it may be, merely in the direction he imagines he would strike him. Whilst administering the curse, he holds the object in the hand of his outstretched right arm.

Both the Arunndta and Aluridja often work in pairs after the following style: One man cowers upon the ground, with or without his pointing-stick in his hand, whilst the second, kneeling at the side of him, holds his pointing-stick at arm's length over the former man's back, and directs it towards the person who is about to receive the evil charm. Vide [Plate XXIII](#).

To make their charm more effective, and the death-penalty more certain, central Australian tribes not uncommonly tie the claws of a bird of prey, the eagle-hawk by preference, to the pointing instrument. It is believed that by this trick the evil magic works like the grip of a bird, by clutching the doomed one's chest and crushing it. If by accident the unfortunate fellow becomes cognizant of this, and it happens that, as actually is frequently the case after a big feast, he suffers from indigestion, he naturally interprets the symptoms of his indisposition as being due to the invisible, tightening girth,

which the charm has laid about him. The fatal termination arrives at a much earlier date in consequence.

A man who discovers that he is being boned by an enemy is, indeed, a pitiable sight. He stands aghast, with his eyes staring at the treacherous pointer, and with his hands lifted as though to ward off the lethal medium, which he imagines is pouring into his body ([Plate XXIV](#)). His cheeks blanch and his eyes become glassy, and the expression of his face becomes horribly distorted, like that of one stricken with palsy. He attempts to shriek, but usually the sound chokes in his throat, and all one might see is froth at his mouth. His body begins to tremble and the muscles twitch involuntarily. He sways backwards and falls to the ground, and for a short time appears to be in a swoon; but soon after he begins to writhe as if in mortal agony, and, covering his face with his hands, begin to moan. After a while he becomes more composed and crawls to his wurley. From this time onwards he sickens and frets, refusing to eat, and keeping aloof from the daily affairs of the tribe. Unless help is forthcoming in the shape of a counter-charm, administered by the hands of the “*Nangarri*” or medicine-man, his death is only a matter of a comparatively short time. If the coming of the medicine-man is opportune, he might be saved.

The medicine-man of Australian tribes is not so much an individual who has the knowledge of medicinal values of herbs and of surgical practices as one who is the recognized sorcerer, capable of rebuking the ills wrought by an enemy or evil spirit ([Plate VIII](#)). He attains his distinction either by heredity or by accidental, but maybe exemplary, craftiness. In the former case, he is looked upon as a favoured son, who has inherited from his tribal and ancestral fathers the magic art of neutralizing the evil charm of a spirit or enemy, which manifests itself in prostration or disease; at the same time he is the official power of the community, who alone can outwit the evil spirit, control the elements, and keep pestilence away from the camp. This hereditary art is recognized as a concrete matter, which is believed to have been deposited within the body of the particular individual by spirit-ancestors or nearer spirit-relations; this matter might have taken the form of a special variety of wood, small bones of animal or man, and a number of sacred stones, all of which the made medicine-man carries about with him in his abdomen, more or less replacing the ordinary entrails originally occupying the cavity. Each tribe has a number of these medicine-men,

whose rank is gauged according to age and the principle they have lived up to. For instance, in the qualification of the early medicine-men of the Adelaide tribe, it was deemed necessary that the candidate should taste human flesh at least once in his life. In the central Australian tribes a medicine-man should not eat of kangaroo which has been feeding upon new green grass; if he does, some of his mystic powers will leak out of his body, and he will immediately drop in the estimation of his tribal admirers. If the offence is repeated a number of times, he is disrespected entirely as a professional sorcerer. There are, of course, a great number of restrictions, which the conscientious practitioner observes most punctiliously.

Every medicine-man of any standing at all has his own history of qualification, which he does not hesitate to make known to the public at opportune moments. Old Kai Kai, the leading Nangarri of the western Arundta on the Finke River, relates how he, as a young hunter, became detached from the rest of the party, and, after tracking a wounded kangaroo for a whole day, he eventually abandoned the pursuit to make for a rock-hole in the stony James Ranges. It was nigh on sunset when he arrived at the hole, tired and thirsty. He threw his wommera and spears upon the ground, and eagerly lay over the cool fluid to still his parching thirst. But when he sipped the water a tadpole entered his mouth, and, before he could spit it out, it slipped down his gullet and dropped into his abdominal cavity with a bump that caused him much pain. When he recovered, he again tried to soothe his burning lips, but met with a similar fate. Several times more he tried, but in vain; as soon as his burning lips touched the surface of the water, a slimy tadpole slipped into his mouth and fell into his stomach with a painful thud. In desperation he made a final attempt to carefully approach the water's level, when he beheld what he took to be the image of his face and body reflected from below. Horror overcame him, however, for the image was that of another man! And, as he looked again, he noticed that the body of the image was transparent, and inside of it there were just as many rounded pebbles as he had swallowed tadpoles! He collapsed at the side of the waterhole and slept like a dead man, for how long he could not say. When at length he woke up, he found himself among the reeds of the flowing sheet of water on the Finke River, which the white people call Running Waters. He now quenched his thirst. And when the recollections of his experience at the rock-hole came back to him, he realized that the man who had looked at him through the water had been a spirit, and he could

still feel the pebbles he had placed inside of him. Now it was obvious to him that he had been ordained a Nangarri, and he returned to his camp, where his relatives were anxiously awaiting him.

Having been called to the side of a “boned” patient, the Nangarri allows a number of the relatives to be present when he applies his weird method of treatment. At first he cuts some ridiculous antics, during which he mumbles or chants some almost inaudible verses. The patient is, in the meantime, laid flat on the ground. The Nangarri approaches the sufferer from the foot end and, throwing himself upon the ground, crawls right on to the chest of the former, biting the skin of his patient at several places as he crawls on to the body. Having “located” the seat of the trouble, the “doctor” slips on to the ground, and, picking up a fold of the skin with the underlying fatty tissue between his fingers over the vital spot, applies his lips, and, perhaps, his teeth, too, to it. He sucks, bites, and kneads the skin, frequently lifting his head and spitting blood on to the ground. The patient is all the while groaning with pain; if he becomes unmanageable, he is called to order by the Nangarri. At length the climax arrives. The Nangarri withdraws from the patient, his cheeks visibly inflated, and, conscious of the expectant eyes of all present, he empties the contents of his mouth into his hands, which he holds like a receptacle in front of him. The fluid, consisting of saliva and blood, is allowed to trickle to the ground or into the fire. Then a triumphant chuckle announces that the malignant element has been discovered! With feigned exaltation, the great healer steps towards the awe-stricken relatives, holding between the index-finger and thumb of his right hand an article, such as a small stick, a bone, a pebble, a meteoric bomb, or a talon, which he avows is the cause of the “boned” man’s affliction, and, having now been skillfully and permanently removed, the unhappy fellow has nothing more to fear.

The good news is immediately conveyed to the prostrate form on the ground. The effect is astounding. The miserable fellow, until that moment well on the road to death, raises his head to gaze in wonderment upon the object held by the Nangarri, which, in all seriousness, he imagines has been extracted from the inside of his body. Satisfied with its reality, he even lifts himself into a sitting position and calls for some water to drink. The crisis has now been passed, and the patient’s recovery is speedy and complete. Without the Nangarri’s interception, the “boned” fellow would have fretted

himself to death for a certainty, but the sight of a concrete object, claimed by the recognized authority of the tribe to be the cause of the complaint, signifies recovery to him, and with its removal comes a new lease of life. The implicit faith a native cherishes in the magic powers of his tribal medicine-man results in cures, which exceed anything recorded by the faith-healing disciples of more cultured communities.

CHAPTER XX

WARFARE

Inter-tribal fights and hereditary feuds—Massacres—Preparations for the fray—On the warpath—Teasing the enemy—Hostilities begun—Treatment of wounded and disabled warriors—Hatred soon forgotten—Blood revenge—Boomerang displays—“*Kurdaitja*” shoes—Recovering the bodies of fallen warriors—Portions of victims bodies eaten.

Aboriginal warfare might be divided into two classes, according to whether it is of the nature of a true and bloody inter-tribal fight, or of a feud arising between two tribal groups or parties. In any case, the hostilities might be of long standing and the enmity might have existed for generations past. The *casus belli* is as multifarious as are those of modern peoples. It might be on account of a natural treasure held by one tribe, such as a valuable ochre-deposit, which is coveted by another. Or it might be simply the result of an elopement or an abduction as between two groups or families belonging to different tribes. The cause frequently determines the method to be adopted during the fight.

When the arch-foe is to be faced, nothing is out of order in strategy, provided the plan is effective, and, above all things, as gruesome as possible. The main objective to be achieved is to make an assault as murderous as circumstances will permit, and to establish a record massacre, in order that the enemy might be thoroughly cowed and taught to long remember the affair.

In former days such battles were, according to all accounts, of fairly frequent occurrence, but nowadays, no doubt largely due to the interference by European settlers, and the smaller numbers of natives, one very rarely hears that any warfare is conducted on a large scale. Indeed, many of the one-time bitterest enemies, such as the Larrekiya and Ponga Ponga, Arunndta and Aluridja, Kukata and Wongapitcha, can now be seen living in close proximity to each other, and apparently on the best of terms.

Where inherited hatred sways, treachery brews. The aggressors know that the most radical method to extinguish the enemy is to take them unawares, and to slaughter them before they can retaliate. For this purpose it is best to

either steal on them in the earliest hours of morning whilst they are sleeping unsuspectingly, or to lie in ambush at a place, like a waterhole, where the enemy is sure to call.

A council of war, consisting of the oldest and most experienced men, is held to discuss the *modus operandi* at length. Thereupon all eligible men are apprised of the decisions arrived at. Considerable time is devoted to the preparation of spears, sharpening of blades, and straightening of shafts over a fire. Any weak spots, where the spears are likely to break when put to the test in battle, are carefully bound with kangaroo sinew, and the blades or stone-heads re-embedded in porcupine grass resin. Quantities of red ochre are ground and mixed with emu fat, with which the bodies of all warriors are covered. Each man looks to his own spear-thrower, and makes sure that it has not warped or split, and that the handle and hook are secure; a warrior with a defective spear-thrower would be next to useless on the battlefield. The men tie their hair back tightly with human hair-string, and go absolutely nude. The work is all done in a thorough and conscientious way, yet there is every indication of humour about the camp, and nowhere does one see any sign of fear or nervousness on part of either the men who are going to fight, or of their near relatives. There is even an indication of frivolity, with much jabbering and laughter. The women assist assiduously while the preparations are in progress, and add considerably to the entertainment of their heroes, who are about to depart. Even when at length the expedition makes a start, some of the old gins run at the side of the men and keep on joking with the men at the expense of the opponents, roars of laughter every now and then announcing a particularly clever sally from the lips of an old hag barely able to control herself with excitement in her endeavour to incite the men.

Before leaving their camp in a body, all warriors congregate in a circle on parade, holding their weapons balanced in the spear-thrower. Two or three of the old men make a careful scrutiny of the group and their outfit and give final instructions as to how the campaign is to be conducted. Quite occasionally, indeed, a general discussion might take place, at which any suggestions of younger warriors are received and weighed by the seniors, prior to departure. Then the official start is made.

As the little troop advances, the men begin to gesticulate and brandish their weapons, whilst they dance, with their beards poked between their lips

and teeth. Occasionally they send forth a loud, piercing yell resembling a war-whoop; then they poise their weapons, at the same time stamping and vibrating their legs, and deriding the enemy with a scurrilous tongue.

If the enemy is to be met in open battle, the place and time of the proposed encounter are arranged beforehand by means of carved wooden missives and special carriers.

Having arrived at the place, and *vis-à-vis* to the foe, vociferation and antics continue to the verge of hysterical jactitation. One or two of the most daring warriors, before hostilities have actually begun in earnest, rush to the fore, and, placing themselves in front of what might appear the strongest of the opposite party, continue their antics in contempt of them. Thus the patience of the enemy is harassed time after time; and in a similar way the enemy reciprocates.

The moment will, however, arrive when one becomes a little too daring or forward, and, with intention or otherwise, touches the person opposed to himself. The insulted warrior, who has thus been called out, responds with a tremendous roar, and drives straight at his opponent with his spear, using it as a lance.

In an instant the scene is changed. All participants retreat to a respectable distance, about fifty feet or more, towards their respective sides, amidst fearful yelling. The moment they reach the new positions, the air becomes alive with spears, and the fight has commenced in reality.

The throwing and aiming of the projectiles are good, but the ducking and dodging of the selected victims are equally so. During the whole term of the encounter, terrifying bawls are heard, which are emitted by the enraged fighters, and responded to by the excited women in the background. Thus the conflict may last for an hour or two, without a casualty being recorded. When a man is hit, the hideous yelling reaches its climax. The wounded fellow, unless he be mortally hit or falls to the ground unconscious, immediately backs out of the "firing line" under cover of his shield, and seeks shelter behind the warriors, where he is attended by the women, who still the flow of blood with a packing of clay, gum leaves, powdered bird excrement, and grease. Should a spear, or portion of such, still be lodged in the wound, the assistance of an old man will in all probability be called upon to extricate the piece. If, however, the spear-head breaks in the

attempted removal, no further operation is undertaken then, and perhaps not until the natural suppurative processes force the fragment near to the surface of the body. The aborigines, although they use a knife freely in the mutilation of their bodies, are not sufficiently skilful as surgeons that they can make an incision into muscle or other tissue to extricate a foreign body, such as a broken spear-head, even if the point is pressing the skin outwards on the side opposite the laceration, and the barbs prevent it being pulled back.

When a man collapses on the spot, as from a mortal wound, a regular tussle ensues, in an endeavour to secure the body, between his friends and foes. The former run considerable risk while they expose their figures to the bombardment of spears; they are, however, covered by others, who come forth to specially shepherd them; often, too, a confused hand-to-hand skirmish follows, during which one or two more are wounded. Should it so happen that a man now falls on the opposite side, a compromise is effected, which permits either side to carry off their wounded in peace. In fact, if the disabled men happen to be of important position or particular valour, the casualty may lead to a permanent armistice.

The aboriginal of Australia does not bear chronic malice towards an ordinary or casual enemy, but soon finds a plausible excuse to throw a damper on his fighting ardour; very frequently, indeed, a bloodthirsty-looking crowd drown their enmity in a combined and convivial festivity, during which the late enemies jest, dance, and sing together. The past is soon forgotten; his revenge appeased, a native immediately reverts to his daily routine and peaceful life. With him revenge is not necessarily individual; the wrong-doing of one tribesman might have to be suffered for by another, maybe innocent, man of the same blood. This blood-revenge, which of course is practised by even the most civilized nations, is often the cause of the death of an innocent white man, who happens to be travelling through the tribal ground, where recently another white man has maltreated or assaulted the natives.

In districts where the boomerang is used, a number of these weapons is carried in the belts of the belligerents. When the parties are within seeing distance of each other, each side begins to throw its boomerangs, making them fly high in the air towards the enemy and return to their respective owner. The demonstration is repeated time after time, as the contending

parties draw near to each other, until at length the boomerangs fly well over the opponents' heads on either side. This is forsooth an awe-inspiring spectacle and has the desired effect of arousing the fighters' ire to a very high pitch. At a later stage, boomerangs are employed in actual battle.

In place of the boomerang, the club or the waddy is not infrequently carried as an auxiliary weapon, but its use is restricted to fighting at close quarters.

During the various encounters, as here briefly described, shields are generally carried to parry the missiles directed against the bodies of the combatants. In the case of the light reed-spears, however, the wommera alone is used for such purpose. The fighters' greatest safety is nevertheless in their wonderful skill at dodging the projectiles. In hand to hand fighting, with club or boomerang, the shield is invariably used to considerable advantage.

Whilst undertaking their reconnoitres, the scouts carry slippers, which they wear when it is necessary to hide the individual tracks of their party. These slippers are generally known as "kurdaitja-shoes"; they consist of a thick pad or sole of emu feathers, knitted together with string and clotted blood, and an "upper" of neatly plaited human hair-string. The wearer of such "kurdaitja-shoes" leaves shallow, oval tracks in the sand, which, if seen by any other natives, occasion much alarm, being immediately recognized as those of an enemy on a treacherous mission; if the enemy is not discovered, the tracks are regarded as those of the "*Kurdaitja*," an evil spirit about to molest the tribe.

At the conclusion of a battle, it depends entirely upon the terms, under which arms were laid down, as to who appropriates the bodies of any fallen warriors. If friendly relations are established immediately after cessation of hostilities, a mutual exchange is effected, by means of which the relatives might come in possession of the bodies of any warriors who fell. If, on the other hand, the hatred has not abated after the battle, whatever bodies were captured during the affray belong to that party who were fortunate enough to secure them.

The natives, who have been in the meantime joined by the women, retreat towards their main camp, and carry the corpse or corpses of their fallen upon their shoulders to a place decided upon. There elaborate

obsequies are instituted. It is, moreover, the custom to cut portions of the soft parts from a dead warrior's body, whether he be friend or foe, and to eat them. The belief is that by so doing the brave qualities of the departed soldier will be kept among the tribe and will not all be taken away by the spirit when it migrates to the ancestral hunting grounds. The pieces which are most commonly consumed by the mourners (or victors) are the kidney fat and the marrow of the long bones; the Gulf of Carpentaria tribes eat pieces of the muscle and occasionally of the liver. This is another reason why the Australian aborigines are often referred to as cannibals; but the title is unmerited. The native of Australia does not go head-hunting and does not organize expeditions, whose object is to slay people upon whom they can feast. We are not justified in calling him a cannibal; the most we can say of him is that opportunity might make him an occasional man-eater.

CHAPTER XXI

SPEARS

Spears used for four different purposes—Technically two divisions recognized—Descriptions of types.

Spears made by aboriginal Australians serve for four distinct purposes—for fighting, for hunting, for ceremonial, and for recreation, but it would never do to make these the basis of classification.

Technically, however, we recognize two main divisions, into which Australian spears can be made to fall, the one including all spears made out of a single piece of hard wood, the other those constructed of two or more pieces. With very few exceptions, the former are projected by the hand alone, the latter by means of a specially designed spear-thrower.

The simplest type of spear, found everywhere in Australia, consists merely of a long stick, more or less straightened artificially, and roughly pointed at one or both ends. Along the north coast of Western Australia, the Northern Territory, and Queensland alike, the spear is made of light mangrove wood; in central Australia it is of acacia; and in the south it is, or was, of mallee. Vide [Fig. 5](#), *a*.

Some of the tribes spend considerable time at straightening these spears. The method in vogue is to place the stick with its curved portion in hot ashes, and, after a while, to bend it over a stone until the right shape is obtained; a little emu fat is often applied to the spot before it is heated.

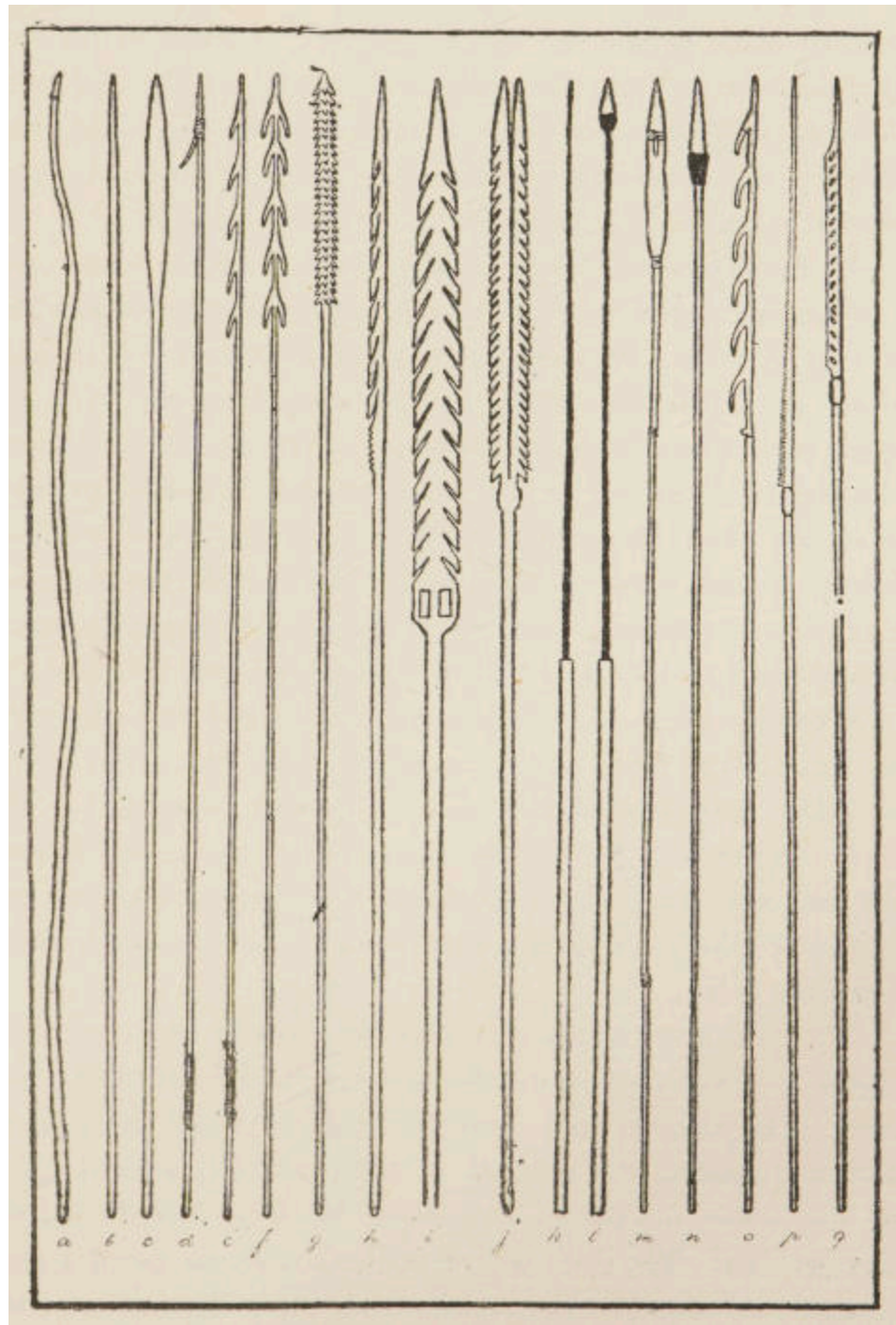


Fig. 5. Types of spears.

Others devote much attention to the shaping of the spear by scraping and rasping its surface. Exceptionally straight and smoothed mulga spears were made by the Barcoo natives of the Durham Downs district and by the Dieri (*b*), whilst on the north coast, the Crocker Islanders' spears are deserving of the same comments; the latter, in addition, are decorated by a few delicate

engravings in the form of circumferential rings and wavy longitudinal bands composed of short parallel transverse lines. The Arunndta groove the spears lengthwise with a stone adze.

An improvement on this type is rendered by the cutting of a pointed blade at one end of the spear (*c*). Some of the best specimens come from the eastern Arunndta in the Arltunga district. The blade is symmetrically cut, sharply edged, and smooth; the remaining portion of the spear is grooved longitudinally throughout its length.

All the above-mentioned types of spear are thrown by hand.

A straight, single-piece, hard-wood spear is made more effective by splicing a barb on to the point with kangaroo or emu sinew (*d*). The barb being directed away from the point, the spear cannot be withdrawn without forcibly tearing it through the flesh of the animal or man it has entered. The natives living along the Great Australian Bight, from Port Lincoln to King George Sound in Western Australia, used to make this the principal weapon; the spear was up to twelve feet in length, perfectly straight and smooth, and was thrown with a spear-thrower.

A rare and perhaps unique variety was found at Todmorden on the Alberga River in the possession of an Aluridja. It was a simple, one-piece, bladed spear, like that described of the Arltunga natives, but it had two wooden barbs tied against one and the same side of the blade with kangaroo sinew, one above the other, at distances of three and six inches, respectively, from the point.

The hard-wood spears may have the anterior end carved, on one or two sides, into a number of barbs of different shape and size. The simplest and most rudimentary forms were to be met with among the weapons of the practically extinct tribes of the lower reaches of the River Murray, including Lake Alexandria. The shaft was of mallee and by no means always straight and smooth; its anterior end, for a distance of from twelve to eighteen inches, had from five to six medium-sized, thorn-like barbs or spikes, which were directed backwards and cut out of the wood, on one or two sides. More rarely one would find spears with a three-sided serrature, consisting of something like two dozen small barbs, directed backwards, extending in three longitudinal lines over a distance of about fifteen inches; at the top the serrated lines merged into a single strong point. Vide [Fig. 5](#), *e*, *f*, and *g*.



A "boned" Man, Minning tribe.

"He stands aghast, with his eyes staring at the treacherous pointer, and with his hands lifted as though to ward off the lethal medium. . . ."

PLATE XXIV

A "boned" Man, Minning tribe.

"He stands aghast, with his eyes staring at the treacherous pointer, and with his hands lifted as though to ward off the lethal medium...."

The most formidable weapons of this kind are those still in daily use as hunting and fighting spears on Melville and Bathurst Islands (*h*). The head of this type has many barbs carved on one side, and occasionally on two diametrically opposite sides. There are from ten to thirty barbs pointing backwards, behind which from four to eight short serrations project straight outwards, whilst beyond them again occasionally some six or more small barbs point forwards. The spears have a long, sharp, bladed point. The barbs are symmetrically carved, and each has sharp lateral edges which end in a point. The size of the barbs varies in different specimens. Many of the spears are longitudinally grooved or fluted, either for the whole length or at the head end only. Usually these weapons are becomingly decorated with ochre, and may have a collar of human hair-string wound tightly round the shaft at the base of the head.

Some of the heaviest of these spears are up to sixteen feet long, and would be more fitly described as lances.

The most elaborate, and at the same time most perfect, specimens of the single-piece wooden spears of aboriginal manufacture are the ceremonial pieces of the Melville Islanders. These have a carved head measuring occasionally over four feet in length and four inches in width, consisting of from twelve to twenty-five paired, symmetrical, leaf-shaped or quadrilateral barbs, whose sides display a remarkable parallelism. The barbs are surmounted by a long tapering point emanating from the topmost pair; and very frequently one finds an inverted pair of similar barbs beneath the series just mentioned. Occasionally, too, the two pairs opposed to each other at the bottom are fused into one, and a square hole is cut into the bigger area of wood thus gained on either side of the shaft (*i*).

The structure may be further complicated by cutting away the point at the top, and separating the paired series of barbs by a narrow vertical cleft down the middle (*j*).

We shall now turn our attention to spears whose head and shaft are composed of separate parts. In the construction of these, two principal objects are aimed at by the aboriginal, the first being to make the missile travel more accurately through space, and in accordance with the aim, the second to make the point more cruel and deadly. Whereas, with one exception, all the single-piece spears, so far discussed, are projected or

wielded with the hand only, in every instance of the multi-pieced spears, a specially designed spear-thrower is used for that purpose.

The native has learned by experience that weight in the forepart of the spear will enable him to throw and aim with greater precision. One has only to watch the children and youths during a sham-fight to realize how well it is known that the heavier end of a toy spear must be directed towards the target whilst the lighter end is held in the hand. Green shoots of many tussocks, or their seed-stalks, and the straight stems of reeds or bullrushes, are mostly used. They are cut or pulled at the root in order that a good butt-end may be obtained, and carefully stripped of leaves; the toy weapons are then ready for throwing. One is taken at a time and its thin end held against the inner side of the point of the right index finger; it is kept in that position with the middle finger and thumb. Raising the spear in a horizontal position, the native extends his arm backwards, and, carefully selecting his mark, shies his weapon with full force at it.

The simplest type of a combination made to satisfy the conditions of an artificially weighted spear is one in which the shaft consists of light wood and the head of heavier wood (*k*). Roughly speaking, the proportion of light to heavy wood is about half of one to half of the other. The old Adelaide tribe used to select the combination of the light pithy flower-stalk of the grass-tree with a straight pointed stick of mallee. The western coastal tribes of the Northern Territory construct small, and those of the Northern Kimberleys large spears composed of a shaft of reed and a head of mangrove; the former being four or at most five feet long, the latter from ten to twelve. The joint between the two pieces is effected by inserting the heavier wood into the lighter and sealing the union with triodia-grass resin or beeswax. The Adelaide tribe used the gum of the grass-tree.

The River Murray tribes used to make the point of the mallee more effective by attaching to it a blade-like mass of resin, into both edges of which they stuck a longitudinal row of quartz flakes.

The Northern Kimberleys natives accomplish the same object by fixing on to the top end of the mangrove stick a globular mass of warm, soft resin, in which they embed a stone spear-head (*l*). In certain parts of the Northern Territory one occasionally meets with a similar type of spear, but such in all probability is imported from the west.

The popular spear of central Australian tribes consists of a light shaft fashioned out of a shoot of the wild tecomia bush (*T. Australis*), which carries a long-bladed head of hard mulga wood. The junction is made between the two pieces by cutting them both on a slope, sticking these surfaces together with hot resin, and securely binding them with kangaroo tendon. The bottom end is similarly bound and a small hole made in its base to receive the point of the spear-thrower (*m*).

As often as not the blade has a single barb of wood bound tightly against it with tendon.

It is often difficult to find a single piece of tecomia long enough to make a suitable shaft, in which case two pieces are taken and neatly joined somewhere within the lower, and thinner, half with tendon. The shoots, when cut, are always stripped of their bark and straightened in the fire, the surfaces being subsequently trimmed by scraping.

A very common type of spear, especially on the Daly River, and practically all along the coast of the Northern Territory, is one with a long reed-shaft, to which is attached, by means of a mass of wax or gum, a stone-head, consisting of either quartzite or slate, or latterly also of glass. The bottom end is strengthened, to receive the point of the thrower, by winding around it some vegetable fibre (*n*).

The natives of Arnhem Land now and then replace the stone by a short piece of hard wood of lanceolate shape.

If now we consider the only remaining type—a light reed-shaft, to which is affixed a long head of hard wood, with a number of barbs cut on one or more edges—we find a great variety of designs. The difference lies principally in the number and size of the barbs; in most cases they point backwards, but it is by no means rare to find a certain number of them pointing the opposite way or standing out at right angles to the length of the head. These spears belong principally to the northern tribes of the Northern Territory.

The commonest form is a spear having its head carved into a number of barbs along one side only, and all pointing backwards (*o*). The number ranges from three to over two dozen, the individual barbs being either short and straight or long and curved, with the exception of the lowest, which in

many examples sticks out at right angles just above the point of insertion. The point is always long and tapering. These spears are common to the Larrekiya, Wogait, Wulna, and all Daly River tribes.



1. Dieri grave, Lake Eyre district.



2. Yantowannta grave, Innamincka district.

PLATE XXV

1. Dieri grave, Lake Eyre district.

2. Yantowannta grave, Innamincka district.

The same pattern of barbs may be found carved symmetrically on the side diametrically opposite, or, indeed, it may be cut in three planes.

An elegant, but rare, type is found among the weapons of the Ponga Ponga, Mulluk Mulluk, and Wogait tribes on the Daly River. Its hard-wood head is long and uniformly tapering from its point of insertion to its sharp tip. On one side there are very many small barbs, diminishing in size from the shaft upwards; as many as one hundred barbs have been counted; they point either slightly backwards or at right angles to the length (*p*).

A spear in use on the Alligator River, and in the districts south and west therefrom, has the barbs along the edge of the anterior moiety directed backwards, whereas those of the posterior portion point forwards. And occasionally one finds the barbs arranged asymmetrically on two sides of the spear-head.

Finally, a rather remarkable type will be referred to, which belongs to the Arnhem Land tribes, or rather to the country extending from Port Essington to the Roper River, including Groote Island and smaller groups lying off the coast. It is a neat and comparatively small spear, about eight feet long on an average. The head, instead of possessing a number of barbs, has a series of eye-shaped holes cut along one of its sides, which give the impression of being so many unfinished barbs, or so many barbs with their points joined together (*q*). The major axes of the holes are parallel and directed backwards; there may be up to thirty holes present. Occasionally there are a few real barbs cut near the shaft end of the head; or a number of incomplete barbs may there be cut with their axes turned towards the front of the spear. The point is always sharp and stands back somewhat from the level of the uncut barbs.

For special purposes, like fishing, two or three of the simple-barbed prongs are frequently affixed to a reed shaft with beeswax or resin, and vegetable fibre. This combination is met with all along the coast of the Northern Territory. The natives know very well that the chances of stabbing a fish with a trident of this description are much greater than with a single prong. As a matter of fact, a barbed spear with less than two prongs is not normally used for fishing purposes, yet a plain, single-pronged spear is often utilized when there is none of the other kind available.

The Australian aboriginals do not poison their spears in the ordinary sense of the word, but the Ponga Ponga and Wogait tribes residing on the Daly River employ the vertebræ of large fish, like the barramundi, which have previously been inserted into decaying flesh, usually the putrid carcase of a kangaroo, with the object of making the weapon more deadly. The bones are tied to the head of a fighting spear. This is not a general practice, however, and the spear never leaves the hands of the owner. The natives maintain that by so doing they can kill their enemy “quick fella.”

CHAPTER XXII

SPEAR-THROWERS

Principle of construction—How held—Some of the common types described—Other uses.

To assist in the projection of a spear, the aboriginal has invented a simple apparatus, which is commonly referred to as a spear-thrower or wommera. In principle it is just a straight piece of wood with a haft at one end and a small hook at the other. In practice the hand seizes the haft, the hook is inserted into the small pit at the bottom of the spear, and the shaft is laid along the thrower and held there with two of the fingers of the hand, which is clasping the haft. In this position, the arm is placed well back, the point of the spear steadied or made to vibrate, and, when the native has taken careful aim, the arm is forcibly shot forwards. The missile flies through space, towards its target, but the thrower is retained by the hand.

One of the simplest types was made by the tribes living along the shores of the Great Australian Bight. It consists of a flat piece of wood, about three feet long, roughly fluted lengthwise and slightly sloped off at either extremity. At one end a mass of resin forms a handle, in which, moreover, a quartzite or flint scraper is embedded. At the other end a wooden peg is affixed with resin against the flat surface of the stick. Both surfaces of the implement are flat or slightly convex; at Esperance Bay they are rather nicely polished, the wood selected being a dark-coloured acacia. Towards the east, however, as for instance at Streaky Bay, the inner side, i.e. the one bearing the hook or peg, becomes concave and the outer side convex. On Eyre Peninsula, the old Parnkalla tribe made the spear-thrower shorter but wider, and its section was distinctly concave.

Northwards, through the territories of the Kukata, Arrabonna, Wongapitcha, Aluridja, Arunndta, and Cooper Creek tribes, the shape becomes leaf-shaped and generally of concave section, with a well-shaped haft and broad flint scraper; the peg is attached with resin and sinew. Within this same area, another type is less frequently met with, which is of similar shape, but flat; it is really used more for show purposes, and for that reason

is usually decorated with engraved circles and lines, which during some of the ceremonies are further embellished with ochre and coloured down.

The last-named is the prevalent type, which extends westwards as an elongate form through the Murchison district right through to the Warburton River, where it is again broader. In both the areas mentioned, the inner surface of the spear-thrower is deeply incised with series of parallel, angular bands made up of transverse notches. In the south of Western Australia, the shape remains the same, but the incised ornamentation disappears.

Yet another variety comes from the old Narrinyerri tribe and from the lower reaches of the River Murray, where it was known as "*taralje*." It is a small, flat, spatulate form, elongated at both ends, the lower (and longer) prolongation making the handle, the upper carrying a point of bone or tooth deeply embedded in resin. The inner side, against which the spear is laid, is flat, the outer surface being convex. The handle is circular in section and is rounded off at the bottom to a blunt point. The convex side is occasionally decorated with a number of pinholes, arranged in a rudely symmetrical pattern.

All through the northern districts of the Northern Territory and the Northern Kimberleys, the principal type is a long light-wood blade, tapering slightly from the handle end to the point and having comparatively flat or slightly convex sides. A handle is shaped by rounding off the ends and cutting away some of the wood symmetrically on each side, a few inches down. A clumsy-looking peg is attached to one of the flat surfaces at the opposite, narrower end with beeswax. The peg is made big on account of the instrument being exclusively used to propel the reed-spears, which are naturally hollow, and consequently have a large opening or pit at the bottom end. This type of thrower is nearly always decorated in an elaborate way with ochre. When used, the thrower and spear are held by the right hand in such a way that the shaft of the latter passes, and is held, between the thumb and index finger, the remaining fingers holding the handle of the thrower. Vide [Plate XIV](#), 2.

A spear-thrower used exclusively for projecting the small variety of reed-spear is known to the Larrekiya, Wogait, Wordaman, Berringin, and a few other coastal tribes of the Northern Territory. It consists of a rod of hard wood, four feet or so in length, tapering a little towards either end. A lump

of resin is attached to one end, and, whilst warm and plastic, is moulded into a blunt point, which fits into the hole at the bottom of the spear. At about five inches from the opposite end, a rim of resin is fixed, and from it a layer, decreasing in thickness, is plastered around the stick to near the extremity. When using this thrower, the hand is placed above the resin-rim, and the shaft of the spear is held by the thumb against the top of the middle finger, without the aid of the index finger. In addition to this, its principal function, the thrower is often used for making fire, the native twirling its lower point against another piece of wood.

A variety of the above type is found in the Gulf of Carpentaria country, on the MacArthur River, which has a tassel of human hair-string tied with vegetable fibre immediately below the rim of resin around the handle.

One of the most remarkable of all spear-throwers is made by the Larrekiya, and other Northern Territory tribes, consisting of a long, leaf-shaped, and very thin, flexible blade, flat on one side and slightly convex on the other. The peg is pear-shaped, and is fixed with vegetable string and beeswax. The handle is thick and cone-shaped, and covered with a thin layer of resin or wax. It is ornamented with rows of small pits, which are pricked into the mass while warm with the point of a fish bone or sharpened stick. The instrument is so thin and fragile that only experienced men dare handle it. At times the blade is curved like a sabre.

In addition to serving as a projecting apparatus, most of the hard-wood spear-throwers with sharp edges are used for producing fire by the rubbing or sawing process; those of concave section also take the place of a small cooleman, in which ochre, down, blood, and other materials are stored during the “making up” period of a ceremony.

Any of the flat types of spear-thrower may be used for making fire by the “sawing process.” The edge of the implement is rubbed briskly across a split piece of soft wood until the red-hot powder produced by the friction kindles some dry grass which was previously packed into the cleft. The spark is then fanned into a flame, as previously referred to ([page 111](#)).

CHAPTER XXIII

BURIAL AND MOURNING CUSTOMS

Customs depend upon a variety of circumstances—Child burial—Cremation disavowed—Interment—Graves differently marked—Carved tomb-posts of Melville Islanders—Sepulchral sign-posts of Larrekiya—Platform burial—Mummification of corpse—Skeleton eventually buried—Identification of supposed murderer—Pathetic scenes in camp—Self-inflicted mutilations—Weird elegies—Name of deceased never mentioned—Hut of deceased destroyed—Widowhood's tribulation—Pipe-clay masks and skull-caps—Mutilations—Second Husbands—Collecting and concealing the dead man's bones—Treatment of skull—Final mourning ceremony.

The burial and mourning ceremonies, if any, attendant upon the death of a person, depend largely upon the tribe, the age, and the social standing or status of the individual concerned. Old people who have become "silly" (i.e. childish), and who in consequence do not take an active part in any of the tribal functions or ceremonies, are never honoured with a big funeral, but are quietly buried in the ground. The reason for this is that the natives believe that the greater share of any personal charms and talents possessed by the senile frame have already migrated to the eternal home of the spirit. As a matter of fact, the old person's spirit has itself partly quitted the body and whiles for the most time in the great beyond. For precisely the same reason, it often happens that a tribe, when undergoing hardship and privation brought about by drought, necessitating perhaps long marches under the most trying conditions, knocks an old and decrepit person on the head, just as an act of charity in order to spare the lingering soul the tortures, which can be more readily borne by the younger members. These ideas exist all over Australia.

When infants die, they are kept or carried around by the mothers, individual or tribal, for a while in a food-carrier, and then buried without any demonstration. The extinct Adelaide tribes required of the women to carry their dead children about with them on their backs until the bodies were shrivelled up and mummified. The women alone attended to the burial of the child when eventually it was assigned to a tree or the ground.

But at the demise of a person in the prime of his or her life, and of one who has been a recognized power in life, the case is vastly different. Both before and after the “burial” of the corpus, a lengthy ceremony is performed, during which all sorts of painful mutilations are inflicted amongst the bereaved relatives, amidst the accompaniment of weird chants and horribly uncanny wails. Before proceeding with the discussion of the attendant ceremonies, however, we shall give an outline of the different methods adopted in Australia for the disposal of the dead.

Cremation is nowhere practised for the simple reason that the destruction of the bony skeleton would debar the spirit from re-entering a terrestrial existence.

The spirit is regarded as the indestructible, or really immortal, quantity of a man’s existence; and it is intimately associated with the skeleton. The natives tender, as an analogy, the big larva of the Cossus or “witchedy,” which lies buried in the bark of a gum tree. As a result of its ordinary metamorphosis, the moth appears and flies away, leaving the empty shell or, as the natives call it, the “skeleton” of the “dead” grub behind. It is a common belief on the north coast that the spirit of a dead person returns from the sky by means of a shooting star, and when it reaches the earth, it immediately looks around for its old skeleton. For this reason the relatives of a dead man carefully preserve the skeletal remains, carry them around for a while, and finally store them in a cave.

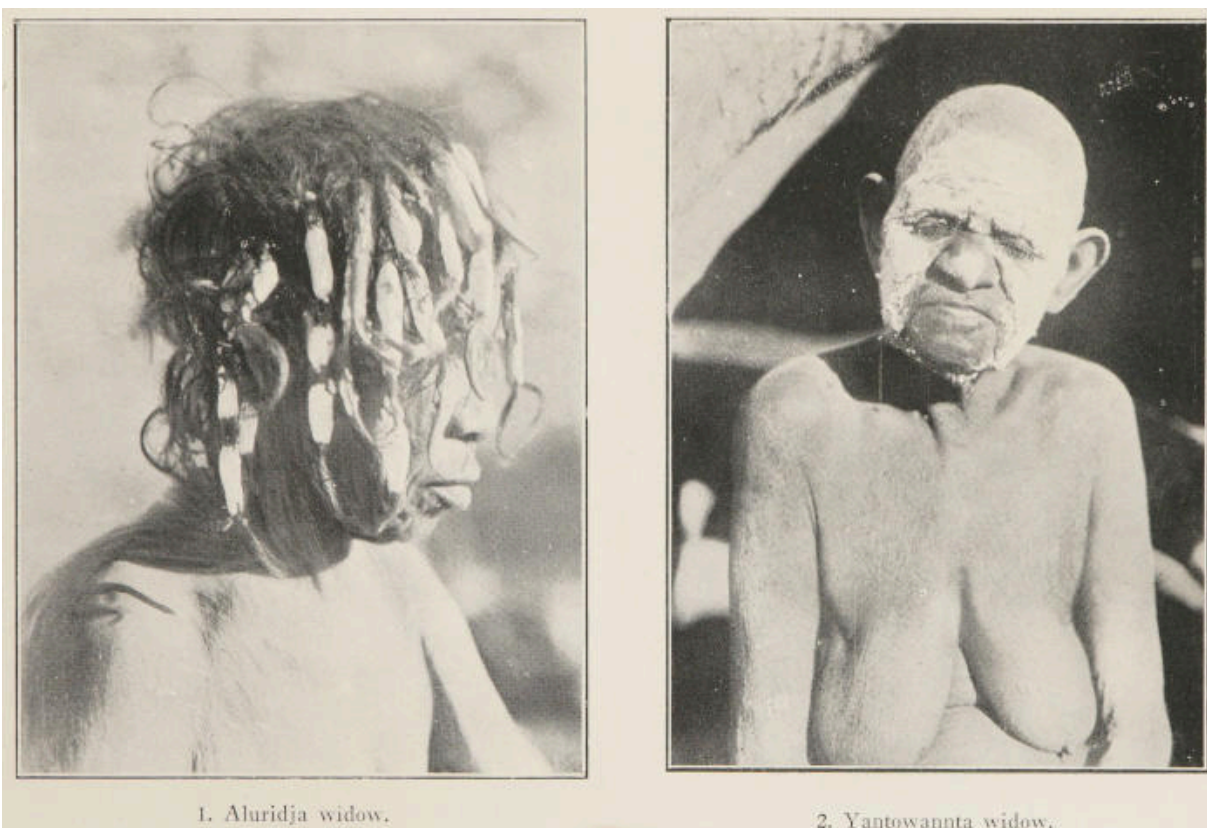


PLATE XXVI

1. Aluridja widow.

2. Yantowannta widow.

Stillborn children are usually burnt in a blazing fire since they are regarded as being possessed of the evil spirit, which was the cause of the death.

The simplest method universally adopted, either alone or in conjunction with other procedures, is interment.

Most of the central tribes, like the Dieri, Aluridja, Yantowannta, Ngameni, Wongapitcha, Kukata, and others, bury their dead, whilst the northern and southern tribes place the corpse upon a platform, which they construct upon the boughs of a tree or upon a special set of upright poles. The Ilyauarra formerly used to practise tree-burial, but nowadays interment is generally in vogue.

A large, oblong hole, from two to five feet deep, is dug in the ground to receive the body, which has previously been wrapped in sheets of bark,

skins, or nowadays blankets. Two or three men jump into the hole and take the corpse out of the hands of other men, who are kneeling at the edge of the grave, and carefully lower it in a horizontal position to the bottom of the excavation. The body is made to lie upon the back, and the head is turned to face the camp last occupied by the deceased, or in the direction of the supposed invisible abode of the spirit, which occupied the mortal frame about to be consigned to the earth. The Arunndta quite occasionally place the body in a natural sitting position. The Larrekiya, when burying an aged person, place the body in a recumbent position, usually lying on its right side, with the legs tucked up against the trunk and the head reposing upon the hands, the position reminding one of that of a *fœtus in utero*.

The body is covered with layers of grass, small sticks, and sheets of bark, when the earth is scraped back into the hole. But very often a small passage is left open at the side of the grave, by means of which the spirit may leave or return to the human shell (i.e. the skeleton) whenever it wishes.

The place of sepulture is marked in a variety of ways. In many cases only a low mound is erected over the spot, which in course of time is washed away and finally leaves a shallow depression.

The early south-eastern (Victorian) and certain central tribes place the personal belongings, such as spear and spear-thrower in the case of a man, and yam-stick and cooleman in the case of a woman, upon the mound, much after the fashion of a modern tombstone. The now fast-vanishing people of the Flinders Ranges clear a space around the mound, and construct a shelter of stones and brushwood at the head end. They cover the corpse with a layer of foliage and branches, over which they place a number of slabs of slate. Finally a mound is erected over the site.

The Adelaide and Encounter Bay tribes built wurleys or brushwood shelters over the mound to serve the spirit of the dead native as a resting place.

In the Mulluk Mulluk, when a man dies outside his own country, he is buried immediately. A circular space of ground is cleared, in the centre of which the grave is dug. After interment, the earth is thrown back into the hole and a mound raised, which is covered with sheets of paper-bark. The bark is kept in place by three or four flexible wands, stuck into the ground at their ends, but closely against the mound, transversely to its length. A

number of flat stones are laid along the border of the grave and one or two upon the mound.

In the Northern Kimberleys of Western Australia, when an unauthorized trespasser is killed by the local tribes, the body is placed into a cavity scooped out of an anthill, and covered up. In a few hours, the termites rebuild the defective portion of the hill, and the presence of a corpse is not suspected by an avenging party, even though it be close on the heels of the murderers.

The Dieri, in the Lake Eyre district of central Australia, dispense with the mound, but in its place they lay a number of heavy saplings longitudinally across the grave. Their eastern neighbours, the Yantowannta, expand this method by piling up an exceptionally large mound, which they cover with a stout meshwork of stakes, branches, and brushwood lying closely against the earth ([Plate XXV](#), 1 and 2).

One of the most elaborate methods is that in vogue on Melville and Bathurst Islands. The ground immediately encompassing the grave is cleared, for a radius of half a chain or more, and quantities of clean soil thrown upon it to elevate the space as a whole. The surface is then sprinkled with ashes and shell debris. The mound stands in the centre of this space, and is surrounded by a number of artistically decorated posts of hard and heavy wood, or occasionally of a lighter fibrous variety resembling that of a palm. Each of the posts bears a distinctive design drawn in ochre upon it; several of the series in addition have the top end carved into simple or complicated knobs; occasionally a square hole is cut right through the post, about a foot from the top, leaving only a small, vertical strip of wood at each side to support the knob ([Fig. 14](#)). The designs are drawn in red, yellow, white, and black, and represent human, animal, emblematical, and nondescript forms.

The Larrekiya erect a sort of sign-post, at some distance from the grave, consisting of an upright pole, to the top of which a bundle of grass is fixed. A cross-piece is tied beneath the grass, which projects unequally at the sides and carries an additional bundle at each extremity. The structure resembles a scarecrow with outstretched arms, the longer of which has a small rod inserted into the bundle of grass to indicate the direction of the grave. Suspended from the other arm, a few feathers or light pieces of bark are

allowed to sway in the wind and thus serve to attract the attention of any passers-by.

When the body is to be placed upon a platform, it is carried, at the conclusion of the preliminary mourning ceremonies, shoulder high by the bereaved relatives to the place previously prepared for the reception of the corpse. A couple of the men climb upon the platform and take charge of the body, which is handed to them by those remaining below. They carefully place it in position, and lay a few branches over it, after which they again descend to join the mourners. The platform is constructed of boughs and bark, which are spread between the forks of a tree or upon specially erected pillars of wood.

The Adelaide tribe used to tie the bodies of the dead into a sitting position, with the legs and arms drawn up closely against the chest, and in that position kept them in the scorching sun until the tissues were thoroughly dried around the skeleton; then the mummy was placed in the branches of a tree, usually a casuarina or a ti-tree. Along the reaches of the River Murray near its mouth, the mummification of the corpse was accelerated by placing it upon a platform and smoking it from a big fire, which was kept burning underneath; all orifices in the body were previously closed up. When the epidermis peeled off, the whole surface of the corpse was thickly bedaubed with a mixture of red ochre and grease, which had the consistency of an ordinary oil-paint. A similar mummification process is adopted by certain of the coastal tribes of north-eastern Queensland.

The Larrekiya, Wogait, and other northern tribes smear red ochre all over the surface of the corpse, prior to placing it aloft, in much the same manner as they do when going to battle. The mourners, moreover, rub some of the deceased's fat over their bodies. When eventually all the soft parts have been removed from the skeleton by birds of prey, and by natural processes of decomposition, the relatives take the radius from the left arm, which they carry away with them. The remaining bones are collected and wrapped up in paperbark, and the parcel buried.



Tooth-rapping ceremony, Wongapitcha tribe.

"The novice lies on his back and rests his head against the operator's thighs, while a number of men sit around in a semi-circle." The operator is seen in the act of applying the rod with his right hand, while he is striking it with a pebble held in the opposite hand.

PLATE XXVII

Tooth-rapping ceremony, Wongapitcha tribe.

"The novice lies on his back and rests his head against the operator's thighs, while a number of men sit around in a semi-circle." The operator is seen in the act of applying the rod with his right hand, while he is striking it with a pebble held in the opposite hand.

In the Northern Kimberleys of Western Australia, the relatives wait until the body has so far decomposed that it begins to drip, at which stage they place a number of pebbles or other articles either in a row or in a circle underneath the platform. Each pebble represents a person who is considered as a likely cause of the death they are bemoaning. Periodical inspections are made of the place, and notice is taken of the drops which have fallen from the corpse. Should it so happen that the wind has blown them in the direction of any one particular stone, which has thereby been moistened, the person represented by that pebble is looked upon as the one responsible for the fellow's death; a resolution is forthwith carried to "bone" him to death.

At the same time the visitors keep a vigilant lookout for any tracks near the grave, which might inform them of the presence of a spirit nearby. Like the Northern Territory tribes they, too, later collect the bones of the deceased and wrap them up in paperbark. These parcels, together with the skull, are deposited in the crevices of rocks outcropping within their haunts, or they are stuck away in a cave, if such be available.

Great is the hullabaloo in a camp when a person of importance breathes his last. Moans and deep sobbing notes are followed by loud yells and spasms of barbarous shrieks, which it is difficult to believe are human; and the yelping hordes of dogs, which are found in every camp, in no small way intensify the din. After a while the pandemonium settles down to a more orderly wailing, although every now and then there might be a spontaneous outburst of the heart-rending yelling again, which can only be likened to a long-drawn canine or, more nearly, a dingo-like howl. The note is taken up by all members of the little community; and the moment the noise is heard by anybody strolling or hunting in the environment he, without deliberation, hastens back to camp to join in the wailing. The men sit with their knees drawn up and their arms thrown around them, covering their faces whilst they are sobbing. The women throw themselves upon the ground, or over the body of the departed, in utter despair; they are later joined by the men. Every now and then the lubras rise, and, seizing a sharp stick with both their hands, they cut deep gashes into the crown of their heads. Then, as the blood pours down over their faces and bodies, the wailing is accentuated with additional vehemence. At times some terrible wounds are inflicted during this part of the obsequies. The widow often cuts a long, median gash right along the scalp. The men, on the other hand, flourish their big stone knives, with which they hack their bodies in a revolting manner. In the Katherine River district, the nearest relatives on the male side not infrequently cut their thighs in such a way that almost the entire mass of muscles on the extensor side is severed, and the man makes himself *hors de combat*. A general mêlée now ensues, during which women deface themselves and each other without restraint, the places of predilection being the head and back. Each mourner submits to the mutilation voluntarily and without flinching. The women, too, make free use of their nulla-nullas, with which they crack each other over the head. But a short while after they will seat themselves in groups about the body, with their arms tenderly thrown around each other, crying bitterly.

Repeatedly I have been present when sad or distressing news has unexpectedly come to hand, or when one of the tribe meets with a painful accident which may be considered fatal, and have noticed with what amount of undisguised sympathy such are received on the part of the women-folk. On one occasion I remember a young gin falling from a high cliff on the Finke River and sustaining a concussion of the brain. As she lay unconscious on the ground, all other women present at the time tore out great quantities of hair from their scalps, and then threw themselves into some spiny tussocks of porcupine grass which grew close-by. The poisonous sting of the porcupine grass is very painful, even when only one enters the skin; but the agony produced by a large number piercing the naked body must be excruciating.

The Larrekiya men lacerate their upper arms and thighs with stone knives, and cut their foreheads with the embedded flints of any handy implements. Both men and women cover their naked bodies with ashes and pipeclay, and, after the preliminary uproar has calmed down somewhat, the females start a doleful chant which sounds something like: "*Nge-e-u, hö-hö-un-un.*" To this the men respond with long-drawn monotones resembling: "*He-e-ö, he-e-ö, he-e-ö,*" the "*n*" and "*un*" above, and the "*ö*" below, sounding like sobs.

The chanting is kept up all the time the corpse is "lying in state," if one be permitted to make use of this phrase in connection with a primitive burial ceremony. Even whilst the body is being conveyed upon the shoulders of the aboriginal pall-bearers, the wailing continues in a systematic manner.

I remember once attending a native funeral at Brocks' Creek in the Northern Territory, when a gin had died who came from a far-distant tribe beyond the Victoria River. Being a stranger, the local tribal honours could not be bestowed upon her remains, but the local natives, who volunteered to bury her, could not let the opportunity pass without singing in a mournful strain as they carried her to rest. The gin's dialect was unknown, and the local tribe had been in the habit of conversing with her in ordinary "pidgin English." Consequently they concocted a little refrain of their own to suit the occasion. It ran "Poor beggar Jinny, him bin die," and was rendered in a sing-song style, like a decimal repeater, throughout the ceremony.

Everywhere in Australia it is the custom among the indigenous people never to mention the name of the person whose death is being lamented. This rule is so far-reaching that should there be more than one tribesman holding the same name, the one surviving his namesake immediately changes his appellation. If, too, the name of the dead one happened to be that of an animal or place, a new word is immediately introduced in the vocabulary of the tribe in place of the former. Thus allusion to the dead man's name is entirely avoided. The reason for this strange custom is that the tribespeople want the spirit of the departed not to be molested; by calling aloud the name of one who has gone beyond, the spirit might be persuaded to come back and haunt the camp; the natives are in constant dread of this. On the other hand, by not addressing the spirit, there is no reason for it to leave the happy ancestral grounds, in which it can consort with all its kin long-departed.

For much the same reason, the hut or wurley of the dead person is immediately destroyed by the relatives of the dead man; if the habitation is allowed to stand, the spirit of the dead will endeavour to come back and occupy it. The natives would be continually encountering the ghost, if not actually then certainly in their imagination, and the fear of such a possibility would make their existence intolerable. Most of the tribes, moreover, so soon as they have destroyed the dead man's wurley, instantly leave the district and select another camping site, well removed from the latter. The only exception to this general rule is the Adelaide tribe who, as previously mentioned, used to build a special bark hut over the grave for the spirit to dwell in.

The person who fares worst is a widow. To begin with, she is required to absent herself and live apart from the rest of the people in a small humpy of her own; and she is not allowed to eat anything during the term of the mourning ceremonies. Quite apart from the general avoidance of mentioning the deceased's name, a widow is positively forbidden to speak to anyone for a term of from a week or two to several months. During all that time, she must observe the strictest rules of tribal mourning; for, if she does not, the spirit will see that her late husband's memory is not sufficiently revered, and it will starve the woman to death.

Directly after the death of her husband, a Yantowannta woman must cut off her hair, short to the scalp, and burn it. In its place, she applies a thick

coating of pipeclay paste, which is periodically added to if there be a tendency for it to crumble away. In addition, she covers most of her face with a similar paste which adheres to the skin like a mask. Vide [Plate XXVI](#), 2.

The early Murray River tribes made a skull-cap of burned gypsum or lime, about three inches thick, which the widows had to wear during the term of mourning. These encumbrances weighed up to sixteen pounds. The hair was generally removed previously by singeing it with hot ashes.

In addition the relict has to regularly cover her body and face with white ashes. In the Daly Waters district, whenever she pays a visit to her late husband's burial place, she will submit herself to the agony of re-opening the wound in her scalp, until it bleeds profusely, to prove how deep her sorrow is.

The Aluridja widows do not cut their hair short, but smear pipe-clay paste and ashes thickly over the scalp, intimately mixing it with the woolly growth. Often the hair is worked up into a large number of locks or strands, round which the white paste is moulded in such a way that the head is surrounded by an array of pendant, cylindrical masses resembling so many candles ([Plate XXVI](#), 1).

A woman, upon the decease of her husband, becomes the property of her late partner's brother; if there are more than one brother surviving, she falls to the senior among them. In the case of no brothers remaining or existing, she is claimed by the dead husband's nearest (male) tribal relative. The law prevails practically all over the Southern Continent. It is not until she is actually received by her new husband that the woman is permitted to speak to anyone. This usually ends the first period of mourning, so far as the gin is concerned, and she returns to live with the others in the main camp; but in most cases she will continue to smear pipe-clay over her scalp for some time longer.

An Arundta woman who survives three tribal husbands is not required by law to marry again, and she is, consequently, left unmolested.

The second period of public mourning is a comparatively short one; it is begun by collecting the dead man's bones from the tree or platform. In nearly every case the bulk of the bones are packed in sheets of paperbark

and hidden or buried. In the north-central and north-western districts, the parcels are either hidden in a cave, buried in an anthill, or stuck into the fork of a dead tree. The cranium is often smashed to pieces or the facial skeleton broken away from the skull-roof. In the old Narrinyerri and certain tribes of the Adelaide plains the calvarium was used as a drinking vessel; a handle was attached by fastening a piece of strong fur-string to the *occiput* through the *foramen magnum*, on the one side, and to the frontal portion, after breaking a passage through the orbital cavities, on the other. Many tribes besmear the skulls with red ochre before assigning them to their last resting place. In the Northern Kimberleys some of the sepulchral caves are so crowded with skulls, arranged in perfect order, that one is reminded of the classical catacombs.

The women are not present when the bones are collected, but, remaining in camp, they start to wail afresh, and even resort to further mutilating the body. When the men return, carrying with them the radius of the dead man, a ceremony is conducted, in which both sexes participate. The women, including the widow, now discard the white cover of pipe-clay and ashes; and in its place they decorate their bodies with red and yellow ochre, and occasionally with charcoal. These colours are applied in vertical bands or lines over the chest and back; whilst across the shoulders there are usually a few horizontal lines. The men have more elaborate designs worked over their chests and backs with ochre and kaolin. After the ceremony is over, the radius is either buried or claimed by the deceased's brother, who uses it as a pointing-bone, it playing a particularly important role during any expedition, which may be undertaken against the tribe suspected of foul play in connection with his relative's death.

CHAPTER XXIV

TRIBAL ORGANIZATIONS

Psychological trend of thought—Primitive and modern systems of nomenclature compared—Multiplicity of Names—Their derivation—Connection with marriage systems—Family crests—“Kobongism” and “Totemism”—Group relationships—Infant betrothals—Business-like courtship—Position of wife—Elopements—Tribal profligates—True wifehood—Hospitable licentiousness—Mutual exchanges of wives.

It is difficult for an European living in the twentieth century to train his mind into the way of thinking like an aboriginal. To require of a person to do so would be like asking his reasoning to slip back through the long ages of progress and mental development, which are primarily responsible for his now being able to hold a foremost position amongst his fellow creatures. The convenience of modern achievements so transcends the awkwardness of primal ways that we might as well suggest to him that the better way of meeting an absent friend would be to walk to a spot he was last seen at, than to catch an electric train bound for the city and despatch a message from the nearest telegraph office to ensure the meeting. The first method would be the primitive, the second the up-to-date. On the other hand, our present systems are satisfied to ascertain a man's individual identity—his Christian and surnames—but to bother little about anything else. Some of our best families certainly pride themselves upon their ancestral history, and honour the crest which once ranked prominently in the social world, but the general tendency, especially of a democratic world, is to meet this sentiment with a satirical sneer. In consequence, our national history is sacrificing much of its constituent, individual charm, and our nomenclature, so far as any original class-systems are concerned, is daily becoming more meagre and commonplace. Nowadays a Smith is simply a Smith, good or bad, with or without genealogical traditions and records. And this is the point we are leading up to, which is so very different with the aboriginal. His vocabulary bristles with a nomenclature so full of ancestral derivations and traditional origins that a single word combines at once identity with genealogy; the English language does not contain a single word which could convey the same amount of meaning; a number are required to explain the sense. There

is no such person as a simple Smith among the aborigines; Smiths there might be, but each is separately described and qualified by his appellation; each person carries his crest, if not his pedigree, embodied in his name. Everybody can understand the significance of the name the moment he hears it; and this understanding is very far-reaching, and as potent as the bonds of freemasonry.

Every individual has a number of names, some of which are never publicly used, but are only known to members of the tribe who stand in very close relationship to him. The names are really more expressions of degree, rank, maturity, and division, rather than personal appellations or addresses. All elders who have officiated during the term of initiation of a novice, or who have instructed or tended a child prior to its attaining its independence, automatically assume a name or title, which, within a restricted circle, explains the social standing of the particular individual.

Then there are factious names, of a religious or sacred character, each of which directly relates to the accepted affinity existing between a living person and an ancestral spirit of semi-human origin. These names are kept very secret, but are embodied in the carved "*tjuringas*," which are only exposed to a limited number of persons upon rare occasions, and then very temporarily.

The names which are most commonly applied to individuals are, in reality, pet or nicknames. A special characteristic, a likeness, a scar, or an abnormality is immediately seized upon as a mark of distinction to embody in the individual's name. Countless examples could be mentioned to illustrate this point. For instance, a man of the Minning tribe at Eucla is called "*Jinnabukarre*" (Lumpy-foot), an old Arunndta man is known as "*Ulgna-bong*" (Blind-eye), and a Wongapitcha man as "*Jinna-Kularrikna*" (Stink-foot). The name I generally travel under amongst the Arunndta is "*Atutannya*" (Big Man), having been thus christened by them on account of my bodily height. Again, circumstances at birth or any presumed causes of conception often determine the name of the offspring. "*Unndulia*," meaning a "shadow," is both the name of a place with a legend, situated in the MacDonnell Ranges, and of a girl, who is supposed to have entered her mother's womb there. Many people have names of animals and birds, with which they have some sort of mystic relationship and legendary connections; in the Arunndta such names as "*Illiya*" (Emu), "*Utnuringita*"

(Witchedy Grub), “*Irridja*” (Eaglehawk), “*Ladjia*” (Yam), and the like are daily met with.

Names are frequently compounded, the resulting word embodying locality, peculiarity, ancestry, animal-relationship, and division all in one. As a person grows older his name usually becomes longer, and in a limited sense recounts his biography. There is apparently no limitation to the length of a name, but the whole word is rarely spoken; yet each possessor of a long name commits it well to memory. The longest personal appellation, which has come before our notice, is that of a very aged woman of the Dieri tribe; expressed phonetically it ran: “*Yangingurrekupulapaiawattimakantana*.”

The intricate and elaborate systems of nomenclature are closely connected with their marriage laws and secret rites. With a few exceptions, these are much the same all over Australia. So far as their marriage systems are concerned, each tribe may be primarily split into two great divisions, between which unions take place; but among members of one and the same division such are forbidden under penalty of death.

Each moiety is further separated into sub-divisions or groups, which are distinguished by a symbol related to a family-crest. The symbol is representative of a natural object, such as animal, bird, or plant, between which and the individual a mysterious relationship is believed to endure. This peculiar belief was first reported to exist among the Australian aborigines by Sir George Grey in 1841, who ascertained that the general name of the mysterious symbol with which an individual identifies himself, was “*kobong*.” Of later years the word has been replaced in works on Australian anthropology by “totem.” No doubt “*Kobongism*” and “Totemism” are closely allied conceptions of crude religious significance; but at the same time the “totem” belongs originally to the American Indian, and it is still an open question whether the imported word completely and adequately embraces all fundamental conceptions of the “*kobong*.”

Among the central tribes the mystical relationship between the present individual and the object (animal or plant) is believed to come through sacred semi-human ancestors which were common to both; the relationship has been handed down from one generation to another. See further, [Chapter XXVII](#).

Each divisional group has a number of such “*kobongs*” or “totems,” which practically control their marriage-systems. Descent in a family is always reckoned from the mother’s side, at any rate so far as the majority of tribes is concerned. To take a simple example: A tribe is composed, say, entirely of families named Jones and Smith. A Smith is only allowed to marry a Jones, and a Jones a Smith. But, in addition, each individual member of the two groups of families named has a separate crest or symbol, such as, for instance, the cat, the dog, the fowl, the rose. A further restriction is that no two members of the same crest are allowed to join in matrimony, so that no Cat-Jones can marry a Cat-Smith, nor a Dog-Smith a Dog-Jones, because they are “related.” But a Cat-Jones may marry a Dog-, a Fowl-, or a Rose-Smith, and *vice versa*, without breaking the law. We will find that there are twelve different combinations possible between the Jones and Smiths of the four crests specified. If there be an issue of the marriage, we have heard that the descent always follows the mother’s side, both as regards family and crest. Consequently, if a Mr. Cat-Jones marries a Miss Dog-Smith, the child will be a Dog-Smith; but should the Dog-Smith children again marry into the Cat-Jones’ family-group, the offspring becomes a Cat-Jones if the child be a boy, but remains a Dog-Smith if a girl.

The north-west central tribes split up each moiety into two sub-divisions, between which marriages can take place, but the progeny always goes to the division different from that of its parents. Let us represent the two pairs of subdivision by A and B, and C and D respectively. A man of the A group is only allowed to marry a woman of the B group, and a B-man only an A-woman. The child resulting from the former union becomes a member of neither of its parents’ groups, but passes over to C, and when one from the latter, that is, when the father is a B-man, becomes a D-member. On the other side, when a C-man marries a D-woman, the issue returns to the A line, and when a D-man marries a C-woman, it goes back to the B. If, for instance, we replace the letters A, B, C, and D by the words “*Pultara*,” “*Kumara*,” “*Panunga*,” and “*Purula*,” in the order specified, we have the general principle of the Arundta marriage system.

Among the Minning, the four subdivisions are called “*Tjurrega*,” “*Menuaitja*,” “*Kakera*,” and “*Ngadeja*.” Let us take a simple illustration. A Tjurrega man marries a Menuaitja woman. A male child is born, which

becomes a Kakera. When he grows up, this Kakera man must marry a Ngadeja woman. The progeny of the last-named union goes back to the Tjurrega, and, assuming it to be a female, she will have to marry a Menuaitja, and her child becomes a Ngadeja.

A few tribes of central and north-eastern Australia further divide each sub-class into two, making eight in all, but fundamentally the rules governing inter-marriage are much the same as those just mentioned.

It must not be imagined for one moment that the above simple outline represents the complete, and, in reality, very complicated, system upon which the aborigines work. There are numerous other restrictions, which are more or less variable according to the locality of any particular tribe. In its broadest outline, the scheme is much the same all over Australia, and it is possible therefore for tribes living on a friendly footing with each other to inter-marry and yet to strictly adhere to the fundamental principles controlling their respective laws. On the other hand, it will be realized how easily any white man, who is in the habit of having indiscriminate intercourse with native women, can make himself guilty of a criminal offence, when he cohabits first with a woman of one group and then with one of another. This applies, of course, only to the uncorrupted tribes, who still adhere strictly to their ancestral practices and beliefs.

Occasionally one finds a further splitting up of the intermarrying divisions, whilst some of the south-eastern tribes appear to have been without any hard and fast system at all. Here and there, too, the descent runs in the male line.

A child is generally allotted to its husband early in infancy, but actual possession is not taken until the girl is of a marriageable age, that is, when she is about twelve years old. In the interim, however, the future husband has certain claims on the child, and can supervise her domestication and instruction. He, on the other hand, is required to make presentations to the child's father and other relatives on the father's side; at this stage, too, he may promise his sister (if she be available) to his future wife's brother.

The relationship between man and girl is not a love-affair, in the modern sense of the word, but the whole transaction is regarded more like a pure business-matter. The sexual element plays perhaps the least important role, the man looking upon the budding woman as his future associate, helpmate

and servant, and also as a valuable asset to his existence wherewith he might be able to barter. Most of his notions in connection with barter are potential, and possibly will never be carried out; in a sense a man's wealth is gauged by the number of women he possesses, the younger they are the more valuable. As a lord of the wilds, it is, indeed, a fortunate position for any man to be in to own a number of wives; having himself, as a hunter, to travel unaccoutred, the women become the machinery of his transport. When in camp, they collect and prepare food for him, while he is resting or roaming the woods after game. Moreover, the women assist in the cosmetics of his person, and are the means of the education of his children in all matters, except those pertaining to the chase, warfare, and certain ceremonies taboo to women.

There are, of course, many cases where the habit of association grows into mutual affection, but one rarely notices the sacrificing love between man and woman such as is everywhere apparent between a parent and a child. In the former case the intercourse is always touched with a suspicion of subserviency on the part of the woman. Indeed, one cannot fail to be led to the belief that in their conjugal relationship club-rule is the dominant factor. One must not suppose that the weaker sex always submits to this rule without demur or without any resistance at all. On the contrary, she is usually not backward in making her voice felt in indignation, even in defiance of the punishment, which must inevitably, sooner or later, be meted out to her. Whilst the chastisement is proceeding, the husband prefers to sit aside in dignified silence, with his face turned away from the querulous gin, until the bombardment of obscene epithets becomes so strong that he considers drastic intervention necessary, if for no other reason than in the interests of peace and the restitution of order in his camp.

In spite of the strict marriage laws, it occasionally happens that a man elopes with a girl who is outside the permissible inter-marrying limits. Such elopements are the nearest to a selective love-match that it is possible to find among the aborigines. The couple are well aware of the fact that they are committing a serious offence, and that every effort will be made by the tribe to capture them so that they may be punished. If they are caught, both man and girl will be severely battered about with sticks and clubs, as a result of which either or both may die. If the man survives, he will be called upon to make heavy payments of foodstuffs, implements, weapons, ochre,

and many other useful commodities to both the girl's father and the man to whom the girl would have passed in the ordinary course of events. If such couples manage to avoid detection, the fear of punishment, which awaits them, keeps them from returning amongst their tribe, and so they might roam about alone or befriend themselves with a strange tribe, and keep away from their own people for years, or, perhaps, never return. If the absconding gin has been married to another, the offence is not considered so serious. There are, for that matter, usually one or more women in each tribe who are habitually lax in morals. These women are scorned by the other members of the tribe, and are publicly recognized as prostitutes. It goes without saying that these women are the legal property of some of the tribesmen, and for that reason any other men, who are not in the relationship of tribal husbands to the women but cohabit with them, are more or less ostracized, even to the extent of total exclusion from any consequential council meetings of the men.

Apart from these public profligates, the aboriginal women are laudably loyal to the moral principles which have been taught them. Ordinarily a woman is true to her individual husband, but there are certain religious ceremonies during which other men, who are of the same matrimonial division, may have legal access to her; these are her tribal husbands.

It is an expression of goodwill and friendship towards a visitor to offer him one or two of the young married women, who might live with him during his stay in camp. If the visitor appreciates the hospitality of the tribe, he receives the women and, in his turn, offers presents to the old men and to the husbands of his temporary consorts. A similar consideration is extended to men who, through a scarcity of the opposite sex in their particular group, have remained single. Under extraordinary circumstances, arrangements are very casually made for a man to associate with a gin who is not of his recognized class; but in this case it is compulsory to obtain the sanction of the medicine-man, who, after administering some of his sorcery, generally consents to the union.

An exchange of wives is not an uncommon event among two tribesmen, provided always that relationships on both sides are within the subdivisions allowed to inter-marry by law. This interchange takes place mostly between childless couples; when there is a family of young children one generally

finds that the parents remain together until, at any rate, the children have grown up.

Those groups of a tribal class-organization, which are not permitted to inter-marry, usually consider themselves in the position of blood-relatives to each other, that is, as brothers and sisters, and fathers and mothers; the forbidden groups of their fathers' and mothers' are, however, not necessarily excluded to the children who look upon the members of those groups as the brothers and sisters, respectively, of their parents.

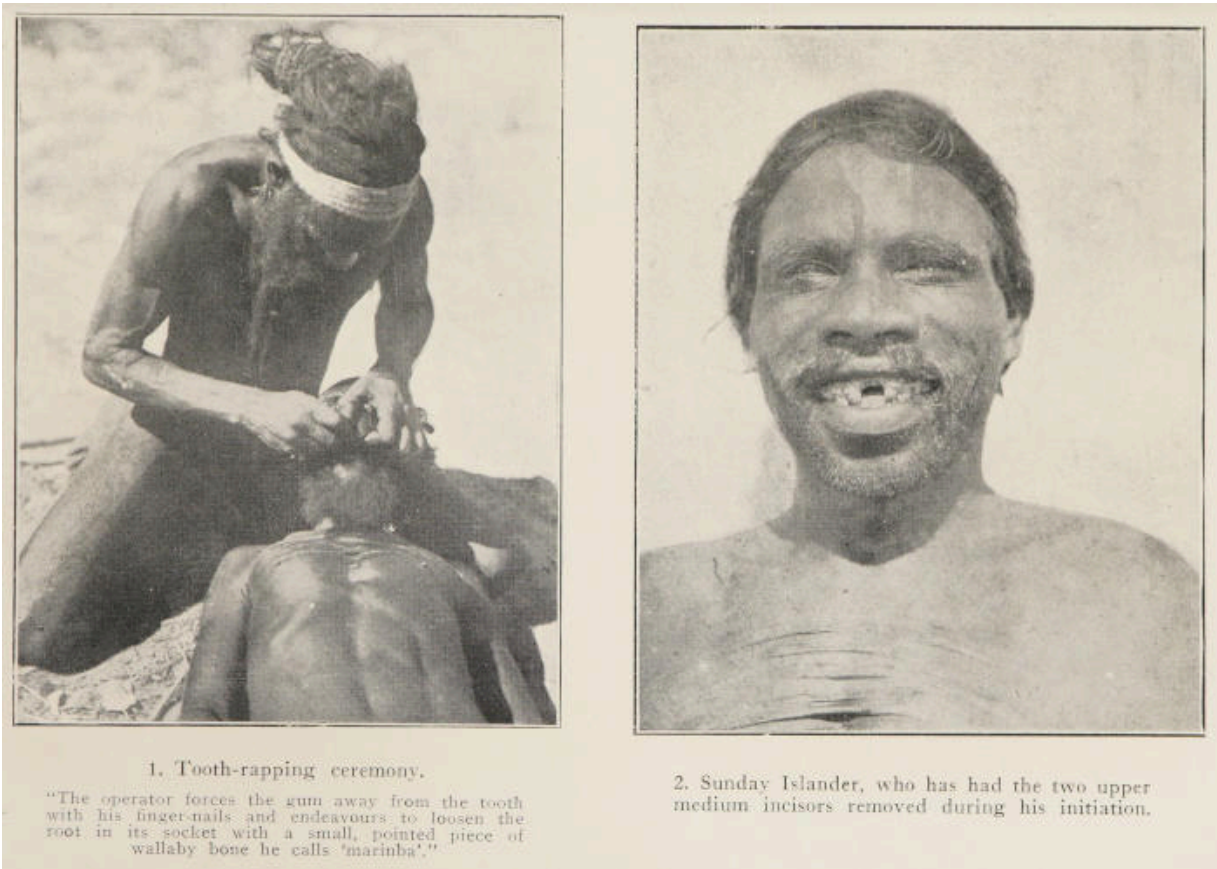


PLATE XXVIII

1. Tooth-rapping ceremony.

"The operator forces the gum away from the tooth with his finger-nails and endeavours to loosen the root in its socket with a small, pointed piece of wallaby bone he calls 'marinba'."

2. Sunday Islander, who has had the two upper medium incisors removed during his initiation.

CHAPTER XXV

TRIBAL ADMINISTRATION

Council of men—Consultations with magicians—Hereditary status of certain men—Strict observance of laws—Dread of supernatural punishment—Europeans apt to ignore primal conventions—Aboriginal sense of justice and humour—Appeal to sentiment—Judge of character—Possibility of erroneous deductions in science and law—Philosophical instincts—Variety of characters composing a tribe.

The affairs of a tribe come under the jurisdiction of a small council of old men who have their discussions in secret conclave, away from the main camp. Seniority and distinction qualify a man for admittance to this council, which is controlled by the strongest personality, much as a prime minister rules over his cabinet. Questions of importance and concern to the community are freely debated at these meetings, and when any decision is arrived at it is placed before a general assemblage of all male members of the tribe who have attained the rank of full manhood. At this meeting questions may be asked, but opposition is immediately squelched and the dissentient ruled out of order. In any case, it is only a senior man who is allowed to raise objection; the juniors are required to maintain a respectful silence, and to listen without interjection. On frequent occasions the medicine-men, though not necessarily members of the supreme council, are invited to attend, especially at such times when the cause of a death is being investigated, or the nature of adverse climatic conditions such as a prolonged drought being probed. These are opportunities at which the sorcerers excel themselves in their wisdom. After an eloquent address they impress their listeners to such a degree by the simplest conjuration that their advice is accepted unreservedly, and is forthwith acted upon.

Such is the executive body of the Australian tribes, who can command, exact punishment, or engage the tribe in warfare. There is no potentate approaching a royal head who ascends by either heredity or self-imposed magnificence, but each tribe has its recognized chief or leader whose position is referred to as "*Ingada*" by the Arundta, "*Jingardti*" by the Aluridja, and "*Judja*" by the Cambridge Gulf natives. The kobong (or totem) is certainly not without significance, so far as the personnel of the

executive is concerned; any heralbic senior, if one may so term the position, has a right to represent his family upon the board of control, and in that capacity his word is respected. But in addition to the orders of these men, the natives obey the laws of the tribe, which have been handed down to them from generation to generation, because they believe in a supernatural power that will punish them for any disobedience. This punishment they maintain would take the form of physical indisposition, disease, or even death, according to the gravity of the breach. They attribute many cases of accidental maiming, blindness, deafness, and paralysis to this cause.

What, perhaps, occupies the attention of the council of old men most frequently is the arranging of corroborees, sacred performances, and initiation ceremonies. Of these the dates have to be fixed and the programmes discussed and outlined in detail, so that the men who are to take part in the proceedings may be in a position to prepare themselves in advance.

In their dealings with the aboriginals, Europeans are too apt to ignore the authority of the old men, and frequently their defiance of primal law has led to disastrous conflicts. When in the MacDonnell Ranges I was desirous of taking two aboriginal children away with me. The circumstance was mentioned to one of the influential old men, who thereupon called together the elders of the tribe; and my request was considered in all its aspects. After a lengthy meeting, during which it was apparent there were two or three dissentient voices, I was finally informed that the children could accompany me under certain conditions which I had to take upon myself to guarantee. This agreement arrived at, the children were given to understand that they were going by the direction of the old men, and I officially received the spokesman's word of honour that, firstly, the children would never desert us *en route*, and, secondly, no attempt would be made on the part of the tribe to interfere with us, or steal the children from the camp at night. Had one attempted this under any other conditions and against the will of the tribe, there would have been serious trouble.

The natives of Australia have a very good idea of justice, which they apply in all their transactions among themselves. If, however, at any time, they find that they have been deceived or unfairly treated, their sense of righteousness almost spontaneously turns to revengeful treachery. They are not susceptible to flattery, and do not look for praise or reward for any

heroic deeds accomplished; to unnecessarily applaud a man is looked upon as a sign of weakness. On the other hand, when one has committed a wrong, he expects to be reprimanded, if not punished, and is disappointed if he receives neither; in fact, he thinks little of the man who neglects his duty by not making him suffer for his offence or wrongdoing. A native has a remarkable memory; and, if once he has escaped chastisement for taking a liberty with another, he will not think twice of repeating the wrong, because he does not consider him worthy of serious consideration. Even should there be resentment upon a subsequent occasion, he thinks that it needs only a little cajolery, when the fellow is easily won over on account of his proven weak character. In this connection his good sense of humour serves him well; he has a natural gift of being able to contort the serious in such a way as to make it appear comic, and, when he wishes, he has a very persuasive tongue.

Provocation, embarrassment, or terror instinctively and spontaneously solicits an appeal to the obtruder's sentiment. The best example I can tender in this connection is an experience our party had in the Tomkinson Ranges when it unexpectedly came upon three women who were cooking a snake. On seeing the white men in close proximity, two of the gins scampered off post haste, but the third was so terror-stricken that her legs refused to move her from the spot. In her bewilderment, however, she had sufficient presence of mind to think of seizing her breasts with her hands and squirting fluid in the direction of the strangers to indicate that she was a mother and on that account had a special claim to beg for mercy.

There is no denying it, the swarthy inhabitants of Australia are excellent judges of character, and they seem by instinct able to explore the trend of thoughts running through each other's brains before they are spoken. This faculty is of fundamental importance to them whilst watching ceremonial dances and other performances; by means of it they are able to appreciate when the outsider fails to understand. Moreover (and this is of considerable importance to the European anthropological investigator), the moment a native under cross-examination grasps the nature of questions which are put to him, he speedily discovers what direction of answer, affirmative or negative, appears to suit the interrogator best; and accordingly he will reply. It is, in consequence, absolutely impossible to decide any matter of importance by soliciting a simple "yes" or "no" in answer to a leading

question. The enthusiasm with which a biased reply is tendered is, further, considerably enhanced by a promised gift or bribe. The significance of these facts, and their possible consequences, will be realized both in a scientific and in a legal sense; erroneous impressions may, in the one case, be gleaned with regard to Australian ethnology, and, in the other, a fellow may be unconsciously incriminating himself in the eyes of our law.



1. Old men introducing a dance during an initiation ceremony, Kukata tribe.



2. Circumcision ceremony, Kukata tribe.

"Beyond themselves with excitement, they lay hands upon the lad and lift him upon the back of two or three of the men who are stooping in readiness to receive him."

PLATE XXIX

1. Old men introducing a dance during an initiation ceremony, Kukata tribe.

2. Circumcision ceremony, Kukata tribe.

“Beyond themselves with excitement, they lay hands upon the lad and lift him upon the back of two or three of the men who are stooping in readiness to receive him.”

Primitive as the natives of Australia are in the scale of humanity, so they appear to us, from a psychological point of view, as but mere children. Being intensely emotional, they are easily moved by rage or grief; they are of a quick temper and the best of friends fall out over the most trivial matter; but in the majority of cases the storm lasts but a very short time and then the friendship is as thick as ever. Happiness seems the essence of their existence, and indeed none but an aboriginal could, or would, keep on smiling under the terrible bondage which our vaunted civilization has imposed upon him.

Among the members composing a tribe, one finds a diversification of character and talent much the same as builds up a modern community—it includes orators, warriors, artists, and clowns. Unauthorized though the action might be, it is not an uncommon occurrence for a man to place himself in a conspicuous position in camp and hold forth on matters of current interest to an enthusiastic and, at times, spell-bound audience; the man of skill and courage becomes the hero of the hour when the tribe goes to war; whilst talents in tune or colour are called upon whenever the occasion demands something out of the common. Some of the men have the reputation of being the jesters; their antics and jokes keep the camp in roars of laughter. Under the last-named category also come the mentally defective and half-witted persons; they are generally followed around by a band of mischievous children, who enjoy, with shameless glee, all the absurd pranks of the imbecile.

CHAPTER XXVI

INITIATION

No person exempt from initiation rites—Piercing the septum of nose—Tooth rapping—Children trained to become hardy—Fire-walking—Body scars—Different patterns described—Sky-shying—Circumcision—Convalescence and return to camp—Deserters drastically dealt with—The sacred “*Wanningi*”—Biting the initiate’s scalp—Subincision—Operation simultaneously performed upon old men—Sacred pole erected at ceremonial site—Initiation ceremonies without mutilation—Mythical origin of mutilation—Female initiation—Enchantment of breast—“Smoking” ceremony—Mutilation of female.

Throughout Australia it is required of a person, before he reaches adolescence, to pass through certain rites and ceremonies, without which he would not be considered mature and would not be allowed the privileges enjoyed by the rest of the adult community. This rule applies to both sexes, and is particularly stringent in the case of the male; we refer to a series of mutilations, which are committed by the elders upon the persons of the rising generation at regular stages of their lives, and always accompanied by pompous, secret, and demonstrative ceremonial. These functions naturally vary in different parts of so large an area as is included in the continent of Australia, but, nevertheless, the differences are not so considerable that one could not consider the transactions collectively under the heading of one and the same chapter.

Broadly speaking, the first, and at the same time the least important, ceremony is the piercing of the lower fleshy portion of the septum of the nose. This is performed at a rather early age in the north and south, whilst in central Australia it does not take place until the child has reached maturity. The Larrekiya pinch a hole through the flesh with their finger-nails when the child is still in arms. Central tribes use a sharply-pointed bone or the blade of a spear. The child is laid flat upon its back and its head placed between the thighs of the operator, who is kneeling on the ground. The child is assured that what is about to be done to it will not hurt, and that, when it is over, its body will develop quickly and become strong. Then the operator seizes the *columna* with the index finger and thumb of his left hand and pulls it well below the nostrils. With the sharpened piece of bone or blade he holds in his right hand, he perforates the tissue beneath the cartilage with

a decisive drive, then quickly withdraws the instrument again. The father, or if the father be dead, the father's brother, usually performs the operation upon a boy, whilst the mother (or mother's sister) attends to a daughter. For some time after the operation, the perforation is kept open by means of a short rod, which is frequently turned about. Among some of the central Australian tribes, this rite is becoming obsolete; the Aluridja and Arunndta, for instance, do not nowadays insist upon the perforation being made at all, and, if it is, then only late in youth. It is at the option of a man to perforate the septum of the gin he takes to himself; if he does so, it is more for vanity sake, thinking that she is better able to decorate herself for corroborees, than with the idea of making her grow robust and womanly.

Next in the ritual sequence comes the tooth-rapping ceremony. This, too, is or was practised practically over the whole of the continent, and is in parts of great importance, ranking with certain tribes as one of the initiation steps. The ceremony extends over weeks and ends with the knocking out of one or two of the novice's incisors. Generally a number of lads or girls undergo the ordeal together; but in the case of the girls no particularly great fuss is made. At times a number of adjoining tribes agree among themselves to hold a monster ceremony conjointly, at which all the youths of correct age are dealt with. Such is a really big occasion in tribal affairs, and many weeks are spent in an endeavour to make the event as successful and as impressive as possible. The decision is made at one of the council meetings of the old men, who, having announced the matter to a general assembly, make arrangements to send invitations to any friendly tribes living around them. In the Northern Kimberleys of Western Australia, a pair of men is selected for conveying each invitation to its destination; they are elaborately painted up and are allowed to wear only a forehead band and a pubic tassel; each carries a message-stick and a bull-roarer of a design which is to figure prominently during the ceremony. The stick and the bull-roarer are to be given to one of the old men of the tribe they are going to. When these messengers arrive at their destination, the nature of their visit is immediately recognized. The old man receives the stick and bull-roarer and hands them to one of his fellow-councillors whose crest or "*kobong*" corresponds to that embodied in the carvings on the bull-roarer; and, in return, the messengers are given similar pieces to take back to the sender. The date for the commencement of the ceremony is fixed about a month in advance, the half moon being favoured; the trysting place is upon the

originator's ground at a place where water and food will be sufficient to supply a great number in attendance throughout the proceedings. In the meantime the ground is cleared and prepared; as in most of their initiation ceremonies, this consists of two large, circular spaces connected by a straight, wide pathway.

While this is being done, the boys are tended by some old men, who keep them at a distance and daily talk to them on matters dealing with manhood's duties, chivalry, courage, and the social position of women. In addition, they are taught some of the principal songs in which they will be required to join during the subsequent ceremonies in order to please the old men.

By this time it has become quite clear to the women that something extraordinary is about to happen. Indirectly they are made aware of the fact that a tooth-rapping ceremony stands near; and then for the first time music is heard; the women commence chanting periodically to the boys' teeth, which they thereby hope to loosen in their sockets.

At this stage a bull-roarer is frequently sounded at night by one or more old men, who walk quietly away from camp and conceal themselves in the bush. At the sound of these, the women are overwhelmed with awe and cover their faces with their hands and quiver hysterically. This is the dawn of that period in a young man's life at which his social status undergoes a complete change; it is the introduction to the series of initiation ceremonies which will ultimately qualify him for acceptance into the inner circle of men who have a voice in the control of tribal affairs. But more than this: it is also the beginning of that time in his life when he must learn to sever himself from his associations with the women and camp apart from his mother, sisters, and other near female relatives. As a matter of fact, this event places him on the threshold of a new life which unfolds to him the secret of the sexes. The sexual significance of the ceremony is clearly indicated through the circumstance that both male and female bull-roarers are sounded during the proceedings, a liberty which is only permitted on rare occasions.

About two days before the expected arrival of the invited tribes, rehearsals are held by the local men. A body of the performers suddenly appears upon the scene, coming from the cover of the forest; they are all richly decorated with red ochre and white pipe-clay, and run in a compact

line up to the cleared space. As they approach the spot, they stamp the ground vigorously and hit their hands together. They halt at one of the circular spaces, and, whilst they continue to stamp and beat time with their hands, some of the oldest among them spring along the track to the opposite space and go through all sorts of mysterious acting, which include tricks of sorcery designed to exhort the men and expel any evil spirits from the ground. The act is repeated every day until the guests arrive, and then it is also produced at night. The actors, who have profited by the rehearsals, are now seen at their best, and the visitors first look on in appreciative bewilderment; at a later stage, they join in and their own sorcerers display their tricks as well. This function lasts long into the night, and, should more visitors arrive, it might be repeated all over again for the late-comers' special benefit. During these demonstrations, the women sit some distance off with their backs turned to the men and keep up their monotonous tune.

The next few days may be spent in convivial sing-songs and food-procuring expeditions, whilst the principal performers busy themselves cutting twigs and leaves from the surrounding trees, with which they completely cover the cleared ground, so that not a track remains visible in the sand. The object of this somewhat tedious process is precautionary, namely, to deny any prowling spirit the opportunity of ascertaining the identity of the persons who have taken part in the ceremony. They fear that, if this information were obtained, the boys would be molested and their teeth stuck tightly into bone of the gums.

When at last the eventful night arrives, a number of the older men decorate their bodies profusely with vegetable down and ochre. The boys' bodies are smeared with red ochre, over which certain designs are drawn in symmetrical fashion, and embodying a number of circles and dashes. The men endeavour to make themselves as awe-inspiring as possible by concealing as much of their normal appearance as they can. They keep aloof from the novices until immediately before the ceremony.

The fires glaring fiercely, the boys are led to the cleared space and told to keep their eyes closed. Upon a given signal, several of the decorated men rush from the darkness, pounce upon the novices, and throw them to the ground. Each man seizes one of them and kneeling behind him places the lad's head upon his lap. The initiate offers no resistance and allows his eyes to be covered by the old man's hands. Whilst some weird chants are being

rendered, the operators appear upon the scene, each carrying a short cylindrical stick and a stone in his hands. These men are not decorated and are near relatives of the boys undergoing the ritual. With much dancing and gesticulating, they draw near to the boys, and, as each of the men kneeling behind raises the head of a candidate, one of the operators steps forward and looks into the boy's mouth as it is being forced open by the assistant. Presently he selects his mark and with his finger-nail presses back the gum from one of the youngster's incisors. Then he places the point of his stick against the tooth and gives it a sharp blow with the stone he is carrying in his other hand.

As a rule the boy does not whimper, but occasionally one may give way and cry with pain. This is immediately resented and forbidden by the old men, who declare that he has been too much in the company of the women and girls.

The stick is re-applied and another blow imparted to the tooth; and the process is repeated until the tooth loosens and falls out. All novices are treated similarly in rotation.

Whilst the elders are continuing their dancing, the boys are taken back to camp by their initiated male relatives, where they are presented with a pubic tassel, a dog-tail necklace, or other article which is strictly peculiar to such as have undergone the ordeal at the correct time. The boys are again given instructions in all sorts of matter becoming of a man, among which discipline and loyalty towards their elders and tribes-people in general are of first importance. The seriousness of the ceremony then relaxes somewhat, and songs and dances are produced to inspire and amuse the boys; the performance rarely concludes before daylight.

In central Australia the operation is nowadays performed without much ceremony and usually in daylight. Among the Wongapitcha it is known as "*Antjuarra*." The novice lies on his back and rests his head against the operator's thighs, while a number of men sit around in a semi-circle ([Plate XXVII](#)). The operator forces the gum away from the tooth with his finger nails and endeavours to loosen the root in its socket with a small, pointed wallaby bone he calls "*marinba*." Then he applies the bevelled point of a short, stout rod ("*tjutanga*") and strikes it with a stone ("*puli*"). The percussion produces a loud resonant note, at which all present cry "*Tirr!*"

After four or five whacks with the stone, the tooth falls from its socket. It is picked up by the operator and shown to the audience, who respond by exclaiming “Ah” or “Yau.”

The Wongapitcha and western Aluridja remove the left central incisor; only the eastern groups of the Arunndta still practise the rite.

One of the chief concerns of an aboriginal father is to make his son fearless and capable of enduring hardship. In all his dealings with his children, he endeavours to avoid favouring and pampering any of them once they have passed out of infancy; but the boys in particular are constantly urged to suppress pain and to make it their special task to under-rate the deprivation of comfort no matter what circumstances might arise. In the same way the boys are trained to be brave; they are told not to be under any apprehension of danger except when it is known to be prompted by the treachery of the evil spirit or by the wrath of the spirit-ancestors of the tribe. With this principle ever before them, most of the initiation ceremonies have been evolved on similar lines. The Kukata even make it compulsory for the novices to walk through a blazing fire, an act they refer to as “*merliadda*.” Other central tribes make the young men lie temporarily upon branches they place over a smouldering fire.



Circumcision of a Wogait boy.

"With his left hand the surgeon seizes the prepuce, whilst a veritable reverberation of short-sounded 'i, i's' meets him from the mouths of all present, and as he draws it well forward a number of hacks severs it."

PLATE XXX

Circumcision of a Wogait boy.

"With his left hand the surgeon seizes the prepuce, whilst a veritable reverberation of short-sounded 'i, i's' meets him from the mouths of all present, and as he draws it well forward a number of hacks severs it."

It is largely during the term of initiation, from early childhood to adolescence, that every individual, boy or girl, receives a number of cuts in different parts of the body, which, when they heal, leave permanent and elevated scars. After the incision is made, ashes, ochre, and grease are usually rubbed into the wound to make it granulate to excess and so produce an artificial keloid. The reasons for making these scars are threefold: spartan, cosmetic, and tribal. The operation is performed with a stone-knife or flint-chip. In addition, a punctate scar is produced by twirling

a fire-making stick until it becomes nearly red-hot, then quickly holding the point against the skin. The process is repeated time after time, each time selecting a new point on the skin, until a chain-pattern results. The latter type of scar is more commonly found on women than on men, and is much adopted by the coastal tribes of the Northern Territory.

It would be futile even to attempt a scheme of classification of the different tribal markings. In some districts, only one or two are made; in others, the better part of the body is covered. As a general rule, the men display a greater number than the women; the latter often only have one or two horizontal cuts across the chest or abdomen, at times, indeed, only a single prominent scar connecting the breasts. Generally speaking, the central Australian tribes do not cicatricize their bodies nearly as much as the northern. The Yantowannta, Ngameni, and other Cooper Creek natives leave the chest clean above the breasts, but cut a few horizontal lines immediately below them on the abdomen; the Dieri add one or two short irregular marks above; at Durham Downs the women have a number of short horizontal lines on each breast. Among the Aluridja, Arunndta, and Arrabonna, one notices principally short horizontal lines across chest and abdomen, with, occasionally, a few vertical bars, less than an inch in length, around the shoulders; a number of the small circular fire-marks are also as a rule noticeable on the forearms of the Arunndta. On the north coast a great variety of marks may be studied. The tribes east and west of Port Darwin have very prominent scars horizontally across chest and abdomen, short vertical bars around the shoulders, sloping bands composed of either parallel vertical cuts or fire-whisk scars, passing from the central point between the breasts upwards to the shoulder on one or both sides, and occasionally a vertical band, consisting of two parallel rows of fire-whisk scars, on one side of the abdomen just beside the navel. An additional pattern is a sagging band across the chest from shoulder to shoulder, consisting of about twenty short vertical cuts.

It must not be supposed that these marks are all regularly observed upon every individual one meets. On the contrary, it is very rare to see a person with all the scars referred to, some having only one or two horizontal lines across the chest or abdomen like the central tribes.

The Melville and Bathurst Islanders imitate the frond of the zamia palm (*Cycas media*) by cutting a series of V-shaped figures, one within the other,

in a vertical row, upon one or both sides of the back, and on the upper and outer surfaces of the arms and thighs. Horizontal lines are cut across the chest, as above described, and here and there a person also has a horizontal band on his forehead, immediately over the eyes, consisting of from eight to twelve short vertical cuts.

The Cambridge Gulf natives, both male and female, cut numerous lines (*"gummanda"*) horizontally across the chest, abdomen, buttocks, and thighs, and long vertical lines down the upper arms, whilst on the back, occasionally, a "waist-band" or *"naualla,"* consisting of numerous short vertical nicks, is added, together with about ten vertical cuts on each calf ([Plate XVI](#), 2). The gins have one or two vertical bands of punctate scars between the breasts.

The Worora at Port George IV further decorate the whole of the back with alternating groups of horizontal and vertical lines, those on the shoulder blade being particularly prominent.

The central tribes have embodied in the curriculum of initiation tests a sky-shying act, called *"Algerrigiowumma"* by the Arunndta, during which the novice is tossed high into the air by a number of older men. The ceremony is somewhat as follows. By agreement, a crowd of men and women assemble at night upon a cleared piece of ground near the camp-fires. After a few songs and ordinary dances, whose object seems to be more to create an atmosphere than that they have any definite significance from an initiation point of view, the whole congregation draws up in one or two lines and settles down to a peculiar dance, shuffling sideways from one edge of the cleared space to the other and back again. As they move they mumble a verse of coarse guttural words which ends in hissing notes resembling the panting of a vicious animal. At this moment two or three men spring from behind and seize the unsuspecting novice. Rushing towards the expectant crowd, they swing him high above their shoulders and throw him among the outstretched arms of the crowd. There he is seized by a dozen or more of the most powerful men, who toss him high up towards the sky. During his flight through the air the position of the boy is usually horizontal. When he returns again by gravitation, he is caught and once more projected skywards; and the same process is repeated time after time. He may then be released, but if the crowd again begin to dance and sing, he will in all probability have to submit to a repetition. The ceremony

amounts to a formal handing over of the boy by the women to the care of the men.

Of momentous importance is the initiation ceremony which includes the circumcision of the novice; it is the first occasion upon which the truth of growing manhood is definitely made clear to the boy. The Aluridja refer to the ceremony as “*Arrarra*”; it extends over a period of several days, during which wild dances and songs are produced. For weeks before the novices are zealously guarded by certain of the old men and kept away from the mixed camp. No women are tolerated within a wide radius of the spot at which the ritual event is to be celebrated; and throughout the vigil the boys remain out of sight of their female relatives and are allowed to converse with none but their guardians. The boys’ diet is restricted to only a few very ordinary items, amounting to a mere sustenance; anything they are able to obtain by assiduous hunting is required to be handed to the men in charge. After having listened to the songs of the men officiating in the ceremony, the boys are allowed on the second or third night to attend in person. They are given to understand that what they are about to see is never to be explained to any of the women or juniors. The performance must be regarded as sacred and most secret; and they must realize that, although invisible, there are many ghost-ancestors present who are following every item of the ceremony with proud appreciation. At such a stage the performers are awe-inspiring spectacles, being richly decorated with white and red down and wearing the cross-shaped “*Wanningi*” in their hair ([Plate XLIII](#), 2).

A great fire is burning at the edge of a cleared space, around which a number of men are sitting and singing and beating the ground with sticks. To the listener the strains sound respectful, if not worshipful; in the Aluridja they run somewhat like: “*Imbinana alla’m binana*,” repeated almost indefinitely. In the glaring light it is noticed that the ground is cleared of the original grass and growth, but is covered with a layer of leaves of the red gum tree which abounds along the watercourses. Upon this matting of leaves the dancers are moving to and fro, madly stamping the ground. Then, as they quicken their pace, each places the point of his beard into his mouth and tries to look as ferocious as possible. The lead is taken by one who is carrying a short hooked stick in front of him. In single file these men

encircle the chorus, the leader touching all present with his crook and shrieking “*Arr, arr, aah!*”

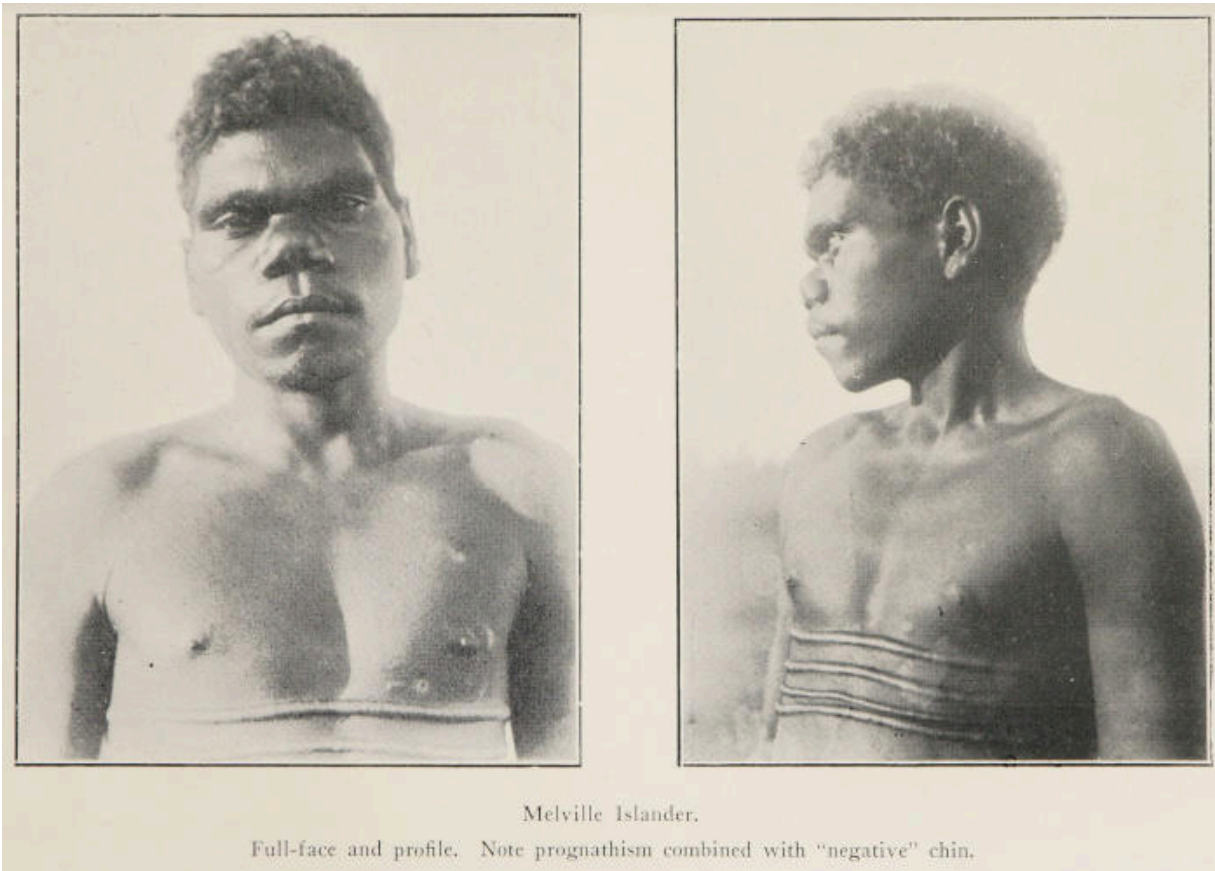


PLATE XXXI

Melville Islander.

Full-face and profile. Note prognathism combined with “negative” chin.

The novice (or novices, as the case may be) is now taken away from the wild scene and again given to understand that henceforth he is not to leave the company of the men, and that he must do as those in authority bid him. He is taken to another fire some distance off and covered with red ochre from head to foot; his hair is tied back with human hair-string; then he is led back to the principal group of performers.

When he re-appears, the dancers rush around in the manner described above, crying “*Arr, arr, aah!*” with husky voices, first more or less confusedly, but gradually in a more defined line which gradually closes upon the youth. Beyond themselves with excitement, they lay hands upon the lad and lift him upon the backs of two or three of the men who are

stooping in readiness to receive him. As he is carried onwards, the other performers, and some of the chorus as well, cluster around the panic-stricken boy, each of them taking hold of some part of his body and helping to make the mass of humanity as confused as possible ([Plate XXIX](#), 2). Thus they tour around the cirque and pull up near to the fire. Two or three of the number now throw themselves beneath the carriers, who, in their turn, immediately pile themselves upon them. The boy is pulled back on to this human operating table full length, and another man immediately jumps on to his chest; others hold him securely by his arms and legs. All the time the men keep up their cries of “*Arr!*” and from different places around the fire the booming, humming, and shrieking notes of bull-roarers rise above the din.

The moment the latter unearthly sounds are heard at a Kukata ceremony, the men respond with an uproarious slogan sounding like: “*Wubbi, wubbi, wubbi, wau!*” This imitates the noise of the bull-roarer, which is supposed to be the voice of the presiding spirit. A perceptible wave of solemnity pervades the atmosphere at this moment, not only in the immediate surroundings of the ceremonial fire, but wherever the piercing chorus strikes the ears of camping groups, who may at the time be many miles away. Women and children shudder with fright and bury their faces in their hands. The initiated men, however, act differently. Where there is only one in camp, he rushes to the fireside and snatches a burning stick which he tosses high into the air. When there are several present, they rush out together into the darkness and unitedly echo the “*Wubbi, wubbi, wubbi, wau!*”

The boy’s mouth is gagged with a ball of hair-string, which serves the double purpose of stifling his voice, should he attempt to cry, and of giving him something to bite his teeth into when he is in pain.

The men at the lower end now force his legs asunder and press them downwards over the side of the bodies below. As this is happening, the operator walks into the space between the thighs, with his beard between his lips and his eyes rolling in their sockets. He carries a knife in his hand—usually a fair-sized freshly broken splinter of quartzite, chalcedony, flint, or quartz, with or without a handpiece of resin—and immediately proceeds to operate. The Wongapitcha at this stage stun the boy by clubbing his head. With his left hand the surgeon seizes the prepuse, whilst a veritable

reverberation of short-sounded “*i, i’s*” meets him from the mouths of all present, and as he draws it well forward a number of hacks severs it ([Plate XXX](#)). The Dieri make use of a short, smooth, cylindro-conical stone, over the rounded point of which the operator stretches the skin, and so pushes back the glans before he cuts.

Among the Kukata the circumcision ceremonial is referred to as “*Gibberi*.” During its period, cicatrices are made on the arms, but the characteristic transverse cuts on the back are reserved for the following rite, which goes by the name of “*Winyeru*.” The prepuse is resected with a stone knife known as “*tjulu*,” and immediately destroyed by throwing it into the fire.

As the surgeon’s hand leaves the boy’s body with the detached skin clutched between two fingers, the act is greeted all round with an appreciative exclamation sounding like “*A ha, a hm*.” The Aluridja refer to the skin as “*banki*,” and bury it shortly after the operation.

The boy, who by this time is usually semi-comatous, is sat up, and the blood which is streaming from him is collected in a piece of bark previously laid beneath him. The Wogait and other northern tribes subsequently dress the wound with paperbark, clay, emu fat, and hot ashes, to stanch the bleeding. When the boy recovers from the shock, he is presented with a spear and spear-thrower, and often a shield as well.

The patient is then taken into the bush to convalesce; and during this period his diet is under strict observation. After about three weeks he has more or less recovered and returns with the men to the main camp, wearing a fur-tassel which covers his pubes. His mother and other near female relatives on the mother’s side, when they behold him, walk towards him sobbing, tearing hair from their scalps and otherwise hurting their bodies as indication of their sympathy for the painful ordeal he has so bravely undergone whilst aspiring towards the status of manhood.

Cases have repeatedly come under my notice in latter years where a circumcision candidate has attempted to evade the operation by travelling away from his tribe and residing indefinitely with some other party, native or European, having no jurisdiction over him. It is usually only a matter of time and he will be ambushed by men of his own tribe and taken back to camp. The operation is then immediately performed, and is made

extraordinarily drastic as a punishment. The skin is stretched forward under considerable tension and severed with a stone knife. In several specimens which are in my possession, the external sheath was cut so high up that a number of pubic hairs were removed with it.

This ceremony in its essential features is much the same among all tribes which circumcise, although there are slight variations in the method of building up the human operating table. In the Wogait tribe, for instance, the boy is thrown on his back over the legs of four men who sit in pairs, face to face, with their limbs alternately placed so that the feet of one are against a thigh of another.

We have already referred to the ceremonial object known as the "*wanningi*," which is produced during this ceremony. This is always constructed specially for the occasion and is destroyed again immediately after; under no circumstances are the women and children allowed to see it, for if they did blindness or some paralysing affliction would strike them for their disobedience. In its simplest form, the "*wanningi*" consists of a cross, from the centre of which a long twine of fur is wound spirally outwards, from arm to arm, and fastened with a single turn round each arm in succession. The object is either stuck into the hair as referred to, or carried in the hand by the functionaries during the final stage of the ceremony. In the northern Kimberleys a similar structure is used which is carried at the end of a spear or long stick behind the back of the performer. The "*wanningi*" is supposed to become inspired by a spirit guardian the moment the object is completed and prevents the boy from suffering too great a loss of blood. It is shown to the boy just prior to the operation and its sacred nature is explained to him.

The partially matriculated man now remains in camp for a while, but is kept under the strictest surveillance and aloof from the opposite sex. He continues to go out on daily excursions with certain of the older men and has to recompense those who took an active part in his initiation with the lion-share of his hunting bag, many items of which he is himself still forbidden to partake of.

Not many weeks pass by peacefully, however, before the excitement starts afresh. An old man, usually the senior among the initiate's group-relatives, quite spontaneously lays hands on the unsuspecting neophyte, by

clutching him between his arms, and bites him on the head. Then he releases him again and darts away like one possessed. Others take the cue and act similarly. Several men participate in this painful but well-intentioned ceremonial, and they may repeat the performance from three to six times in succession. The youth tries to make light of the injury done him but often cannot altogether suppress crying with agony. The blood pours freely from the numerous wounds in his scalp. He will in all probability have to undergo this painful ordeal on two or three different occasions.

There is yet another big test to be passed by the man-in-the-making before he is finally admitted by the controlling council to mature membership. And this is again accompanied by an additional mutilation of his person, which is commonly referred to as the "mika operation." With few exceptions it is performed by most of the central and north-western tribes and in much the same manner.

The ceremony does not take place until some time after the young man has thoroughly recovered from the effects of his previous sufferings. It follows, in many instances, as the natural climax to a lengthy religious demonstration during which a series of sacred songs and dances has been produced. Needless to say the proceedings are inaugurated at a spot remote from the general camping ground, and preference is given to a moonlight night. No woman or child is tolerated near, and the novice is told that he must regard his presence among the old men as a decided favour. Ordinary songs are rendered for a while which are designed to fecundate the mother-stocks of their game supplies; then follow more dramatic incantations which are believed not only to attract certain spirit-ancestors to the spot but also inspirit the sacred but ordinarily inanimate tjuringas which are lying before the celebrants at the fireside. The excitement grows and with it the men wax enthusiastic. It is decided to operate upon one or more young men who happen to be available; and now everybody seems beyond himself with frenetical animation. The suggestion is received with applause. No secret is made about the matter; and the nomination is made in public. The atmosphere is so tense with hysterical veneration that the mere mention of a likely function enjoining the spilling of blood is received with enthusiasm and general appreciation. Arrangements are made forthwith.

Three or four men who are to be intimately concerned in the affair dodge into the darkness of the bush and return a short time later besmeared with

ochre and pipeclay, and decorated with eagle-hawk's down. As they approach the fire, they stamp the ground with their feet and balance their arms in a horizontal position. Thus they encircle the fire two or three times and, in doing so, make a peculiar hissing noise like that produced when wood is being cut. All the time the other men are beating time to the movements of the performers by smacking their hands against their buttocks.

Presently the dull, humming sound of a bullroarer breaks upon the performers' ears. At this, all men throw up their arms above their heads and yell, while the notes of the bullroarer reach a higher pitch. A jumble of wild, vociferating men ensues. The novice stands in their midst. The bullroarer's noise is stopped, and with it the voices of the men die away. Only the painted performers continue to stamp and again make the peculiar hissing noise as they move around the fire. By this time a number of men have placed themselves behind the novice who is made to stare into the brightest portion of the fire and told not to move his eyes from the spot.

Upon a significant nod or other signal from the leading performer, some of the men at the back of the youth lie upon the ground, while others place themselves on top of them again. The moment this has happened, the youth is tripped backwards over them by some who have been standing at his side, and by them he is also held down and gagged.

The leading performer now moves straight towards the victim and in his hand one notices he is carrying a stone cutting knife. His attendants spring to each side of him and hold the young man's legs apart. Now the leading man, who has been ordained with the surgical duties, follows the ancient practice of the "gruesome rite" by splitting the urethra for a distance of about an inch down its length. The withdrawal of the surgeon from the scene meets with the approval of all participants and eye-witnesses; and this fact is made known by the combined exclamations of "A, A, Yah!"

The young man is promptly pulled up from the improvised living table and his gag removed. He is ordered to sit over a wooden receptacle, usually a shield but occasionally a food-carrier, in which the blood pouring from the wound is allowed to collect. In a crude but hearty manner, more by action than by word, he is congratulated. Now he is a man, a real man, and he is at

liberty to join in the discussions of the rest of the men and to ask any pertinent questions relating to the affairs of his tribe.

During this time the commotion around the camp-fire continues without abatement. The excited throng, spurred on by what has preceded, seems to have become intoxicated by the sight of blood. The men who lay beneath the initiate during the operation figure prominently among them and display the clots and congealed patches of blood which cover considerable portions of their naked bodies. This provokes an appetite for seeing more and ere long one of the senior men, who has been initiated some time ago, volunteers to have the subincision of his urethra extended. He of his own free will calls upon an old man to perform on him and, when the nominee steps forward, he submits to the torture without flinching. Others follow his example. There is no doubt that the brave demonstration of masculine fearlessness stimulates the newly initiated member who is sitting aside and recovering from the shock which the trying ordeal has given his system.

The Kukata men at this stage, knowing that the candidate has now successfully passed the second great initiation rite, known as "*Winyeru*," and is henceforth to be allowed to mix with the women, betray extreme sensual excitement. And it is whilst they are in that state that some of the oldest men approach them, carrying spears, with the points of which they extend the previous slit in the urethra by a further short distance. It is only natural that the blood which follows the cut is squirted in considerable quantity. The custom is to allow it to do so until the excitement abates, and then to stop the bleeding by holding a fire-stick near the wound. After submitting themselves to such treatment on numerous successive occasions, it could only be expected that the whole external length of the urethra is eventually slit. The old men maintain that thereby their carnal powers are increased, and for that reason their forefathers introduced the corresponding female operation.

The after-treatment of the patient is much the same as described in connection with the circumcision ceremony. When eventually the newly made "man" returns to the main camp, a great sensation is caused among the women who wail as if there had been a bereavement, and cut deep gashes into different parts of their bodies.

Many of the northern coastal tribes, such as the Wordaman of the Wickham and Victoria Rivers, erect a sacred pole at the site of the enactment of some of their religious and initiation ceremonies. This pole is called "*Djundagalla*" and stands six or seven feet high. It is painted in alternate bands of black and white, occasionally red and white, and resembles a barber's pole more than anything. The "*Djundagalla*" stands in the centre of the cleared space and the rites are performed around it. In the northern Kimberleys, we find a stone phallus taking the place of the pole.

It is not every tribe that submits its young men to these mutilations at the initiation ceremonies. There are some which institute great graduation-festivals without the infliction of bodily harm to the virile aspirants. Notably among these are the Larrekiya, Melville Islanders, and the tribes living along the coast from the King River to the heads of the Roper and East Alligator Rivers.

As an illustration of a tribe which celebrates the coming of manhood without resorting to operative measures, the Larrekiya perhaps serve best. The boy, when definite signs of adolescence manifest themselves, is decorated with the kapok of the silk cotton-tree (*Bombatt malabaricum*) and birds' down. A straight band passes below his eyes from ear, and the ends thereof are connected by means of a horseshoe-shaped figure traversing the cheeks and having its closed end at the chin. Another horizontal band extends from shoulder to shoulder, above the nipples, and from this two symmetrical lines are constructed down the abdomen and on to the thighs, where each terminates in a circular band around the knee. A white line is also drawn down the outer surface of each upper arm and is made to end in the plaited armlets worn above the elbow. His forehead is decorated with a broad band consisting of a number of parallel strands of opossum fur thickly besmeared with white pipeclay; in the middle of this is stuck a plume of emu or heron feathers, and fur-tassels pend from either side of it. He also wears a coiled bark belt and, over it, a human hair girdle supporting a large pubic tassel.

The initiates are made to sit in a row before the old men and are instructed to keep their eyes closed with their hands. The old men stamp the ground wildly and brandish their spears poised in the spear-throwers. Every now and then they utter harsh cries of "*Arr-re! Arr-re!*" and "*Gora!*" Whilst this pandemonium is in full swing, the boys are ordered to open their eyes

and behold their elders performing; then they are led away into the bush and have to wait on the men, having especially to collect for them many things that are good to eat. During this period they are often cowed by being struck between the shoulder-blades, and threatened with violence if at any time they talk publicly about anything that has transpired or in any way betray the trust which the old men have placed in them. Upon their return to camp, the young men have additional scars cut into the skin of their chest and are then entitled "*Böllier*" which signifies that the first stepping stone to maturity has been passed.

A second ceremony takes place some years later. Each youth is then under the individual charge of an old man and is decorated much the same way as the Böllier candidate described above, with the distinguishing features of four red ochre stripes across the white forehead band and an extra plume of white cockatoo feathers stuck into his hair. The proceedings start soon after sundown and last till about midnight; they include much gesticulation and vociferation. At the solemn moment when the "conferring" of the maturity-degree takes place, the youth, still tended by the old man, remains motionless, with downcast eyes, and listens to the melancholy chant rendered by the old men in low lagging accents:

"Makolär manga, malolär, ä, är, maktär, immanga."

No beating of sticks or clapping of hands accompanies this tune, and no further ceremonial dance follows.

The youth has now been elevated to the status of "*Mollinya*" which qualifies him to the full rank and privileges of manhood. Further cicatrices may now be added to either side of his abdomen. The cuts are horizontal but do not extend right up to the median line.

During the period intervening between Böllier and Mollinya festivals, bustard, flying-fox, and yam are forbidden articles of diet, but after the latter event the fledgelings are invited to eat with the old men. They honestly believe that if any of the young men, while undergoing initiation, ate one of the forbidden articles secretly, the medicine man would be able to detect the food in his stomach; and having thus disobeyed, the medicine man would be justified in running a spear through the offender, or at any rate compel him to swallow certain things which would poison him. These rules are strictly observed, and, whenever some of the privileged members

have eaten flying-fox or bustard, they take the precaution to collect the bones and burn them.

The tribes on Nullarbor Plains will tell you that the initiation ceremonies originated in the following way. Many, many years ago, the emu and the kangaroo were more or less human in appearance and possessed of mighty powers. One day the emu caught the kangaroo with the object of making a man of it. But the great struthious bird had no hands wherewith it might have performed an operation; all it possessed was a “finger” on each side of its body. It might be explained that the emu, because it cannot fly, is not regarded as a “bird” in the generally recognized sense, and consequently the wings are looked upon as “fingers.” In most of the vocabularies, indeed, no distinction is made between “finger” and “hand,” the south-western tribes of central Australia referring to one or the other as “*marra*.” Nothing daunted, however, the emu removed the *præputium* from the kangaroo by clutching it between its wings and pulling it off. Thereupon the emu said to the kangaroo: “Will you make me a man?” And the kangaroo replied, “Yes.” The kangaroo had the advantage over the emu because it possessed five “fingers,” with which it could perform the operation the right way. The animal caught hold of the bird and circumcised it with a sharp splinter of flint. But the emu requested to be further operated upon and so it came about that the kangaroo decided upon a subincision. To the present day the emu retains the marks of this operation. Some while after these happenings, the tribal fathers ran across the sacred emu and noted the change in its anatomy; they forthwith mutilated each other in a similar way, and only then did they realize that they were men.

Not boys alone are required to submit to the various initiation ceremonies here mentioned, but in most tribes young women are “made” marriageable by having to submit themselves to ordeals which are quite similar to those of manhood’s approbation.

While discussing the female breast, we noted that when it begins to develop a girl is taken away by the men and the breast anointed and sung to, to stimulate its growth. This procedure is the forerunner of initiation. The girl’s development is forthwith watched with care, and when the unmistakable signs of ripening are detected the event is celebrated with dance and song.

Men and women attend, and the items rendered are more or less of the nature of an ordinary corroboree, although occasionally some special feature characterizes the performance. For instance the Larrekiya and Wogait tribes pass the girl through a “smoking” ceremony after the following fashion. An old gin places herself behind the girl and lays her hands upon the latter’s shoulders. Then all the other women taking part form a continuous chain by standing in a single row behind each other and “linking up” in a similar way. They begin to sing “Ya, Ya, Ya,” in a long-drawn melancholy note, and the old-gin immediately stamps her feet, and, moving forwards, pushes the girl along in front of her. All the other performers follow her, stamping in unison and holding on to the shoulders of the person in front. Quite unexpectedly the monosyllabic “Ya” is changed to “*Yen da min*,” and at this the old gin stops short and strikes the girl’s back thrice with her hand. The same performance is repeated time after time during the night. Early in the morning of the next day, the girl is led to the sea, and the whole party wades out to about hips’ depth. Here a grotesque dance is started during which they strike their arms, bent in the elbows, against the sides of their bodies under water, the splash producing a peculiar hollow-sounding note. The process reminds one of a goose flapping its wings while enjoying a bath. At this stage, the wording of the song sounds like “*A-lö-lö-lö*,” and when its final syllable has resounded, all bathers duck under the surface of the water.

Next a fire is kindled upon the shore, and, when a good blaze has been obtained, a heap of grass and leaves previously steeped in water, is piled upon it. Upon this the old gin seats herself and makes the girl sit upon her lap facing her and with her legs astride. The volumes of smoke which are generated completely hide the two from view. The idea is to allow the smoke to thoroughly play upon the parts of the novice, the process being facilitated by the manipulation of the old gin. When the ceremony is concluded, the girl is led into the bush by the old women and for some time to follow she is not allowed to partake of certain articles of diet, such as for instance snake, dugong, and goanna.

Several of the northern and north-eastern coastal tribes mutilate the hand of a young gin during the period of her initiation by removing two joints from a finger. The forefinger of either hand is generally chosen by the former tribes, the latter favouring the small finger. The Ginmu at the mouth

of the Victoria River make the amputation with a stone knife. In this district a singular case came under my notice which is of considerable interest from an evolutionary point of view since it suggests a phenomenon usually only met with in crustaceans, reptiles, and other creatures whose position is very much lower in the animal kingdom. A young girl had had two end phalanges of a finger imperfectly removed, and yet upon the mutilated stump a horny growth resembling a diminutive finger-nail had formed anew. The Daly River tribes remove the bones by tying a ligature of cobweb which they find in the mangroves very tightly around the joint. The end phalanges of the finger, thus deprived of the circulation, gradually mortify and drop off. Occasionally the joints may be bitten off by a parent of the child.

As a general rule, it may be said that wherever mutilations of the male are undertaken during initiation ceremonies, a corresponding operation is performed upon the female; and, *vice versa*, where the former practice is not indulged in, the latter is also unknown. Generally speaking, too, the female mutilation ceremonies are much the same wherever practised in Australia, but the implements or devices employed for the actual mutilation vary in different localities.

Invitations to the event are sent by special messengers to adjoining groups and neighbouring friendly tribes. These messengers are of mixed sexes and are decorated by having their bodies covered with ochre. The common method is to make the ground colour of the body a rich red and to draw upon it concentric circles of white and black. The men carry a "female" tjuringa, whilst the women, apart from numerous necklaces and armlets which they wear, are unadorned. The latter are near group-relatives of the young woman concerned. Their mission is readily understood by the people they look up during their walk-about, and, without much interchange of words, acceptance is indicated by the recipients of the message by resorting to an intimacy with the feminine emissaries. Although considerable liberality is shown during this indulgence, the privilege is by no means stretched to beyond the bounds of a tolerable promiscuousness, even though the messengers may be entertained at the distant camp for two or three days before they return home.

The celebrating camp in the interim has been busily preparing for the approaching event. Nightly corroborees have been held at the chosen spot

by both the men and the women, and the novice has repeatedly appeared before the performing crowd richly decorated and besmeared with emu-fat and ochre. At no time, however, even after the invited guests have arrived, does the excitement become anywhere near as great as during the initiation ceremonies of the opposite sex; in fact, at its best, the performance is extremely dull and monotonous.

When at length it becomes apparent that even the principal actors themselves are tiring, it seems as though the moment had arrived when only a desperate decision could revive the enthusiasm. A number of men, who stand in the same group-relationship to the novice as her future husband, lead the girl away without any ado, except perhaps that the remaining members slightly spur their acting. This stage is mostly reached at daylight, as often as not early in the morning, after the whole night has been spent in dancing and singing.

Away from the din of her tribespeople's celebration in honour of the occasion of her stepping from girlhood to womanhood, the silent victim is told to squat on the ground whilst the men surround her. Her oldest "group-husband" produces a flat, wooden tjuringa, of the "male" type, with which he several times touches her person, whilst he mutters incoherent and garbled words. This is done to dispel from her all possible pain and likely loss of blood during the operation she is about to be submitted to.

Then she is requested to lie flat on her back, and her head is placed upon the lap of one of the men who squats to keep it there. It follows the act which is destined to make her marriageable; her virginity is doomed to mechanical destruction.

The instruments, if any, which are used for the operation vary according to locality. In the central areas (Aluridja, Wongapitcha, Kukata), an ordinary stone-knife with resin haft is used. The Victoria desert tribes employ cylindro-conical stones from six to eight inches long, and from one and a half to two inches in diameter. Among the tribes of the northern Kimberley districts of Western Australia no real instrument is used at all, but the operator winds the index and middle fingers of his right hand together with a long piece of fur-string; and this device answers the same purpose as the above-named instruments.

The tribes indulging in this practice admit that their action is prompted by a desire to offer the girl's pudicity to one of her spirit-husbands. We might indeed look upon this rite as the equivalent of sacrificing the *jus primae noctis* to a mythical or legendary tribal relative who is supposed to be living in the astral form and who is likely to come back to earth at any day.

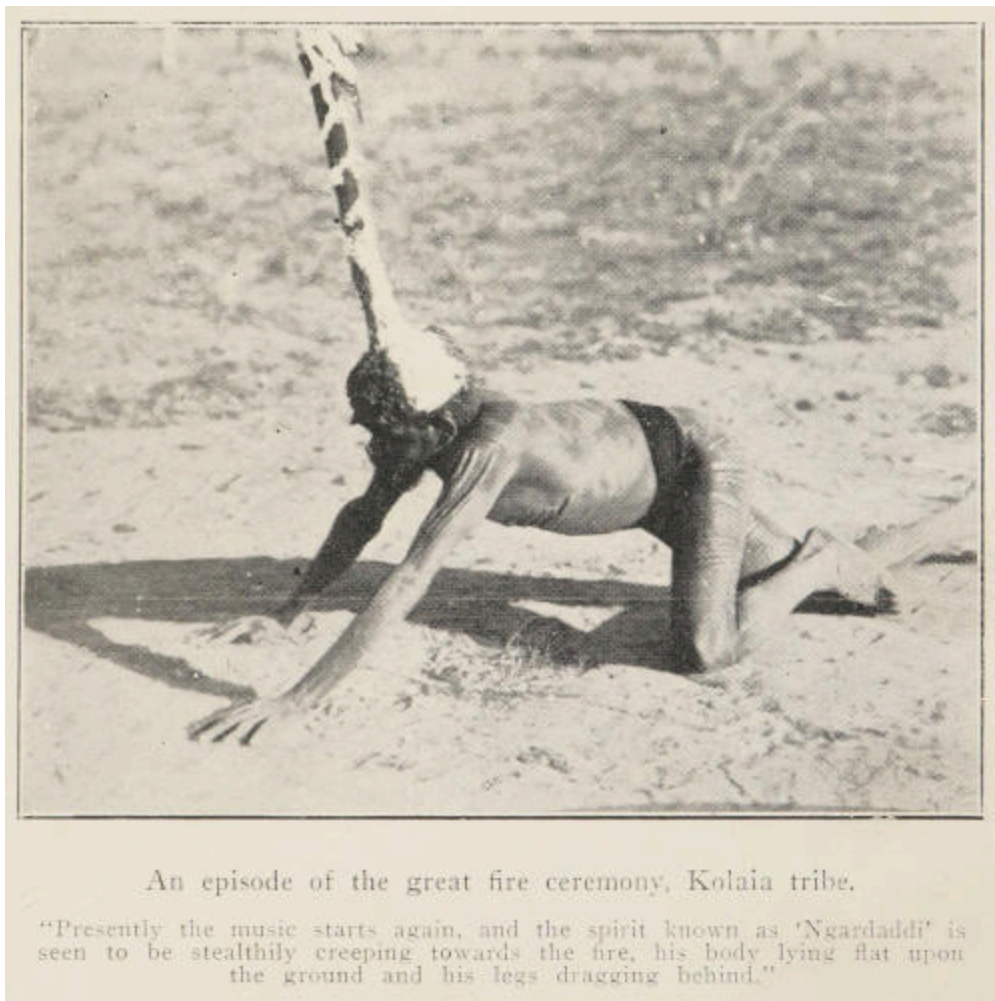


PLATE XXXII

An episode of the great fire ceremony, Kolaia tribe.

"Presently the music starts again, and the spirit known as 'Ngardaddi' is seen to be stealthily creeping towards the fire, his body lying flat upon the ground and his legs dragging behind."

CHAPTER XXVII

RELIGIOUS IDEAS

Religious instincts of aboriginal—Nature worship—Fire ceremony—Fire legends—Mythical fire thief called "*Ngardaddi*"—Water legends and ceremonial—Sun worship—Sun myths—The moon man—The mythical serpent—The kobong and totem—The tjuringa—Tjuringa legend—Ancestor worship—"Knaninja" or "Totem" deities—The significance of the tjuringa—Sacred tjuringa caves—"Totemic" diet restrictions—Gradation of sacred ceremonial—Great emu ceremony—The "*Altjerringa*"—The sacred yam or "*Ladjia*" ceremony—The "*Etominja*" design—Sex worship—The phallus—Mythical origin of phallus—Ideas concerning procreation—Grey hairs blackened artificially—A phallic monolith known as "*Knurriga Tjilba Purra*"—Foetal elements or "*Rattappa*"—The "*Tjilba Purra*" embodied in the headgear—"Waraka," a phallic stone on the Roper River—Similar Kukata legend—Phallic ceremonial on Cambridge Gulf—Cylindro-conical stones of phallic significance—Matronal chasm of Killalpaninna—"Arrolmolba," a sacred stone possessing stimulating principles—Phallic drawing of "*Mongarrapungja*"—Evil spirits—Disenchanted enclosures—Aboriginal belief in Supreme Being—Etymology of His name—The eternal home of all deities and spirit ancestors.

It has often been written that the Australian aboriginal is without religious ideas and without religious ceremonies. Such assertions are grossly incorrect and by no means portray the psychological side of the primitive man in its true light. He has, on the contrary, religious institutions and obligations which verge on the basis of all modern conceptions and recognition of divine supremacy. If we can class Nature-worship, Ancestor-worship, and Sex-worship as the beginnings of all religious teachings, then the Australian aboriginal has certainly inherited by instinct and tradition a very solid foundation from which we might trace the origin of many, if not most, of our most sacred beliefs in Christianity. At the same time, it must not be forgotten that it is really a difficult matter to distinguish clearly between mythological beliefs and what we class as religion. Religious thought has fluctuated with the advance of civilization and science to such a degree that, even within the short space of time covered by the more reliable records of our history, several revolutionary modifications have come about. As time advances, man becomes more sceptical and more exacting in his demand for proofs, and in his despair over finding nothing tangible to worship, he resorts to the recognition, by instinct or persuasion, of a God who is a Spirit. But all the while, as this secular metamorphosis is proceeding, he keeps his innermost feelings and faith alive by appealing to

his knowledge of the gospel or his belief in salvation, in the manner it was presented to him by myth, by legend, or by the Scriptures. His principal guide is his intellect; the less it is trained the stronger his inherited conviction; the more scientific it becomes, the greater his desire to probe the truth.

The modern man has so accustomed himself to an artificial environment that he takes the so-called “elements” of Nature, especially water and fire, in a strictly matter-of-fact sort of way. But the primitive man, who realizes that his very existence is dependent upon these factors, has learned to respect, preserve, and worship them as legacies he imagines to have been left him by some of his illustrious forbears who, he supposes, have gone to an unknown realm where they live in peace and can only return temporarily to their former haunts in the invisible form or through the medium of some other object which is related to the individual in some mysterious way.

The aboriginal looks upon fire as one of the great indispensable quantities of his social existence; it is the element which dispels the evil spirits from his camp; it is the means by which comfort and friendship are made accessible to him; it is his universal companion. More than this, it is the fire, with its warmth and its light, which draws individuals, families, groups, and tribes together and through its agency and influence that social concourse is established which lies at the bottom of all conviviality, oracular discussion, and ceremony. How well this sentiment agrees with the knowledge we possess of the origin of civilization! Indeed the appreciation of fire together with the knowledge of its preservation is perhaps the mightiest factor responsible for making our species human. Once man learned to nurse an original flame he found through accidental cause and kept it constantly by his side, his progress became an established fact. His crude camp-fire talks developed into discussions which he further expanded by means of drawings on the walls of caves he occupied. The free exchange of thought brought about by congregation round the cheerful flame could not fail to incite the intellect; and thus he ascended to the high road of civilization and gathered the fruits of culture he now enjoys.

The Aluridja, Wongapitcha, and some of the north-western coastal tribes believe that many years ago, a party of ancestral creatures, more animal than human, came down from the sky through the branches of tall gum-trees to confer with the spirits which roam about at night and conceal

themselves in inanimate objects during the day. These monsters brought a fire-stick with them and when they reached the earth, they lit a fire to cook some grubs which they had taken from the bark of the trees during their descent. As they were feasting, the spirits called them and they went with them to a cave where the bones of the persons rested, originally occupied by the spirits themselves. Whilst they were away, the fire which had been left unguarded, decided to run into the bush and, being in a mischievous mood, started an enormous blaze which burned down much of the forest and the tall gum-trees as well. The spirit-ancestors and the heavenly monsters beheld the disaster with consternation and called upon the fire to come back. This it did. But it so happened that some of the tribes' fathers were hunting in the area, and when they saw the fire, which was strange to them, they snatched portion of it away and ran with it to their camp, where they kept it and fed it with dry grass and sticks. The spirits and their visitors were very angry and never left the fire out of their sight, lest it might abscond again; they were compelled to live on earth for a very long time until the trees grew up again to their lofty domain. The hunters, on the other hand, zealously guarded their prize fearing that it might run away from them. Even to the present day, this belief exists among the older folks, and they always take great care that the ground is cleared of inflammable matter to stop the fire from bolting; to be on the safe side, they invariably carry or keep near to them a fair-sized, glowing fire-stick.

Among the Minning this legend is circulated in a slightly modified form. Two ancestral spirits had their fires burning in the sky at points represented by the pointers of the Southern Cross constellation, when one day they decided to come down to the earth to hunt opossum. They took their fires with them, but while engaged in the chase they left them at their camp. When they had obtained a sufficient number of opossums to make a good meal, they returned to their camp, where they noticed six young men sitting around the fires, who immediately made off, and, in doing so, each took a fire-stick away with him. The spirits gave chase and re-captured five of the thieves, but the sixth, who was named "*Warrupu*," reached the camp of his tribe and handed the fire-stick to his mother, "*Wenoinn*." The woman ran with it to the white sand hills about Eucla in which she intended hiding it. But the spirits had noticed her and came towards her from above with a spear. In her predicament, the woman threw the fire-stick away, which immediately set the whole of the country ablaze between Eucla and Israelite

Bay. All the tribes were thus enabled to seize some of the fire which they have carefully watched over ever since.

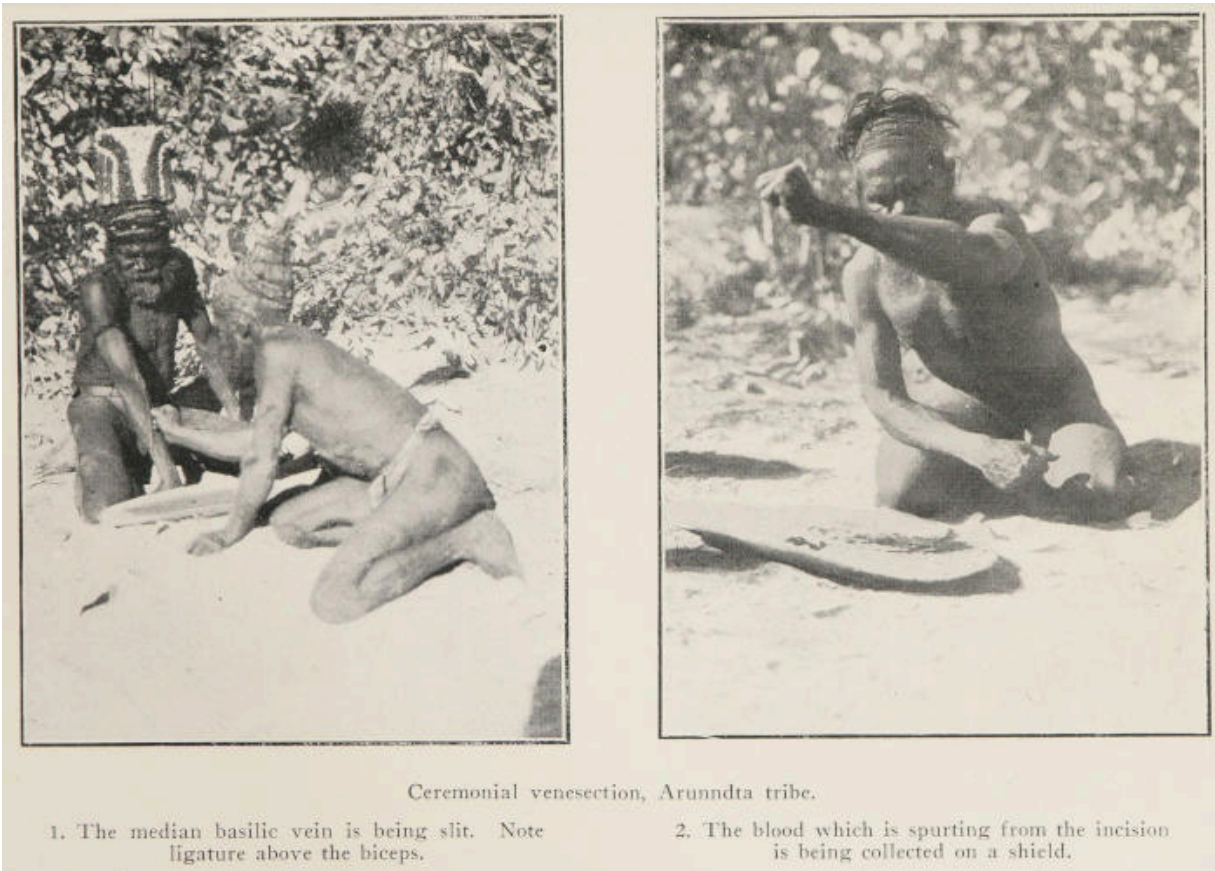


PLATE XXXIII

Ceremonial venesection, Arunndta tribe.

1. The median basilic vein is being slit. Note ligature above the biceps.
2. The blood which is spurting from the incision is being collected on a shield.

A similar tradition is perpetuated by the north-western tribes referred to and affords the motive of one of the most earnest and sacred fire-ceremonies known in Australia. The performance takes place during the night. It is introduced by two men; the one represents a mischievous spirit trying to steal back the sacred fire which is being carefully guarded by a number of men impersonating the ancestral tribesmen who originally discovered it; the other is a warrior who has accidentally come upon the would-be thief and overpowered him. The spirit crouches at the feet of the warrior, sitting upon his heels, with his head drooping upon his chest and his hands hanging loosely between his thighs. The warrior stands erect

behind his supposed captive, with his legs apart, and continues striking the fellow with small bundles of brushwood, one of which he holds in either hand. The beating is done regularly, both hands rising simultaneously, high above the warrior's head, and falling together upon the spirit's head.

Some two chains away, the tribal ancestors are grouped by the fire-side and are chanting the following lines:

*“Wai dang bunnai,
Inna dinna dulla ngai.”*

The men sit in a row at the back of the fire, with their thighs asunder and their legs bent in the knees; their chins are resting upon their chests whilst they beat the backs of their heads with small bundles of brushwood, keeping time with their song and with the performance of the warrior.

When, after a while, the music ceases, the warrior is seen to be lying asleep beside his captive. The ancestors become restless and begin to move sideways, first in a body to the left and then to the right; then they move backwards and forwards. This movement is peculiarly weird since the performers do it by shuffling over the ground in the sitting posture, with their arms held erect, but bent in the elbow.

Presently the music starts again, and the spirit known as “*Ngardaddi*” is seen to be stealthily creeping towards the fire, his body lying flat upon the ground and his legs dragging behind. He advances very slowly, turning his face towards the ground, in search of the fire which escaped from heaven. He wears a tall head-dress quite thirty-two inches long, which consists of a tightly fitting hemispherical cap carrying a column in its centre, at the top of which a bundle of split black-cockatoo feathers is attached. The feathers are from the male bird's tail, and the brilliant red patches in them are representative of fire. The whole structure is made of paper-bark and human hair-string, the outer surface being decorated with ochre, pipeclay, charcoal, and vegetable-down. Vide [Plate XXXII](#).

All the time the men at the fire-side are beating time with their hands and simultaneously turn their heads from side to side, to all intents and purposes quite unconcerned about the *Ngardaddi* who is gradually crawling near to them. This is done to entice the thief nearer and lead him to believe that he is unobserved. All of a sudden, however, when the spirit is about to touch

the fire and is in the act of snatching it from the tribesmen, one of the group on either side of the fire throws a handful of dry grass upon the smouldering heap. The flame responds immediately and casts a bright light all around.

Alarm is raised by the tribesmen by clapping their hands together violently. The spirit collapses and lies flat upon the ground at full length. Two or three of the men nearest by seize some of the burning grass and hit the prostrate figure over the head. The spirit jumps to his feet and treads the ground as if endeavouring to make his escape. Seeing this, the men at the fire rise quickly and treat their victim most unmercifully with bundles of burning grass and twigs. Eventually each of them seizes a fire-brand and digs the burning end deeply into the spirit's back and the unfortunate fellow eventually decamps into the darkness amidst the bellowing whoops of his victors.

The air is fouled for some distance around by the smell of the burned skin, reminding one of the stench in a smithy when horses are being shod. The back of the spirit-impersonator is naturally severely scored by the cruel treatment it is subjected to, but the fellow takes it all in good faith and without flinching.

The object of the ceremony is twofold. Firstly all members of the community who are present, men, women, and children, are taught to appreciate the value of fire, and secondly it is believed that the exemplification of so harsh and drastic a treatment for attempted theft will tend to make abortive any schemes of the evil spirits.

The Arundta are quite convinced in their own minds that in the days of their tribal fathers there was no water on the surface of the ground they occupied; their ancestors in those times were compelled to live on grass and succulent plants, no consideration being given to the fact, as we have learned, that the vegetation derives its moisture from outside sources. But it happened one day, when their forefathers were out hunting, that they met with a number of strange-looking men who were sitting around a pool of pure water from which they were drinking. At the sight of the men, the strangers fled, leaving the water behind. The hunters gave chase but all except one disappeared and he made for a cave in the hills. The hunters closed the mouth of the cave with a big stone and went back to the pool of water to quench their thirst, but when they reached the spot, the water had

turned into a massive, round stone. The men made back to the cave and removed the obstruction, but imagine their surprise when they found the cave empty. Upon making a careful search, however, they discovered a long cylindrical stick which had some peculiar markings on it. They took the stick and walked once more towards the petrified pool, and, lo, they beheld the stranger they were looking for walking in the sky. When he saw the stick in the hands of the hunters, he took the form of a cloud, and as he bent his body towards the stick, his long matted hair fell forwards and from it water poured upon the earth beneath. The hunters drank freely of the precious fluid and when they looked skywards again the cloudman had vanished.

From that day onwards the Arunndta medicine men ("*Nangarri*") have kept that spot sacred and taboo to the women and children; they call the big stone "*Imbodna*" which means "the hailstone." The man who fled to the cave and then escaped from the hunters as a vapour they call "*Nangali*," the name for a cloud. The tribe has never since been without water because Nangali left his magic wand in the hands of their ancient sorcerers and whenever the country was suffering from drought they could call upon him to appear in the sky and bring forth rain.

Nangali is one of a group of celestial beings who have been termed "*Atoakwatje*," that is Water-Men; they are now looked upon as Demigods who control all terrestrial supplies of water from their abode in the clouds. The Atoakwatje are believed to have certain mysterious connections with some of the tribal sorcerers who in a sense parade on earth as their disciples and attend to the rain-making ceremonies through which they are able to commune with each other.

When the people are in need of water, the rain-makers assemble around the Imbodna and one or two of them produce the sacred stick, known to the Arunndta as "*kwatje-purra*," literally meaning "the reproductive organ of water," and to the Aluridja as "*kapi-wiyinna*." Nowadays these sticks, which strictly speaking are of phallic significance, are flat and more like a tjuringa in shape, and have a number of peculiar markings on them. For a time the stick is laid beside the great water-stone, and the sorcerers kneel while they chant with a barely audible voice. They rise to their feet and the most influential individual who is decorated with stripes of yellow vegetable-down and wears a dog-tail tassel on his belt, lifts the stick

towards the sky and continues mumbling. The other members kneel again and all present chat together. The man who is standing poises the stick horizontally between his hands and rocks it one way, then another; and this performance is frequently repeated.

When at length the principal performer sits down, the other men leave the spot and run in a single file towards the camp, loudly crying “*kurreke ta ta*” in imitation of the call of the spur-winged plover.

In the evening a general corroboree is indulged in; and all grown-up persons, male and female, are allowed to join in. Several refrains are forthcoming which are connected with ordinary rain or water festivals. The principal rainmaker does not attend but joins the camp again during the night. It appears that in the interim he has visited the sacred cave, in company of one or two of his brother-sorcerers, to hide the magic stick and preserve it for future use. Any representative of the Atoakwatje group inherits the power to fashion and use the rain-stick, but it is imperative that he learns the art under the direction of a senior and duly qualified nangarri.

A ceremony directly connected with sun-worship belongs to the old Arunndta people and is known as “*Ilpalinja*.” When the weather has been and continues to be unpleasantly cold, and the mating season of birds and animals has on that account been long delayed, the men construct a large colored design upon the selected ceremonial ground. Radiating from a point upon a cleared space, many lines are drawn with red and white vegetable-down to represent the rays of the sun; and these are intersected at different distances from the central point by a number of concentric circles which represent the fathers of the tribe. The centre of the design is occupied by a stick which is supposed to incorporate some mystical and sacred sun-creature known as “*Knaninja Arrerreka*.” The same *Ilpalinja*-design is occasionally carved as the crest of the *Knaninja* upon a sun-tjuringa. Vide [Fig. 6](#).

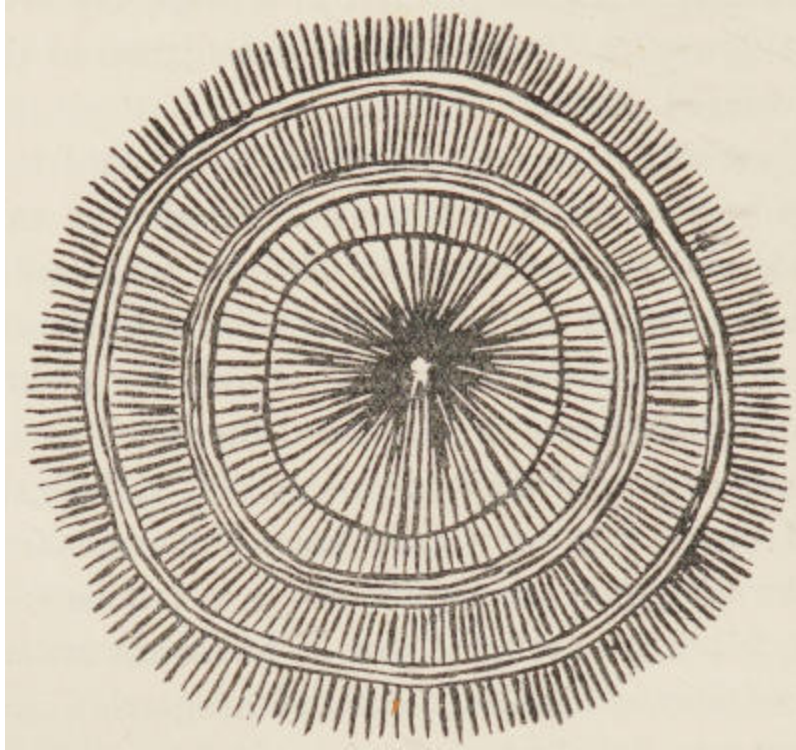


Fig. 6. Sacred sun-design of the “*Ilpalinja*” ceremony ($\times 1/20$).

A most impressive function might occasionally be witnessed on the north coast, which is associated with the setting sun; it is known to at least two tribes, the one living on the upper reaches of the Victoria River and the other on the western shores of Carpentaria Gulf, including some of the islands. It is usually performed in conjunction with demonstrations calling upon a fabulous being which lives in the sky to fecundate certain species of plants and animals necessary for their daily life. The Carpentaria tribes, moreover, keep their sacred poles, akin to the *tjuringas* of central Australia, not in caves but in special huts which they construct upon chosen spots absolutely taboo to the general public. These slabs of wood are up to five feet long and are covered with peculiar carvings and markings; they are of the two sexes. Ordinarily they are kept “asleep” by laying them on the floor of the hut side by side, and covering them with sand. When the hour of the ceremony arrives, they are brought out by the “Sun-Men” and stuck in the ground in the full light of the sinking sun. Just as the orb is about to touch the horizon, the tenders of the sacred implements kneel, with their faces turned towards the sun and, lifting their hands, bend their bodies to the ground much after the fashion of an Eastern salaam. We have before us a true form of worship recognizing the supreme powers of the sun, but aimed

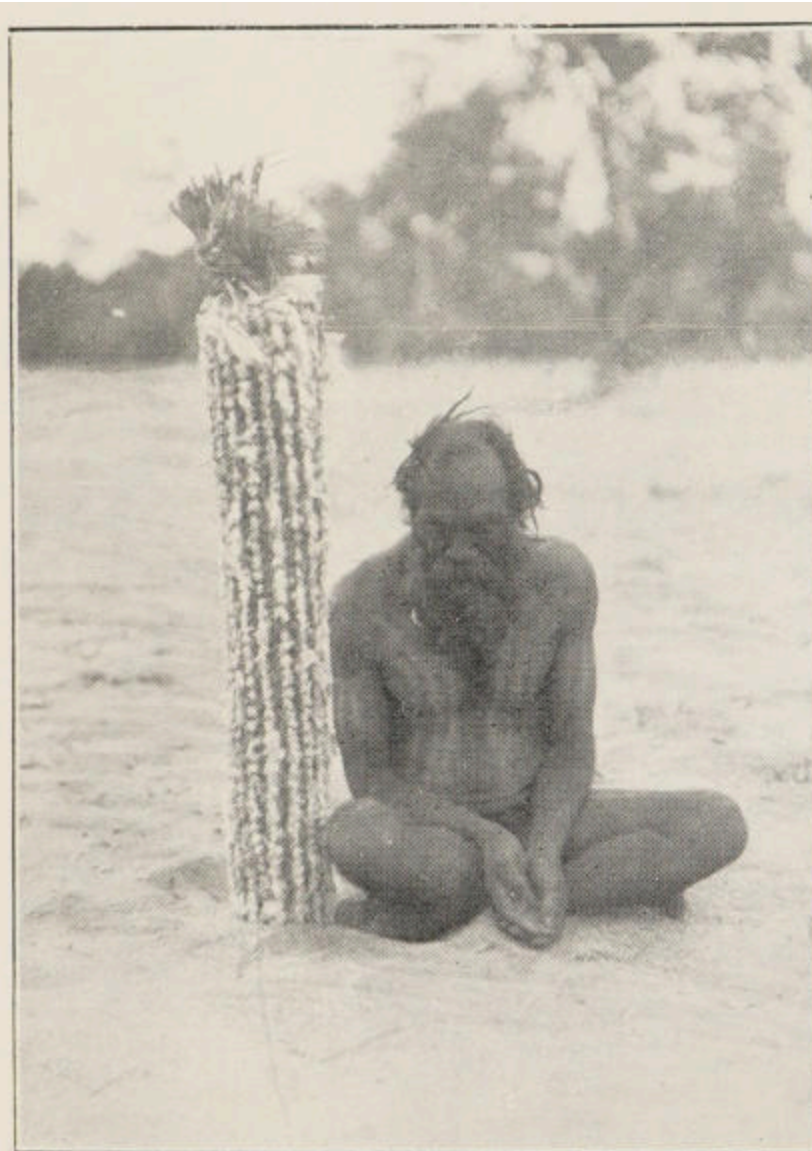
primarily at calling upon a demigod or Deity in supplication for making a needed article of diet, animal or vegetable, fruitful or prolific.

Mythologically the sun is regarded as a female having human form and a fiery exterior, who walks daily across the firmament and returns at night to rest at her sacred haunts on earth. Some of the central tribes, like the Aluridja, split the sun's identity into an indefinite number of such women, a different one of which makes the journey every day.

The moon on the other hand is thought to be a man who originally inhabited the earth but was one day chased off it by a gigantic dog the Aluridja call "*Tutrarre*." The man jumped into space and walked among the clouds until he reached the earth again. His long walk had made him so hungry and thin that he ravenously ate a great number of opossums which he found in the trees at night. In consequence he swelled out, and became fat and round. Then it was his bad fortune to fall in with the dog again, and this time his obesity prevented his escape. The dog tore him to pieces and swallowed him, bone and all. But it so happened that one of his arm-bones flew from the dog's jaws and found its way to the sky. There it floated from east to west as a luminous sickle and gradually swelled until it was perfectly round. The dog stood looking up at the bone and howled in anger, but the moon-man reappeared in the sky and converted the dog into stone.

The Kakatu natives believe in a moon-man who lives in the sky and controls the clouds. On a certain day, very long ago, this man was seen by the ancestors of the tribe. It happened thus: Just about dusk, a cloud was observed descending from the sky which came to rest upon the summit of a hill; it was glowing red. A big man, a woman, and two girls stepped upon the earth, and the man took a fire-stick from the cloud which then became black and ascended again. It was the moon-man and his family. The party walked down on to the plain and camped, the old man making a fire with his torch whereby his feminine escort could warm themselves. The moon-man left, taking a new fire-stick with him. In a deep, green water-hole lived a monstrous snake whose colour was much like that of the slime which covered the surface of the water. A lengthy and secret interview took place between the moon-man and the snake on the bank of the lagoon, and the snake produced many tubers of water-lily, and mussels also, for the moon-man to eat. Then the two heard a rustling noise. The snake exclaimed: "What is that? Who dares approach our trysting place?" The moon-man

snatched a fire-brand and held it high in the air; this made it light as day. The moon-man's daughters could be seen creeping towards the men to hear the secret discussion! With a curse upon his mouth, the angry father hurled the fire-stick at his deceitful daughters. The stick struck the ground and sent a shower of sparks over the girls. In an instant everything became dark as night, but every now and again there came from the spot the girls had last been seen at long-drawn growls; from the same spot flashes of light shot forth and illumined the clouds. The snake and the moon-man had disappeared, but the daughters remained just where they had last been seen, for they had been turned to stone and had assumed the rigid form of a dog whose head was directed skywards as if to rebuke the moon-man for the curse he had brought upon them. For a long time the clouds remained dark; then the moon-man re-appeared among them and cast a mournful beam upon the canine image of his daughters. From then till now he has continued to appear periodically in the sky, and his repentant daughters gaze at him; but at times, when the sky is covered with heavy black clouds, the daughters become angry and growl aloud. At these times, too, bright flashes dart from their eyes across the clouds like spears of fire threatening revenge. The stone has remained to the present day, and is known by the name of "*Koreno kardjo* (dog) *gambi*" (stone).



The "*Tjilbakuta*" of the great emu ceremony,
Arunndta tribe.

"The moment the sacred object is completed, the
Illiyakuta delegates one of his group to act as its
attendant or guardian."

PLATE XXXIV

The "*Tjilbakuta*" of the great emu ceremony, Arunndta tribe.

"The moment the sacred object is completed, the Illiyakuta delegates one of his group to
act as its attendant or guardian."

The snake is an important character in the mythology of practically every tribe of Australia; in fact most of the permanent water-holes are supposed to be inhabited by great serpents which guard the supplies, destroy unlawful consumers or polluters thereof, and frequently communicate with those spirit ancestors of the tribes who are descended from the original snake-man still living in the sky. In many cases the mythic snakes can be recognized in some characteristic features of the landscape. Take, for instance, the great artesian spring near Coward Springs Station which is known as Blanche Cup. This is looked upon as the mouth of a snake, while the hill immediately at the back of it (Mt. Hamilton) is its head. In consequence the formation is called "*Worma-Kadiabba*" (snake's head) by the local Arrabonna tribe. The natives have a dread of these imaginary snake-monsters and prefer not to visit a water-hole at night; in fact, at any time, day or night, they feel safer in the company of a man who is "related" to the snake, because he can protect them and give them the right of approach. The snake is possessed of evil and will molest any but its totemic "relatives."

The fundamental conception of the kobong (or totem), so far as the Australian aboriginal is concerned, is of a religious nature. In the beginning of all things, the Aluridja say a number of exalted creatures of human form came out of the earth and were gracious to their tribes-people. Then appeared a menace in the shape of a gigantic dog which chased the good people from one place to another, until they decided to adopt the forms of various animals and plants, and thereby became either too fleet for the dog or were not recognized by it. Other good people now descended from the hills and drove the dog back to its hiding place in a cave where the evil spirit dwells. The newcomers kindled a fire at the mouth of the cave and kept the evil beings in captivity whilst the original Deities re-assumed the human form. Ever after, however, these good creatures were able to alter their appearance from human to animal at will; but each individual in his choice adhered to the particular animal or plant which had saved him from the ravages of the great evil dog. Eventually they formed themselves into flat slabs of stone or wood, upon the surfaces of which they scratched the emblems of their animal representation and the traditions of their long wanderings on earth. The spirits of these Deities now live in the sky but can return at any time to re-enter the slab generally known as the "tjuringa." Among the Minning at Eucla the larger of these objects are known as

“*wagal-wagal*,” the smaller as “*bobi*,” whilst further west, in the Laverton district, “*kaidi*” is the prevailing word. It is true, the *tjuringa* is not known to all tribes; in which case the Deities are supposed to have entered such natural objects as rocks, hills, and conspicuous trees.

The Roper River natives believe that their deified forbears were molested not by a dog, but by a hideous old woman or witch, who, by the influence of evil, entrapped them and subsequently ate them. On one occasion, however, a party of warriors were successful in decoying her away from her haunts and slaying her. The jubilant victors decided to cut out the old woman’s tongue as a trophy, but as they were thus engaged, the tongue flew out of the mouth and spun round in the atmosphere above them, making a terrible noise as it did so. The men chased the tongue, but it flew towards a beefwood tree and embedded itself deeply in the butt; in vain they looked for it and tried to cut it out; it had become part of the tree. Before returning, however, the men took a piece of wood out of the tree, shaped like the woman’s tongue, which they tied to a piece of human hair-string and swung round their heads with joy. Behold their mixed feelings of delight and fear when the piece of wood began to howl with a voice like that of the slain witch! The tribe retained that piece of wood as a sacred memento of their victory, and they gave to it the name the witch was known by, namely “*Kunapippi*.” Nowadays this object is the equivalent of the central Australian *tjuringa*.

All tribes recognize the existence of deified ancestors, now real or spiritual, whom they regard as sacred and worship accordingly. All ancestors stand in a definite, intricate, and intimate relationship to some animal, plant, water-hole, or other natural object which they have at some time or other represented; some indeed in the first place appeared as animals and later took the human form. They are now looked upon as being those powers who by virtue of sacred ceremonial can produce the species they have at some time incarnated, in plenty or allow it to proliferate. As a matter of fact, some of the sorcerers of the tribes often declare that they can see the inside of a sacred rock or *tjuringa* teeming with young, ready to be produced.

The Arunndta refer to their “*Knanimja*” (i.e. “totem” Deities) as “*Altjerrajara*,” meaning the Supreme Number; the Aluridja as “*Tukurata*” or “*Tukutita*”; and the Dieri as “*Muramura*.”

Just as the “totem” ancestor is connected with an animal, plant, or other natural object, and is embodied in the sacred form of the tjuringa, so the individual who traces his descent from such ancestor recognizes a close and mysterious affinity between himself and the tjuringa which has become his by heredity; henceforth it becomes his sacred talisman which protects him from evil and procures for him the means of maintaining his existence.

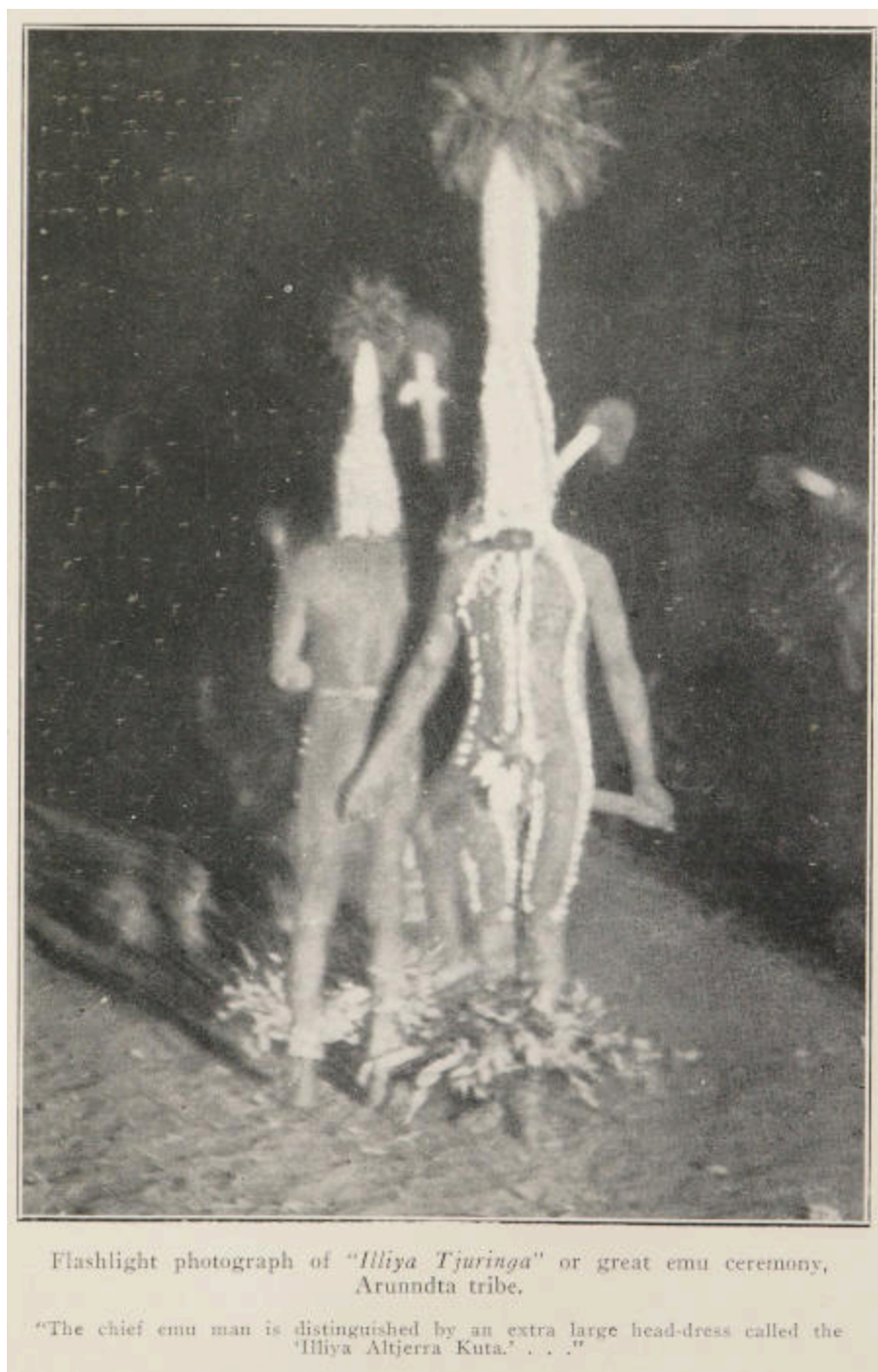
The emblematic representation of the deified ancestor, based upon the form of an animal or plant living to-day and in some way “connected” with the individual, is the “*kobong*” of the north-western tribes first referred to by Sir George Grey.

The “totem” is very dear and sacred to the native, and is religiously protected by him. I well remember on one occasion on the Alberga River I discovered a small black and yellow banded snake which I killed. An Aluridja man who was attached to the party at the time was greatly shocked at this, and, with genuine sorrow, told me that I had killed his “brother.” Turning to an Arundta he lamented aloud: “*Kornye! Nanni kallye nuka kalla illum,*” which literally translated means: “Oh dear! This brother of mine is dead.”

One thing is always essential and that is that a native performs frequent, prolonged, and reverential ceremonies, remote from the women and children, and in the presence of his tjuringa. Under these conditions the tjuringa is believed to have powers similar to those of the Deity it embodies.

When not in use, the tjuringas are stored in caves, the entrances to which are small and not easily discernible; the ground is proclaimed taboo to any but initiated tjuringa holders and is strictly regarded as a *sanctum sanctorum*. Although the sticks and stones are the individual property of the tribesmen, the objects are generally kept together, and only brought out during a religious ceremony. The old men are the authorized custodians of the sacred collection. The female tjuringas are included, because even though a woman may possess one, she must never see it; if she does, accidentally or otherwise, she is in imminent danger of being killed. No unauthorized hunter is allowed near the prohibited area under any pretext at all; even if an animal he has wounded should by accident make for the sacred ground to breathe its last, the hunter is required by tribal law and

usage to sacrifice it to the divine factors incorporated in the tjuringa, by leaving it on the spot.



Flashlight photograph of "*Illiya Tjuringa*" or great emu ceremony, Arundta tribe.

"The chief emu man is distinguished by an extra large head-dress called the '*Illiya Altjerra Kuta*.' . . ."

PLATE XXXV

Flashlight photograph of "*Illiya Tjuringa*" or great emu ceremony, Arundta tribe.

“The chief emu man is distinguished by an extra large head-dress called the ‘Illiya Altjerra Kuta.’...”

When on the warpath, a warrior always craves to carry his tjuringa with him, even though this is not always possible. He firmly believes that with the talisman kept on his person, or at any rate knowing that it is nearby, no deadly missile thrown by an enemy will penetrate his body. The mere knowledge of the fact that his opponent has a tjuringa with him, and he not, is sufficient to make a coward of the bravest fighter. Should he be wounded or take ill, one of his “totem-brothers” endeavours to produce a tjuringa, from which, if the medicine man considers it necessary, a little powder is scraped and handed to the patient to swallow with water.

With regard to “totem” animals which form the objects of hunting expeditions, a man is allowed to kill and eat thereof with some restriction. He must kill only one animal at a time, and only in accordance with the method prescribed by the tribal fathers and handed down to them by tradition. This is usually a straightforward hunting method, with as little loss of blood as possible. If much blood should flow, the hunter is obliged to cover it without delay with sand. If possible, other men should cut up the carcase, and only certain portions be handed to the “brother” of the slain animal.

Each tribe has an endless variety of objects (animal, vegetable, terrestrial, meteoric, mythic, and so forth), which may figure as a “totem.” Any one of these may be the primary motive of a separate cult or sacred ceremony, but here again the variety is usually reduced to the number corresponding to the most sanctimonious and most useful creations affecting the affairs of the particular tribe.

The ceremonies take the form of either a direct worship or a prayer for increased productivity of a certain plant or animal, either being offered to the Knaninja or “totem” ancestors living as spirits in the sky. Usually the two ideas are embodied in one grand ceremony, and the method of procedure is governed by tradition. Such ceremonies have been particularly elaborated by the Arunndta tribe, who refer to them by the same name as that of the sacred object, namely “*Tjuringa*”; less frequently they call them “*Intitjuma*,” the latter name being applied more to ceremonies without worship.

The Tjuringa ceremonies are divided into grades according to their importance and sacredness. The water ceremony is ordinarily called "*Kwatje Tjuringa*," but if the "totem" spirit ancestor is invoked to attend, it goes by the name of "*Kwatje Tjuringa Knaninja*"; if the principal spirit ancestor is assumed to be present, the title becomes "*Kwatje Tjuringa Knaninja Knurrendora*"; and finally the most sacred water ceremony of all is the "*Kwatje Tjuringa Altjerra Knaninja Knurrendora*."

As a typical illustration we shall discuss the "*Illiya Tjuringa*" or Great Emu Ceremony of the eastern Arunndta groups. The date of the performance is decided by the senior emu "brother" of the tribe, the oldest member who claims to be related to the Illiya Knaninja. Somewhat extensive preparations are made beginning a few days prior to the opening event. Only fully initiated men take part, but the women are allowed to witness certain of the most awe-inspiring stages from a distance. Whilst the younger men are out collecting leaves, out of which they make the down later to adorn the bodies of the performers, the older men prepare the sacred ground. Others slay a number of brown hawks, off which they pull the feathers and then pluck the down. A suitable site having been selected, the old men clear it by removing all grass and bush from the surface and smoothing the sand with their feet. The "brothers" who claim relationship alike to the great Emu-Man, the Emu-Knaninja, and the emu itself, thereupon proceed to anoint the sacred ground with their blood, for which purpose they puncture the median basilic vein of the forearm with a quartzite chip and allow the fluid of kinship to sprinkle upon the sand. It is surprising to see the amount of blood sacrificed by the men on occasions like this; and time after time, when such is required, the process is repeated. By examining the forearms of an old stager, one can usually count a number of small scars along the course of a vein indicating places where a perforation has at different times been made. A venesection is made after much the same manner among the various tribes.

The following Arunndta method will serve as an example. A ligature of hair-string is in the first place tied tightly round the upper arm, a little above the biceps muscle, after the style of a tourniquet to check the flow of blood in the veins and thereby distend the vessels. The man then makes a small longitudinal cut through the skin and punctures the vein beneath it lengthwise; the blood spurts forth immediately and is collected in the

handle-pit of a shield. When the flow is to be stopped, the native removes the ligature, and this in most cases is all that is needed. Should, however, the blood continue to come, he places a small amount of down over the incision and presses it against the vein, or winds three or four strands of fur-string around it. The little pad of down is usually left on the arm until it dries and falls off. None of the women are allowed to witness this operation, which is called "*Ilgarukna*." The blood, when it is to be used as an adhesive for the down-decoration, is applied with a small brush ("*ipinja*") made of twigs tied together with fur-string. Vide [Plate XXXIII](#).

The principal among the emu group is called "*Illiyakuta*," and it is he who directs the performance. He takes his followers to a secluded place, such as a clump of timber or down a creek-bed, and there the wooden tjuringas belonging to the ceremony are produced and painted afresh with red ochre and emu fat.

Down is made out of the white, felty leaves and twigs of *Kochia* bush, which the Arunndta call "*kemba*." Small quantities of these are placed upon a flat slab of stone and pounded with a pebble. The fluffy material which results is next mixed and rubbed by hand with powdered kaolin or ochre according to the colour required, the white being known as "*wadua*," the red as "*wanjerra*."

A sacred object is now constructed which encloses the painted tjuringas and is called the "*Tjilbakuta*." It is about three feet high and is made in the following way. The tjuringas are laid one on top of the other and bound together with many lengths of human hair-string, which completely obscure the shape of the separate pieces. A thick layer of the stalks of the kangaroo grass (*Anthistiria*) is laid around the parcel and kept in position with a few lengths of twine, and then the whole structure is covered with great masses of human hair-string wound spirally from top to bottom. A cylinder results which is decorated with alternate vertical bands of red and white vegetable down. Into the top of this Tjilbakuta one bundle of emu feathers and one of black cockatoo tail-feathers are stuck; and often additional plumes are hung beneath them. The moment the sacred object is completed, the Illiyakuta delegates one of his group to act as its attendant or guardian. For the time being his body is decorated with symmetrically placed, curved ochre bands upon the chest and vertical bands down the arms; at a later stage he ornaments his body more elaborately, prior to taking part in the principal

performance; but all the time he remains in his place of hiding beside the Tjilbakuta. Vide [Plate XXXIV](#).

At the sanctified place close by the other men have been stacking firewood at different points to illuminate the proceedings during the evening. Occasionally, too, the Illiyakuta group of men cover a portion of the ground with a coloured emblem of the traditional emu.

Early in the afternoon of the festive day the men who will take part in the ceremony at night begin to prepare themselves. Many of the non-performers help them.

Large quantities of down, both vegetable and birds', are used to decorate the bodies. The design is shaped much like a cobbler's apron, extending from the neck down the front to the level of the knees. The greater part of this surface is red, but it is lined with white and split along the centre by two parallel lines of white. The back is not decorated at all. The entire surface of the face, including the eye-lids and beard, is thickly covered with down which is white, except for an oval red patch around the mouth.

The principal attraction, however, of the sacred emu ceremony is the head-dress, which is both elaborate and imposing. To prepare it, the attendant combs back the actor's hair with his fingers, and interlaces it with stalks of grass and small twigs in such a way that a tall conical structure results right on top of the head. This is made secure and of a uniform exterior by winding much human hair-string around it, at the same time taking in a plume of emu feathers at the apex of the cone. The headgear is completely enveloped in red and white down, extending upwards from the head as alternate vertical bands. The chief emu-man is distinguished by an extra large headdress called the "*Illiya Altjerra Kuta*"; this measures a good three feet in length, and it embodies, between the apex and the emu plume, deeply enshrouded with hair-string and down, the sacred "*Illiya Tjuringa*." Other members who are of the same rank as the "Tjilbakuta" guardian, wear their insignia beneath the emu feathers in the form of a sickle-shaped rod, which carries at each of its points a tuft of white cockatoo feathers. All performers cover their person with a dog-tail appendage which hangs from a thin waistband of human hair-string. And lastly, they all tie bundles of eucalyptus twigs, with the leaves attached, to their legs just above the ankles. If possible, old or half-dried leaves are selected in order that a more

pronounced rustling is produced when the men move about; the noise is made to imitate the rustle of the wiry feathers of an emu. Vide [Plate XXXV](#).

At nightfall the Tjilbakuta is removed from the hiding place and planted on the edge of the ensanguined patch. The guardian is thus given an opportunity to slip away and to attend to his ceremonial toilet, which is similar to that of the rest of the Tjilbakuta group. When he returns, the performance is about to begin, and all except he leave the ground.

The stacks of wood are set fire to by invisible hands, and, so soon as the flames flare upwards, the silence is broken by the booming note of a bull-roarer, which is produced some distance off in the bush.

The Tjilbakuta guardian sits beside the object like a statue, with his eyes rivetted to the ground immediately in front of him. From behind him the thud of stamping feet and the rustle of dry leaves announce the coming of the official performers, while from the other side the non-performing members step from the darkness and take up their position by squatting between two fires. When the decorated men come into view, the latter start beating their boomerangs together in perfect time to the stamping of the feet of the advancing actors. They come as a body of five or six rows, one behind the other, each man holding his hands locked behind his back and uttering a deep guttural note resembling a pig's grunt. The folded hands held over the stern represent the tail, the guttural noise the call of the emu.

The Illiyakuta, wearing the tall Illiya Altjerra Kuta, is in the front row, and he is attended on either side by a Tjilbakuta man. The chief now starts a chant: "*Immara janki darrai*," and all the others, including the sitting men, join in; the same is repeated several times. When the two parties are opposite each other, the performers quicken the pace of their stamping and extend their arms sideways, thereby widening their ranks. After this they retreat to behind the Tjilbakuta and one hears a shrill chirping note resembling the cry of a young emu.

The interpretation of this act needs no special elucidation. The decorated performers are those of the tribe's manhood who, in all matters pertaining to the emu, have a right to communicate, through the Tjilbakuta, with the astral emu ancestor living in the great celestial domain of the ancestral spirits, which is known as "*Altjerringa*." They are invoking the benign Knaninja or originator of their particular "totem" species to increase the

numbers of emu on earth for the exclusive benefit of their tribe. It is the Illiyakuta who imagines that he receives the favourable response from above, and, when it comes, it is he who imitates the cry of a young emu. It often happens, however, that the chief persuades himself to believe that the Great Spirit had not heeded the appeal, in which case the last-mentioned cry is wanting. The ceremony is repeated time after time.

Altjerringa, it will be observed, is a compound word consisting of “*Altjerra*,” the Supreme Spirit, and “*inga*,” a foot or trail. The implied idea is that Altjerringa is the “walk-about” of the spirit ancestors, where they walk, and have always walked, and where the spirits of all tribes-people eventually hope to find their way.

After this act, the performance becomes less restrained and takes more the form of a corroboree. Some of the men seize firebrands from the burning stacks and hurl them in the direction of the women’s camp. From the moment of the sounding of the bull-roarer at the beginning of the ceremony until now the women sat huddled together, with their faces buried in their hands, thoroughly cowed by the portentous happenings. When the firebrands come whizzing through the air and crash into the branches of the trees around them, sending sparks flying in all directions, they are almost beyond themselves with fear. But just at this juncture the men call upon them to look towards the festive ground and behold them dancing. In obedience to the order, the women’s fears are dispelled and soon superseded by a noticeable enravishment. They feast their eyes upon the array of manhood in gala dress, and it is not long ere they pick up the rhythm of a dance by beating time to the step. Provided the Tjilbakuta has been removed to a place of secrecy, well out of reach of accidental discovery, the men entreat the women to come up and join in the song. Thus the sublime is eventually reduced to commonplace, and the remainder of the night passes in joviality.

To refer briefly to a vegetable ceremony, we shall select the yam or “*Ladjia Tjuringa Knaninja*.” The preparations are much the same as those of the emu ceremony. An enclosure is first made in a secluded spot with branches, in the centre of which the “totem” or Knaninja “stick” is erected. Several men immediately set about to decorate it with vegetable down as previously described. The design in this case consists of vertical rows of red circles upon a yellow ochre background. In addition, a large plume of split

eagle-hawk feathers is stuck into the top of the stick. All ordinary performers wear conical head-gears or “*tdela*” made of *Cassia* twigs, into the apices of which tightly bound bundles of grass stalks (“*gortara*”) are fixed carrying plumes of emu feathers (“*mangalingala*”) ([Plate XXXVI](#), 1). Other men have squat, cylindrical bark structures called “*elbola*” placed over their heads, which are elaborately decorated with vertical coils of human hair-string and coloured down.

One of the principal actors represents the “*Kuta Knaninja*.” His head-gear consists of two long kutturu, tied together with hair-string and completely covered with gum leaves, the whole being subsequently besmeared with blood and decorated with coloured down. As the assistants are dressing this character, they keep up a chant sounding like “*Winni kutcherai*.” Vide [Plate XXXVI](#), 2.

The leading figure is the “*Ingada Ladjia Knaninja*,” who wears a tall vertical head-piece which contains the tjuringa of the Ladjia Knaninja. The tjuringa is, however, not visible, but is covered with pieces of bark, securely tied over it with hair-string, the whole being richly decorated with vertical bands of red and white down.

The Great Spirit of the Yam, called “*Knaninja Tjilba Ladjia*,” when he leaves Altjerringa, takes up his abode in a cave near Mount Conway, where the tjuringas are kept, but at night, before the fires are lit, he is supposed to come to the ceremonial ground and occupy the decorated “totem” stick described above. During the performance he is surrounded by all the ordinary performers, who are known as “*Tjilba Ingarrega*,” and are directly under the guidance of the Ingada and Kuta Knaninja.

A group of men who are not decorated sit near one of the fires and sing while the performers are thus encircling the Ladjia stick:

“*Imbanai yinga*
Wi ma bana Ladji di bana
Yammana wi ma bana
Jai ra ja ja
Jai ja ja na
Wi ba na na
De a re a ja betja,”

the voices finally fading away to an almost inaudible whisper.

If the Great Spirit, Ladjia Altjerra Knaninja, is gracious, the tap-root of the yam will be sent deep down into the earth near the Jay River and from there spread its laterals all over the country to supply the needs of the tribe.

When some of the most sacred ceremonies are performed, the oldest “relatives” of the presiding Knaninja often construct a coloured drawing upon the consecrated ground, whose purpose is similar to that of the “totem” stick above described. The drawing is executed in coloured down, both vegetable and bird. A space of suitable size, often measuring many feet in length, is cleared of grass and stones, and sprinkled with water, when it is ready to receive the down. In the case of, say, the “*Ladjia Tjuringa*,” the design takes the form of a number of concentric circles alternately red and white, from the outermost of which six equally spaced groups of red and white lines stand out radially. The enclosing border of the design consists entirely of white down. Vide [Plate XXXVII](#).

Once constructed, this drawing, which is known as “*Etominja*,” is zealously guarded by one of the old men. If, peradventure, an unauthorized person happens upon the sanctified place, he is killed and buried immediately beneath the spot occupied by the design; thereupon the ground is smoothed again and the *Etominja* re-constructed. Nobody in camp ever hears what became of the person, and should any relative track him in the direction of the area known to be tabooed, he is horror-stricken and runs away.

While the old men are re-constructing the *Etominja*, they sing to the Knaninja as follows:

*“Yedimidimi
Dakarai pa ma taka,
Pa mitu min jai, jin tarai,
Ja ra nai malgada, ja ranai.”*

The next, and probably the most important, group of religious ceremonies is that dealing with Sex-Worship. For years past peculiarly shaped stones have been found in caves and among the possessions of the Australian aborigines whose shape was strikingly suggestive of a phallus, but hitherto no actual phallic ceremonies have been observed. It was my

good fortune to witness such among the Aluridja, Arunndta, Dieri, and Cambridge Gulf tribes. From enquiries made of the old men, it appears that in former days this form of worship was practised considerably more than it is nowadays. New stone phallus are rarely made by the present tribes; those in their possession have generally been inherited from previous generations. The old men have the phallus in their keeping, and they are very loth to either produce or part with them.

The natives of the King Sound district in the north-west believe the origin of the phallus to be as follows: In the early times a scourge was raging among their forefathers, from the effects of which many were daily dying, when a hairy man and his mate, a woman of ordinary human form, came to earth from above. The evil was due to the exhalation of poisonous breath from the gaping jaws of a green monster resembling a crocodile. The stranger relieved the sufferers from the awful curse by showing them how to perform an operation upon their person which taught them to endure pain and protected them against future ravages of the pestilence. This great and benevolent stranger then took his departure, but left his name to designate the surgical operation which to the present day is performed upon the male members of the tribe; the name, strange though it may seem, is "*Elaija*"; and it is known, at any rate, as far east as Port George IV. But the tribe had become so weak through the terrible havoc the disease had wrought that the old men called him back and entreated him to stay. Elaija, however, took from a dillybag his female companion was carrying, a stone carved after the shape of a mutilated member, which he gave the name of "*Kadabba*." When the old men gazed upon this object, they took fright and appealed to Elaija, but the good fellow had vanished. The stone has remained with the tribe ever since, and through the divine property Elaija endowed it with, their threatened extinction was eluded. Moreover, they continued to practise the operation on all young men because it made their members like the Kadabba of Elaija, which they knew had the power of multiplying their kind. And so the Kadabba became a sacred object whose procreative power they have learned to worship, thinking that by such observance they would augment their own capacities of sex. Vide [Fig. 7](#).

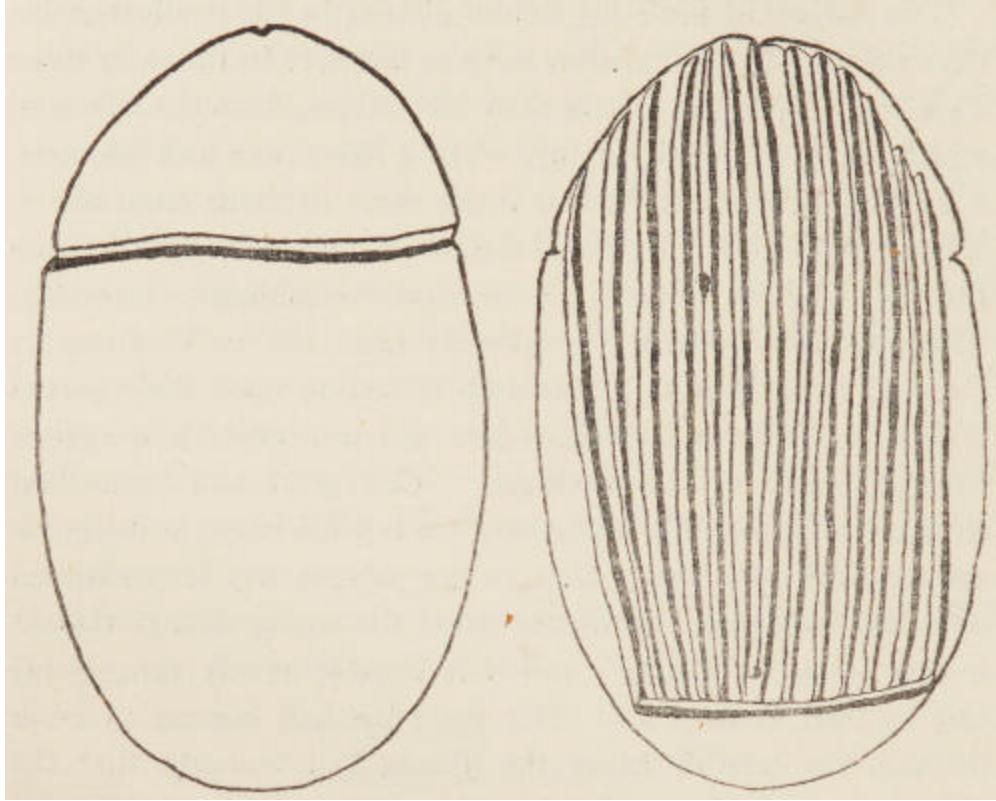


Fig. 7. Stone phallus, Northern Kimberleys, Western Australia ($\times 1/2$).

One often reads, and I was under the same impression myself until I became better acquainted with the tribes, that the Australian natives do not connect the knowledge of conception with any intercourse which might have taken place between the sexes. This I find is not altogether correct, although usually the younger people are kept in complete ignorance on the subject. No doubt strangers are treated similarly when they put any pertinent questions to the old men on matters of sex. The old men believe in the duality of human creation, the spiritual and the material; sexuality is regarded as the stimulus of corporeal reproduction, but the spirit quantity is derived through mystic and abstract influences controlled by a “totem”-spirit or Knaninja. Under these circumstances, it is not surprising to note that the ceremonies of the phallus are transacted principally by the old men of the tribe who aim at the rejuvenation of their waning powers.

It is interesting to see the old men preparing for a ceremony which is to be dedicated to a Knaninja or Spirit of Sex, because they all endeavour to conceal the white hairs of their beards by rubbing powdered charcoal into them. The bark of the cork tree (*Hakea*) is used for the purpose; pieces of it

are charred, crushed between the palms, and applied where needed. It is astounding what a difference this process makes to the appearance, and some of the old grey-beards really look as though they had been made twenty years younger by magic.

In the eastern MacDonnell Ranges stands a cylindro-conical monolith whose origin is believed to be as follows: Many generations ago, the paternal ancestors of the Arunndta walked from a district situated, as near as one can gather, somewhere in the neighbourhood of Ediowie; they were known as the “*Kukadja*,” and were characterized by the enormous dimensions of their organs. These old men or Tjilba of the tribe migrated northwards to beyond Tennant’s Creek and settled in the productive “*Allaia*” country which surrounds the Victoria River. In that same district one finds, even at the present day, cave drawings of human beings with the anatomical peculiarities referred to ([Fig. 8](#)). At a later time, the head-man of the Kukadja, named “*Knurriga Tjilba*,” returned southwards to the Macdonnell Ranges. While roaming the hills, he espied two young women sitting on the side of a quartzite cliff, and without deliberation began to approach them. He was in the act of making lewd overtures when the guardian of the girls, a crow ancestor, caught sight of him and hurled a boomerang at him. The missile struck the great man and cut off the prominent portion of his body, which in falling stuck erect in the ground. The force of the impact was so great that the man bounced off the earth and fell somewhere near Barrow’s Creek. He bled so profusely that a clay-pan soon filled with his blood. Thus his followers found him, and overcome with sorrow they opened the veins of their arms to mix their blood with his. Then all the members of the party jumped into the pool and disappeared for ever.

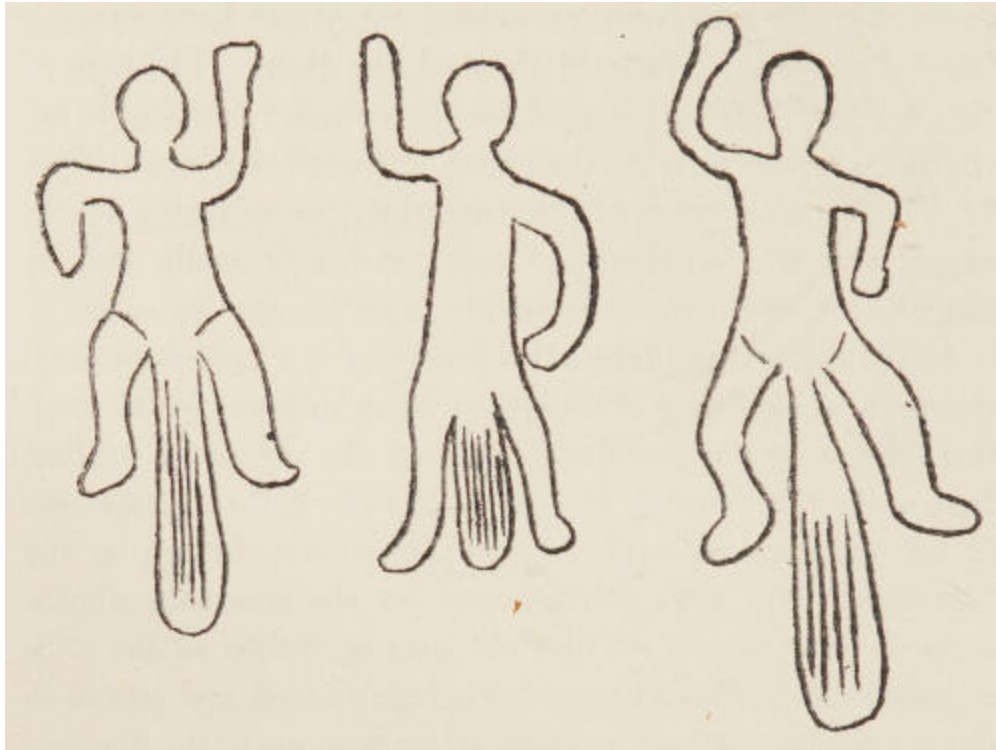


Fig. 8. Ochre drawing of “Kukadja” men, north of Wickham River, Northern Territory ($\times 1/3$).

The severed portion of the old man’s body, however, remained just where it fell and turned to stone. It has long been known as “*Knurriga Tjilba Purra.*”

The two young women can also still be detected in the cliff as prominent rock formations.

The stone has been protected by the tribe as long as the old men can remember, because they realize that it contains an inexhaustible number of unborn tribes-people. These mythic, foetal elements are generally recognized to exist in certain objects of phallic significance, and are called “*rattappa.*” The medicine men maintain that they can at times see the dormant living matter in the stone. It is on that account that it is regarded as sacred, and every now and then very secret and worshipful ceremonies are transacted near its base, the main objects of which are to multiply the future membership of the tribe and to preserve the sexual powers of the old men.

The Tjilba Purra naturally figures prominently in some of their ceremonies. In fact, it is reproduced and worn upon the head of the leading man during the functions. The sacred effigy consists of an upright column, about two feet high, composed of a stout bundle of grass stalks, in the

centre of which the tjuringa is contained. It is decorated with alternating bands of red and white down throughout its length. This upright column represents the "*Tjilba*" or revered ancestor whose spirit is invoked to "sit" in the tjuringa; at the top of it a plume of wiry emu feathers, well powdered with charcoal ("*unja*") to give it a youthful appearance, takes the place of the forbear's hair and beard. Standing at an angle with the central column, a similar though slightly smaller structure is intended for the "*Purra*" or phallus; it carries a plume of white cockatoo feathers at its end to represent the glans. Vide [Plate XXXVIII](#), 2.

A landmark, of similar significance as the Tjilba Purra of the Arunndta, exists on the Roper River in the Northern Territory; it is a pillar of sandstone known as "*Waraka*." Waraka is also the name of the great Spirit Father of the tribe. In very early times this man came to earth in a semi-human form, and made the country abound in game, animals, birds, and fish. Then he found a woman on the shores of Carpentaria Gulf who remained with him as his wife. Many children came of the union; and Waraka's mate has since been looked upon as the mother of the tribe. The woman's name was "*Imboromba*," and to this day the tribe takes its name after her. Warraka had an enormous sex characteristic which was so ponderous that he was obliged to carry it over one of his shoulders. Eventually the organ became so huge that Warraka collapsed and sank into the earth. His burden remained, but turned to stone, and is now looked upon by the local natives as the great symbol of Nature's generative power which first produced their game supplies and then the original children of the tribe; it is revered accordingly.

The Kukata have a somewhat similar legend of the origin of a stone of phallic significance, the name of the possessor of the large organ being "*Kalunuinti*."

In the extreme north-western corner of Australia, in the Glenelg River district, the natural stone is replaced by an artificially constructed one which possesses the true shape of a phallus. The stone is about three feet long and stands in a vertical position in the ground commanding a ceremonial cirque as if intended to watch over the proceedings which are instituted there.

On the shores of Cambridge Gulf, a grotesque dance is performed by the men, during which a flat, wooden phallus is used, shaped almost like a tjuringa, about seventeen inches long and three inches wide at the middle. It is painted in alternate bands of red and black, running transversely across the two flat surfaces, which are, in addition, decorated with the carved representations of the male organ of generation. The dance takes place at night and is too intricate to describe in detail. It is introduced by the following chant:

*“Wa, la, ja-la-la
Wa, la, ja-la-la
Wa la gori wau!”*

The verse is repeated three times, and then the performers stamp the ground with their feet, about ten times in quick succession, the action suggesting running without making headway. Presently, and with one accord, the whole party falls upon the knees. The phallus is seized with both hands and held against the pubes in an erect position, and so the party slides over the ground from left to right, and again from right to left. An unmistakably suggestive act follows, when the men jerk their shoulders and lean forward to a semi-prone position, after the fashion generally adopted by the aborigines. Still upon their knees, the men lay the phallus upon the ground and shuffle sideways, hither and thither, but always facing the object in front of them. After several repetitions of this interact, the performers raise their hands, in which they are now carrying small tufts of grass or twigs, and flourish them above their heads, while their bodies remain prone. Then follow some very lithe, but at the same time very significant, movements of the hips. When, presently, they rise to their feet again, the phallus are once more reclaimed and held with one of the pointed ends against the pubes in an erect position. A wild dance concludes the ceremony, during which the men become intensely agitated and emotional; very often, indeed, their excitement, verging on hysterical sensibility, evokes an orgasm.

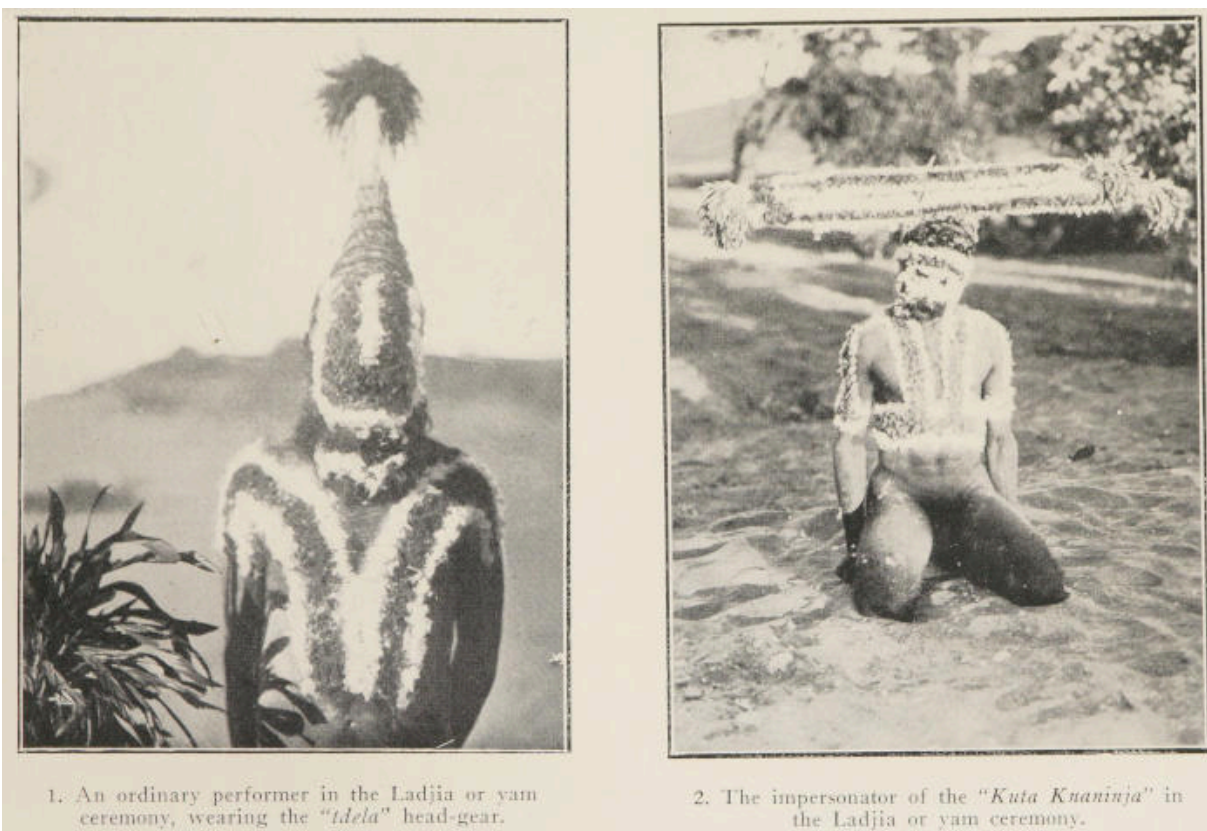


PLATE XXXVI

1. An ordinary performer in the Ladjia or yam ceremony, wearing the "tdela" head-gear.
2. The impersonator of the "Kuta Knaninja" in the Ladjia or yam ceremony.

These occurrences must not be confused with the mixed intercourses which occasionally take place at the climax of friendly corroborees to celebrate the meeting of neighbouring tribes. In this case we merely have to do with an inter-sexual embrace following an animated orgy, in which those members of both tribes standing in the general relationship of husband to wife take part.

The Dieri have a number of long cylindro-conical stones in their possession which are supposed to temporarily contain the male element of certain ancestral spirits now residing in the sky as their recognized deities. These are on an average about fifteen inches long and an inch and a half in diameter, circular in transverse section and pointed at one end. The old men have these phallus in their custody, and are very unwilling to let them get out of their reach because they believe the virility of the tribe is dependent upon the preservation of the stones. Should one of them be accidentally

lost, the mishap is calculated as little short of disastrous; should a stranger find the object, the old men maintain that evil will come to him, and if he keeps it he will die. The stone is used principally during religious ceremonies connected with sex-worship, but it is also produced during some of the initiation practices. After he has submitted to the “gruesome rite” in his initiation, a novice is required to carry the stone, firmly pressing it against his body with his arm, until he is overcome by the exhaustion occasioned by the painful ordeal. By so doing, the young fellow’s virile powers are supposed to receive considerable stimulation through the agency of the phallus he carries. The object drops into the sand beside him; and, when he recovers, he returns to the men’s camp without it. Two of the old men thereupon track the lad’s outward course and recover their sacred stone to take it back to a place of safety.

The tribes inhabiting the great stony plains of central Australia and those adjoining them, and also the Victoria Desert tribes, are occasionally in possession of nodular ironstone and concretionary sandstone formations, of the “natural freak” kind, which simulate the *membrum virile* to a marked degree. These are believed to have been left them by a deified ancestor and are kept by the old men as a sacred legacy; they answer in every way the purpose of an artificially constructed phallus.

Closely allied to the phallic significance given to natural pillars of rock and smaller imitative specimens, is the idea that natural clefts in the earth represent a female character. Killalpaninna is the name of a small lake lying about fifty miles east of Lake Eyre in central Australia, it being the contracted form of the two words, “*killa*” and “*wulpanna*,” which stand for that typical of woman. It is the conviction of the Dieri tribe that when a person, especially one stricken with senility or enfeebled by sickness, at a certain hour passes from the water of the lake into the open, and is not seen doing so by the women, he is re-born and rejuvenated, or at any rate cured of his decrepitness. In this sense Killa-Wulpanna has from time immemorial been an aboriginal Mecca, to which pilgrims have found their way from far and wide to seek remedy and solace at the great maternal chasm which has such divine powers to impart. This fact is of particular interest, since a native, generally speaking, is superstitious about entering any strange water, and does so very reluctantly, thinking that, by doing so, the evil spirit will foist disease upon him through the medium of the water.

A singular stone exists in Ellery Creek, a short distance south of the MacDonnell Ranges, which is called "*Arrolmolbma*." It was at this place that a tribal ancestor, named "*Rukkutta*," a long time ago met a young gin, "*Indorida*," and captured her. The stone at the present time shows a cleft and two depressions which are supposed to be the knee-marks of Rukkutta. On account of the intimacy which took place, the stone is believed to be teeming with rattappa, which entered by the cleft. The ancient Arunndta men used to make this stone the object of special veneration, and during the sacred ceremonies which took place at the spot, they used to produce carved slabs of stone they called "*Altjerra Kutta*" (i.e. the Supreme Spirit's Stone or Tjuringa). These inspired slabs of stone, being of the two sexes, were allowed to repeat the indulgent act of Rukkutta and Indorida, while the natives themselves rubbed red ochre over the sacred stone of Arrolmolba, and engaged in devotion. The act of rubbing red ochre over the surface of the stone was supposed to incite the sexual instinct of the men and to vivify the virile principle of the tribe. By this performance the men believed they took from the pregnant rock the embryonic rattappa which in the invisible form entered the wombs of the gins and subsequently came to the world as the young representatives of the tribe they called "*kadji kurreka*."

Among the cave drawings of Australia, designs are here and there met with depicting scenes from ceremonies having to do with phallicism and other sex-worship. In the picture reproduced from the Pigeon Hole district on the Victoria River ([Fig. 9](#)), one notices a man of the Kukadja type who was named "*Mongarrapungja*" in the act of dancing around a sacred fire with an ancestral female. The organ of the Kukadja, it will be observed, passes into the flame, whence a column of smoke is rising to find its way to the body of a gin which is drawn in outline above the dancers. Here we have the representation of a traditional ceremony associating the Kukadja's phallus with the impregnating medium supplied by fire, which, we have already learned, may be looked for in the column of smoke.



The sacred "*Etominja*," Arunndta tribe.

"If, peradventure, an unauthorized person happens upon the sanctified place, he is killed and buried immediately beneath the spot occupied by the design; . . ."

PLATE XXXVII

The sacred "*Etominja*," Arunndta tribe.

"If, peradventure, an unauthorized person happens upon the sanctified place, he is killed and buried immediately beneath the spot occupied by the design; ..."

The Australian tribes, without exception, believe in the existence of an evil spirit, or Demogorgon, who prowls their camps at night. It is on this account that natives are loth to leave the glare of the camp-fire, fearing they might be caught and injured by the spirit. In reality, they picture this fellow much as we do a ghost. Ordinarily he is invisible, and during the day haunts the graves of the dead tribesmen. At nightfall he opens the grave and walks about under cover of the dead man's skeleton; the natives often imagine they can hear the bones rattling as he passes near them; and they shudder with fear. It is on this account that they imagine the spirit to be not brown like themselves, but white, the colour of bone. If he catches anybody roaming alone, he will bite his victim in the abdomen and the latter perceives an unpleasant feeling or pain in that region. So far as one can ascertain, the symptoms of this damage are those of an acute attack of dyspepsia. The Arunndta call this ghost "*Iltdana*."

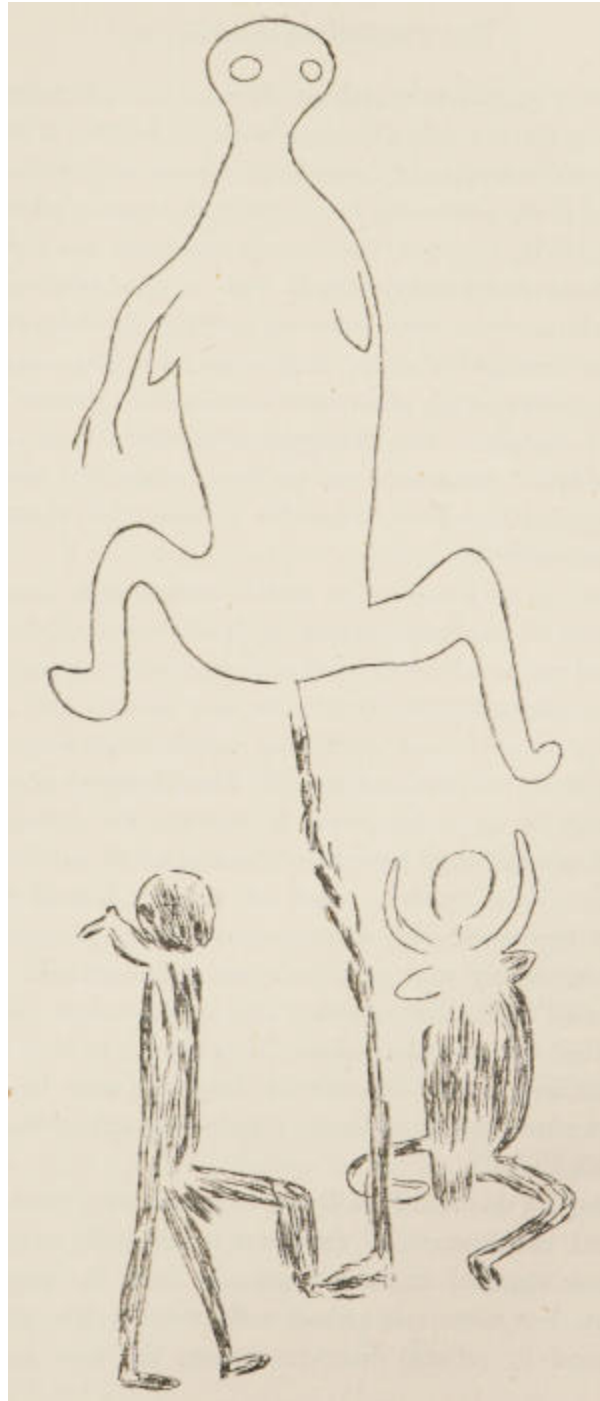


Fig. 9. Charcoal drawing of a Kukadja man named "*Mongarrapungja*" dancing at a sacred fire with an ancestral female, Pigeon Hole, Victoria River ($\times 1/3$). Tracing.

There is another evil creature which is at large after dark; it is known by the Arunndta as "*Erinnja*." It is supposed to have fleshy parts, a tail, and large projecting teeth; it runs like a dog, and possesses only one large toe on

each foot. If it catches anybody unawares, it will turn the captive's head back to front. The natives claim that this monster has a predilection for opening people's navels with a bone nose-stick. It takes children from their mothers and ties them to trees for the great mythical dogs to feed upon; for this reason the terrified mothers often place meat outside their huts to appease Erinnja's hunger. The Nangarri or medicine men imagine they can detect the presence of the brute when it is about, and they immediately exhort the people by informing them of the impending mischief.

At times it is necessary to protect a person or party from likely harm or witchcraft emanating from the supposed presence of an evil spirit. It may be that a warrior is indisposed and is anticipating molestation, or a number of men may wish to perform a sacred ceremony during which the evil spirit may be thought an undesirable listener. The Nangarri or medicine man alone has it in his power to exorcise the demon. And after the surroundings have been clarified of all deterrent contamination, he proceeds to mark off an area around the man or party concerned with selected stones which he consecrates. This area, mostly oval or circular in shape, remains a strict taboo to all beyond its confines, and is believed to disenchant any morbid influence the evil spirit may bring to bear upon it. If, in the former case, he wishes to further protect his patient, he places some green gum leaves over the sufferer's body. Vide [Plate XXXIX](#).

Although a spiritual Evil Being is feared more than a Good is revered, the existence of the latter is faithfully admitted, so far as our personal experience goes, by both the central and northern Australian tribes; and such a belief has, moreover, been found by reliable observers among the now practically extinct tribes of southern Australia. According to the late Dr. A. W. Howitt, the natives of Victoria and New South Wales used to speak with bated breath of a great Supernatural Being which once inhabited the earth and now lives in the sky. The belief is original and not in any way due to missionary influence. The oldest myths, such as only the untaught and unsophisticated grey-beards can tell, contain references to the existence on earth in ancestral times of convivial beings, kindly disposed towards the people, who eventually found their way to the sky by way of towering trees, since destroyed by fire. The Supreme Being is called by the Arunndta "*Altjerra*," by the Fowler's Bay natives "*Nyege*," and by the Aluridja "*Tukura*." The benign Altjerra roams about the sky ("*Alkurra*") and keeps a

watchful eye upon the doings of the tribes beneath him. The natives are so convinced of his ubiquitous presence that the Arunndta, for instance, have a favourite exclamation when committing themselves on oath in the form of "*Altjerr'm arrum*," meaning something like "*Altjerra* hear it"; that is, an appeal is made to Altjerra as witness to what is said, much after the way a school child endeavours to convince one of the truth of an utterance by exclaiming: "God strike me dead if I tell a lie."

The etymology of the Supreme Being's name is often really poetical. The Sunday Islanders, for instance, recognize such a Super-Being whom they call "*Kaleya Ngungu*." This name embodies in it the ideas of the past as well as of the future. "*Kaleya*" in that locality means the "finish" of anything, or even "Good-bye," while "ngung" stands for that which is to come or is to be. The implication is that this Great Being has emerged from the obscurity of bygone days and continues to live into the still greater uncertainty of times ahead. From a religious point of view, then, we note here a recognition of a spiritual quantity whose influence has been exercised uninterruptedly from the earliest past, and will be continued again, and forever.

So, too, the name of the Aluridja God, "*Tukura*," is composed of two words expressing the ideas of genesis and eternity. In the same sense, the Arunndta regard their demi-gods, or "*Altjerrajarra namitjimma*," as being the creators of life, although as such they themselves continue to live uninterruptedly and *ad infinitum*. And we have already noted that they roam for ever in the great celestial "walk-about," which is known as "*Altjerringa*" by the Arunndta, "*Talleri*" by the Kukata, and "*Wirrewarra*" by the Narrinyerri.

To this consummate home all spirits of the dead find their way and there join their spirit ancestors. In fact, so many have gone to the happy ground that most tribes look upon the stars as the camp-fires and fire-sticks of their departed relatives and friends. It is on this account that many stars have been named after notable tribesmen, the natives imagining that they can discover the new addition in the firmament after the deceased has been interred. Whenever a shooting star is observed travelling towards the earth, it is taken to be the spirit of one returning temporarily to its terrestrial haunts.

CHAPTER XXVIII

ABORIGINAL ART

Psycho-analysis of the conception of art—Oldest records appear as rock carvings—Descriptions from several localities—Evidences of great antiquity—Tree carvings—Carved grave-posts of Melville Islanders—Carvings on wooden weapons—Engraved boabab nuts—Carved pearl-shell—Bone not carved to any considerable extent—Charcoal, kaolin, and ochre drawings—Cave drawings—Ochre mines and legends—Barter with ochre—Preparation and application of pigment—Hand and foot marks—General picture of cave drawings—Ochre drawings on sheets of bark—Ochre body decorations—Coloured down and ground designs—Study of designs in detail—Subsidiary sketches of determinative character—All features of model shown on one plane—Keen observers—Natural features of rock surface embodied in design—Celestial phenomena artistically explained—The emu—Perspective—Action—Animation—Composition—“Totem” designs—Sacred designs—Remarkable drawings from the Glenelg River—Conventional drawings and patterns—Tracks of man, animal, and bird—Images in the sky—Fish—Flying fox—Fleeing kangaroo—A fight—Time symbols—Circle-within-circle and U-within-U designs—Kangaroo, caterpillar, and native pear-tjuringa drawings—Anthropomorphous designs.

No truer insight could be obtained into the mind of a primitive man than by means of a psycho-analysis of his artistic productions and predilections. Like most other peoples, past or present, the Australian aboriginal has developed his talents to an astonishing degree. Just as is true of his dances and musical performances, the real value of a production can only be appreciated by one who can throw himself, heart and soul, into the responsive, reciprocative, and assimilative mood and atmosphere the artist endeavours to create. The psychological factor is the more important; an aboriginal's design may be crude, but his imagination is, nevertheless, wonderful; we see the line, but he sees the life; we behold the image, he the form. In the absence of such a psychological reciprocation, however, the effect and quality of any artistic production may fall flat. Indeed, we ask, what is contained in a mere line? With due deference to the Euclidian definition, a line, from a primitive point of view, might represent almost anything, provided the necessary imagination is there; and it is just this imagination which is particularly cultivated by the primitive man. The question arises: “Who is the better artist: the man who can satisfy himself or the man who can satisfy others?” The former is unquestionably the more primitive idea of satisfaction. Simplicity of design is by no means indicative

of a deficiency in talent, provided the inventor has evolved the necessary imagination which permits him to behold in a design before him the reality of the original. This instinct is common to all mankind; we have only to observe our own children to appreciate the excellence of their creative talent and imagination when they are at play among themselves. Modern adult man is too realistic, perhaps too unnatural; he cries for something tangible, something concrete to appease his tastes. Just as one often sees youngsters, European as well as aboriginal, playing at their favourite game of "father and mother," assuming certain inanimate objects, like sticks, to represent their children, so the aboriginal artist's imaginative mind can actually see the real living picture contained in the crude diagram before him; he conveys his thoughts to his fellows by means of a lengthy verbal explanation, and, when they have caught the idea, they can wax as enthusiastic over the thing as the artist himself. This sentiment we have sacrificed to a certain extent. It is difficult to say exactly who is an artist and who is not. By nature we are all artists, and most of us can satisfy our individual needs in a more or less conventional way. But to pose as a modern artist, it is required of us to be able to take the inspiration from Nature and reproduce it in such a way that others are able to grasp the significance or beauty of the design without more explanation than a mere title. Yet the conception of "art" is a variable and relative measure, and we know well there have been different tendencies and schools, some of which are so highly specialized that the untrained mind cannot grasp, or even admire, the quality of the reproduction, because it cannot see with the same eye as the artist specialist. Here, then, we have a reversion to the primitive instinct, combined with all the perfections and skill which culture and training have evolved during the long space of time lying between a primitive foundation and the high standard of modern excellence.

The oldest records of primitive art in Australia are preserved in the form of carvings upon rock surfaces, akin to those found in parts of Europe, South Africa, and Egypt. The technique of these carvings is twofold. At Port Jackson, and elsewhere along the coast of New South Wales, and to a small extent at Port Hedland in Western Australia, a great variety of representations of fish, animals, and men have been cut and scraped in outline into the surface of the rock; in the Mann Ranges also, certain nondescript designs were found lightly scratched upon the surface of some diorite outcrops with a fragment of rock. At different points in the northern

Flinders Ranges, at Yunta, at Eureowie in New South Wales, on the Flinders and Burnett Rivers in Queensland, and also at Port Hedland in Western Australia, an extensive series of designs occurs which has been chipped into the rocks with the aid of pointed stone chisels. The carvings at Port Hedland and those in the Flinders Ranges are very old indeed. In fact, they are so old that none of the tribes now living remember anything about them, and refer to them as being the handicraft of the Evil Spirit.

The most striking feature about the Port Jackson carvings is the large size of individual designs, some of the fish measuring nearly thirty feet in length, and some of the kangaroo over ten feet in height. That the carvings have been made for generations past is evident from the fact that in places a practically obliterated design has been covered, and re-covered, with new designs.

At Port Hedland there are many acres of a low limestone plateau literally covered with carvings, many of which are badly weather-beaten; several of the carvings are of the scraped type found at Port Jackson. The accompanying illustrations depict a chosen number of the Port Hedland designs drawn to scale.

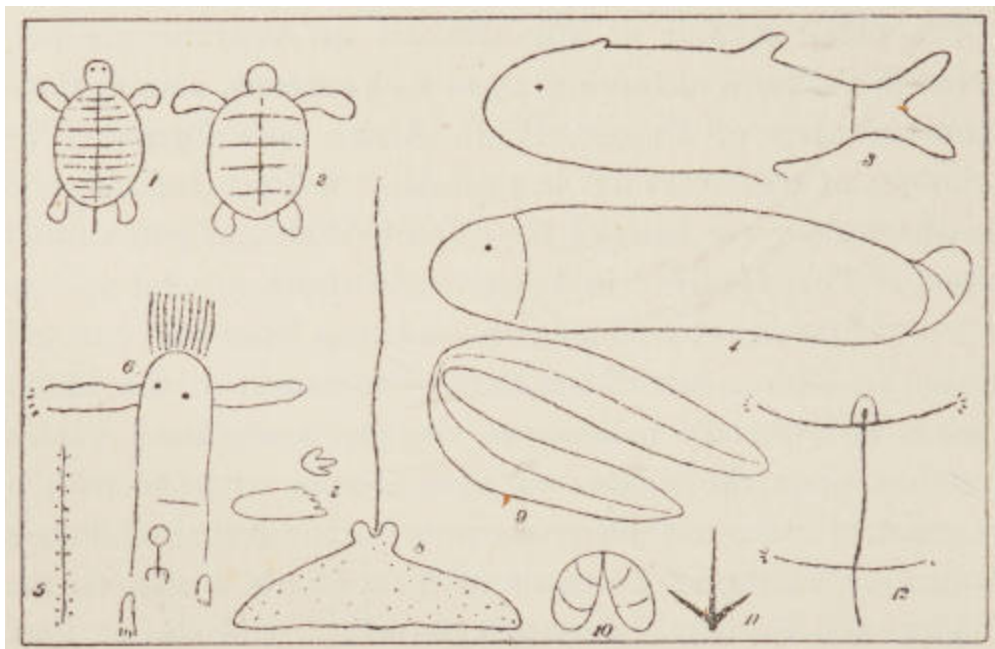


Fig. 10. Rock carvings at Port Hedland ($\times 1/20$).

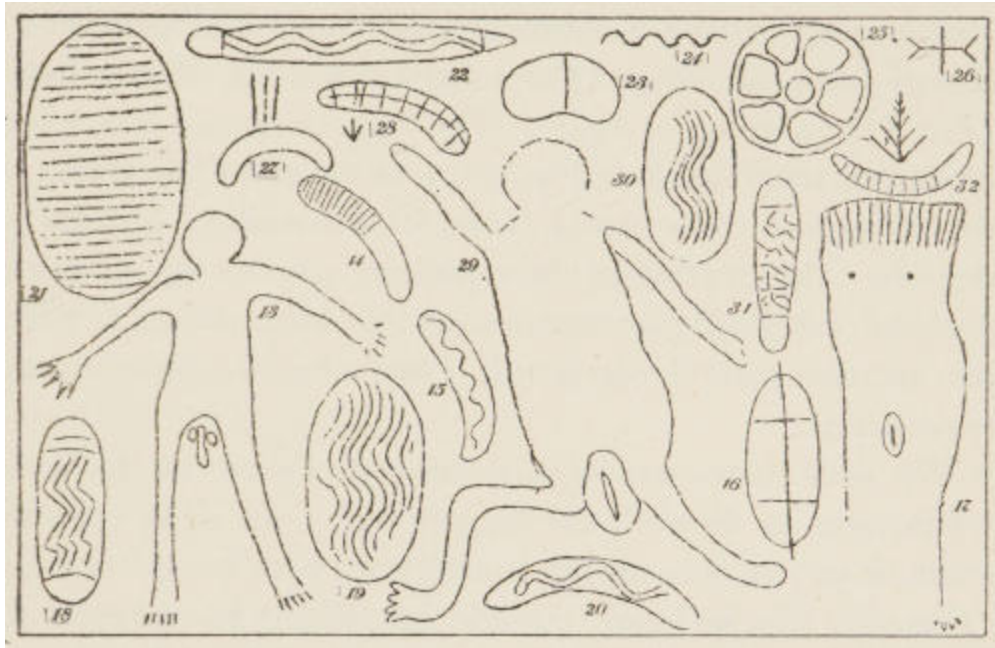


Fig. 11. Rock Carvings at Port Hedland ($\times 1/20$).

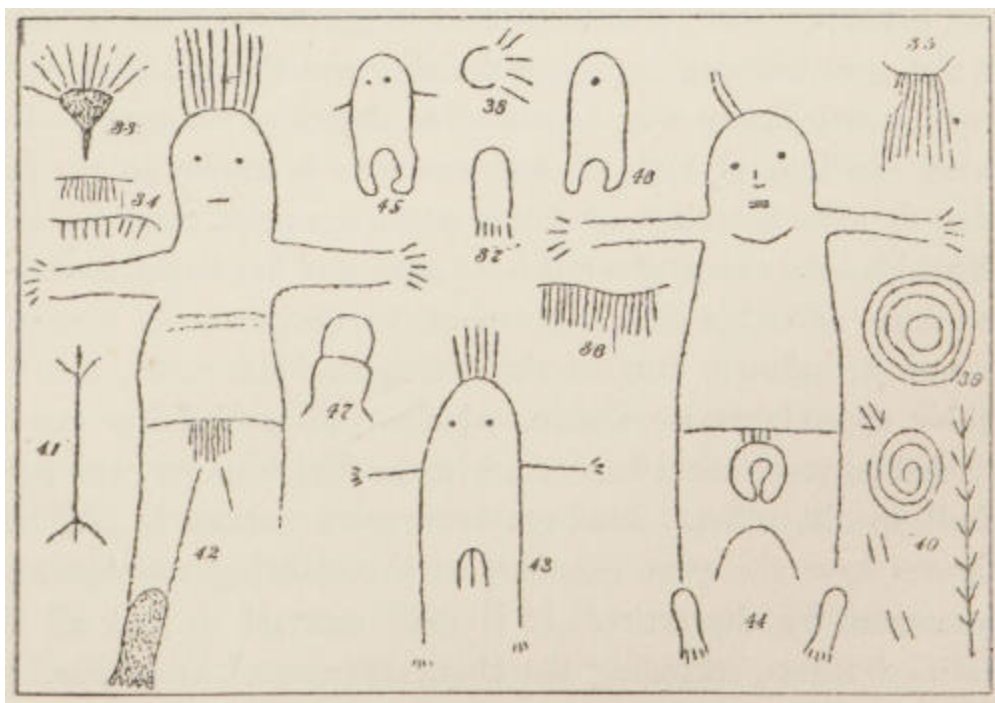
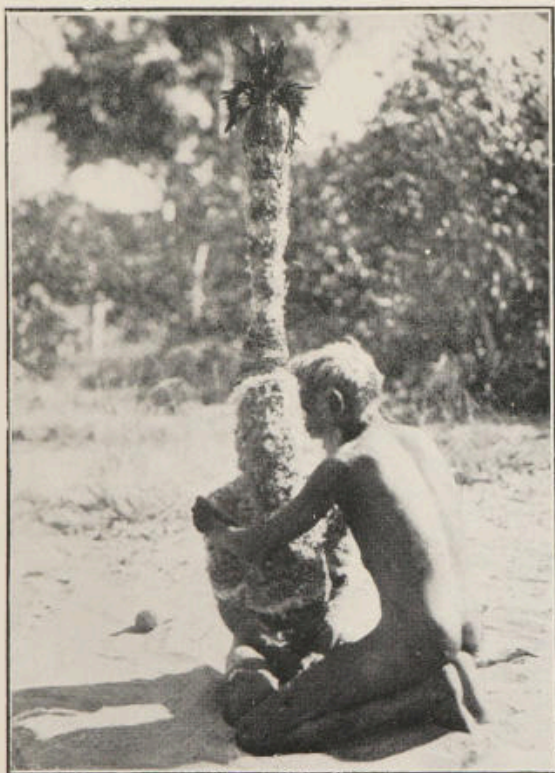


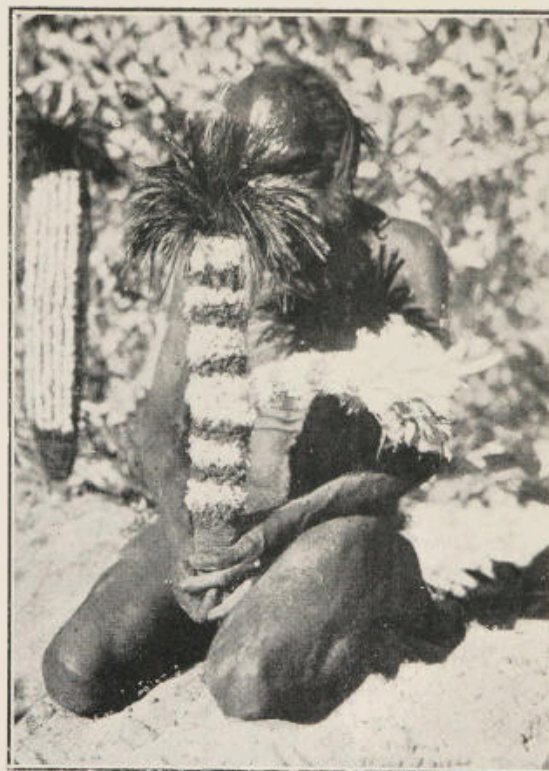
Fig. 12. Rock carvings at Port Hedland ($\times 1/20$).

Among them we observe: Two turtles (*Figs. 1 and 2*), the one of which (*Fig. 1*) being shown in a dorsal aspect, the other (*Fig. 2*) apparently in a ventral; two large fish (*Figs. 3 and 4*) resembling the Port Jackson type; a lizard track (*Fig. 5*), showing the trail of the tail in the centre; a human foot-

print (*Fig. 7*), with an emu track above it; a large stingray (*Fig. 8*), with protruding eyes and long tail standing erect; a shark's liver (*Fig. 9*), at any rate, it was described as such by a native; a stingray's liver (*Fig. 10*); an emu track (*Fig. 11*); a lizard (*Fig. 12*); shields (*Figs. 16, 18, 19, 21, 30*), variously decorated; boomerangs (*Figs. 14, 15, 20, 27, 28, 32*), variously decorated; spear-throwers (*Figs. 22 and 31*); a corroboree plume (*Fig. 33*), such as is worn in the hair or stuck in the armlets during many of the ceremonial dances; pubic tassels (*Figs. 34, 35, 36*), used as a covering, suspended from a hair belt; a "circle-within-circle" design (*Fig. 39*), often figuring conspicuously among emblematic and ceremonial patterns; a human foot-print (*Fig. 37*); a dog track (*Fig. 38*); a lizard (*Fig. 41*); a group of figures (*Fig. 40*), consisting of a spiral, of similar significance as *Fig. 39*, a chain of turkey tracks on the right, and three paired tracks of a bounding wallaby on the left. One or two of the remaining smaller figures are of doubtful meaning, the most remarkable being *Fig. 25*, which for an aboriginal ideograph is extraordinarily complex and symmetrical.



1. Singing to the presiding spirit or Knaninja of the old women or "*Arrekulja Tjuringa*."



2. Ceremonial head-gear ("*Tjilba Purra*") of phallic significance.

1. Singing to the presiding spirit or Knaninja of the old women or “*Arrekutja Tjuringa*.”

2. Ceremonial head-gear (“*Tjilba Purra*”) of phallic significance.

We shall next consider a number of figures in this series which have to do with the human form. In some cases the attempts at representing the human body have been very crude (*Figs. 17, 45, 46*), and it would be difficult to recognize the results if it were not for the association of these figures with others in which the human form is more evident. In *Fig. 13* we behold a man whose attitude suggests that he is either walking or dancing; in all probability the former, because his body is not in any way decorated to suggest a ceremony. But when we look at *Fig. 42*, our attention is drawn to the fact that the man there represented is wearing a pubic tassel hanging from his belt, carrying a tall head-gear, and has tribal markings or scars upon his chest; moreover, in this case the eyes and mouth are shown, but not the nose. In *Figs. 6, 43, and 44*, pubic appendages are shown, which in all probability are intended to represent a hair tassel in the first instance, and pearl shell in the others; head-gears are also indicated. All the figures have the arms extended as if a dancing attitude were attempted by the artist. It is most unusual to find all the facial features, including the chin, represented as in *Fig. 44*. There is no neck shown in any of the last-mentioned figures, although in other cases (*Fig. 13*) it is clearly indicated. It will be observed that in several of the figures (*Figs. 6, 42, 44*), the feet are represented by their tracks. The only remaining figure is that of a dancing gin (*Fig. 29*), the characteristic position of whose arms suggests the dance during which they hold a skein of hair or fibre-string between their hands to jerk it to and fro.

A real illustration of the Port Hedland type of carving is presented in [Plate XL](#), 1, but owing to the carving occurring upon a horizontal plane, the tilting of the camera, which was necessary, has rather seriously distorted the design. A correct copy is, however, included among the designs reproduced on [page 300](#) (*Fig. 10, 6*).

Turning now to the pre-historic carvings of the Flinders Ranges and other localities cited, it would be impossible to discuss the enormous number of designs in anything like detail, without unduly burdening the dimensions of this volume. At certain places, like Deception Creek and Yunta, the carvings are so numerous that it is difficult to find a space entirely devoid of any; the places are deserving of being ranked as primitive art galleries, and every

endeavour should be made to preserve them as such. Considerable care must have been taken in the execution of these carvings, since either the outline or the entire area of every design has been diligently chipped away. And the work of hours must have run into the labour of years, yea, of generations, because at one or two spots like Yunta one design has been carved over the top of another, time after time, until eventually the ground appeared as though it were covered with an elaborate carpet. A few illustrations only will be selected to serve as types of the handiwork to be found in such great profusion.

In the first picture ([Plate XL](#), 2), a design will be observed near the centre consisting of seven parallel vertical lines lying beneath a horizontal band. From what we have learned above, we shall have no difficulty in recognizing the shape of a pubic tassel or apron, so commonly used throughout Australia. Immediately to the left of this is the image of a long lizard clearly and wholly intagliated into the rock; whilst just above the tassel, and a little to the right of it, stands the form of a creature with two legs, a plump body, two large eyes (appearing like circles at the top), and an elevated beak between them—this is the representation of an owl! About midway between owl and lizard is a small mark resembling an inverted broad arrow which stands for the track of a bustard, commonly called wild turkey.

The next illustration ([Plate XLI](#), 1) may, for convenience of discussion, be divided into two halves by the joint-fissure which is seen running vertically down the middle of the picture. In the centre of the left-hand half, the dark, and deeply pitted, shape of a human foot-print, with the toes pointing upwards, will be apparent. Upon the brightly illuminated surface above it, the vertical strands of a pubic tassel are indistinctly visible; whilst on the left of the foot-print a small wallaby track has been cut, in which the long median toes, as well as the lateral, have been truly imitated. Beneath the latter, upon the elevated portion of rock, is a single corroboree circle. But the most interesting figure is near the centre of the right-hand half of the picture; the creature it most resembles is the duck-billed platypus. The “duck-bill” is very conspicuous, whilst the head, the plump body, and the stumpy tail all agree with the appearance of the platypus, even to the claws, which are clearly seen on the right hind leg. It is, of course, possible that the carving was intended to stand for a native “hedge-hog” or *Echidna*, but the

former explanation would seem more true to Nature. The remaining carvings are principally representations of the paired spoors of wallaby and kangaroo.

The following two photographs ([Plate XLI](#), 2, and [Plate XLII](#), 1) present similar subjects, the carvings consisting of an array of corroboree circles, human foot-prints, and tracks of wallaby, kangaroo, and turkey. A new feature is contained in the upper illustration in the shape of the claws of kangaroo or possibly of the human hand; these are seen in the top right-hand quarter.



PLATE XXXIX

A disenchanted area, Victoria River district.

"This area, mostly oval or circular in shape, remains a strict taboo to all beyond its confines, and is believed to disenchant any morbid influence the evil spirit may bring to bear upon it."

It will be admitted that these primitive carvings or petroglyphs of the northern Flinders Ranges have more than a passing resemblance to the

ancient graffiti of Egypt. It will be noticed, too, that the last illustration includes a slab of rock upon which some of the designs only occur in part. This is because the slab has fallen from its original position, since the carvings were made upon it; and so, at several points, portion of a design can be seen upon a rock lying in the valley, whilst the piece belonging to it might be detected, still *in situ*, up in a cliff. This in itself seems to suggest that the work of ages has been going on since the ancient artists put their talents to a test.

It is astounding to what height above the level of the ground the natives must have climbed to decorate the rock faces with their carvings; in many instances atmospheric denudation has so altered the shape of the cliffs that it would be impossible at the present day to reach some of the designs, to say nothing of finding a footing to undertake the carving.

But more, several cases were observed where a design was bisected by a gaping fissure; all the evidence was in favour of the separated portions of the design having originally been contiguous, but subsequent earth movement had forced them asunder. In addition to the actual cleft existing between the two portions of a design, cases were noted where a slight faulting had occurred along the fissure, through which one side of the disrupted design stood at a measurably higher level than the other.

The strongest geological evidence in support of the great antiquity of these carvings, however, is in the presence of a dark rust-coloured *patina* or glossy surface film which everywhere covers the exposures of the rock and carvings as well. These protective films are characteristic of all desert and arid regions. In Egypt, it has been ascertained that the presence of the patina is a ready means of distinguishing primitive carvings from those made within historic reckoning. Professor G. Schweinfurth, the famous Egyptologist, has pointed out that whereas the prehistoric carvings are covered by the patina, the incised hieroglyphics of even the fifth and sixth Dynasty at El Kab are as fresh-looking as though they had been carved yesterday. In some of the Australian examples there is no appreciable difference between the thickness of the film on the rock and that covering the design; in others the designs look quite fresh, or, if they show anything at all, it is but a rudimentary glaze. The conclusion is that some of the Australian designs must, upon this evidence alone, be regarded as extremely old.

But there is further evidence. If we could definitely claim the platypus design as authentic, it would mean that the ancient artist was familiar with the form of an animal which at the present time is quite unknown in central Australia. But geology tells us that in times gone by, in the Pliocene period, perhaps even later, all the great lake systems of the Australian interior were not salt, as they are now, but fresh water. Under those conditions, it is quite feasible that the duck-billed monotreme might easily have lived in that region; if so, it might have supplied prehistoric man in Australia with a model he perpetuated in the rocks.

The platypus design is by no means unique. Among the carvings at Yunta there are several depicting the spoor of a very large animal, which are rounded at one end like a heel and have four or five serrations resembling toe-marks at the other. The picture is not unlike that of a wombat track, but the dimensions are far too great. The Yunta “tracks” measure nearly ten inches in length and are practically the same in width. There is no animal living in Australia at the present time whose track would be anything like as large; the nearest known animal which might answer the form of the carving would perhaps be that of a hippopotamus. This animal is, of course, not indigenous to Australia, but we know that an extinct animal, probably not unlike a hippopotamus, used to roam the fertile plains of central Australia in Pliocene times; that was the *Diprotodon*. In the accompanying sketch I have placed a tracing of the Yunta carving beside one of the reconstructed *manus* of the *Diprotodon*, and one must admit that there is a plausible agreement between the two. Vide [Fig. 13](#).

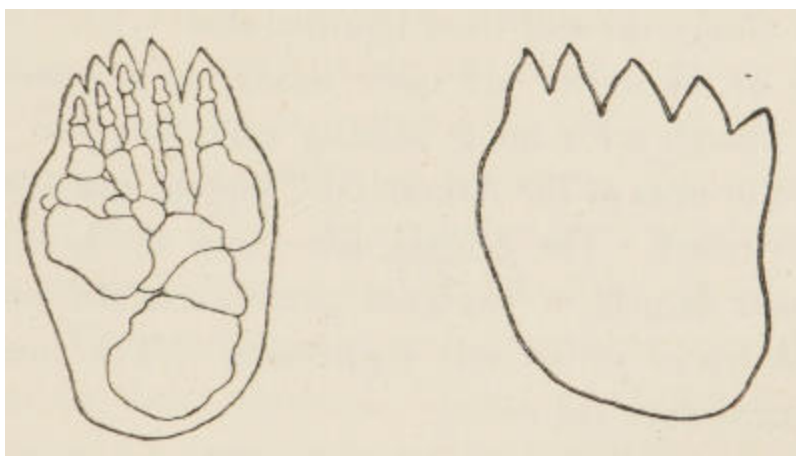


Fig. 13. Sketch of reconstructed manus of *Diprotodon* compared with tracing of carving of supposed *Diprotodon* track at Yunta ($\times 1/8$).

An aboriginal never exaggerates the dimensions of a track when drawing from Nature. Upon this point, indeed, he is most exacting, because upon an accurate knowledge of and familiarity with such things his very livelihood depends, and the acquisition of accuracy represents part of the recognized standard of his educational system. He would never, for instance, draw the track of a wallaby larger than it is in reality, because confusion would immediately arise as to whether it might not be that of a kangaroo or euro; and if he drew it smaller than it actually is, doubt would be raised as to whether it might not represent the track of a kangaroo-rat, or even of a marsupial mouse. So, too, when he wishes to draw a turkey track, he has to be careful, because if he makes it bigger than it should be, it might be mistaken for that of an emu; and if he makes it smaller, people who see the track might take it to be that of a curlew, or even of a plover.

For the above reason alone, a native would never draw the track of a wombat on the elaborate scale of the Yunta carving. Furthermore, there are some exceptionally large bird tracks carved into the rocks at Balparana, in the Flinders Ranges, which seem too big to be intended for those of an emu; the question might reasonably be asked whether they could not have been made by a primitive hunter at a time when the now extinct “moa” or *Genyornis* still lived in Australia.

A type of decorative art quite similar to the ancient rock carvings, though on a much smaller scale, is to be found on the stone-tjuringas of the Arunndta, Tjingali, and other central Australian tribes. The designs, like those on the wooden objects, consist largely of engraved circles, straight and sinuous lines, and tracks of totemic significance. The intaglios are usually tinted with red ochre.

Leaving the discussion of carved rocks and stones for the time being, and directing our attention to the subject of carved trees, we find that this interesting cult flourished principally in the tribal territories which are now included in the State of New South Wales and the extreme southern portion of Queensland, the haunts of tribes now practically extinct.

The design or decoration was either carved straight into the bark, or the latter was previously removed and the cutting done in the sapwood or heartwood. Great variety was displayed in the choice of designs. Some were crude and unshapely, others neatly and cleanly cut and of intricate pattern;

among them were perfect geometrical designs, consisting of groups of circles and quadrilateral figures, usually concentric and often combined with parallel, wavy, zig-zag, or spiral lines; yet another class of design would consist of some animal or human form, more or less grotesquely modified.

The object of this elaborate tree-carving was of a twofold nature. Firstly a certain number of trees would be so treated, within the immediate surroundings of the grave of a notable tribesman, to permanently mark the place of sepulchre; and, secondly, the butt of a tree commanding the ground of an initiation ceremony might have been so distinguished, if the importance of the occasion warranted it.

Along the north-west coast of Australia, where the boabab tree flourishes, the tribes often carve animalistic and other designs into the bark, which, on account of its softness, lends itself admirably for the purpose. The carvings are usually to be found near a camp or at the site of an ordinary corroboree ground. The designs, once they have been cut into the bark, remain there during the life of the tree. The accompanying illustration ([Plate XLII](#), 2) depicts an emu, which stands three feet high, carved into a boabab off King Sound, Western Australia. The whole of the bark within the area occupied by the design has been removed. The feet, it will be observed, are portrayed in a perfectly free way in order to leave no doubt in the mind of an observer as to the track the bird actually makes. Two other figures stood beside the emu, cut in a similar way into the bark of the same tree; one was a snake, a shade over five feet in length, the other an emu track.

In place of the carved trees, the Melville and Bathurst Islanders erect carved and painted posts around the graves of their people—men, women, and children alike. These pillars, which are of hard and heavy wood, are from four to six feet high, circular in section, and have the top carved into a “head” of one or other of the patterns drawn in the accompanying figure. The top of the pillar is either flat, rounded, or pointed, the “head” being formed by cutting one or more circumferential grooves of chosen width, and at chosen distances, below the top. At times an oblong hole is cut transversely through the post five or six inches from the top, leaving only two narrow strips of timber, one on either side, to support the “head” thus formed. Vide [Fig. 14](#).

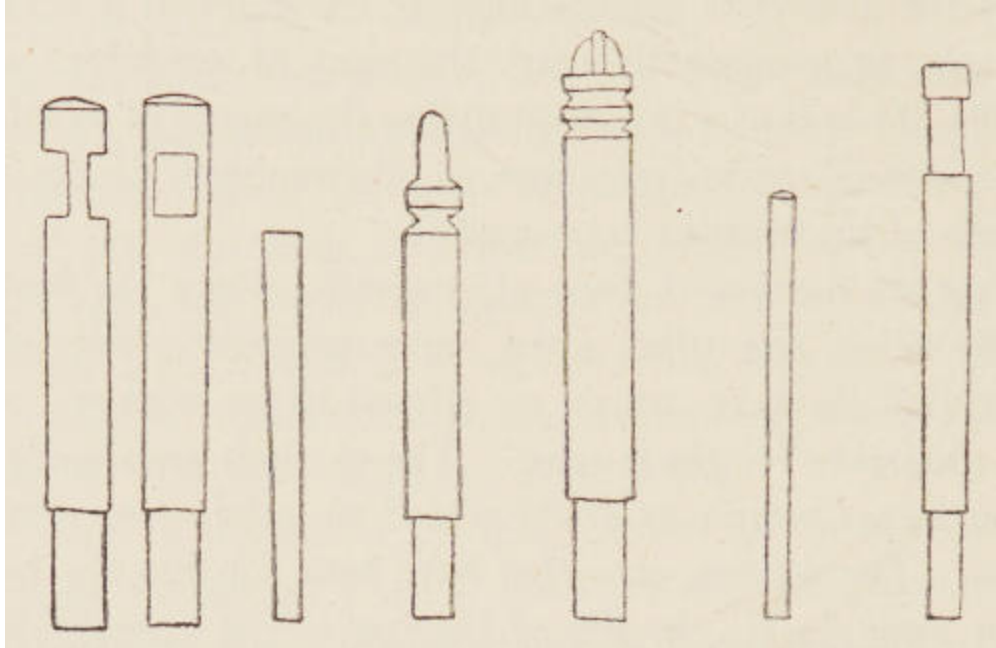


Fig. 14. Carved grave posts of Melville and Bathurst Islanders ($\times 1/40$).

Designs and patterns quite similar to, but on a smaller scale than those cut on trees, are found carved upon weapons of some of the tribes. Shields and spear-throwers are those most commonly found decorated with incised patterns. Some of the hardwood shields of the River Murray tribes are richly incised with parallel, zig-zag, and geniculate lines, and with squares standing point to point in a longitudinal line, all the spaces between the squares being filled in with parallel "elbows." The Worora tribe at Port George IV, on the other hand, decorate their shields with fantastic representations of snakes, emus, and tracks of various animals. The light-wood shields of central Australia are destitute of any ornamentation except wide and shallow longitudinal grooves, which are also characteristic of the bark food-carriers in use all over the continent of Australia.

So far as spear-throwers are concerned, the handsomely carved specimens produced by the natives of the Warburton and Gascoyne Rivers, and of the King Leopold Ranges in Western Australia, deserve special mention. The favourite pattern in that region appears to be longitudinal geniculate bands, alternately incised lengthwise and crosswise, together giving the effect of a false herring-bone motive. A new element is introduced in wood-carving in the ceremonial spear-throwers of central Australian tribes by the inclusion of the concentric circles pattern already referred to; the Arunndta in particular produce some very showy specimens

on gala occasions. The decoration is very finely graved upon the inner flat surface. The old Victorian types occasionally had pictures of animals, birds, and men carved upon them.

Boomerangs are often decorated with incised patterns, but more frequently the decoration is only lightly graved into the wood with the point of a stone-knife or with the sharp cutting edge of a shell, tooth, or bone. From an art point of view, the finest productions come from the north of Western Australia. The King Leopold Ranges natives cover one whole side of their boomerangs with an incised pattern, consisting usually of parallel geniculate lines, false herring-bone, or concentric rhomboids posed along a median line. Some of the most attractive specimens, however, come from the Pidunga tribe at Broome. These natives covered both surfaces of the missile with a wonderful variety of designs, which included excellent representations of emu, kangaroo, snakes, crocodiles, turtle, tracks of every description, dancing men, corroboree circles, and many decorative designs. Other articles, such as adze-handles, tjuringas, and message sticks, are carved after much the same fashion.



Fig. 15. Ochre drawing, Glenelg River, Western Australia ($\times 1/8$).

There is yet another class of incised decorative art to record which is found in the far north of Western Australia. The King Sound and other natives of the northern Kimberley district have developed a cult quite peculiar to themselves, in that they carve ornate designs upon the brown surface of the large nuts of the boabab. The method they have adopted is to hold the nut firmly in the left hand and work the designs into the dark, outer layer of the shell with the sharp point of a bone, or, as is the case nowadays, with the point of a piece of iron wire or of a pocket knife. The instrument is held in the right hand, with the four fingers against the palm, while the thumb is laid straight along it on top. The nut is steadied against the body whilst the point of the instrument is applied from the distant side. By applying semi-rotary movements with the hand, the point is made to plough forwards, and by so doing the thin, brown surface-skin is broken and falls away, leaving a white, and slightly jagged, line upon a dark background. Many are the designs which cover the surface of a boabab nut; and it must be admitted the artists exercise considerable judgment in the grouping of the subjects displayed. Among the more important figures are included animals, birds, reptiles, fish, and human beings, besides many of a more complicated and less apparent nature. Vide [Plate XLIII](#), 1.

The north-western tribes, from Broome to Wyndham, and to a lesser extent those of the Northern Territory, artistically decorate the pearl-shell coverings they wear suspended from the belt by cutting designs into the smooth surface of the inner shell of the oyster. By rubbing powdered red ochre into the portions thus roughened, the carvings stand out in bold contrast against the nacreous background. The designs are largely conventional and often embody the human form; a few tracks of animals or of birds are also occasionally added.

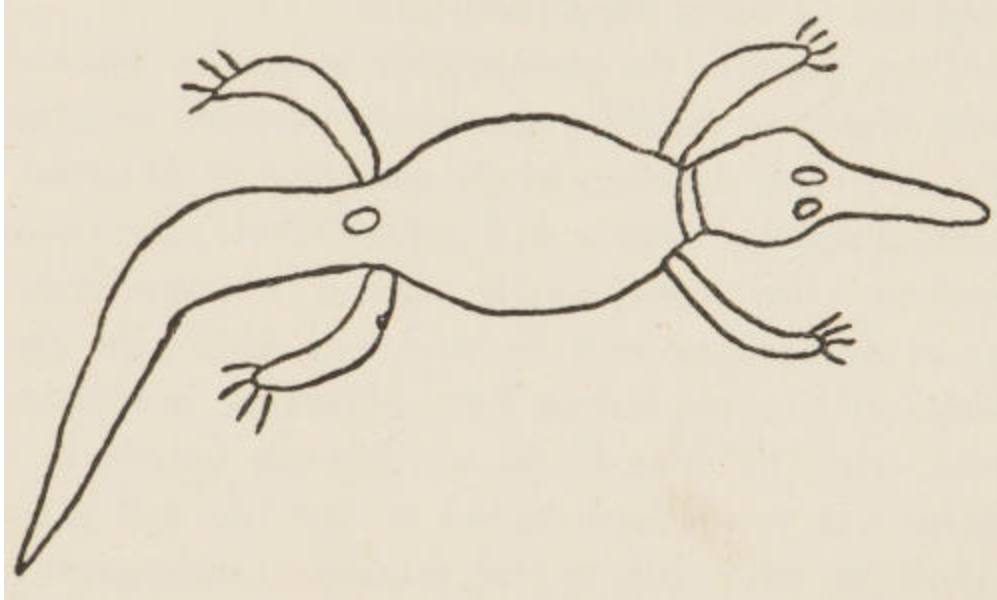


Fig. 16. Carved crocodile design on boabab nut, Derby district, Western Australia ($\times 3/5$). Tracing.

Unlike his racial relative, who used to live in Europe during the Stone Age, the Australian aboriginal does not pay much attention to the carving of bone. The little he does, in fact, is more utilitarian than artistic. We have had occasion to note that the old Murray River tribes used to make the points of their spear-throwers of bone. In central Australia a wing bone of the pelican is cut at both ends and worn through the septum of the nose; occasionally one end is plugged with triodia resin whilst the hollow in the opposite end carries a plume. Not infrequently the slender ulna of a kangaroo serves a similar purpose; the shaft is cut about six inches from one end and sharpened by scraping it with a stone fragment; the condyles are left intact to represent the head of the pin. Two types of bone fish-hooks have already been referred to.

The central tribes make a useful gouge out of a strong hollow bone of the kangaroo or dingo by splitting it longitudinally and grinding down its ends on the slope. The implement thus fashioned has a sharp, bevelled, semi-cylindrical cutting edge at either one or both ends.

We now come to the consideration of another big and important branch of primitive art, which comprises the charcoal, kaolin, and ochre drawings of the aboriginal of Australia. As did his palæolithic relative in the Old World, the aboriginal during the rainy season spends much of his time under the cover of overhanging rock shelters, well within the cheerful

influence of his never-failing fire. Moreover, in mid-summer months, when the heat of the sun becomes intense, he often finds his way to the same haunts to have the full benefit of the shade the solid walls of rock produce. Congregated under these conditions, there are always some who spend their time in decorating the surrounding walls of the cave. This may be done for purely æsthetic reasons, or as the result of a discussion, or, indeed, to sanctify the abode and so to make it impregnable to the Evil Spirit. Whatever the reason may be, the mere act of drawing a figure upon the wall by a recognized artist always solicits the patronage of many, who will follow the different manipulations of the entertainer with considerable interest. There are usually a few men in every tribe who have established a reputation as artists; and their work is prized by the heads and protected by tribal law from the hands of vandals who would at a frivolous moment deface or disfigure a work of art which the tribe is proud to look upon as their own. It is gratifying to observe that there is very little tendency on the part of the aboriginal, humble as he is, to destroy wantonly or deliberately a work designed to create an environment for him during his leisure or to protect his body and kin against aggression by evil during the darkness of night.

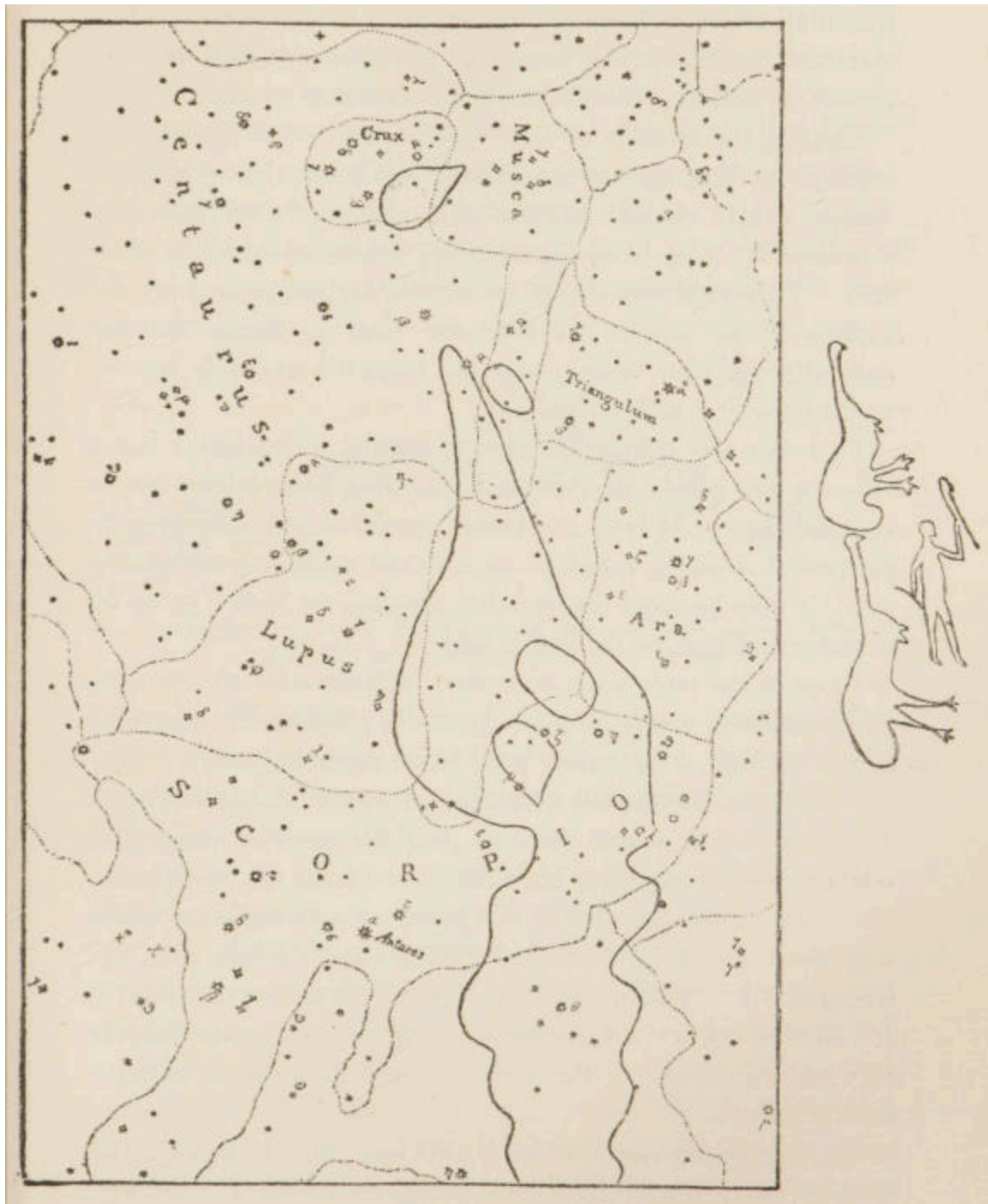


Fig. 17. "Dangorra," the great emu in the southern sky.

Usually the interior of rock shelters and caves has, in consequence of long ages of weathering and mineral precipitation, become deeply stained and dark in colour. In addition, the continued burning of fires within has helped to smut the stony roof. This condition makes an admirable background for the application of colour. Where the wall is black, charcoal

naturally finds no favour, and the pigments available are reduced to white, yellow, and red.

Pipe-clay and ochres are always stocked in quantity; not a tribe in the whole of Australia has ever been known to be without them. Where the tribal ground is not in possession of natural deposits, supplies are obtained from a neighbouring tribe, it may be from considerable distance, by barter or by an actual expedition to the ochre mine.

In quite the same sense as modern peoples refer to red ochre as blood-stone, the natives of Australia connect the formation of the natural, red pigment with blood mythologically.

There is a fine deposit of red ochre in the Flinders Ranges, near Parachilna, which for ages past has supplied more than one tribe with pigment; the mine was known as Yarrakinna. The ochre was regarded as the blood of a sacred emu which was there killed by a horde of wild dogs. From time immemorial the "Salt-water Tribe" used to send a number of its men across from Queensland to obtain a large supply of the precious stone at the spot and return with it to their native ground.

The expedition would be under the leadership of an old man, and his party would consist of young men who had recently been initiated. The journey was a long and arduous undertaking, and young fellows were selected in order that their strength and powers of endurance might be put to the test. When they reached to within a certain stage of the mine, the old man ordered everybody to discard any belongings he might be carrying, and, upon a given signal, the party, led by the old man, began to run towards a big hill which stood before them. Many a hurdle was in their way, but it was imperative that they kept running. Then they took the sloping ground, and presently a large boulder lay in front of them; this the young men were informed was the petrified dog which had killed the emu. Each in his turn was asked to throw a stone at it as he skipped by. Next, they came upon a group of stones which they were told represented the cursed remains of the female dog and pups. As the names of these were mentioned, each of the men again threw stones.



Fig. 18. Boomerang with a number of emu designs carved upon it, Pidunga tribe, Broome ($\times 1/7$.)
Tracing.

Suddenly the party was ordered to halt. Just ahead of them was the platform of rock upon which the sacred emu had expired. The men were requested to remain where they stood and to keep silent whilst the old fellow made for a chasm below the platform. He returned not long after, with his hands full of rich red ochre, which he rubbed over the bodies of his young attendants. Thereupon all present cut off their beards, which they had been wearing long purposely for the occasion, and walked in a body to the chasm. "The great Emu wants feathers," they exclaimed, "we offer her the token of our manhood"; and, as they spoke, they threw the beards into the chasm below. Then each man was allowed to fill his bag with red ochre, and, placing it upon his head, he ran down the hill to the place at which he left his belongings. Nobody was allowed to look back; and should, by accident, a bag of ochre be dropped on the way, it had to be left just where it fell, and under no circumstances picked up again.

According to the Kukata, there was once an old man who had several wild dogs, which were ferocious in habit, generally, but obeyed their master. One day, when he was out hunting, he saw the track of a kangaroo which he made up his mind to follow. He had his little daughter with him, and, not thinking it advisable to take her with him into the scrub, he decided to leave her to play on a clay-pan while he followed his prey into the hills. He captured the kangaroo and returned to the clay-pan, but imagine his disgust when he found that his dogs had, during his absence, killed his child and devoured her flesh.

In his wrath, the old man chased the dogs into the hills at the point of his spears, until eventually he drove them into a cave, the entrance of which he closed with a number of large stones. The wounded dogs in their plight attacked one another and tore themselves to pieces. In consequence, their

blood poured freely into the cave and soaked deeply into the rocks. Ever since, the tribe have gone to that cave to collect supplies of the “blood-stained” rock—the red ochre—which they require for their ceremonies and corroborees.

Ochre and pipe-clay, which form an article for inter-tribal barter, are carried from one tribe to another in oblong parcels contained in bark wraps, which are folded at the ends and kept together with fibre-string. Small quantities are always carried by the men in their chignons and dilly bags, and, when larger supplies are required for special occasions, they are consigned to the care of the women.

When ochre is required for decorative purposes, it is necessary to reduce it to a fine powder. This is done by placing a measured amount upon a level surface of rock and grinding it with the aid of a medium-sized pebble. Hand-mills of this description are to be found at any sites which natives have been in the habit of embellishing with their drawings; they may be recognized as small, shallow depressions scooped into the surface by the continued abrasion of the hand-piece. Where there is no flat, natural surface available, the grinding is done upon portable slabs the women carry around from one camp to another.

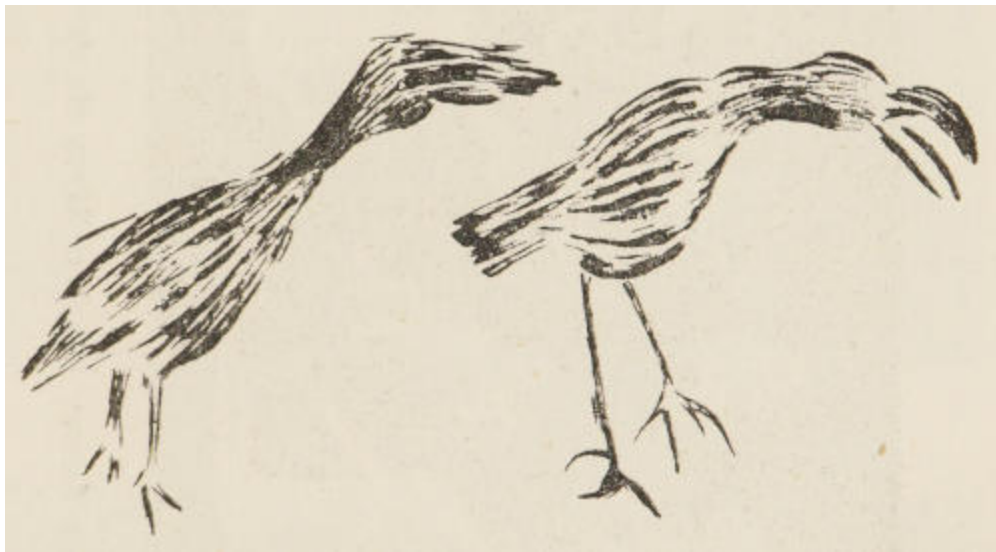


Fig. 19. Charcoal sketch of crows, Pigeon Hole, Victoria River ($\times 1/3$). Tracing.

The ochre (or pipe-clay), having been finely ground, it is collected upon a piece of bark, or in a bark food-carrier, and mixed with sufficient water to make a thick paste; and it is ready for application. The Bathurst Islanders

use the large, concave shells of *Cyrena* in much the same way as European artists formerly used the valves of fresh-water mussels (*Unio pictorum*) for mixing their pigments in.



Fig. 20. Pipe-clay cave-drawings of dancing figures, Humbert River, Northern Territory ($\times 1/12$).

The native spreads the paint with his fingers, where larger surfaces are concerned, and with a short stick where finer lines or details are to be added. The Bathurst Islanders cut short pieces off the green shoots of the lawyer-cane (*Calamus*) and chew one or both ends of the sticks until all the fibres have been separated; these then fulfil the same purpose as the paint brushes of a modern artist.

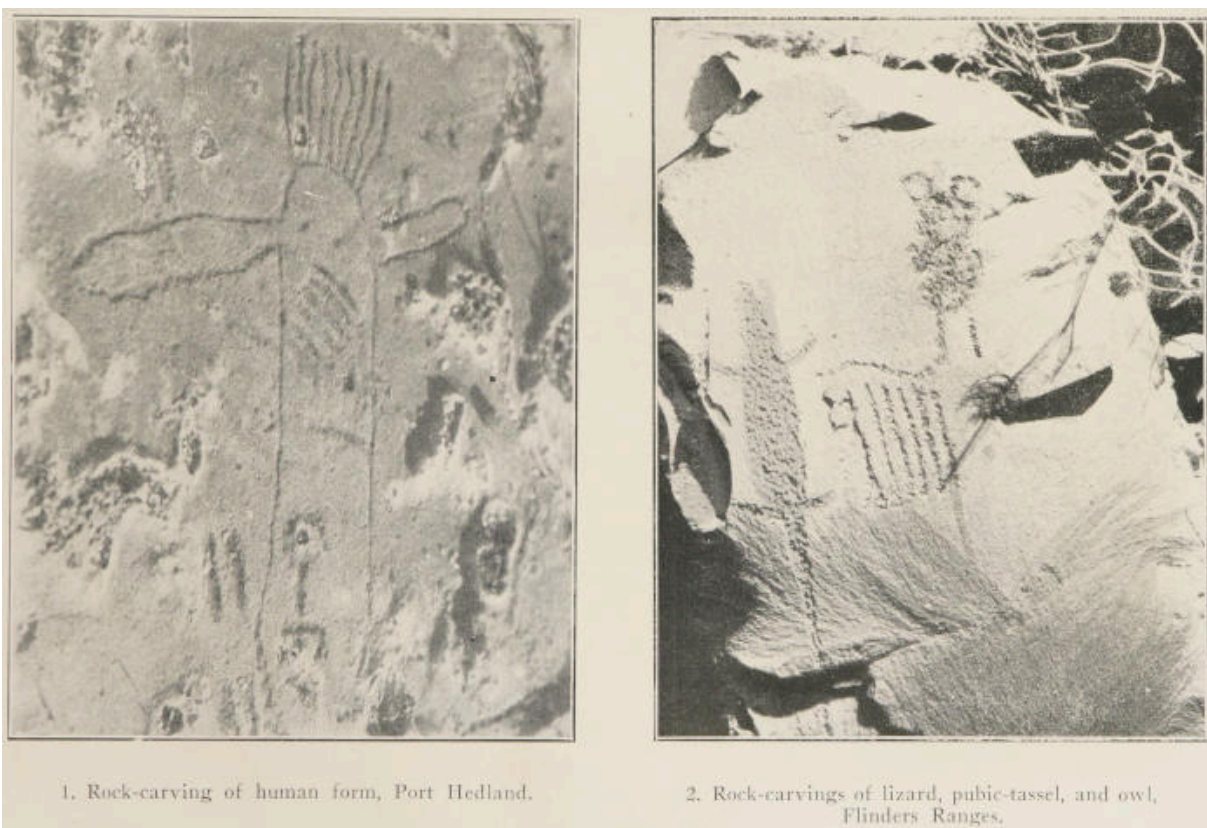


PLATE XL

1. Rock-carving of human form, Port Hedland.

2. Rock-carvings of lizard, pubic-tassel, and owl, Flinders Ranges.

A favourite practice, and one which is met with all over the continent, is to obtain a “negative” shape of a person’s hand. This is done in the following way: The person puts a small handful of ochre or pipe-clay into his mouth and crunches it to a pulp; then he fills his mouth with water and thoroughly mixes the contents. He holds the hand he wishes to stencil against a flat surface, spacing the fingers at equal distances, and spurts the contents of his mouth all about it. A short while after, the hand is withdrawn. The area which it covered remains in its natural condition, whilst the space surrounding it has adopted the colour of the ochre or clay. Very often the “hand” is subsequently painted over with a colour different from that of the surrounding area. The Arunndta refer to the hand-marks as “*ilja imbadja*.” Vide [Plate XLIV](#).



Fig. 21. Charcoal drawing of hopping kangaroos, Pigeon Hole, Victoria River ($\times 1/6$). Tracing.

A native attaches considerable importance to his identity being thus recorded and preserved in some of the caves, believing the brand to stand for his individuality with as much certitude as, say, the European who leaves his card or carves his name in stone or wood. It is compulsory for members of a certain rank in the Worora tribe to have their “hand-shadows” perpetuated upon the walls of caves in which the bones of their ancestors are reposed, because the spirits of the dead are thus supposed to be apprised of any visits which have been made to their last earthly resting places.

It is beyond dispute that the natives possess the faculty of being able to recognize the hand-marks of their relatives and tribesmen, even though they may not have been present when they were made.

Less frequently, the negative imprint process is applied to a person’s feet or any of his private belongings, such as stone tomahawks. At times, too, the hand is smeared over with ochre and then smacked against a surface to obtain a positive. Vide [Plate XLIV](#), 2.

The number of ochre drawings one finds on rock surfaces is naturally unlimited, especially when extending observations over the whole of Australia; and the variety in regard to subject, colour, composition, and execution is correspondingly large.

Without at this stage attempting to delve deeply into the consideration of individual designs and their artistic merits, let us cast our eyes for a moment upon an illustration, which might be considered typical, to familiarize ourselves with the general appearance of Australian cave drawings ([Plate XLV](#), 1). We have before us a rock shelter or *abris* on the Forrest River in the north of Western Australia, a more or less vertical wall at the base, overhung at the top by a solid ledge or “roof” of quartzite. For the better

part of the day this spot is protected from the intense heat of a tropical sun; and during the “wet season,” also, it provides a shelter from the prolific rains. But a casual glance at the picture suffices to convince one of the presence of a number of conspicuous designs drawn in ochre upon the rock faces. One in particular, that of a huge snake, immediately catches the eye, but if a careful scrutiny of the lower portion of the escarpment is made, many other smaller designs will be discovered.

When the rocky background is light-coloured, and not sooty, the artists often include charcoal drawings among the coloured ochre designs. The native applies the charcoal either in the form of a powder with the aid of his fingers, or he selects a piece of charred wood, big enough to hold between his fingers, and this he uses after the fashion a modern artist does a crayon. According to the method employed, the result is either a figure blackened all over, or a line drawing, shaded here and there. Some particularly creditable charcoal drawings were obtained near Pigeon Hole on the Victoria River in the Northern Territory; these are described later.



Fig. 22. Bark-drawing depicting an eagle-hawk clawing and tearing the carcass of a wallaby, Port Darwin.

An aboriginal not only paints the sides of the caves he temporarily occupies, but he also delights in decorating the sheets of bark which in certain districts, such as the north coast of Australia and Melville and Bathurst Islands, are used for making his huts waterproof. The method he adopts in applying ochre to bark is precisely similar to that already considered in connection with his cave drawings. And we might at the same time extend these remarks to the decorative designs which appear in such profusion upon his spears, shields, boomerangs, spear-throwers, waddies, clubs, food and water carriers, dilly bags, ceremonial objects, personal ornaments, and, in fact, anything he has occasion to manufacture and handle.

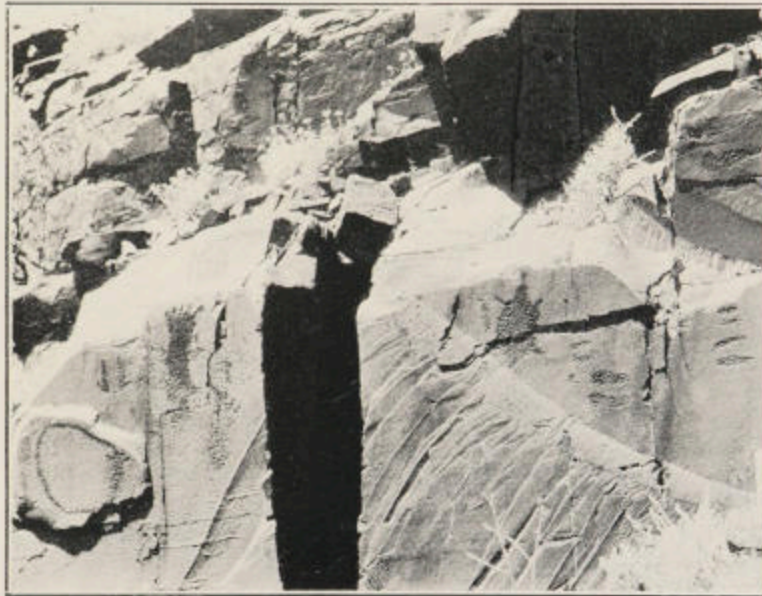
A special variety of ochre drawing which may justify a few remarks is the tribal body decoration. We know that as a simple, but effective, means of protecting his skin against the weather, an aboriginal periodically anoints his body with emu fat; moreover, to evade detection by the game he is stalking, he often covers his body with ochre, earth, or clay to simulate the colour of his surroundings as nearly as possible. But for reasons, to him entirely cosmetic, he finds occasion to rub red ochre powder (and charcoal also) over his face and body.



Fig. 23. Pipe-clay drawing of man and dogs, Humbert River ($\times 1/12$).

He has a distinct liking for the beautiful and does not hesitate to avail himself of anything which might tend to make his person more attractive looking by the application of colour. Not only the sire, but the whole family endeavour to improve their swarthy appearance by painting ornate designs over different regions of their bodies. Longitudinal, parallel bands of red, yellow, or black, extending up the legs, back, and abdomen, together with transverse lines on the chest, shoulders, upper arms, and outer surfaces of thighs, are symmetrically drawn, and connected here and there (as, for

instance, on the chest or back) by lattice patterns and concentric circles. Parents are very proud of their children thus decorated. On the Forrest River, a favourite mode is to draw a broad step-ladder-like pattern from the ankles up the front surfaces of the legs, continuing this up the trunk to about the level of the nipples, and then circling outwards, down an arm on either side, to run out at the elbow. This design is usually painted in yellow.



1. Rock carvings (including platypus design), Flinders Ranges.



2. Rock carvings, Flinders Ranges.

PLATE XLI

1. Rock carvings (including platypus design), Flinders Ranges.

2. Rock carvings, Flinders Ranges.

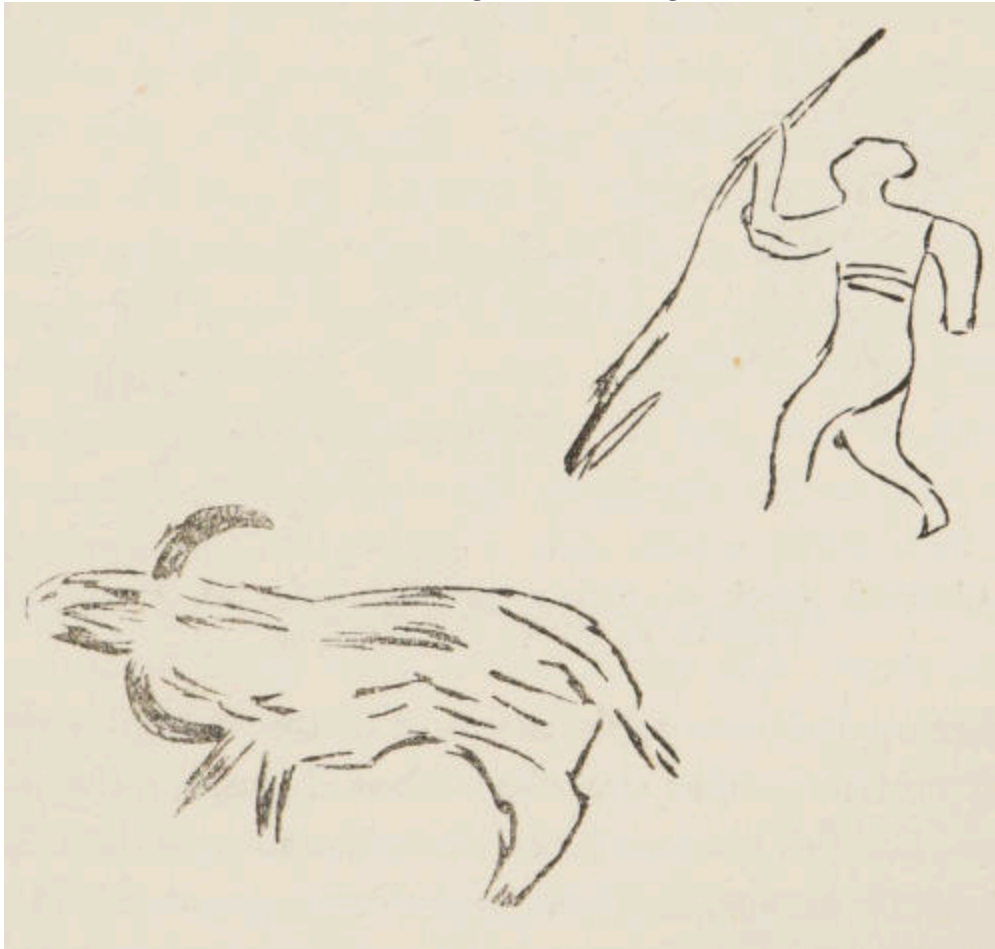


Fig. 24. Charcoal sketch of native hunting buffalo, Pigeon Hole, Victoria River ($\times 1/3$). Tracing.

We have frequently referred to the fact that during corroborees and ceremonial dances, the bodies of the performers are decorated. White pipe-clay is one of the principal pigments used, although red ochre is also much in evidence. In the performance of ceremonies, we learned that a common motive embodied in the decorations was the human skeleton. The quickest, and perhaps most effective, way of whitening the face in the representation of the skull is for the performer to literally “wash” it in finely crushed pipe-clay. To accomplish this, he scoops a quantity of the material with his hands, and, closing his eyes, rubs the stuff all over his face and possibly his head, too. By this process even the eyelids are thoroughly whitened. The

other lines, horizontal and vertical, which are to represent the bones, he rubs on to the body with his fingers ([Plate XLVI](#)).



Fig. 25. Charcoal sketch of native spearing kangaroo, Pigeon Hole, Victoria River ($\times 2/5$). Tracing.

Another method is to apply the paint in the form of a water mixture, similar to that described when discussing the ochre drawings. For this purpose, especially when an important event is pending, a number of men are chosen to attend to the “make up” of the performers. The assistants kneel beside those who are to act, and apply the paste with their fingers. The most delicate parts to handle are the eyelids. The actor is required to close his eyes whilst the artist carefully applies the paste to the lids; but it occasionally happens that some of the material slips on to the eyeball and is rubbed against it before the sufferer can give the alarm. Vide [Plate XLV](#), 2.

We have already referred to the coloured down decorations which are attached with human blood to the bodies of the performers taking part in sacred and other ceremonies, and we have also mentioned a ground drawing known as “*Etominja*” ([Plate XXXVII](#)), which is constructed in a similar way. Some of the latter (e.g. the “walk-about” of the “*Tjilba Purra Altjerra Knaninja*”) are very large; others, as for instance that connected with the “*Erriakutta*” or yelka ceremony, are constructed over the entire surface of mounds which cover many square feet of ground.

Having briefly reviewed the different methods of art production in vogue in Australia, we shall proceed to consider a number of the designs in greater detail, deduce their origin, trace their evolution, and, where possible, give

their interpretation. It will be realized at the outset that some of the designs are crude in the extreme, whilst others are undeniably shapely and quite up to the standard of an average European's artistic proficiency. The latter remarks apply best to actual representations of natural forms. It must be remembered that the artistic reproductions an aboriginal makes are invariably from memory; the primitive artist never draws with a model in front of him. If we were to ask a number of Europeans to draw, say a horse from memory, there is no doubt we should receive a great variety of results in response to our request. So, among the aboriginal artists, there is a great diversity of talent which is more individual than tribal.

If, for instance, we study the different attempts at representing the form of one of the most familiar subjects we could ask an aboriginal to experiment upon—the ubiquitous kangaroo—we should find by comparison of the productions placed before us, a very marked difference in quality. Compare, for instance, the two pictures of kangaroo on [Plate XLVII](#). They are the works of men of the same tribe, are all similarly drawn, and come from the same locality. Yet, in the upper picture, the outline and proportions of the two animals are so incorrect that it is very doubtful whether many people not acquainted with the locality would guess what animal the pictures are intended to represent. In the lower picture, however, anybody acquainted with the shape of a kangaroo would have no hesitation in pronouncing his diagnosis. The characteristic attitude, the large tail, the disproportion between the front and hind limbs, and the shape of the head are quite true enough to nature to permit of correct identification.

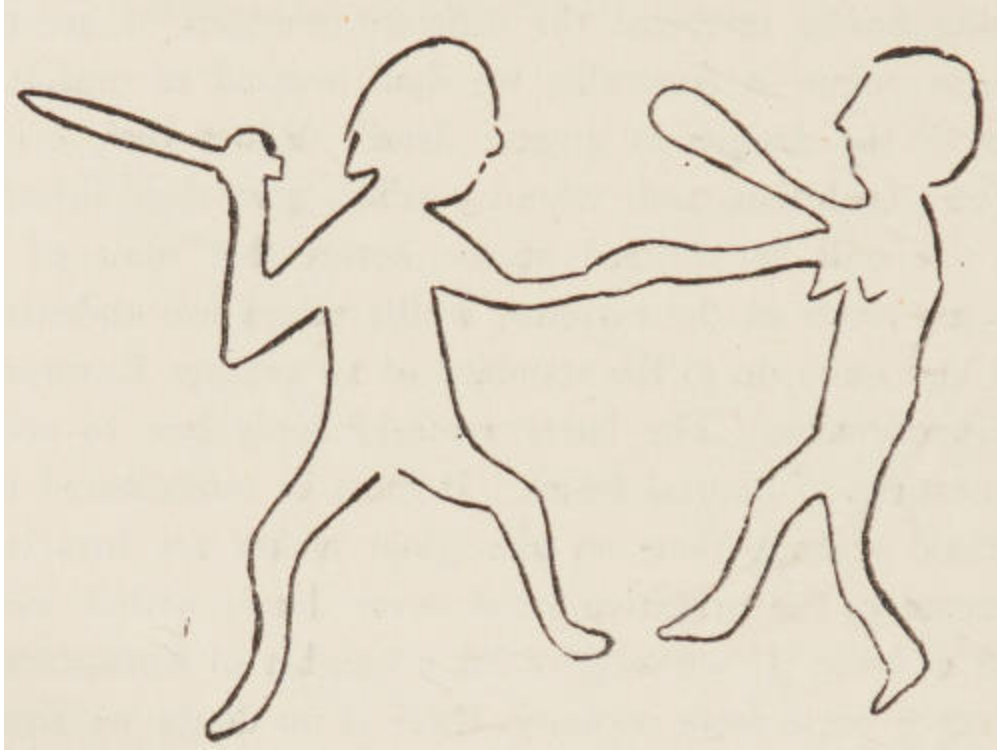


Fig. 26. Carving depicting a quarrel between a man and his gin. Arunndta tribe ($\times 1/2$). Tracing.

The three designs are all drawn in charcoal, the figures in the first two cases being outlined with a white pipe-clay line, and in the second case with one of yellow ochre. If we wish to go one better still, we need only study the pipe-clay drawing on bark by a native of the Katherine River district shown on [Plate XLIX](#), 1—a very creditable picture of a dead kangaroo.

Some of the designs one meets with are so accurately drawn that a scientific determination of the species becomes possible. Look for a moment at the fish, portrayed in pipe-clay, shown in [Plate XLVIII](#). The piscine nature of the form, here depicted on rocks, is not only apparent, but it is possible to say with some certainty that the two shown swimming belong to the *Toxotes*, which are commonly called Archer Fish. The form shown in [Plate XLIX](#), 4, is unquestionably meant to be one of the *Therapon* species. Both kinds of fish are known to be living in the Katherine River, not far from the site at which these pictures were drawn.

But if, on the other hand, some of the designs are so poor as to be barely recognizable or even quite unrecognizable by us, how does the aboriginal manage? When the artist is present, he can explain. But he is not always available!

If, by way of illustration, we were asked to say definitely what the meaning of the central figure on [Plate L](#), 1, was we should in all probability want to know more about it before committing ourselves. But an aboriginal can give us a correct reply immediately. The locality at which the photograph was obtained is north of the Musgrave Ranges in central Australia. But that does not give us any clue. After studying the picture more closely, we might be able to distinguish the outline of a quadruped, the four legs being shown, one behind the other, in a row, and a big head on the right-hand side, in a position suggesting that the animal is feeding. But these are characteristics common to many animals!

So far, therefore, we have seen nothing to suggest the class of animal we are dealing with. When we look again, we might note that there is a crude image of a human being shown on the back of the animal; and behind this is a structure which might stand for a saddle. We guess the answer and claim that the group is a very poor drawing of a man on horseback.

But there are other animals a man could ride! And when we look again, we observe that the second leg of the animal, counting from the right, has a peculiar enlargement attached to its lower end. That structure is the key to the riddle; it represents the track of the animal! Those familiar with the great beast of burden, now used extensively in central Australia, will recognize the two-toed spoor of a camel.

This method of pictorial elucidation is by no means exceptional. We have already noticed something similar in the ancient carvings at Port Hedland, where the human foot-print is added to disperse any doubt which may be entertained in so far as the correct interpretation of the figure is concerned. A similar device is well exemplified in the accompanying sketch of an ochre drawing of a human form from the Glenelg River district in the northern Kimberleys of Western Australia ([Fig. 15](#)). In the carving of an emu from the King Sound district, which is reproduced in [Plate XLII](#), 2, we noticed the same sort of thing.

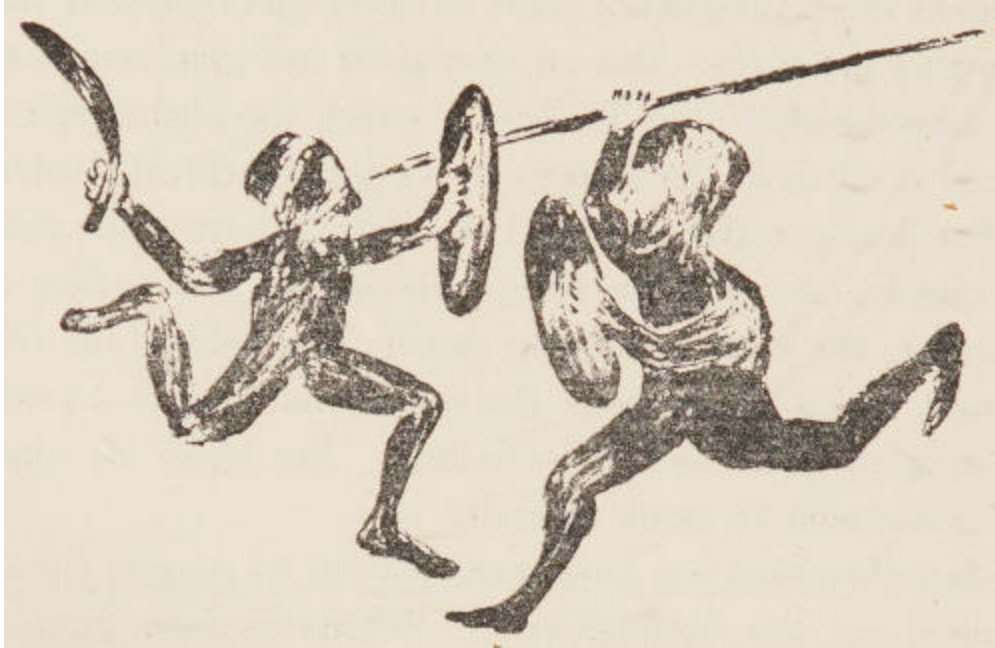


Fig. 27. Ochre-drawing of spear-boomerang duel, Arunndta tribe ($\times 1/2$). Tracing.

The cases before us are not accidental, but we have acquainted ourselves with the recognized determinative system of Australian pictographs which is quite analogous to that known to have been practised by the ancient Egyptians. Consider, for instance, the character signifying “to love”—a human figure in profile with one hand lifted to the level of the mouth. The same figure, with a few parallel wavy lines, signifying water, drawn against it, means no longer “to love,” but “to drink.” The wavy lines in this instance are the determinative. In the Australian illustrations given above, we have selected samples which are easily followed, but there are many cases where the reading would be quite impossible if it were not for the presence of the little, subsidiary, determinative sketch.

In his endeavour to make the meaning of some of his designs clear, a native often embodies as many features as possible, quite regardless as to whether in reality they would all be visible in the one plane he is drawing. In the picture of a crocodile appearing on a boabab-nut from the Derby district in Western Australia, shown in [Fig. 16](#), it will be observed that the reptile, in spite of having its dorsal surface represented, has its vent indicated. The long, slender muzzle of this figure, by the way, makes it clear that the smaller species of the two northern Australian crocodiles (*C. Johnstoni*) is intended. The human figure, too, very often appears half in full and half in profile.

The aboriginal is a keen observer, and takes careful note of many things besides a kangaroo, a snake track, or other similar natural objects which may lead him to his daily bread. When travelling in the Buccaneer Archipelago in the far north-west I remember one of the natives drawing my attention to a peculiar formation in the clouds, and saying, in the Sunday Island dialect: "*Arrar ninmiddi*," which means, literally: "Cloud knee." My instructor proceeded to draw the extraordinary shape he could see with his finger upon the hatchway of the pearling lugger we were sailing in, after which he completed the figure of a man. I was struck with this man's faculty of observation, because the cloud effect he referred to was rather out of the common and projected from a cirro-cumulus like the bent limb of a swastica.

It is in this way that many inspirations come to the cave artist. Repeatedly one has occasion to notice how a pre-existing feature or defect in the rock face—a crevice, a floor, a concretion—becomes the centre piece of a design drawn to suit it. The feature one finds most commonly embodied in a cave drawing is a small hole. This often figures in the place of an animal's eye, or a hole into which a snake is disappearing. A local bulge in the rock may also be taken in as part of a design and represent portion of a head or body.

Not only does the artist embody suitable natural features in his designs, but, conversely, he also applies his knowledge of form to explain already existing phenomena in the world about him. The embodiment of his artistic ideas in his poetical explanations of Nature's wonders plays, as might be expected, an important role in his mythology. These remarks apply especially to any striking characteristics in the sky. When among the tribes of the Musgrave Ranges, I ascertained that the black-looking gap in the Milky Way, close to the Southern Cross, which is commonly known as the Coal Sack, was referred to as "*Kaleya Pubanye*," that is, the "Resting Emu."

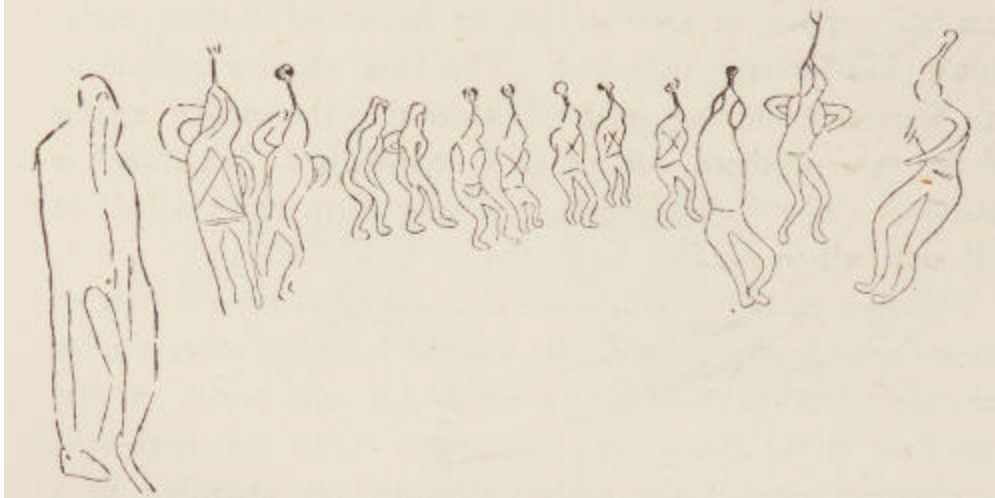
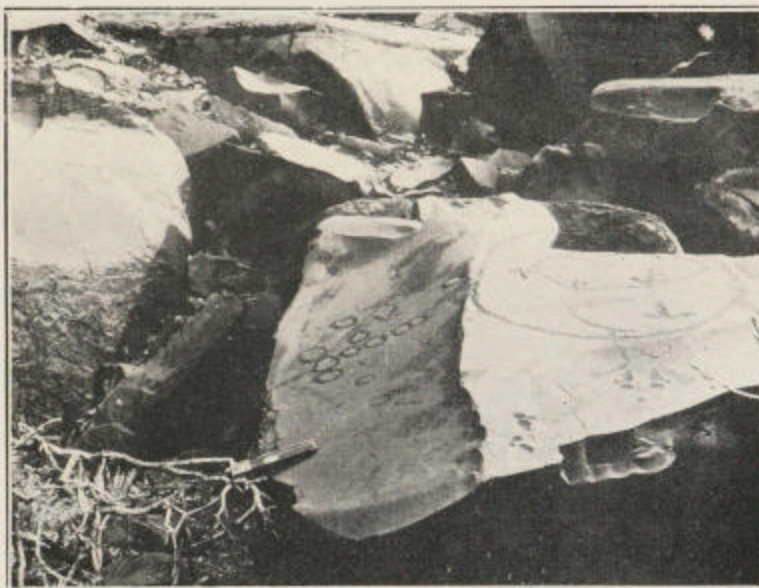
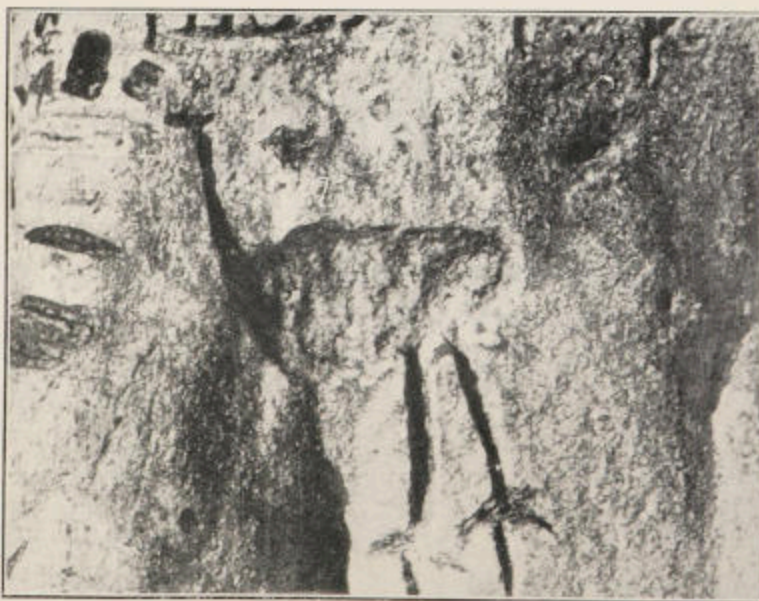


Fig. 28. Charcoal sketch of ceremonial dance, Pigeon Hole, Victoria River ($\times 1/6$). Tracing.

In the north of Australia, the Larrekiya, Wogait, and other tribes have adopted a similar designation for a series of dark spaces along the Milky Way. But they have extended the idea considerably in that the Coal Sack represents only the head of a gigantic emu, the beak of which is pointed towards the *Musca* constellation (*i.e.* towards the south). A small star of the Southern Cross group very appropriately stands for the eye of the bird; the nebulous effect usually surrounding this star gives it an extra life-like appearance. The neck is but faintly discernible near the head, but becomes clearly visible in the neighbourhood of the nearer Pointer; it passes between the two Pointers and curves slightly towards the constellation of *Lupus*. Within the constellation of *Norma*, the dark space widens considerably and represents the body of the emu. The blunt tail turns sharply towards, and into, the constellation of *Scorpio*. A nebulous patch lying practically on the point of junction between the imaginary areas of *Ara*, *Scorpio*, and *Norma* affords a good division between the legs of the bird, whilst another lying between μ and ζ of the Scorpion group separates the tail. The lower portions of the legs are not very clear, but some of the more imaginative natives maintain that they can distinguish three toes on each extremity. There is no doubt the primitive eye has herein discovered a striking similarity between an optical phenomenon in the southern sky and a living creature, which is of great importance in the hunting field, and at the same time plays a prominent role in tribal folk-lore. They refer to this emu by the name of “Dangorra.” Vide [Fig. 17](#).



1. Rock carvings, Flinders Ranges.



2. Emu design carved into the butt of a boabab tree, King Sound.

PLATE XLII

1. Rock carvings, Flinders Ranges.

2. Emu design carved into the butt of a boabab tree, King Sound.

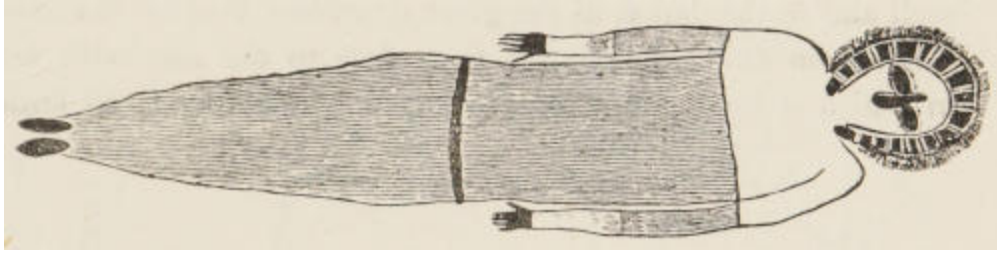


Fig. 29. Remarkable cave drawing, Glenelg River, N.W. Australia

As affording a means of comparison, a hunting scene is reproduced carved upon the surface of a club by aborigines of Victoria. The little group is composed of an aboriginal hunter who in one hand is poising a spear and in the other is carrying a boomerang; behind him are two emus standing in much the same position as that assumed to be the case in the heavenly image just described.

The Minning at Eucla recognize only the long neck of the emu in the sky, and refer to it as “*Yirrerrri*”; on the Nullarbor Plains the same portion is looked upon as the heavenly tjuringa of the emu.

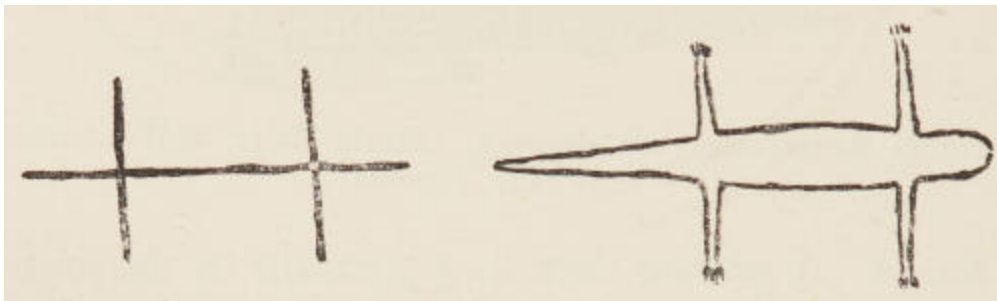


Fig. 30. Pictograph of lizard, natural and conventional form.

Speaking generally, there is perhaps no other creature living which figures so frequently in aboriginal art, both on the cave wall and in the dance, as the great struthious bird of Australia. This is no doubt due in the first place to the admirable way in which it lends itself for the purposes mentioned; its antics in the field suggest many tricks for mimicry at a corroboree, and its distinctive form supplies the artist with a model which never fails to attract the attention of the artistically inclined among his people. In [Plate XLIX](#), 2, we have a pipeclay drawing of an emu from the Katherine River which is rather exceptional in that it shows the bird more *en face* than is usual; the proportions are, on the whole, good, except that the head is screwed upwards in a rather strange way. On a boomerang from

Broome ([Fig. 18](#)), we have a series of engraved emu pictures, all in profile, and in different attitudes.

On the whole, an aboriginal's pictures are flat and without perspective. He takes the inspiration direct from nature and reproduces the subject singly, and as a separate entity; a number of such designs are drawn side by side with or without pictographic sequence. But there are countless occasions upon which artists, especially the more gifted, prefer to draw a real scene from life, combining subject with action. Environment or surroundings rarely, if ever, receive attention.

Take as a very simple illustration the lizard shown in the pipeclay rock drawing from the Katherine River ([Plate XLIX, 3](#)). The general shape of the body, together with the large and well-differentiated head, strongly suggests a species of the large monitor which is common throughout the district. The interesting feature about the picture is, however, the life which is indicated by the fact that the reptile is drawn in the act of shooting out a long, split tongue.

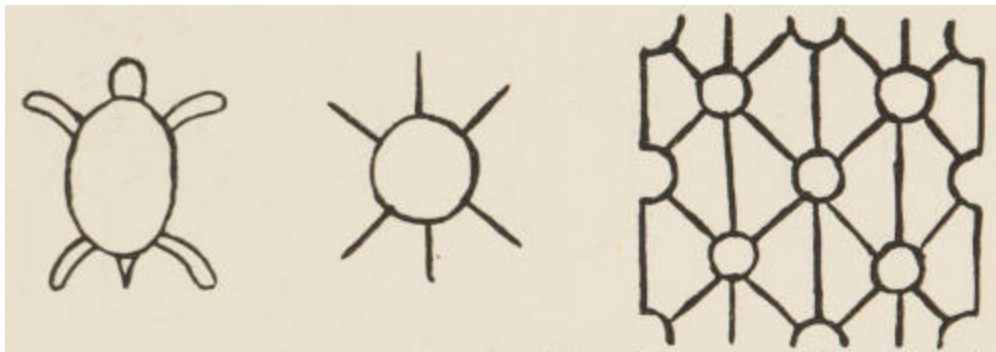


Fig. 31. Normal, conventional, and emblematic representations of turtle.

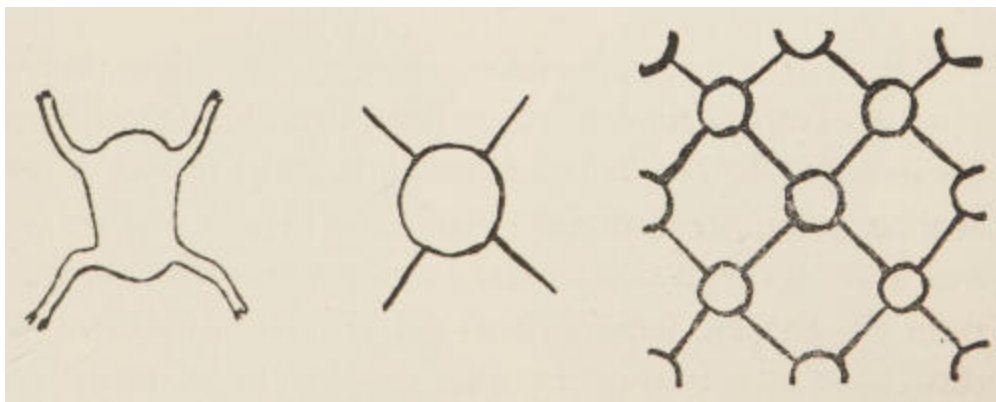


Fig. 32. Normal, conventional, and emblematic representations of frog.

Again, in the charcoal sketch of two crows from the Pigeon Hole district ([Fig. 19](#)), one bird is represented in an attentive attitude, as though on the point of flying away, while the other is very characteristically shown in the act of cawing.

One could produce an almost endless variety of decorated figures, representing men and women performing at ceremonial dances and corroborees to illustrate the life and action which is embodied in aboriginal art. In [Fig. 20](#) a selected number of pipe-clay drawings from the Humbert River, Northern Territory, have been grouped together to serve this purpose.

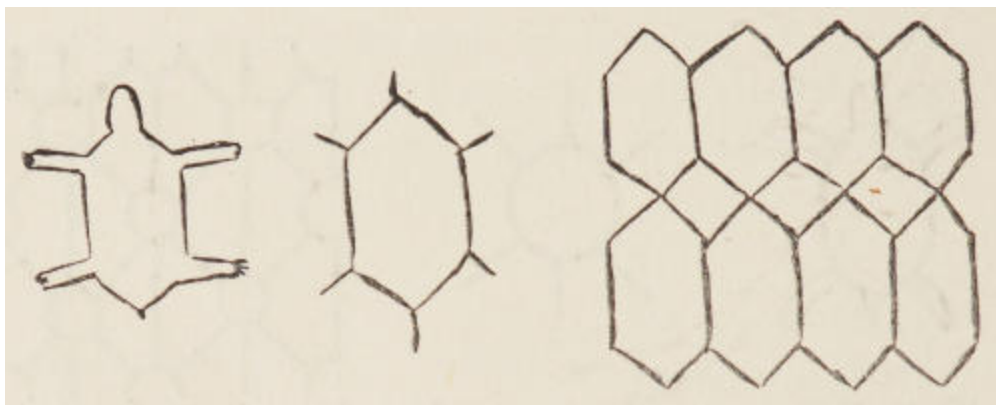


Fig. 33. Normal, conventional, and emblematic representations of echidna.

The most interesting effects, however, are those brought about by a combination of two or more figures. How different, for instance, the two kangaroo shown together in [Fig. 21](#) seem to those previously discussed ([Plate XLVII](#)). These are charcoal drawings from Pigeon Hole on the Victoria River, and in them the hopping movement of the animals is indicated very clearly. The animal in the rear is in full flight, as the erect position of the tail and the general holding of the body betray; but the one in the lead is on the point of drawing up and is turning its head back towards its mate.

How realistic, too, the little bark drawing is from east of Port Darwin ([Fig. 22](#)), in which a bird of prey is shown mounted upon a wallaby or kangaroo, with its claws and beak embedded in the flesh of its victim.

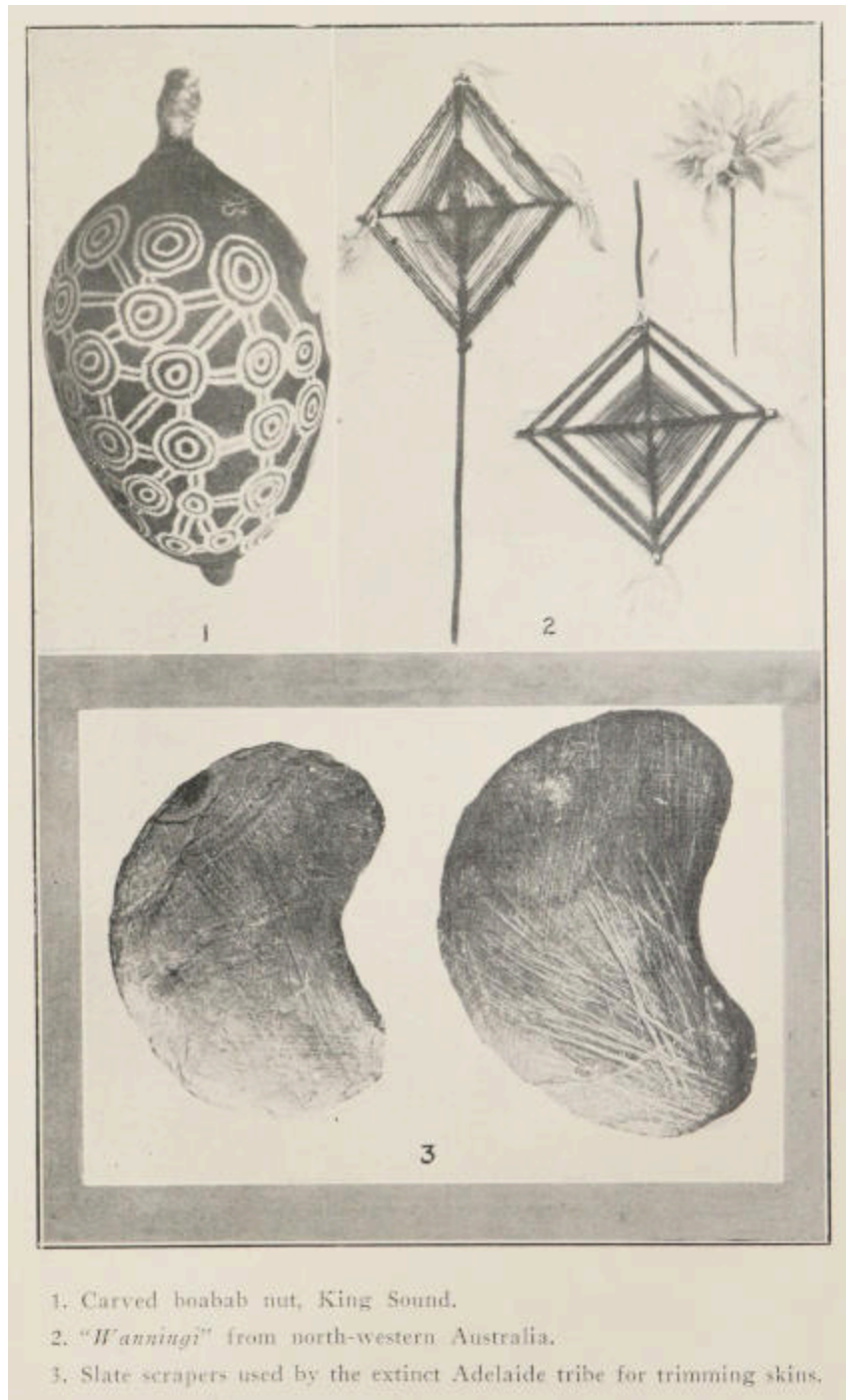


PLATE XLIII

1. Carved boabab nut, King Sound.

2. "Wanningi" from north-western Australia.

3. Slate scrapers used by the extinct Adelaide tribe for trimming skins.

A neat pipe-clay drawing from the remote Humbert River district is presented in [Fig. 23](#). The group, which is three feet in length, is composed of a central figure of a man who is holding one arm on each side towards a dog, as if offering them something to eat or for the purpose of patting them. The dogs seem to be giving their attention to the man.

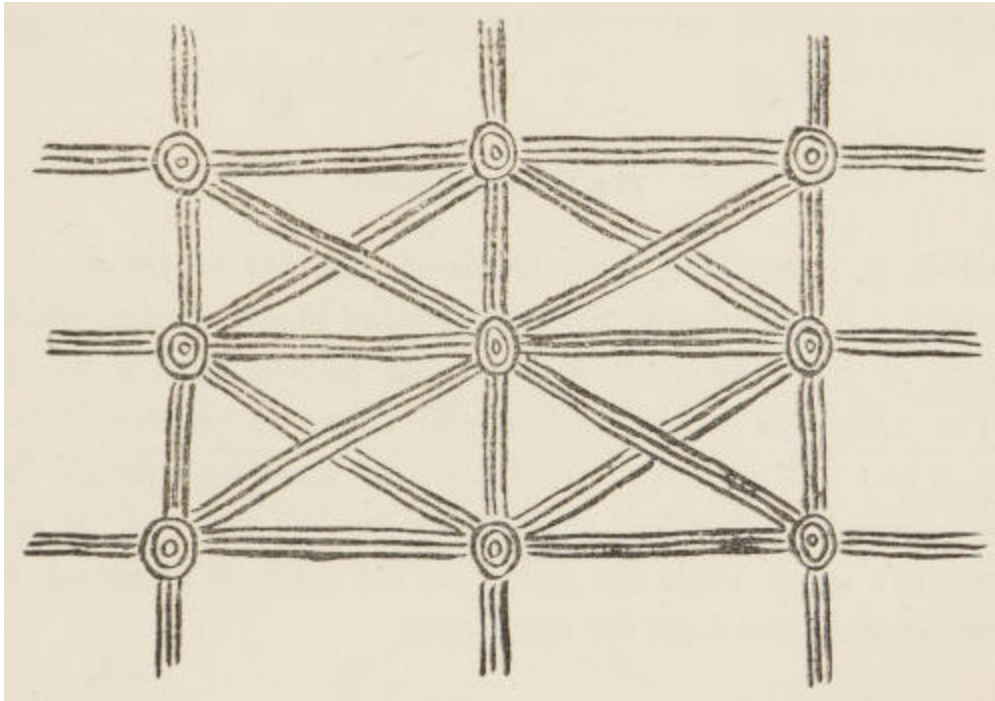


Fig. 34. Conventionalized “*Ladjia*” or yam Tjuringa pattern.

Two more charcoal drawings from Pigeon Hole, though roughly sketched by the artist, depict very graphically scenes from the hunt. In one ([Fig. 24](#)), the hunter is in the act of stalking a buffalo or bullock with his spear held in readiness to throw, while in the other the attitude of the hunter indicates that the spear has just been thrown and is entering the body of the prey, a kangaroo ([Fig. 25](#)).

The carving of an Arunndta man, reproduced in [Fig. 26](#), is most effective. An angry husband has been caught by the artist in the act of punishing his wife with a waddy. The placement of the legs of the two persons indicates stability on the part of the man engaged in the flagellation, and a swinging movement on the part of the woman who is being held back by her hand.



Fig. 35. A dog-track.

We have already seen the carved representations of two stages in a stone-knife duel by an Arundta tribesman ([Fig. 4](#)), and here, in [Fig. 27](#), an ochre drawing is reproduced which is, if anything, more animated than any previously discussed. A spear-boomerang duel is being fought, during which each of the combatants is protecting himself with a shield. The artist has evinced considerable talent in portraying the men just at the moment when both are bounding through the air towards each other, the one on the left parrying his opponent's spear, while the other, on the right, is preparing to receive the blow from the boomerang.

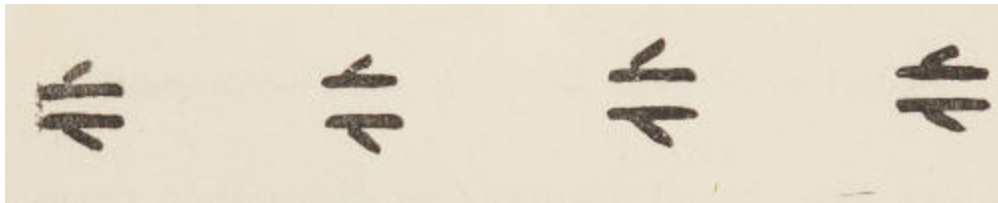


Fig. 36. A kangaroo-track.

One might now go a step further in analyzing aboriginal art. The productions we have studied so far embody the ideas of form, life, and action; and, it might be added, occasionally one finds a very fair sense of composition as well. Such, indeed, might already be said to be true of several of the pictures discussed above, but a finer specimen lies before us in the charcoal drawing from Pigeon Hole ([Fig. 28](#)). This faithfully portrays a scene from a gala ceremony, in which the body of performers, fully "dressed" for the occasion, are acting before the leader, who, in his turn, is being supported by two others in the foreground. It must be admitted that the composition of this group of figures is remarkably good, and, what is quite exceptional, a very successful attempt has been made at perspective. All figures are shown in different attitudes of dancing. The impression this charcoal drawing gives one, at first glance, is that of a rough sketch in crayon resembling the outline a European artist might make on his canvas prior to starting upon the actual painting.



Fig. 37. A rabbit track.

Leaving that section of aboriginal art which deals essentially with designs copied directly from Nature in a sense more or less purely artistic and æsthetic, we shall turn our attention to a few types which are more specialized.

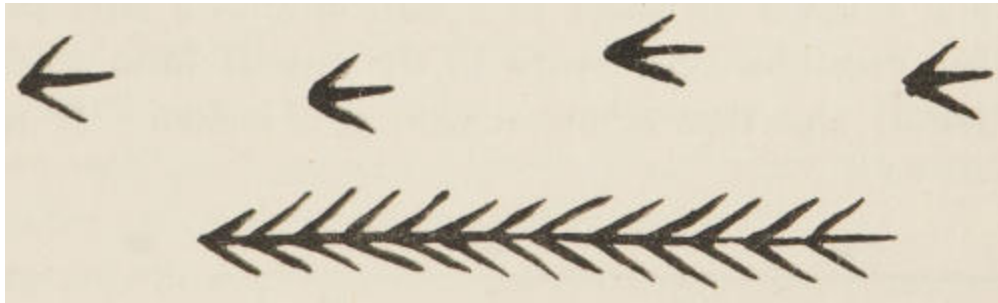


Fig. 38. Emu tracks.

From a study of his religious ideas, we have learned that the aboriginal identifies himself with some mystic, natural creature or object, which he adopts as his “totem.” It would only be reasonable to expect, therefore, that some of the drawings represent these objects; and that they are recognized by the natives as having particular personal or family significance. Looked at from a modern standpoint, these designs are really the equivalent of a family crest, and are claimed only by those rightfully entitled to them. This explanation must be given for many of the naturalistic designs appearing on rocks, trees, grave posts, and personal belongings. These “totemic” crests or symbols being hereditary, we have before us a primitive form of heraldry, a conception we have already learned to be covered by the word “*Kobong*,” originally introduced by Sir George Grey from the north-west of Australia.

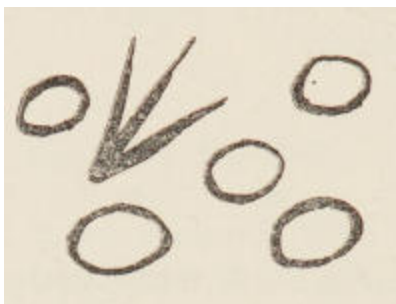


Fig. 39. Pictographic representation of nesting emu.

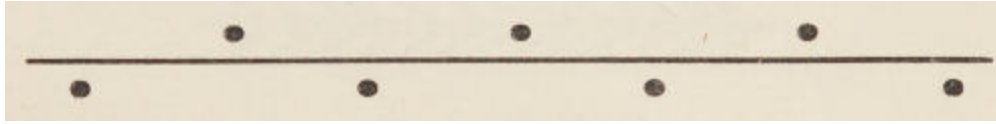


Fig. 40. A lizard track.

We have also ascertained that some of the central as well as north-western tribes of Australia believe that the earliest tribal ancestors originally were more animal than human in appearance, and adopted the shape of a man only at a later period; that they can, however, return to the animal form whenever they desire; and that others remain semi-human. It is not surprising, therefore, to find amongst their drawings and carvings representations which are partly human and partly animal in outline; these are honest attempts at perpetuating the traditional appearance of the ancestral beings of the tribe. In the photograph attached hereto ([Plate LI](#), 1), taken at Forrest River, two pictures of such creatures are to be found which are drawn in ochre. There were many others, from three to five feet in length, reptilian in shape, some with human hands and feet, others with hair shown upon the head, and in most of them the sex unduly prominent. These remarkable designs are, therefore, not naturalistic, but have been evolved on purely fictional or mythological lines, based upon the tradition of the tribe and upon the imagination of the artist.

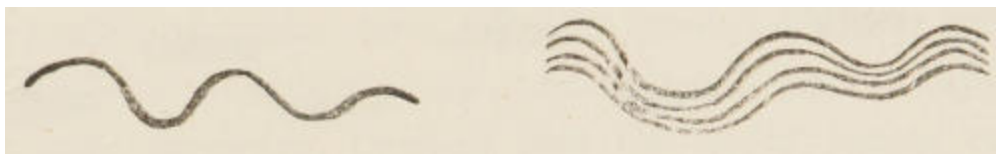


Fig. 41. A snake or snake-track.

From the consideration of these artistic effigies of their Demigods, it is not a big step forwards which brings us face to face with the sacred tribal drawings. During initiation ceremonies, especially of the now practically extinct south-eastern tribes of Australia, gigantic figures resembling a human being were moulded into the surface of the ground and subsequently tinted with ochre, which were supposed to conceal the Great Spirit or Deity, which, like the “*Altjerra Knaninja*” of central Australia, watched over the proceedings as the young men passed from a condition of adolescence to

that of permanent manhood; numerous carvings and ochre drawings were also made upon the trunks of any trees nearby.

Not only during the initiation ceremonies are these practices resorted to, but when a sacred observance is contemplated, especially those having to do with the “totem,” elaborate designs are painted in ochre upon the surrounding surfaces of rocks and trees which depict an act connected with the traditional origin of the sacred object.

A classical illustration is to be found in the MacDonnell Ranges, at Emily Gap. According to Arunndta belief, it was at this spot that the early semi-human ancestors of the witchedy grub or “*Utnguringita*” alighted from Altjerringa. They brought with them large numbers of the grub, which they cooked and ate. The territory dominated by these ancient beings extended from Heavitree Gap to Emily Gap, and across to Jessie Gap. On the western wall of the first-named gap, known by the natives as “*Ndariba*,” an inclined slab of rock, not high above the level of the sandy bed of the Todd River, contains a series of peculiar concentric iron stains which are regarded as the impressions of the stern of an *Utnguringita* Altjerra who sat there, and, as he collected grubs, moved forwards. The *Utnguringita* came into frequent conflict with the Dingo or “*Knullia*” people whose country lay immediately west of Heavitree Gap, but, nevertheless, they blessed the land with many eggs, which developed into larvæ and supplied the tribe with food.



Fig. 42. Human foot-prints and trail.

Eventually the *Utnguringita* ancestors returned to Altjerringa, but they left a number of stone tjuringas in Emily Gap, which are supposed to be occupied by the spirits whenever a sacred ceremony is performed on the spot. On the eastern stony wall of this gap some rather imposing designs are to be seen, which originally must have occupied most of the area available. The drawings are very old; their origin dating back long before the

recollections of the present generation. It is wonderful how well the work has withstood the denuding action of the weather for so long. The natives tell you that the old Altjerringa men applied the pigment to the rock and that they mixed it with the “*knudda*” (fat) of the grubs. It is more likely that the ochres were mixed with emu fat; in places the pigment seems as though it were chemically combined with the rock, and it could only be removed by chipping the surface. The designs in their present condition ([Plate LI](#), 2) consist of a series of parallel, vertical lines, alternately coloured red and white, and capped by horizontal bands of the same colours, the white of which containing three or four red dots. What the original designs may have been like, it is now difficult to say, but the natives maintain that they included the images of some women they call “*Aluggurra*,” who were waiting at the foot of the cliff while the men were concealing their tjuringas in the rocks and nooks above. To the present day, the old men of the local Arundta group store their ceremonial objects in the same sanctuary, thinking that the sacred figures on the wall will protect them from the hands of inquisitive intruders.

There remains yet another class of ochre drawing which deserves mention. I allude to the famous discovery of Sir George Grey in 1837. There is perhaps no other Australian drawing, old or modern, which has been so freely discussed and criticized. During an expedition in the northern Kimberleys of Western Australia, it was my good fortune to re-discover several drawings of this type in practically the same locality as that recorded by Sir George Grey, near the Glenelg River. One figure was perfect, others were partly obliterated or incomplete. The best design was in a cave near the top of a prominent bluff the local Worora people call Berrial; it was drawn in ochre upon a steep face of rock immediately under an overhanging ledge of quartzite. The figure was unquestionably that of a human being, although it measured fully nine feet in length. It lay fully extended, upon its left side, with its arms placed straight against its sides. It reminded one forcibly of a Buddha in a Ceylonese temple. What made the figure seem un-Australian was that it was clothed in a long, striped garment, resembling a priestly gown, from which only the head, hands, and feet were excluded. A loosely-fitting belt is also shown. As seems common to all these drawings, the facial features are only indicated by the eyes and nose, the mouth being omitted. Another characteristic, which is shared by all other drawings, is that the head is surrounded by a number of peculiar,

concentric bands, through which, and from which, many lines radiate, giving the structure the effect of a halo surrounding the head of a saint. The picture bore an unmistakable likeness to the type illustrated by Sir George Grey, and was drawn in red, brown, black, and white. Vide [Fig. 29](#) and [Plate L](#), 2.



Fig. 43. "A man is tracking a rabbit." Simple example of pictography.

There is no doubt about these curious drawings, now more or less adopted by the local tribe, having originated under some exotic influence. It is historically known that for centuries past excursions have been made to the north of Australia by Macassans and other eastern people, who may have been responsible for the first drawing of a figure of so sacerdotal an appearance, which the aborigines have since learned to copy so perfectly. It has also been speculated that shipwrecked sailors might be responsible for the representation of the clothed human form; in fact, an American scientist, John Campbell, claims to have deciphered an ancient Japanese inscription upon one of Grey's figures referring to the "hopeless number," presumably of the castaways.



Fig. 44. Pictographic representation of emu-hunt.

In regard to the quaint head-gear which distinguishes these designs, one need not go far from Australia to find something quite analogous, for the Papuans wear an article which is quite similar to what might be suggested by the drawings.

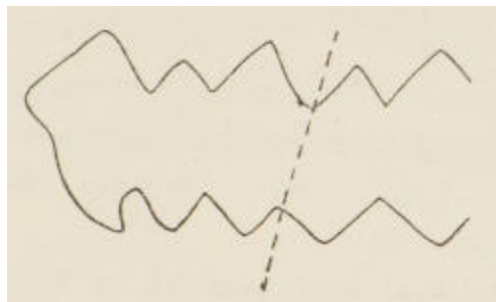


Fig. 45. Flying fox pattern.

We now turn to a more psychological aspect of primitive Australian art, which includes such factors as convention and imagination. These processes lie behind the symbolization of thought which has evolved a means of pictographically conveying messages from one individual to another, or, collectively, from one tribe to another. Through long usage, the artist has learned to reduce the complexity of a familiar, naturalistic design in such a way that, while still retaining its intrinsic interpretation, he is able to demonstrate by a few lines what ordinarily would require an intricate drawing. The ultimate aim of such a system is, of course, to reduce the execution of a design to a minimum of energy and time, without imperilling the correctness of its interpretation. But whilst the artist is designing to simplify the complexity of his thought symbol, the reciprocative factor of assimilation on the part of his fellows is being stimulated. And by means of this joint education, a design, with a simple motive from Nature behind it, might gradually become so conventionalized that the uninitiated fails entirely to grasp its significance.



1. Hand marks in cave, Port George IV, Worora tribe.



2. Foot marks in cave, Port George IV, Worora tribe.

PLATE XLIV

1. Hand marks in cave, Port George IV, Worora tribe.

2. Foot marks in cave, Port George IV, Worora tribe.

During this evolution, or, should one say, metamorphosis, in the perception of art, the tendency must necessarily be to smooth out the

irregular and unsymmetrical contours of Nature and bring the design down to as nearly geometrical as possible. For the same reason, a single design is often repeated indefinitely, so that a single form derived from a simple motive may expand into a continuous chain or ornate pattern covering a considerable area. A large series of devices has been established, the majority of which are known all over the Australian continent; but, as the same time, there are many signs which are entirely of a “totemic” nature and can only be understood by a person belonging to that particular “totem.”

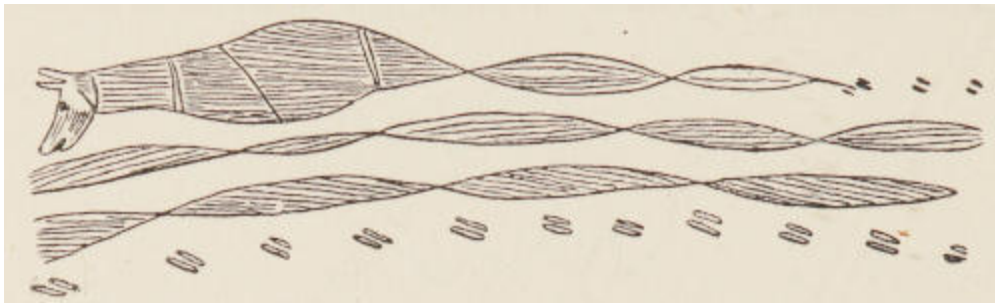


Fig. 46. Conventional representation of hopping kangaroo.

Let us take a few simple cases illustrating the different points enumerated above.

A simple drawing of a lizard would include a substantial body, with rounded head and tapering tail; the legs are usually extended, and at times have the claws marked. Vide [Fig. 30](#).

The conventionalized form consists of a long straight line crossed at right angles by two shorter bars, at points about one-quarter and three-quarters distance respectively from one end.

A turtle design consists of an oval, representing the shield, from which extend the head, tail, and paddles. In its modified form this becomes a circle with six short lines radiating from the circumference at equal distances apart ([Fig. 31](#)).

The picture of a frog in its simplified form becomes, like that of a turtle, a circle, but has only four radiating lines ([Fig. 32](#)). When designs like these are expanded symmetrically into patterns, the result is after the style of that shown in the accompanying sketches. These patterns are extensively used in totemic devices upon tjuringas and implements.

The drawing of an echidna, or native hedgehog, ordinarily is like the sketch shown above ([Fig. 33](#)), but as a result of its conventional transformation it becomes a simple hexagon, from the corners of which the limbs may or may not be shown as simply projecting lines. The pattern obtained by linking up a number of hexagons is not uncommonly found engraved upon weapons and implements.

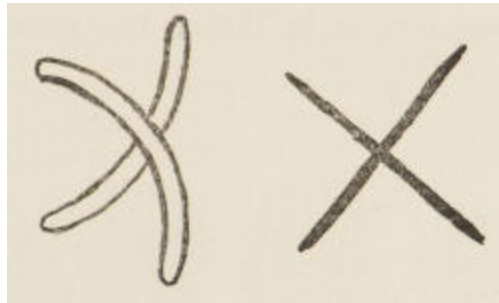


Fig. 47. Crossed boomerangs, the symbolic representation of a fight.

Many of the conventional patterns are not so apparent as the few just mentioned. Let us take, for instance, the “*Ladjia*” or Yam Tjuringa pattern of the Arunndta. Only the initiated would be able to recognize in a number of groups of small, concentric circles, regularly placed at the corners of rectangular figures, as the stems of the yam plant, and a system of parallel lines connecting the circles both straight and diagonally as the roots. Vide [Fig. 34](#).

Tracks of animal, bird, and man are conspicuous among the designs, generally, whether they be true copies from Nature upon the walls of a cave, more or less modified gravures upon weapons, or conventional patterns incised upon a message stick. A dog track is represented by a larger dot, suggesting the imprint of the ball of the paw, to which are attached four smaller dots, which lie in a row in close proximity to the former; the smaller dots stand for the impressions of the claws ([Fig. 35](#)).

A kangaroo track shows the two long parallel median impressions, with a lateral at an acute angle to the former at either side. The same figure is repeated a short distance away from the one described, and in a straight line with it. The same design on a smaller scale denotes a wallaby track. Occasionally the lateral lines are dispensed with ([Fig. 36](#)).

Small oblong dots drawn in pair at equal distances from each other indicate the hopping of a rabbit ([Fig. 37](#)).

The characteristic broad arrow-like footprint of an emu has already been referred to; the smaller variety of the same design implies that a wild turkey is meant. When a number of birds are to be represented collectively, the archetype is developed into a continuous pattern by linking one track up with another into a chain ([Fig. 38](#)).

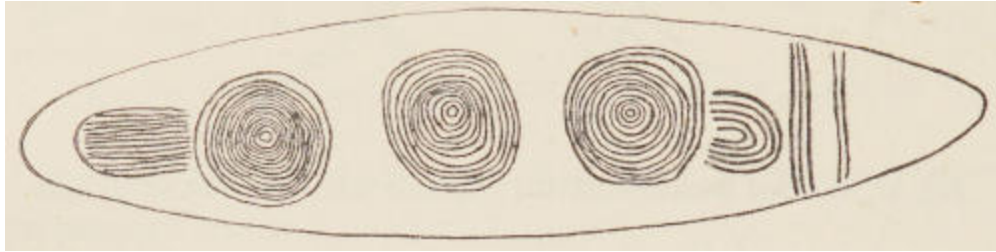


Fig. 48. Witchedy grub Tjuringa, Arunndta tribe ($\times 3/10$). Tracing.

When attention is to be drawn to the fact that the birds are laying, or sitting on eggs, a number of small circles are drawn about the track ([Fig. 39](#)).

A lizard track is indicated by drawing a number of dots, equally spaced along a straight line, and on alternate sides of it. The dots are the claw impressions, the line the trail left by the tail ([Fig. 40](#)).

A single wavy line, or a number of parallel wavy lines, represents a snake or a snake track ([Fig. 41](#)).

The human footprint is either correctly shown in detail, or is reduced to a short, straight line at one end of which five (occasionally only three or four) dots are drawn in a sloping line to indicate the toes. When walking is to be implied, these footprints are either shown one behind the other in a straight line, at equal distances apart, or they stand alternately right and left by an imaginary central line. The common way of showing the last-named system conventionally is to connect the alternate footprints, whether they be actually shown or not, by a zig-zag line ([Fig. 42](#)).

In all cases mentioned above, the track may stand equally well for the object itself.

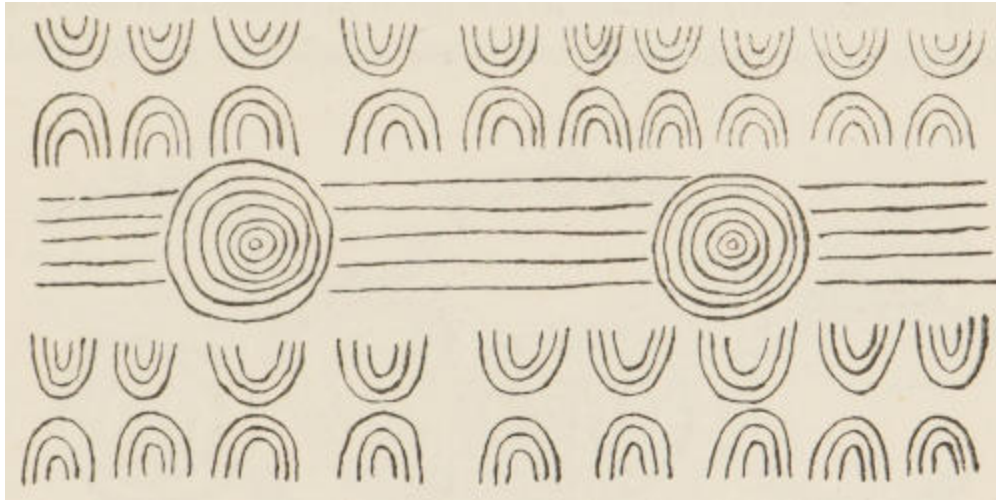


Fig. 49. Symbolic pictograph of kangaroo Tjuringa, Arunndta tribe.

It is a common thing to see two or more of the above systems combined, after the style of the figures shown on [page 344](#).

The interpretation of the first of these messages would be: “A man is tracking a rabbit.”

The second ([Fig. 44](#)) would read: “A hunter is pursuing an emu and is accompanied by his dog.”

In the same way as the natives of the Northern Territory have applied their artistic talents to deciphering images of earthly objects amongst the celestial bodies, and point with pride upon the great emu, “*Dangorra*,” which nightly watches over them, so the Wongapitcha and Aluridja of central Australia recognize in the constellation we know as the Southern Cross the shape of an eagle-hawk’s claw, and call it in consequence “*Warridajinna*”; the Milky Way they consider to be a creek bed and assign to it the name “*Karru*”; the northern Kimberleys tribes believe the Milky Way to be the track of a great carpet snake they refer to as “*Womma*.”

In the representation of a fish, the scales often take the form of a cross-hatched pattern; but there are many cases in which the form of the fish is not shown at all, yet the cross-hatched pattern remains and is nevertheless indicative of the fish.

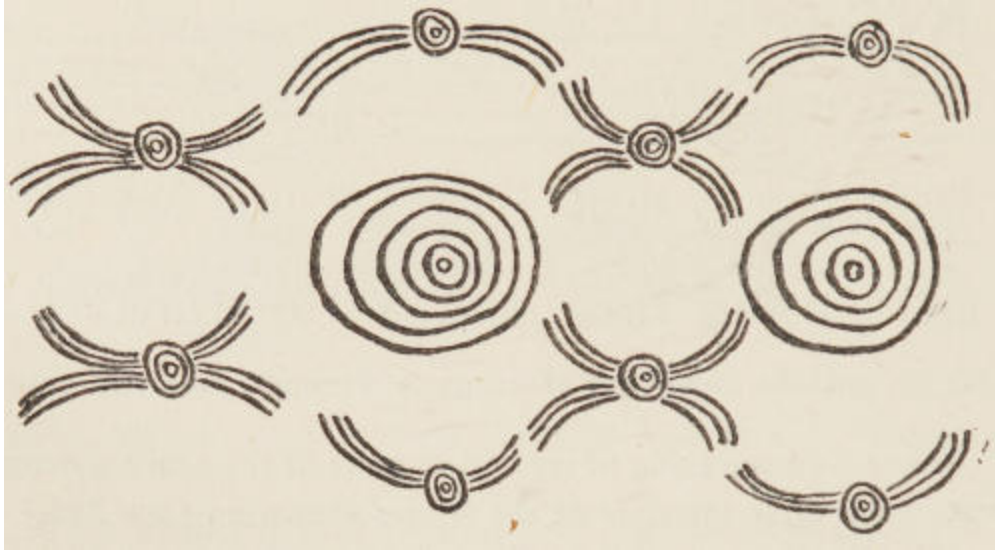


Fig. 50. Symbolic pictograph of caterpillar, Tjuringa, Arunndta tribe.

A design, fairly common in the north of Western Australia, consists of two wavy lines which are parallel, in the inverted or reflected sense, and joined at one end. The true significance of this pattern does not become evident until one hides from view all but three of the “waves,” say that portion lying to the right of the dotted line in the accompanying sketch. When this is done, the form of a “flying fox” is immediately recognized ([Fig. 45](#)).

In central, northern, and eastern Australia, a pattern frequently met with on boomerangs, fighting sticks, and message sticks, consists of strings of lenticles longitudinally striped and generally associated with kangaroo tracks. This device is analogous to the one standing for the walking of a man, viz. the zig-zag, in so far as it stands for the hopping of a kangaroo. Here and there one finds the pattern “finished off” at one end with the head of a kangaroo ([Fig. 46](#)).

A duel or tribal fight of any description is graphically recorded by two crossed boomerangs, but the conventional derivative of this design is simply supplied by two crossed lines ([Fig. 47](#)).

Time is chronicled by two phases of the moon; a crescent standing for new, and a circle for full moon.

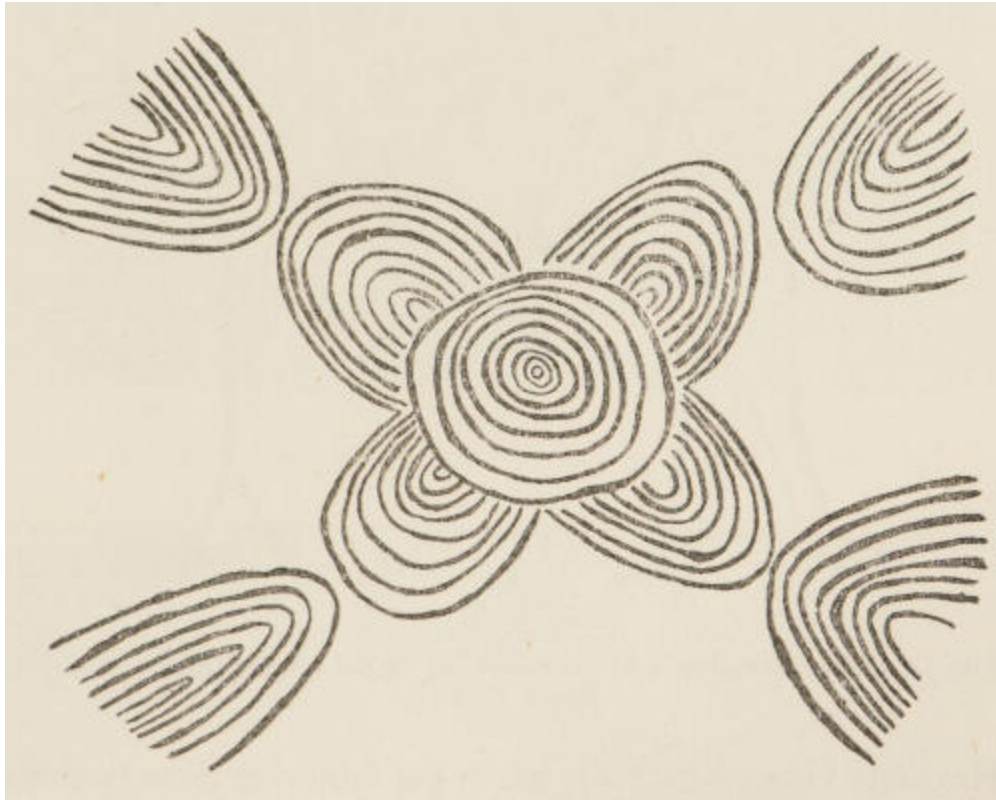


Fig. 51. Symbolic drawing of “native-pear totem,” Arundta tribe.

The substance, origin, home, or habitat of any creature figuring in a drawing or gravure is particularized by the addition of a circle-within-circle design. For instance, in the tjuringa of the witchedy grub shown in [Fig. 48](#), the parallel, straight lines, enclosed within a “U,” at the left-hand side, represent the grub, the three concentric circle groups in the centre are the gum trees in which it lives, the “U within U” pattern at the right is an ancestor whose “totem” was the witchedy grub, while the parallel lines at the extreme right of the tjuringa are markings on the grub which have been adopted by the man who owns the tjuringa, in the form of cicatrices, he cuts on his chest.

The “U within U” pattern is frequently met with engraved upon tjuringas, and in most cases it conventionally conveys the idea of a “sitting” person or animal. We have already noted something similar in the peculiar concentric iron stains of Heavitree Gap ([page 342](#)), but in the following three tjuringa drawings of the Arundta additional illustrations are given.



Fig. 52. Ochre drawing and tree-carving of man with shield, Humbert River ($\times 1/10$).

In the first instance, an “*Arrera Knaninja*” or Kangaroo Tjuringa, the meat and the fat of the animal are represented by the two series of concentric circles in the centre, whereas its sinew is indicated by the horizontal lines connecting the circles. The numerous U groups on both sides of the central figures stand for a great number of kangaroo, each of which is sitting or lying on the ground ([Fig. 49](#)).



1. Cave drawings, Forrest River, north-western Australia.



2. Decorating the body with pipe-clay, Humbert River, Northern Territory.

PLATE XLV

1. Cave drawings, Forrest River, north-western Australia.

2. Decorating the body with pipe-clay, Humbert River, Northern Territory.

The second example is from the “*Yeapatja*” or Caterpillar Tjuringa. The large “circle-within-circle” groups in the centre are generally recognized to

be the bushes upon which the caterpillars were born, while the U groups with smaller concentric circles in their centres are intended for caterpillars attached to smaller plants upon which they are feeding ([Fig. 50](#)).

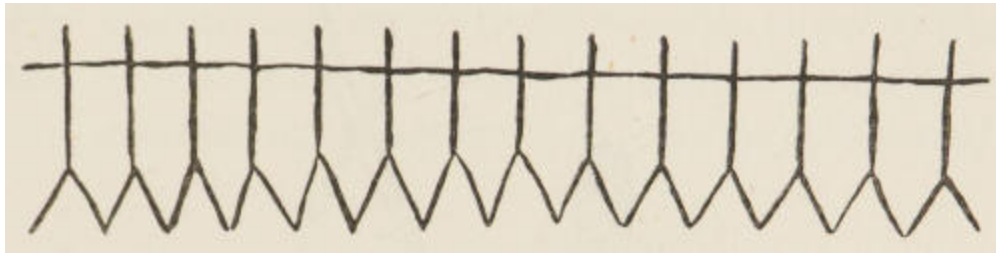


Fig. 53. Human chain-pattern.

Lastly, a sacred drawing of the “*Alangua Knaninja*” or Native Pear (*Marsdenia*) “totem,” which belongs to the Altjerringa women, is composed of a central “circle-within-circle” group representing the “totem,” whilst surrounding it a number of U groups are supposed to be the mythic women seated on the ground ([Fig. 51](#)).

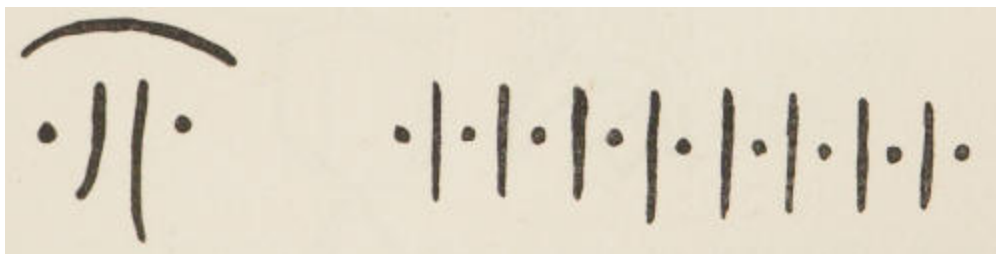


Fig. 54. Camps consisting of a man and his wife (left) and of eight men.

We shall now turn our attention to the consideration of the representation of the human figure and its derivative forms. Several more or less obvious designs have already been discussed. The first step towards conventionalism is seen in the two figures from the Humbert River district, the first an ochre cave drawing, the second a carving on a boabab tree ([Fig. 52](#)). We notice in the ochre drawing, which was one foot six inches high, a fairly shapely and solid figure of a man holding a shield in his left hand; in the carving, which measured one foot nine inches in height, the solid and shapely outline has been reduced to a matter of just a few straight lines; that is, if we neglect for the present the consideration of shield and boomerang which the figure is holding. The result is, therefore, a design resembling a Latin cross, at the lower end of which is attached an inverted V-shaped

symbol representing the legs; a small circle may or may not be added to the top end to stand for the head.



Fig. 55. Anthropomorphic designs, carved on spear-throwers. Tracing.

This design is often repeated indefinitely in a lateral sense, so that a rather ornate pattern results in which the individual figures “join hands” and

their “toes touch” below. A chained pattern, such as is shown in the accompanying sketch ([Fig. 53](#)), may be noticed in [Plate XLV](#), 1, near the centre of the picture, below the ledge on which the bold drawing of a snake appears, and on the same level as the semi-human design on the extreme left.

The ultimate stage of this conventionalism so far as the human figure is concerned is a simple, straight line, the upright arm of the cross; this is extensively employed in carved representations of people on message sticks. We might now hark back to the question at the beginning of our disquisition on aboriginal art “what lies in a mere line” and supply one answer at any rate.

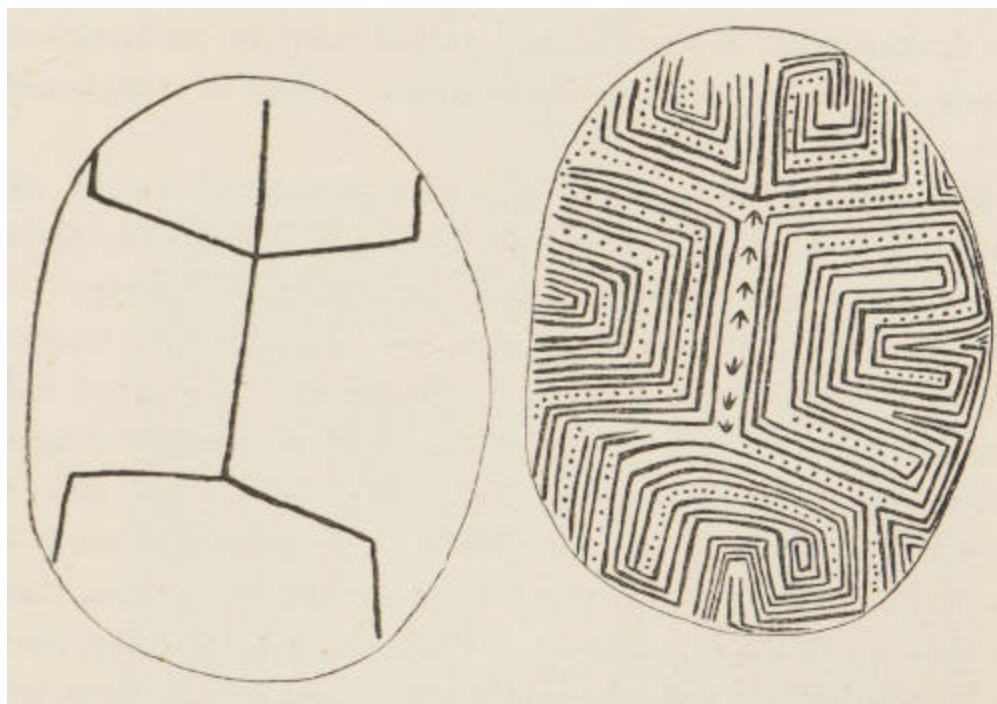


Fig. 56. Anthropomorphic design, carved on pearl-shell, Sunday Island. Tracing ($\times 1/4$).

Any number of people might be represented by the same number of short, straight, vertical lines. Followed by a zig-zag line, such people are represented as being on the march. When a small dot stands on each side of the straight line, or perhaps a number of dots in intermediate positions between the lines, the design conveys the idea that the people are camped, the dots standing for camp fires. Moreover, when a horizontal line lies over the upright lines, the last-named indicates that a hut, shelter, or breakwind was used during the encampment ([Fig. 54](#)).

When there is an obvious, and intended, difference in the lengths of the upright lines, the longer represent men, the shorter women.

Reverting now to the cross, and looking more closely into its development, expansion, and embodiment in anthropomorphous designs, we meet with one or two points of considerable interest.

We have had occasion to note that throughout central and northern Australia, the tribes during the final acts of initiation ceremonies make use of a sacred cross, called "*Wanningi*" or "*Wanninga*" in the former region and "*Parli*" in the western portions of the latter. These wanningi are constructed only immediately before they are required and are destroyed again the moment the ceremony is over. Women are not allowed to see them under any consideration. A wanningi is made by fixing two pieces of wood together in the shape of a cross, then, by starting at the intersection of the arms, a long string, made of human hair, is wound spirally round and round, from arm to arm, until the whole space between the arms is filled in. The size of these crosses varies from three or four inches up to two feet or more. This object is produced by the Aluridja just before the mutilation of the neophyte is to take place. At this critical moment of the youth's life, when he is stepping from adolescence across the great gap which will lead him to manhood, the spirit of the Great Tukura presides invisibly concealed within the wanningi, but returns to his high abode again when the function is over. Vide [Plate XLIII](#), 2.



Wordaman native with his body and head decorated in imitation of skeleton and skull, Victoria River, Northern Territory.

PLATE XLVI

Wordaman native with his body and head decorated in imitation of skeleton and skull, Victoria River, Northern Territory.

The spiral winding of hair-string around the arms of the wanningi associates the idea of the rhombic outline of the string with the arms of the cross. In the representation of the human form, one often finds the two patterns combined, or, it may be, the rhomb replaces the cross.

In the example before us ([Fig. 55](#), a), we have an engraved pattern appearing on a spear-thrower, the motive of which, were it not that the artist had added a human face, would have been very difficult for the untrained eye to recognize. As it is, we have the unmistakable evidence of an anthropomorphous design. Not only does a modified rhomb represent the

body of a man, but the figure itself is flourishing the crossed arms of a “Wanningi” in its right hand. The principal design thus identified passes, at the bottom, into a pattern composed of several polygonal figures which may, no doubt, be looked upon as derivatives of an original rhomb.

In the other illustration ([Fig. 55](#), b), which is also a carving upon a spear-thrower, the intricate association of the rhomb with the human form is again apparent. The figure of a man, with face in profile, is represented in a plain and more or less conventional way; the straight trunk with the two arms resting upon the hips, symmetrically on each side, in itself suggests the rhomb, but, in addition, most of the intervening spaces have been filled in with parallel lines and a cross-hatching pattern which embodies the rhomb.

Conventionalism in the representation of the human figure has thus gone further than the mere inclusion of the cross or the rhomb motive, by working in with the original design a derivative or new pattern which fills up all the surrounding spaces.

By means of this system, a new element is introduced in the shape of symmetry. If a vertical line, drawn through the centre of the trunk, be taken as a median line in a simple design of the human figure, all subsequent patterns which are drawn will be grouped symmetrically about it. The most popular pattern used to fill up the available spaces with, is one of a “concentric” type. By this method a distinctive, bi-laterally symmetrical pattern is evolved, which after prolonged usage may actually take the place of the original, and have a true anthropomorphous significance.

Take the illustration of conventionalism of this kind shown on the carved pearl shell covering from King Sound reproduced in [Fig. 56](#). The original motive was a simple line drawing of a human being after the style of the one on the left-hand side. The next stage in its evolution was brought about by blocking the areas between the limbs, and between the head and arm, on either side, respectively, in a manner which made the resulting pattern appear equally balanced in respect of a median longitudinal line running through the back and head of the original figure.

Very numerous designs of this nature are constantly met with in all tribal areas of Australia, but in most cases the stranger who is not aware of the intermediate or transitional stages may fail entirely to grasp their meaning or origin.

CHAPTER XXIX

STONE IMPLEMENTS

Survival of Stone Age in Australia—Stones used in their natural shape for throwing, pounding, cooking, and grinding purposes—Hand-mills—Rasps—Stone tomahawks—Scrapers—Operating knives—“Cores” or “nuclei”—Stone knives—Spokeshaves—Awls—Concave scrapers—Slate scrapers of Adelaide tribe—Scrapers embedded in resin—Adzes—Bladed spears and knives—Stone spear-heads—Method of manufacture described.

There are not many places left in the world where the man of the Stone Age can still be seen roaming the wilds of his inherited possessions. Even in Australia, although there remain one or two areas where comparatively little havoc has been wrought among the primitive institutions of the indigenous man, yet the influence of civilization is slowly, but very surely, encroaching indirectly upon his ancient cults by such means as inter-tribal barter, if not actually by the hand of the white intruder. Especially do these remarks apply to the manufacture and utilization of stone implements; it is, of course, only to be expected that the superiority of the metal blades of the white man's implements would appeal to the native who formerly had to spend hours making a crude cutting edge which only too often broke when applied to the test for the first time. We shall, however, treat the subject regardless of the alterations which have been brought about by our appearance upon the scene, and without attempting to draw up a hard and fast scheme of classification.

At the present time, whilst there are not only some of the primitive men alive still, but also a limited number of observers who have had the good fortune of seeing them at work, it is of vastly greater importance to record the living facts than to write exhaustively, nay, even speculatively, upon the comparative shapes and embodied techniques of artefacts whose stony composition will ensure their keeping, even fossilized, long after the men who made them, and the scientists who lived among them, have passed into oblivion.

The Australian aboriginal makes adequate use of any suitably shaped pieces of stone he happens to find whilst in pursuit of game; both in the Musgrave Ranges and the northern Kimberleys stones are used in their

natural shape for hurling into a flying flock of birds, for shying at a bounding wallaby, for bringing down nuts of the boabab, and for precipitating fledgelings out of a nest.

Other stones, usually oblong and rounded pebbles gathered in a river bed, are used for pounding and cracking purposes. At any camping ground these pebbles can be picked up in great numbers, showing one or two places, usually the points, at which the percussions have worn the stone away; pounding stones and hammers of this description are equally plentiful in the sandhills on the plains of Adelaide, all over central Australia, and along the north coast. They are used for pounding seeds and foliage (the latter of which is to serve for corroboree decorations), for pulverizing ochre, for cracking nuts and hollow bones containing marrow. Vide [Plate LIV](#), 1.

The underlying surface consists either of a level portion of an outcrop or another, but larger, stone, which takes the place of an anvil. Some of the coastal tribes of eastern Australia used to shape their heavier pounding stones by chipping away material at one side until a stout, cylindrical handle was formed, the whole resembling a pestle; dumb-bell shaped pounders were also made, but were rare.

We have already learnt that natural pebbles or rock fragments are also used, together with a wooden rod, for knocking out teeth during initiation ceremonies.



1. Cave drawings (kangaroo, etc.), Forrest River, north-western Australia.



2. Cave-drawing of kangaroo, Forrest River, north-western Australia.

PLATE XLVII

1. Cave drawings (kangaroo, etc.), Forrest River, north-western Australia.

2. Cave-drawing of kangaroo, Forrest River, north-western Australia.

When cooking game, many tribes, both in central and northern Australia, select a number of large, irregular slabs, which they place into a shallow

hole they burn a fire in. After the oven stones have been thoroughly heated, the fire is removed and the meat cooked on the hot stones. The Worora at times cut the carcase open and place a number of heated pebbles inside.

River-worn pebbles, measuring four or five inches in diameter, are also extensively used by all central tribes, such as the Dieri, Aluridja, Wongkanguru, Ngameni, Arunndta, Wongapitcha, and Kukata, in conjunction with a large flat slab, as a hand-mill. The slabs or nether stones, which are generally known as “nardoo stones,” are longish-oval in shape, and up to two feet in length.

The Wongapitcha use slabs of no particular shape, which they call “*tchewa*.” The upper surface is flat or concavely worn through constant use. It is the gin’s lot to work the mill. She kneels in front of the slab, with its longer axis pointing towards her, and places some of the seed she wants to grind upon it; then she starts working the pebble forwards and backwards with her hands, rocking it gently in the same direction as she does so. When ground to a sufficient degree of fineness, the flour is scraped by hand into a bark food-carrier, and more seed placed upon the slab. On account of the rocking motion, the hand-piece, which the Wongapitcha call “*miri*,” eventually acquires a bevelled or convex grinding surface. Fine-grained sandstones or quartzites are most commonly found in use, but occasionally diorites and other igneous rocks might be favoured. The women usually carry the hand-stone around with them when on the march, but the basal slabs are kept at the regular camping places.

Along the Darling River, and in the west-central districts of New South Wales, the nether stones consist of large sandstone pebbles, in the two less convex surfaces of which perfectly circular and convex husking holes have been made in consequence of the daily use they are put to.

Haphazard rock fragments, usually of sandstone, with at least one broken surface, are extensively made use of for rasping and smoothing down the sides and edges of boomerangs, and of other wooden articles during the course of their construction.

Any suitable, flattish-oblong pebbles of hard quartzite, diorite, dolerite, and other igneous rock of homogeneous and finely crystalline texture, which have been symmetrically worn by the weather, are collected by the natives during their excursions and subsequently worked up into hatchet

heads. This is done by obliquely chipping or grinding that of the smaller sides which is considered the more suitable, on one or both faces, until a straight or convex cutting edge results. The chipping is done with another fragment of hard rock, the grinding against an outcrop or slab of sufficiently hard stone which happens to be handy. The shape of the pebble is in most cases improved by chipping it before the cutting edge is ground, according to whether it is going to be ovate, triangular, or elongate-oblong when completed. Some patterns, such as those of Victoria, New South Wales, and the eastern-central region of South Australia (Strzelecki Creek), have a transverse groove cut right around the piece, at about two-thirds the whole length from the cutting edge, which is designed to hold the wooden haft when the implement is in use.

In many of the tribal districts igneous rocks do not occur naturally, but they are nevertheless obtained by barter from adjoining friendly tribes. The Dieri, Wongkanguru, Ngameni, and other Cooper Creek tribes obtain all their stone axe heads from New South Wales; the south-eastern tribes of South Australia used to receive their supplies from the hills tribes of what is now Victoria; and the Aluridja, Kuyanni, Arrabonna, and Kukata were regularly supplied from the MacDonnell Ranges and from Queensland through Arunndta agency. The fortunate tribes who owned outcrops of suitable stone carried on a regular trade with the surrounding districts and opened up quarries to meet the demand. The supplies were, however, not tribe-owned, but usually the property of a limited number of men who came to them by hereditary influence. Similar conditions are met with on the north coast; Sunday Island, consisting essentially of coarse-grained granitic rock, the natives have to import most of the material they use for making their stone implements from the mainland opposite; in consequence, they are loth to part with their weapons.

The stone axe head is fixed to a wooden handle after the following fashion. A long, flat piece of split wood or wiry bark is bent upon itself and tied together at its ends. The stone is thickly covered at its blunt end with hot porcupine grass resin and inserted into the loop of the haft, which is firmly pressed into the resin against the stone and tied together with human hair-string as near to the stone as possible. The free ends of the handle are then also tied together; after which the resin is worked with the fingers to fill up any gaps which may remain between the handle and the stone. The

handle, and often the axe head as well, is decorated with punctate and banded ochre designs.

The size of the stone axes varies considerably; as two extremes, a large Arunndta specimen from the Finke River measures nearly eight and a half inches in length, by three in breadth, and weighs three and three-quarter pounds, whilst one from King Sound, in the north of Western Australia, measures three and a half inches by two and a half, and weighs only six ounces, the handle of the latter being only six and three-quarter inches long.

The flakes and splinters which fly from the pebble during the making of an axe head are not all discarded as useless by the native; among them he often finds one or two pieces which have a strong sharp edge with a butt opposite, suitable for holding between two or more fingers. Flakes of this type make useful scrapers with which he can work the surfaces of his wooden weapons and implements.

The same flaking and chipping process is purposely applied to rocks of a particularly hard and brittle nature, such as a fine-grained, porcelainized quartzite or chert, to obtain flakes for cutting, scraping, and holing purposes. Many of the best operating “knives,” with which initiation mutilations are performed, are derived in this simple way; as might be imagined, some of these implements are as sharp as a razor.

One frequently finds a fair-sized block of suitable stone among the paraphernalia of a native in camp, from which he chips pieces as he requires them. These blocks have been termed “cores” or “nuclei”; they are six inches or more cube in the beginning, but by the time a goodly number of flakes have been removed, the parent piece becomes much smaller and gradually assumes the shape of a truncated cone whose surface shows many faces from which flakes have been knocked off.

When deciding upon a place for removing a flake, a native always selects a corner, in order that the detached piece might be triangular in transverse section, and, therefore, without exception, lanceolate in shape. Thus the simple flakes are obtained which make stone knives and spear heads. To serve as a knife, the flake is fitted with a handle in one of the following ways. It may be attached by means of porcupine grass resin in the bend of a folded haft of wood, as described of the axe above, or its thick end may be held in a cleft, made at the top of a stick, and secured by a good quantity of resin.

The simplest form, however, is one common throughout the central and northern regions; it consists of a blade of quartzite embedded at its blunt end in a round mass of resin. The largest stone knives come from the tribes immediately north of the MacDonnell Ranges. The Kaitidji make quartzite blades up to seven inches long and two and a half inches wide, which they embed in a ball of resin and attach to the top of a short, thin, and flat slab of wood. The blades of these knives are protected by keeping them in sheaths of bark when not in use.



Rock-drawings of archer fish (*Toxotes*), Katherine River, Northern Territory.

PLATE XLVIII

Rock-drawings of archer fish (*Toxotes*), Katherine River, Northern Territory.

The coastal tribes of the Northern Territory, such as the Wogait, Mulluk-Mulluk, Ponga-Ponga, and Sherait, break similar flakes of quartzite from a

core, which they insert into the split end of a reed spear and make secure with a mass of resin or wild bees' wax.

A narrow, oblong fragment, with the two long edges bevelled on the same surface, such as would be obtained by removing two flakes from the same spot, and keeping the lower, finds considerable application in the sense of a spokeshave. The implement is specially prized when it is slightly curved. Much of the trimming, smoothing, and rounding of wooden surfaces is accomplished with this tool. The native sits with his legs straight in front of him and holds the object he is shaping (like for instance the boomerang shown in [Plate LV](#), 2) tightly between his heels. He seizes the stone flake with the fingers of both hands, leaving a clear space of about an inch in the centre, and laying the cutting edge against the wood, pushes it forwards at an angle. This process planes down the surface very effectively, and the ground soon becomes covered with the thin shavings produced.

In former days the River Murray and south-eastern tribes used pointed splinters of stone for making holes through the skins of animals they made up into rugs. Nowadays the northern tribes make awls out of bones which they sharpen at one end; they are used principally for holing the edges of their bark implements prior to stitching them together with strips of cane.

By additional chipping, the main flake, whether obtained from a nucleus or otherwise, is often altered considerably in appearance, without necessarily improving its effectiveness as an implement or its deadliness as a weapon.

The south-eastern natives, as, for instance, those of the Victorian Lakes district, as well as those of central Australia, used to select a flat fragment of hard rock, into one straight side of which they chipped a shallow concavity; this instrument answered the purpose of a rasp when finishing off such articles as spears, waddies, and clubs which had cylindrical, convex, or curved contours to bring into shape.

The old Adelaide plains tribe were in possession of scrapers which they constructed out of thin slabs of clay-slate. The implement was more or less semi-circular, but had a concave surface on the inner side; occasionally its corners were rounded off, producing a reniform shape. On an average the diameter was something like four or five inches. This implement was used exclusively to scrape skins of animals, after the following fashion: The

convex surface was pressed against the palm of the right hand and securely held between the body of the thumb and the four fingers. The skin was laid around a cylindrical rod and held firmly against it with the opposite hand, while the implement was placed over the skin with its concave surface so adjusted as to fit over the convexity of the rod. In this position, the scraper was worked downwards, or towards the native, with its concave surface running ahead of the hand and shaving the skin. Thus the skins were thoroughly cleaned, and all adherent pieces of fat and flesh removed. Slate scrapers of this type are still to be found in large number in the drift sands along the shores of St. Vincent's Gulf, especially in the neighbourhood of Normanville. Vide [Plate XLIII](#), 3.

Most types of spear-thrower carry a scraper embedded in a mass of resin or wax at the handle end. The scraper most favoured is either of quartzite or of flint, about an inch or slightly more square, and chipped on one or both sides of the cutting edge. It is almost wholly embedded, with perhaps only the chipped portion showing below the binding mass which helps to form the handle.

Similar stones are fixed at both ends of a curved or straight piece of wood, of circular section, which is then used as a scraping or chopping tool commonly referred to as an adze. Used as a scraper, the wooden handle is gripped at about one-quarter its length from the bottom, both hands being at the same level, with their fingers overlapping and the thumbs lying against the wood on the opposite side; but when used as a chopper the hands are held one over the other, each clenching the handle separately. Occasionally one hand only is used to direct the tool, while the other holds the object to be worked ([Plate LV](#), 1).

The small, sharp flakes which are chipped from a bigger piece during the construction of a scraper are carefully examined and the most shapely of them are collected for the purpose of sticking them lengthwise, one behind the other, to the two edges of a bladed, wooden spear head. This type of spear was common along the lower reaches of the Murray River and Lake Alexandrina.

In Western Australia a special type of knife called "*dabba*" is constructed in a like way, but the flakes are larger, three-quarters of an inch long, and

embedded in resin along one side only of the stick. The implement measures about two feet in length.

The long, single flakes, obtained from a quartzite core, may be further chipped along the edges to sharpen them. This process is seen typically along the coastal districts of the Northern Territory, the Daly and Victoria River districts in particular.

But where the manufacture of stone spear heads is seen to perfection is in the northern Kimberleys of Western Australia. The north-western tribes are expert at making lanceolate spear heads with serrated edges and beautifully faceted sides; some of the specimens are up to six inches long and are delicately chipped all over. People who have not had the opportunity of witnessing the method employed in making them are perplexed to understand how it is possible to accomplish such delicate work without breaking the object; the point in particular of these spear heads is often nearly as fine as that of a needle.

The way it is done is briefly as follows. A rough flake or fragment is broken from a core, or rock *in situ*, by holding a bone chisel or stone adze in much the same way as one clasps a pen or pencil, and stabbing the block near the sharp edge, or by striking it with another rock fragment. The size of the flake thus detached will depend largely upon the purpose to which it is to be put; the fractured surface is always plane. The fragment is now taken in the left hand, its flat surfaces lying full length between the thumb and fingers, and its edges chipped by striking them from above with a sharp stone hammer held securely in the right hand. The flake is frequently changed about, so that what is now the bottom surface later becomes the top. The edge is always struck nearly at right angles to the flat surface whilst the chips break away into the hand underneath. The Yampi Sound natives call the rough primary flake "*munna*," and the small chips resulting from the trimming of its edge "*aroap*."

The original shaping aims at obtaining a roughly symmetrical leaf-form, truncated at the base where it is subsequently to be embedded or held at the end of a spear or haft. The flake is left thick at its base and made to taper towards its point.

At first the chipping is done by fairly strong, but well-directed blows from above; later by quicker and lighter taps. Occasionally the edges are

rasped with a flat slab of sandstone at right angles to the plane of the flake—a process which breaks away small chips from either side of the edge which is being rubbed. The flake at this stage is called “*ardelgulla*” by the Yampi natives, and “*arolonnyenna*” by the Sunday Islanders.

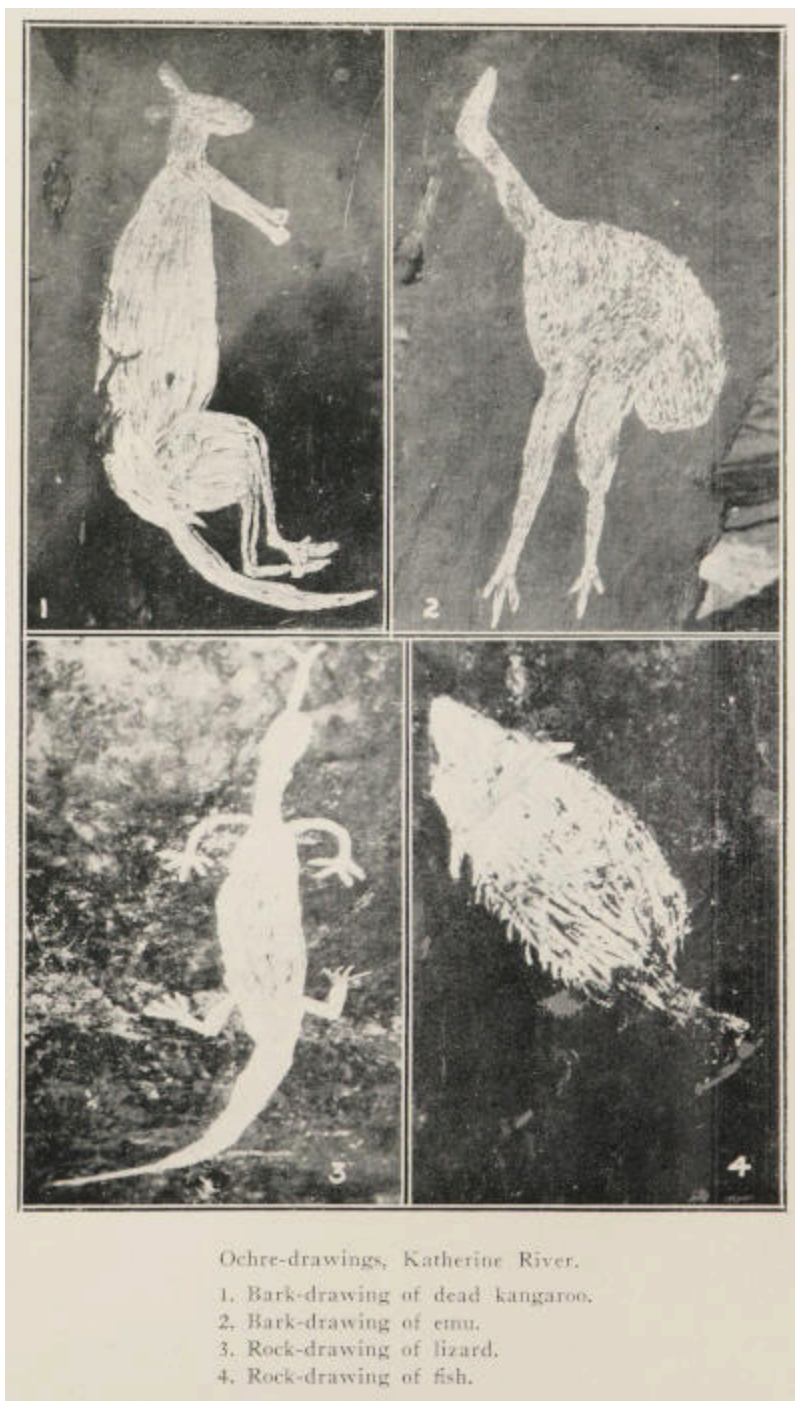


PLATE XLIX

Ochre-drawings, Katherine River.

1. Bark-drawing of dead kangaroo.
2. Bark-drawing of emu.
3. Rock-drawing of lizard.
4. Rock-drawing of fish.

When the preliminary shaping has been completed in the way described, the native first strives to obtain a perfect point, then to trim the sides. The former item is a very delicate operation which requires much patience and skill; the latter takes many hours to complete.

The method adopted for this finer secondary chipping process is after the following principle. A block of stone, about a foot cube, is used as a working table or anvil, which the Worora call "*muna*," and upon this they lay a cushion consisting of two or three layers of paper-bark, called "*ngali*." The native sits with the stone in front of him, and in his left hand grips the unfinished spear head ("*ardegulla*") with one of its flat surfaces lying upon the cushion at the near, left-hand corner of the anvil stone. His thumb, index, and middle fingers hold the flake, the thumb being on top, the index finger against the edge, and the middle finger beneath; the two remaining fingers press against the edge of the block below to steady the flake upon the cushion. In his right hand he seizes a short rod of bone, which is sharpened at one end, and is known as "*onumongul*," in such a way that the unsharpened end is securely gripped between the thumb and index finger, whilst the principal portion of the rod is pressed with the remaining fingers against the palm of the hand immediately below the body of the thumb. The sharpened point of the bone thus points towards the native's body. Holding the implement in this position, he rests the small-finger side of his hand near the further right-hand corner of the basal block of stone, and, after carefully adjusting the point of the bone against the edge of the flake, he presses it down with the body of his thumb and skilfully snaps off a chip. The process is repeated again, and time after time, the position of the bone being constantly changed as he works along the edge towards the point. Then the flake is turned on its other side and the same method applied. As the native works, the whole of his attention is absorbed. He bites his lips together, and, when he applies leverage with the bone against the flake, he stiffens his body from the hips upwards, his eyes being rivetted to the spot

from which the chip is to be removed. He frequently sharpens the point of his instrument upon the basal block of stone. Vide [Plate LIV](#), 2.

The most delicate final chipping of both the point and margins is executed with a thinner and more finely pointed bone, which is usually made out of the radius of a kangaroo. In districts coming under the influence of European settlement, the bone is often substituted by a piece of iron, and the stone by bottle glass or porcelain.

During the operation the native often cuts his fingers on the flake or razor-sharp splinters; the blood which follows he removes by passing his fingers through his hair. Even at this stage, when the flake is assuming a symmetrical, lanceolate shape, and goes by the name of “*tanbellena*,” its edge might occasionally be very carefully rubbed on the basal stone; but the final *retouche* is invariably given to it with the bone implement.

At no time during the making of the spear head does the native use his wrist, the whole of the pressure or movement coming from his elbow or even from his waist, while his body is kept in the rigid position referred to above. The finished spear head is called “*ngongu nerbai*” or “*kolldürr*.”

The process described is of such a delicate kind that the point not infrequently breaks just when the spear head is practically ready for use; this necessitates not only the construction of a new point, but the margins on both sides of it have to be chipped back in order that the point may be a projecting one.

One has to admire the industry of these men, when it is realized that the spear head in ninety-nine cases out of a hundred will be good for only one throw, the brittle stone shattering immediately it comes into contact with a solid body such as the bone of the prey or the ground.

CHAPTER XXX

MUSIC AND DANCE

Talented mimicry—Association of sound with music—Beating time to dance and music—Musical instruments—Skin drums—Rattles—Clanking boomerangs—Music sticks—Bamboo trumpet—Artificial fireflies—Vocal productions—Inflection of voice—Rhythm—Corroborees and boras—Imitative notes—Crocodile—Emu—Crow—Frog—Wailing women—Jungle fowl—Clever acting—Kangaroo—Fight—Man-of-war—The hunting gin—Killing the bandicoot—Slaying the enemy—Envious of chirping insects—The effects of singing ensemble—Conversation by song.

An aboriginal is a born mimic. Nothing delights him more than to reproduce from Nature incidences and scenes before an appreciative and visionary audience. But in the same sense as detail of design in his drawings or carvings is often deemed unnecessary by his vivid imagination, so to the outsider his acting might seem tainted with an air of becoming vagueness which makes it appear pantomimic. Yet, as a conversationalist, an aboriginal is usually so animated by the recollections of his experience that he unconsciously becomes a dramatist, and his narration an epic. Granted the necessary perception, however, the feelings and emotions which actuate the performer are readily grasped by those observing his dances, and whose sympathy he is courting. He lets himself go, without mock-modest constraint, endeavouring by every action to interpret with his body the impulse he has received. Lured by the glint of an inspiration, his nearer vision is blinded, his musculature quivers involuntarily, and his only desire is to catch, to imitate, and to give expression to his exalted feelings. Held in a rapture, his feelings transcend anything he ordinarily perceives, his staid personality has vanished, and all that the inner individual attempts, or can attempt, is to externalize by his movements those sensuous, but illusive, impressions his soul is imbibing.

To many the real interpretation of such movement would be impossible; but the aboriginal lives for his dances, of which he possesses an almost inexhaustible variety, the outcome of tradition and invention. He has learned to make his dance a medium of sensual expression, and to combine an instinctive impulse with movement. By his dancing he impersonates both friend and enemy, he copies the hopping of a marsupial, or the wriggling of

a serpent, or the strutting of an emu, and he emulates the legendary practices and sacred ceremonials of his forefathers. In his dances lives the valour of his warriors, and dies the evil magic of his foes. Through his dances he endeavours to commune with the spirits of his dead, he hears the voices of his mythical demigods, and he beseeches his deities to protect his person and to bless his haunts with an abundance of game.

The magnetical charm about a tribal dance lies in the rhythmic motion of the performers, in the harmonious way their naked bodies sway to the accompaniment of crude but effective music, and in the clever association of sound with motion and silence with rest. The dancers are mute during their performance, the music being supplied by a band or chorus of either men or women, or both, who squat near by. A performance without musical items is practically unknown. The dancers keep their movements and steps in such remarkably true accord with the vocal and instrumental parts that it is difficult to dissociate one from another; in addition, the rhythm for each new dance is usually set by the audience and followed by the dancers to the instant.

The beating of time is usually done by hand, especially if women are attending the performance. In most tribes, the person squats on the ground, holding the thighs together, and strikes the cleft thus produced with the palm of a hand. More commonly both hands are used together, with the inner side of one laid over the back of the other, and the fingers of the lower one placed together in such a way as to form a concave surface. By this means, loud, explosive sounds are produced.

On Melville and Bathurst Islands, and on the Victoria River, the palms of both hands are struck against the buttocks, one on either side of the body, while the person is standing. Along the coast of the Northern Territory, the natives, as often as not, simply clap the hands in rhythmic order, or they slap the palms of one or both upon the ground; occasionally one even notices mothers gently slapping the buttocks of their babies-in-arms, all under the impulse of a catchy air which is striking their ears.

A peculiar sort of sound accompaniment is rendered by the women dancers of the Katherine and Victoria River districts of the Northern Territory. As each of the dancers hops forwards in a straight line, with her heels together and her feet turned outwards, she jerks her body in mid-air

and whacks the muscles of her thighs together, an act which produces a loud, sharp sound. In this way she moves both forwards and backwards, making a similar noise with every hop, whilst her feet make a track in the sand which is to represent the female turtle coming on shore to lay its eggs.

In the same districts, as well as on the Daly River, the dancing gins use skeins of string stretched between the thumbs of their hands, which they sway to and fro like the bow of a fiddle. Although this manoeuvre does not produce a sound, it is here mentioned because the movement takes place in perfect rhythm and in unison with the singing which is going on; and one is reminded of a modern conductor using his stick.

If we now turn our attention to the consideration of the accompaniment produced with musical instruments, there is a small choice at our disposal. We find that certain of the southern tribes, along the River Murray, made use of skins, which they stretched across their thighs, as they sat upon the ground, and struck with their hands or a stick like beating a drum.

In the Kimberley district of Western Australia, the large nuts of the boabab, when dry, are used after the style of the European toy known as a baby's rattle by the children, but curiosity soon leads to the destruction of the shell, when the pithy matrix and the seeds are eaten. Occasionally these nuts are introduced into ceremonial dances by the men; they are then elaborately and beautifully carved as previously referred to.

In the same district, and in fact all along the north coast, large convoluted sea-shells with a small pebble inside of them, or even a number of smaller shells threaded upon a string, serve the same purpose of noise-making.

Bundles of gum leaves, fresh or dry, tied round the ankles or arms of the performers, produce a rustle which imitates the noise produced by the wiry feathers of a romping emu. Most of the tribes adopt this scheme, especially in connection with sacred festivals and ceremonies having to do with the emu.

Among the central and northern central tribes, the boomerang is extensively used as a musical instrument. The operator, taking up a squatting position, holds a boomerang at half-arm's length in each hand, so that the concave edges are turned towards his body. Then by bringing the instruments near each other, with their surfaces parallel, he claps their ends

together in quick succession, and by so doing produces rhythmic clanks to suit the step of any dance or the time of any song ([Plate LII](#)).

The Larrekiya, Wogait, Berringin, and other Northern Territory tribes make use of “music sticks.” Two of such are required. One is of hard “iron-wood,” about nine inches in length, flatly cylindrical, and bluntly pointed at one end; the other, which is the beating stick, is simply a smaller rod, of circular section, made of light mangrove wood. The former stick is held firmly in the left hand, whilst it is struck by the latter, not far from its end. The beating stick is held in the right hand with one end of it pressing either against the third or fifth finger. The sounds produced by the percussion are ringing, sub-metallic clanks; and any alteration in the length of the free end of the beating stick naturally tends to vary their pitch.

The instrument which is capable of producing the loudest, and, at the same time, most weird sound, when correctly manipulated by an aboriginal, is the bamboo trumpet, otherwise known as the drone-pipe or “*didjeridoo*.” This consists of a piece of bamboo, of the stout, tropical variety, from four to five feet long, the septa of which have all been burnt out with a fire-stick. The outside surface is decorated with engraved designs. Drone-pipes are used by all coastal tribes living between the Gulf of Carpentaria and Cambridge Gulf, and as far inland as Wave Hill on the Victoria River. Where the bamboo is not available, the instrument is made out of a long hollow limb of the woolly-butt eucalyptus; this is the prevailing type in the western portion of the area mentioned. To serve the requirements of a single night’s performance, a green stem of a native hybiscus bush might be cut off and the thick bark removed *in toto* in the form of a pipe.

When using the “trumpet,” the operator blows into the end having the smaller diameter, with a vibratory motion of the lips, and at the same time sputters into the tube indistinct words which frequently sound like “*tidjarudu, tidjarudu, tidjaruda*” ([Plate LIII](#)). The effect, though rhythmical, is a monotonous, plaintive, and humming sound which is continued uninterruptedly throughout the proceedings. The native, while he is blowing into the pipe, continues to breathe normally through his nostrils, after the same style as one does when using a blow-pipe in the laboratory. In the stillness of the tropical night the droning noise can be heard for miles around. The wording of the accompaniment on the bamboo trumpet during

a Larrekiya performance sounds much like the following refrain:
“*Didnodiddo diduadu didnadiddo diduadu ... didnarib.*”

In addition to music and dance, a unique, and certainly most effective, pyrotechnical embellishment of a nocturnal ceremony is supplied by the Dieri. Along the Cooper Creek, travellers have occasion to notice the great number of large beetles which fly towards the camp-fire at night-time. When a dance is on, a collection of these beetles is made and short glowing embers inserted into their anal apertures; whereupon they are released again. As the naked figures of the men are moving to the sway of song, these little fire-balls buzz and flit in among them, and, cruel as the invention may seem, greatly add to the weirdness of the din. The Dieri call these artificial fire-flies “*turapitti.*”

Vocal productions consist of recitals of notes which are frequently encased in articulations without definite meaning or significance, the notes alone expressing the sentiment which prompts the song. It is the combination of these notes which gives rise to the simple melodies, and the repetition of the melodies in regular sequence makes the song. In his songs the aboriginal portrays the hate for his foe with vehemence, the love for his child with affection, the spirit of the chase with lustfulness, the cunning of his prey with counter-deception, and the dignity of his forbears with veneration. As the pulsations of his temperament and passion sway his mind, so his voice rises or falls in harmony with the flush of joy or the gloom of sorrow.

When singing in chorus, the monotony of a melody is frequently re-animated by one of the principal singers, who, with a stentorian inflection of his voice, leads off anew. In this way, the pitch of a melody is repeatedly altered by one or two of the recognized vocal experts; but at all times the pitch relations remain in perfect concord with each other.

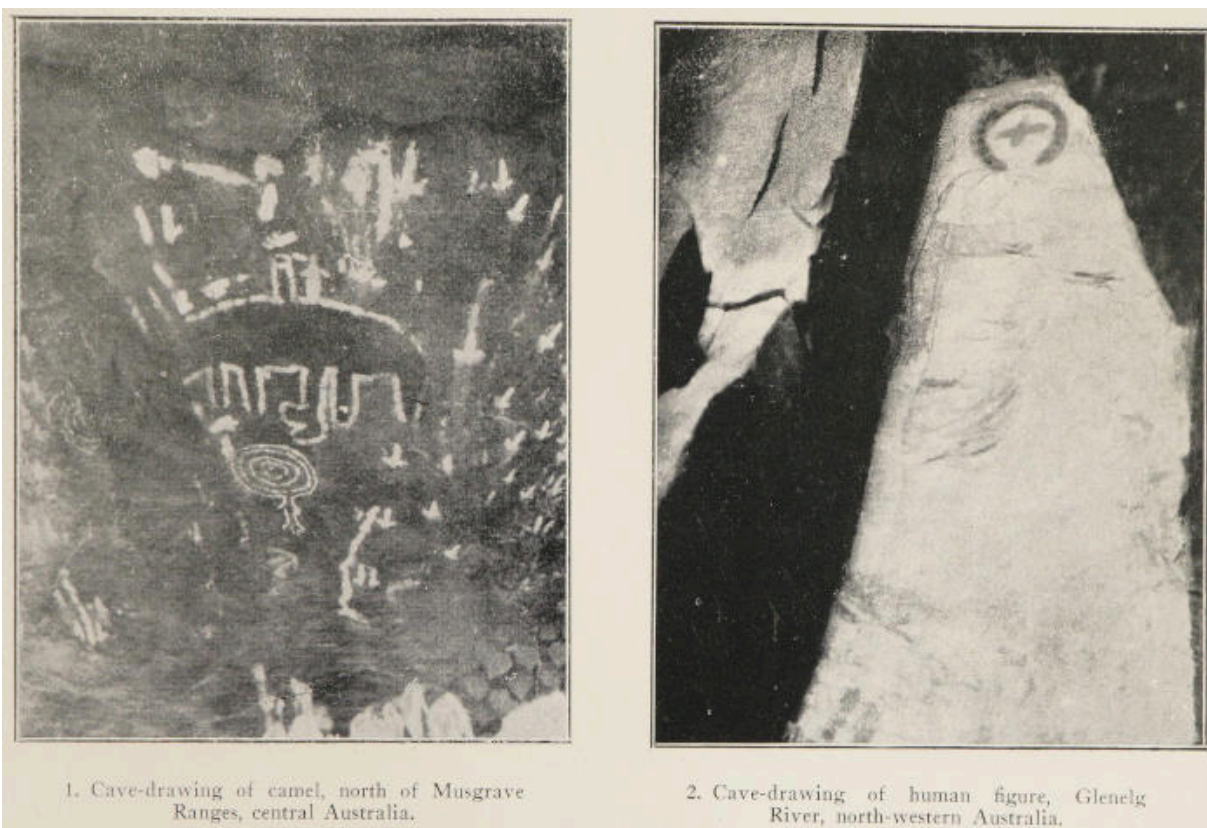


PLATE L

1. Cave-drawing of camel, north of Musgrave Ranges, central Australia.

2. Cave-drawing of human figure, Glenelg River, north-western Australia.

It is considered distinctly artistic to be able to frequently change the pitch of the voice from a deep bass to a shrill falsetto at will, and only the most experienced singers attempt it. A new tune is introduced by one of the older men, and the same person will later infuse new vitality into it by picking up the strain at different stages by a clever inflection of the voice, after the style of a rondo.

The rhythm throughout the proceedings remains excellent, but great variations are met with during the rendering of different items; it is always in keeping with the dance, if the latter is indulged in, even at the risk of running away momentarily from the time of the music.

Performances which include dances as well as songs in the way of entertainment are generally called corroborees; events of a ceremonial, ritual, or religious nature are termed "*boras*."

The notes included in the songs of tribal performances are often imitative of the voice of Nature, and among them we find allusions to the calls and cries of birds, animals, reptiles, and mythical creatures. At the same time, any characteristic actions or attitudes are faithfully reproduced as special features of the dances.

In the crocodile ceremony of the Cambridge Gulf natives, a number of men stand in a row, one behind the other, with their arms extended and their legs asunder, whilst the individual impersonating the crocodile ancestor wriggles along the ground between their legs. When he comes abreast of the foremost man, he lies flat on the ground, with his legs and feet held closely together to imitate the reptile's tail. To further mimic the crocodile, he extends his arms sideways, strongly bent at the elbows, and with the hands flat upon the ground. Retaining this position, he next elevates his body by straightening his arms, and, when fully erect, opens his mouth and emits a harsh, booming note resembling that of a crocodile.

In the same district, the great emu man, during his ceremony, walks within a human circle, his body prone from the hips, with one arm held forwards to represent the emu's neck and the hand of the other held over his stern to indicate the tail. As he walks around bowing his body, after the fashion of a strutting emu, he eructates deep, guttural noises, resembling the grunting note of the bird.

How the caw of a crow is embodied in the musical programme of a ceremony will be apparent from the following episode which was transacted at the Forrest River. A number of men stood in a ring, and, at a given signal, lowered their bodies between their knees. They let their heads fall forwards, and at the same time lifted their arms, which they bent in the elbow to resemble wings. The latter they moved lithely to and fro after much the same way as a young bird does when it is being, or wants to be, fed. At this moment a chant was started in imitation of the crow's call: "*A wa, a wa, a weh!*" and was oftentimes repeated. Then they all hopped around like so many birds in search of food, and two men entered the ring. Still in the same posture, these two hopped towards each other and extended their arms until each pair crossed the opposite pair. In that position they swung their bodies backwards and forwards, whilst their arms sea-sawed in front of them. Then they re-joined the group, and all continued the hopping. In the next act, an old man lay flat on his back, in the centre of the ring, with

his arms and legs stretched from him. He represented a carcase. The “crows” hopped around him and cried: “A, a, a, la, la, la-la-la,” and it sounded very much like the caw of a crow. This item was repeated. One of the crow men then hopped to the “dead” man. He lifted one of the arms from the ground, held it up, and let it go. The limb fell “lifelessly” to the ground. Immediately this had happened, all performers jumped into an upright position, rushed towards the man feigning death, and carried him from view. Apart from the imitation of the crow’s call, no regular song accompanied the act, but all the onlookers were beating time, to correspond with the hopping, by slapping their hands against their thighs.

The vocal accompaniment at a ceremony of welcome on Bathurst Island is in the form of a trill, a rapidly repeated “i, i, i, i, i ...,” changing occasionally to “hi, hi, hi, hi ...,” which is very cleverly reproduced in imitation of the note of the great stone plover.

In the corroboree of a frog, the Larrekiya sing the following refrain:

“Iwoggiana wogien woggiana wogien, immangana jellerrima wogien.

Imbaka kwuerakwa wogien. Imbelluoluodö wogien.”

Then they pause for a moment, and, continuing immediately, mimic the croaking by crying:

“Yi kwa e kwa, yi kwa e kwa!”

The singing is assisted by an accompaniment on the bamboo trumpet which sounds like: “*Tip bu, tip bu, tip bu, tip pau-a, tip pau-a, tip pau-a.*” Music sticks are used by all taking part in the corroboree, the percussions being equally timed and of uniform strength.

A little performance, melodramatic in a way, which is very popular with the Larrekiya and Wogait in the Northern Territory, is designed to scoff at the wailing of women. It is introduced by the following refrain:

“Arada kadji mara uda kayan yan; arada kadji mara uda kayan mölle ulpululu mölle, jangaji karra kanjin mada da nga la ulmin ja jandadbi karra kun kun mukale la enbulukdi bana mölle mara kanjinmada da buluk di bölle arada kadji balla.”

Then the wailing begins. With a pathos and sincerity so cleverly feigned that one might easily be led to believe they are genuine, the men strike up a lament which is thrilling. Monosyllabic sounds pierce the air, which terminate with a sob:

“Lö -e, lö, lö -e, lö -ö, lö -ö la.”

Then a burst of laughter ends the solemnity; and the refrain is continued amidst much hilarity:

“Kanjin mada danö wüdningi juan madji karra kunkul mukana mölle enbuluk di bana mälle kanjin mada.”

At the conclusion, a general criticism takes place of the skill displayed by the different performers in the reproduction of the effeminate sounds. Throughout the proceedings an accompaniment is played on the bamboo trumpet sounding like: *“Tib bu, tib bu, tib bu-a”*; whilst a number of men beat their music sticks in a rhythm represented by one long and three short taps.

The Bathurst and Melville Islanders are famous for their impressive ceremonies. At most of them, a group of men stand in a semi-circle and supply the accompaniment by clapping one or both hands against thigh or buttock, producing dull, drum-like sounds by the impact; they further supplement the rhythmic sounds by crying, *“brr, brr, brr”* in unison with the beating. Any actors who enter the ring in front of these men frequently ejaculate a sound like a short *“poop,”* and this immediately solicits a concerted *“iah”* from all taking part.

In their corroboree of the jungle fowl, an old man rushes into the ring, bending his arms at the elbow and holding them close against his body, while he rests his closed fists upon his chest. His attitude resembles that of a professional runner, although in reality his arms are supposed to represent the wings of the bird. The man looks about the ground, from side to side, as though in search of something. Presently he imitates the familiar call of the bird, with a shrill penetrating voice, which sounds like *“he-r-r-l.”* In answer to this, the chorus surrounding him spontaneously gives utterance to a number of short, non-articulate noises which sound like *“brrl, brrl, brrl,”* and are claimed to be the call of the bird’s mate. From now on the performer intermittently cries *“he,”* and, every time he does so, the rest of

the men respond with a loud “*yeh*.” Then he throws sand and clouds of dust behind him with his feet, with the object of imitating the way a jungle fowl throws up earth into large mounds in which its eggs are deposited.

When the kangaroo ceremony is announced, one or two men jump into the ring, with their legs slightly bent at the knees. Their arms are held forward, bent vertically at the elbows, their hands being kept prone and partly closed. The whole posture is to resemble that of the hopping marsupial of Australia. In jumping round the space, the actor turns his head from side to side, face downwards, as though in search of food. When such is supposed to have been found, his hands may assist to support his body and the hopping ceases. His legs are then bent strongly in the knees, so that his body sinks and his buttocks come to rest immediately above the heels, or, indeed, he may squat entirely upon the ground. He now holds his hands together and passes them between his thighs to the ground, from which he scoops a quantity of sand and lifts it to the level of his mouth. Suddenly dropping the sand, the performer bounds to his feet and runs round the ring, vigorously stamping his feet and making a peculiar hissing sound. The act is frequently interrupted by cries of “*poop*” from the mouths of the saltant men, which are immediately responded to by the crowd with a boisterous “*iah*.” Then the hopping re-commences, the actor frequently stopping to scratch his leg with one of his hands held like a claw. Occasionally he lifts an arm to his mouth to lick it just like the animal itself does. The pace of the time-beaters at this stage quickens, and the actor begins to scratch all accessible parts of his body as if possessed. An unexpected “*poop*,” followed by a hearty “*iah*,” terminates this part of the performance amid loud shrieks and applause.

It is not long before two performers again leap into the ring, each armed with a light reed in his hand, resembling a spear. The above demonstration is partly recapitulated, being, however, slightly modified in that one man, at intervals, takes the part of a hunter endeavouring to drive his spear into the enraged “kangaroo” on the opposite side of the ring.

The corroboree of a fight is presented by an imaginary spear duel between two angry men. They are “armed” with a light reed each, and upon entering the arena, composed of the time-beating chorus, place themselves in defiance of each other and cut all sorts of antics. They brandish their imaginary weapons, roll their eyes, and throw their bodies about wildly.

Then they start running around the ring, slapping their thighs together, with their heads thrown back and their arms upwards.

As an instance of the clever way in which aborigines mimic special happenings or striking episodes, and also hand on the records to subsequent generations, the ship-of-war corroboree is here briefly outlined. This is a reproduction of events which happened at Fort Dundas about a century ago, but the present day natives still enjoy looking at it and include it among their favourite turns. After the usual preliminaries, a performer rushes into the ring and stands with one leg in front of the other and slightly flexed in the knee. Throwing his head back and looking upwards, he starts a vigorous "hauling action" by alternately throwing one arm out in the direction he is looking, closing the fist and making an imaginary pull right down to his loins, while at the same time the other arm is thrown out and returned in a similar way. This act is in imitation of the hauling in of a sail on a European man-of-war, and is rhythmically followed by the time-beaters. Running to the opposite side of the ring, the same act is repeated. Then the same hauling motion is continued, but this time pulling from below upwards towards the body, in a manner suggesting the heaving of an anchor. During this performance, several sharp and harsh interjections are heard, which suggest a skit on the orders of the commanding officers; in fact, the whole ceremony is to be looked upon as a travesty of naval discipline.

During several of the performances described above, the following chant was heard, in perfect accord with the rhythmic smacks produced by the hands of the chorus:

*"Tupera monan la jerra
Tupera monan la jerra,"*

and

"Tupatunan tenakomejona."

At Fowler's Bay, the men like no dance so much as one which caricatures a woman on a collecting excursion. The performer walks from the darkness of night into the bright glare produced by the camp-fire, with an exaggerated bend of the knees and a loudly perceptible stamp of the feet. Over one of his shoulders he carries a collecting bag, and as he moves

forwards and sideways, his eyes are rivetted to the ground in front of him; he is supposed to be looking for food-stuffs. Suddenly he stops, stoops, and feigns to be picking up something, which is assigned to the bag. Then he hurries on and repeats the same performance. Whilst he is acting thus, the audience, grouped at one side of the fire, is diligently beating time to his steps by knocking two boomerangs (“*kaili*”) together.

In another act he walks quickly into the light, stops suddenly, and looks hard at the ground in front of him. Presently he lifts his right foot, only to immediately bring it back to the ground again with a thud. Emitting a squeal like that of a dying or wounded bandicoot, he stoops and makes out he is lifting his prey triumphantly into the air.

Yet another performance portrays the slaying of an enemy. A second actor, who represents the vanquished foe, is requested to lie in the centre of the arena, whilst the victor dances around him, wildly flourishing and swinging his club. Every now and then the ground is bashed with the heavy weapon as near to the prostrate figure as possible; and after every blow the fallen warrior is seen to writhe his body between the legs of the victor as if he were really suffering terrible agony.

An aboriginal often sings to himself for no other purpose than for his own entertainment, in which occupation he manifests considerable pleasure, and repeats the tune to his heart's content, at times almost to the verge of physical exhaustion. In his endeavour to become a noted singer amongst his tribespeople, a man at every opportunity stimulates his ambition. He listens with envy to the tireless chirping of a cricket and suggests to himself the advantages of acquiring such powers of vocal endurance (as he imagines them to be). The Larrekiya youths admire the large *Cicadae* for similar reasons; and they do not hesitate, upon occasions, to catch one or two of the winged music-makers and suck their viscera, hoping thereby to acquire increased musical talent.

When singing ensemble, the musical productions have widely varying purposes and meanings. The performance might be purely convivial and entertaining, when a number of refrains are hummed or sung, solo and collectively, to the combined accompaniment which is rendered by most of the others present. Any attempt at harmony is wanting. Boisterous music can always be counted upon at tribal war-dances, when the excited and

infuriated mobs almost lose control of their reason, and by singing to their ancestral fathers endeavour to bring destruction upon the enemy by the wildest imprecatory acts.

At initiation ceremonies, the old men, at the time of spilling the novice's blood, are no less excited; and the musical items are reduced to hoarse, rude utterances of a decidedly disquieting flavour. Under these conditions the soul of a true savage unmask itself. It speaks in coarse, disconnected sounds which are hardly recognizable as human, but, at the moment, none the less in sympathy with his inner feelings.



1. Ochre-drawings of mythic semi-human creatures, Forrester River, north-western Australia.



2. Sacred "*Utnguringita*" or witchedy grub drawings, Emily Gap, MacDonnell Ranges.

PLATE LI

1. Ochre-drawings of mythic semi-human creatures, Forrester River, north-western Australia.
2. Sacred "*Utnguringita*" or witchedy grub drawings, Emily Gap, MacDonnell Ranges.

On the other hand, the music which is produced at religious ceremonies is quieter and of a more ding-dong style. The natives present their sacred chants, which they have inherited from their forefathers, for the purpose of getting into communion with the spirits surrounding them; and, although such proceedings may be extremely monotonous, the solemn colloquy is nevertheless musical.

It is a common practice, for that matter, among the tribes of Australia, for one individual to carry on conversation with another by singing the words. When, for instance, it is the intention of the persons engaged in conversation to make the matter as little noticeable as possible, or when they want to impart information to each other without attracting the attention of a third party, they clothe their words in song. And the same is also done when a third party is to be criticized. Moreover, it is for exactly similar reasons that all communications which are supposed to be carried on with superhuman beings are chanted or crooned, in order that the Evil Spirit's attention may not be drawn to the fact.

CHAPTER XXXI

LANGUAGE

Aboriginal dialects euphonious and full—Words largely imitative of natural call-notes—Derivations from other characteristics—Linguistic generalizations impossible—Vocabularies replete in certain directions—Gesture and sign languages—Passionate conversationalist—Reckoning of time—Computation of numbers—Word expressing action—Terminal syllables—Affirmation and negation—Exclamations—Specific suffixes—Diminutive expressions—Verb endings—Word with a variety of meanings—Personal pronouns—Aspirate sounds rare—Consonantal combination “*ng*” very frequent—Greetings—Exclamation and Interrogative—Curious Similarities.

As full as the aboriginal legends are of poetry, so their dialects are full of music. For euphony, fullness, and vowel quality, few modern languages equal the primitive tongue; and much of the charm is derived from a natural source. Even in our own language, the vernacular name of a bird or animal is often derived from the particular call made by the creature. In Europe we have christened the cuckoo, the peewit, and the boo-book owl because of their distinctive cries; in a like sense, the central Australian natives call the bell-bird “*ban-ban-balele*,” the mopoke “*kore-gore*,” and the spur-wing plover “*kurreke-tar-tar*,” all three names being sounded like the natural cries of the birds themselves.

As further illustrations we might mention the Dieri and Arunndta word for kangaroo, viz. “*arre*,” which, one is told, is what the animal “says” when brought to bay, and the Sunday Islanders’ name for a grasshopper, “*dingi-ding*,” which alludes to its chirrup.

In regard to the coining of words, quite apart from the sonorous qualities the object named might possess, the aboriginal is no less inventive. The Arunndta call thunder “*kwatche-ingoma*,” i.e. “roaring water,” and a sharp shower of rain pattering on the ground “*kwatche unndoma*,” i.e. “dancing water.” What is commonly known as the milk-bush in Australia (*Sarcostemma*) is called “*epi-epi*” by the Aluridja because it exudes a milk-like sap when broken and reminds them of the female breast (“*epi*”). Kangaroo grass (*Anthistiria*) is referred to by the Arunndta as “*arre-arre*” because, they maintain, the red colour of its seed stalks resembles that of

the marsupial. The Arunndta for the familiar beef-wood tree of central Australia is "*iltjija*," meaning long fingers, the word drawing attention to the length of the leaves.

Whereas in our own language words like "choleric" have originated because the bile in a human system was supposed to beget wrath, and whereas one bushman might accuse another of being "livery" when he is ill-tempered, the Arunndta have a verb, "*lunbatunbatterama*," meaning to grumble or to be sulky, in which "*unba*" stands for bile.

In view of the great number of Australian tribes and the multiplicity of their dialects, any attempt at linguistic generalizations would seem to be futile, when often even adjoining tribes have adopted totally different vocabularies for the most common commodities of life. As a matter of fact, the groups which build up one and the same big tribe often have considerable differences in their vocabularies. For instance, the eastern groups of the Arunndta make use of very many words and expressions which are quite foreign to the western. Yet in respect of certain words, it is known that with slight modifications their significance has carried far beyond the borders of a single tribe and has been accepted by a group of tribes living up to a few hundred miles apart.

Although the aboriginal tongue is crude in its construction, it is, nevertheless, wonderfully rich and scientifically exact. Whereas a modern language becomes very commonplace in the ordinary course of conversation, and is inclined to handle subject matter somewhat flippantly, the aboriginal system of nomenclature is both profuse and incontrovertible. During his descriptive narrations from Nature, a European rarely bothers about discriminating between objects composing one big class. If he talks of a forest, he is usually satisfied to convey the idea of a number of trees standing collectively at the site his story is dealing with; if he wishes to be exact he might specially describe the trees as pines or gums. But the aboriginal wants, and gives, more; his vocabulary, in fact, rarely includes such words as "tree," "animal," "meat," or "seed"; he tells you immediately, without being specially asked, that the "tree" is a gum, the "animal" a wallaby, the "meat" that of kangaroo, and the "seed" that of the water-lily. Moreover, his verbal supply is so copious that in a single word he can tell one the name, the age, the habitat, and many characteristics.

A language without words is known to most Australian natives; thoughts and messages are communicated by means of gestures from individual to individual, and from tribe to tribe. This system is so perfect, and the code so well understood by all, that important tidings are transmitted from one centre to another in incredibly short time. The method might to a certain extent be compared with the flag-signalling of a marine; it is too complicated to be discussed in detail, but we shall select a few code signals by way of illustration.

Halloa! To attract the attention of a person whom conversation is to be taken up with, the native, standing as erect as possible, and with his legs astride, lifts his hands to his head and swings them outwards (in opposite directions), downwards, and upwards again, time after time. Whilst doing so, he calls aloud, with a shrill, piercing note, even though the other fellow be well beyond hearing distance.

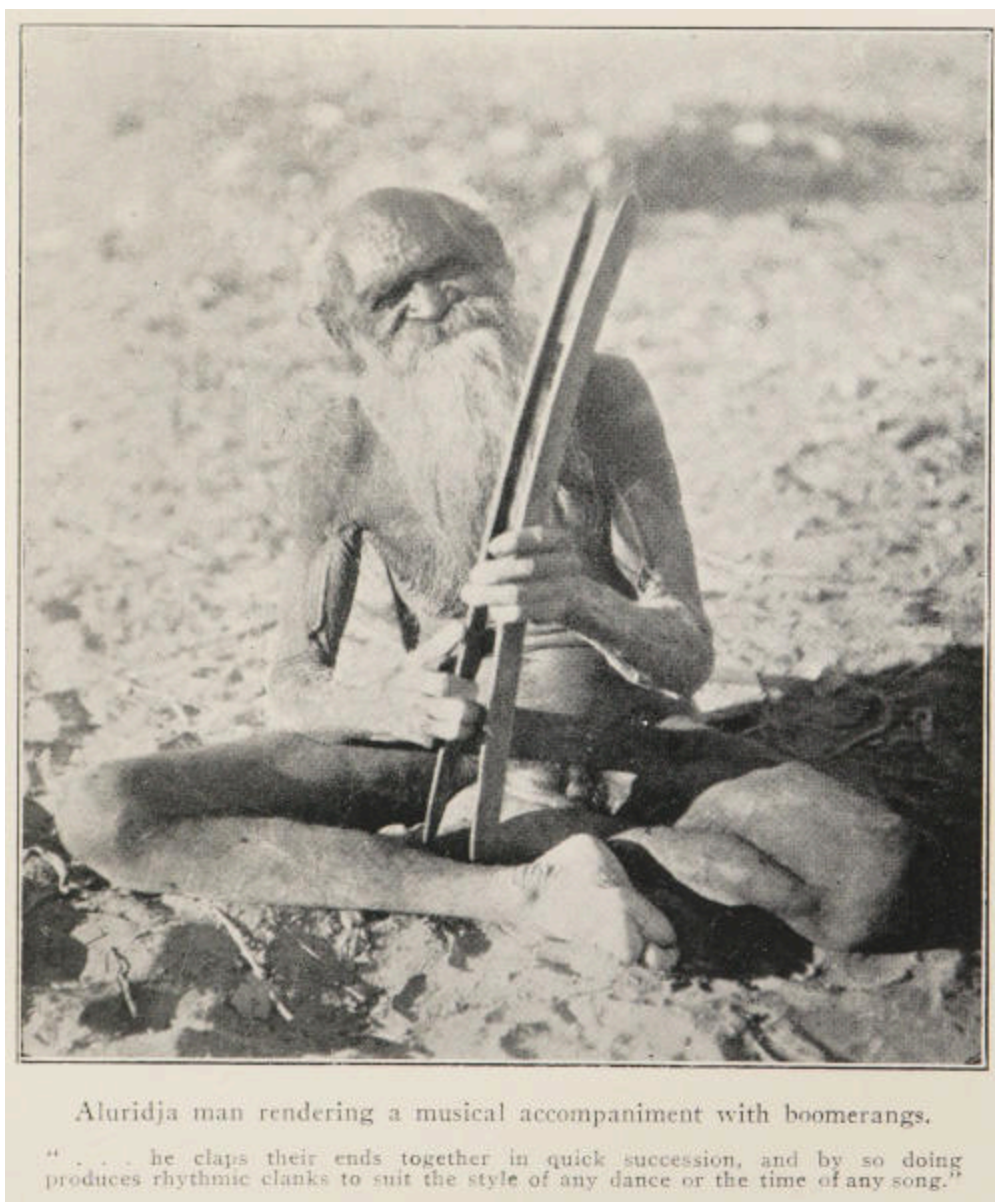


PLATE LII

Aluridja man rendering a musical accompaniment with boomerangs.

"... he claps their ends together in quick succession, and by so doing produces rhythmic clanks to suit the style of any dance or the time of any song."

Come here! Maintaining the erect position, he throws both hands upwards, to above a shoulder on one side, then sweeps them (extended) in front of his body to well behind it on the opposite side, at the same time bending his body forwards from the hips.

I am coming to you. The person places his hands upon his chest, and throws them towards the stranger.

Follow behind me. He slaps his right hand against his buttock.

Stop or stay there. The arms are held, bent at the elbows, at each side of the body, with the open palms of the hands turned towards the distant man, and vertical circles are described with them in that position.

Go that way. He points the index finger in the required direction and jerks the hand forwards several times.

A woman. He places his hands over his breasts.

Kangaroo. The hand is passed, with successive semi-circular movements, indicating the hopping, from one side of the body to the other.

A snake. The hand is moved forwards in a wavy line.

I can see. He touches one or both eyes with his hand.

Dead man. Throws his head back and extends his arms laterally upwards.

A man. He strokes his chin.

I can hear. He points to his ear.

To kill. The palms of the hands are placed together and moved as if striking an object.

Emu. Bends the fingers of one hand at right angles to palm and sways the arm to and fro.

Sleeping. He lays his head upon the palm of a hand.

Sitting. He points to the ground.

Running. His legs are moved in a running fashion.

Hungry. He draws in his abdomen and shows his ribs.

An intricate system of sign language is carried on by movements of the hands and fingers when the natives are at close enough range for the signals to be visible. The following are a few of the more common signs used every day by the Arunndta tribe:

To call the attention of a person, a man holds his open hand in front of his face, with the palm towards him, and drops it again. The sign having been acknowledged, a mutual exchange of gesture correspondence is indulged in, by means of which an almost inexhaustible number of ideas can be communicated in the form of coherent “speech.” The following few signs (most of which are sketched on [page 391, Fig. 57](#)) may be taken as typical of a vocabulary which in its entirety is both elaborate and intricate:

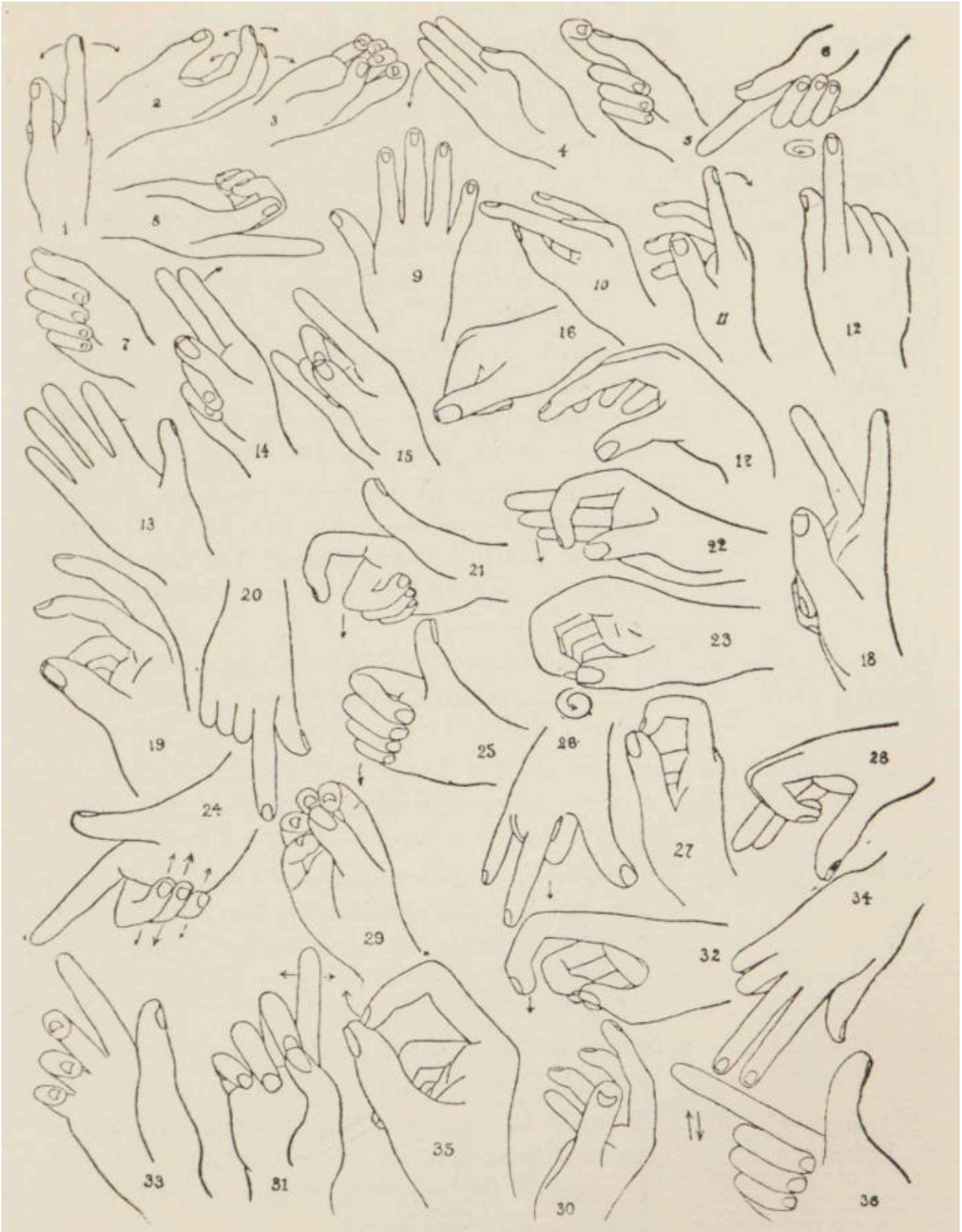


Fig. 57. Sign-language of Arundta tribe.

<i>Which way?</i>	The forefinger is held erect (the other fingers of the hand being closed) and several times moved downwards or away from the face (No. 1).
<i>Big!</i>	The hand is held supine while the fingers are first extended and then closed in again upon the palm. The thumb is kept more or less erect (No. 2).
<i>Little!</i>	The hand is held with its palm upwards, and the fingers moved towards the thumb, the middle one touching it (No. 3).
<i>To rest ("I am going to sit down")</i>	The hand is held semi-prone, with the fingers extended and lying closely together, the thumb resting at the top. It is jerked downwards two or three times (No. 4).
<i>To camp ("I am going to camp")</i>	With the hand in the same position as in No. 4, the middle, fourth, and fifth fingers are closed in upon the palm, while the index is made to touch the thumb (No. 5).
<i>To camp (collectively)</i>	The hand remains in semi-prone position, but with the forefinger pointing towards the ground at half a right angle, and the same three fingers as mentioned in No. 5 remaining closed (No. 6).
<i>To walk</i>	The hand is held supine with only the forefinger extended; it is moved from side to side (No. 8).
<i>Come this way quickly!</i>	Holding the hand nearly horizontal and semi-prone, the fingers are closed in upon the palm and the thumb placed over them; in that position the hand is jerked sideways (No. 7).
<i>Listen!</i>	The hand, with fingers fully extended, is held with its palm turned towards the face and moved to and from the face a number of times (No. 9).

<i>Hear him!</i>	The hand is kept prone, with the middle and fourth fingers closed and the remaining three digits extended. Thus it is several times jerked towards an ear of the man who is “speaking” (No. 10).
<i>What did you say?</i>	The hand is held erect, with the palm towards the face; the thumb and forefinger are kept straight, but the other fingers flexed; the forefinger is jerked to and fro (No. 11).
<i>Fighting</i>	With the hand in an erect position, but the palm turned away from the face, all digits except the index are closed; the last-named, moreover, is jerked with a circular motion (No. 12).
<i>On the summit of a Hill</i>	The sign-maker extends his hand and with the small finger border of it taps the crest of his head (No. 13).
<i>Running up the Hill</i>	He extends the index and middle fingers upwards, while the hand is more or less prone, and jerks them up and down (No. 14).
<i>It is no good!</i>	The hand in a prone position (with the index and small fingers extended, the fourth finger closed in, and the tips of the thumb and middle finger touching) is moved in a circle from the wrist (No. 15).
<i>He is dead</i>	Change the position of hand from that of No. 7 to that of No. 16.
<i>Moving up the Valley</i>	The hand is held partly extended in a prone position and moved horizontally, from side to side, at the side of the man’s head (No. 17).
<i>Behold the ears (of a Kangaroo)</i>	He holds the hand in a vertical position with the index and middle fingers extended, moving them in imitation of an animal’s ears (No. 18).

<i>Steady!</i>	Hold the hand in position indicated in No. 7 and move it vertically with a striking action.
<i>A Euro</i>	Much the same position as shown in No. 18, but the index and middle fingers are only semi-extended; the other fingers are made to touch the thumb (No. 19).
<i>An Echidna or Native Porcupine</i>	The hand is held prone and horizontally, with only the forefinger extended, which is moved regularly from side to side (No. 20).
<i>An Opossum</i>	Hand semi-prone and horizontal; thumb extended and forefinger bent in second joint from the tip; the remaining fingers closed in upon palm. The hand is moved downwards (No. 21).
<i>An Iguana</i>	Hand semi-prone and horizontal. The tip of forefinger is brought nearly into contact with that of thumb, the remaining fingers being fully extended. The hand is moved vertically up and down (No. 22).
<i>A Carpet Snake</i>	Hand semi-prone and horizontal. The points of the fingers are bunched against the thumb, and circular movements made with the hand (No. 23).
<i>A Tiger Snake</i>	Hand semi-prone. The forefinger extended and directed downwards at about half a right angle; thumb extended; the remaining three fingers are moved alternately away from and towards the palm of the hand (No. 24).
<i>A Black Snake</i>	Hand semi-prone and horizontal. All fingers closed in upon palm, thumb extended upwards. Slight downward movements (No. 25).
<i>The Evil Spirit</i>	Hand in prone position and directed downwards. Thumb, index and small fingers extended, the remaining fingers closed. Frequent down and up movements (No. 26).

<i>Native Pear</i> (<i>Marsdenia</i>)	Hand held erect; forefinger flexed to touch the tip of thumb; the remaining fingers closed (No. 27).
<i>Native Orange</i> (<i>Capparis Mitchelli</i>)	Hand prone; index and middle fingers flexed; thumb, fourth and fifth fingers extended. Forward and slightly upward jerks (No. 28).
<i>Marsupial Mice</i>	The hand is held in a supine position, while the points of the fingers are grouped around the thumb and jerked towards the sign-maker's mouth (No. 29).
<i>Native Plum</i> (<i>Santalum</i>)	With all fingers semi-extended, the hand is held in a supine position and jerked towards the face (No. 30).
<i>A Dingo or Wild Dog</i>	The hand is held in a vertical position, with the forefinger extended; the other digits are flexed, the tips of the middle finger and thumb touching. Side movements are made with the forefinger (No. 31).
<i>A Bustard or Wild Turkey</i>	Hand prone; thumb, middle, fourth, and fifth fingers bunched together, the forefinger bent downwards. Vertical movements (No. 32).
<i>A Lizard</i>	Hand semi-prone; forefinger and thumb fully extended, the remaining fingers flexed. Upward and downward movements from the wrist (No. 33).
<i>A Crow</i>	The thumb, index and middle fingers extended, the remaining fingers flexed. The two extended fingers are moved against each other (No. 34).
<i>Wild Bee Honey</i>	The tip of the forefinger is repeatedly flicked from the thumb, while the three remaining fingers are kept in a flexed position (No. 35).
<i>A Wallaby</i>	Hand semi-prone; the forefinger and thumb are extended while the remaining fingers lie

against the palm. Small, jerking, upward movements are made with the forefinger, in which the hand also takes part (No. 36).

My Son

The sign-maker taps his chin with the extended forefinger of his hand.

My Sister's Daughter

He holds his right nipple between the thumb and forefinger of his right hand.

*My Sister's Daughter's
Daughter*

He surrounds his right nipple with the tips of his fingers and alternately opens and closes them with a scratching action.



Wordaman youth playing on the "drone pipe" or "bamboo trumpet."
"When using the 'trumpet,' the operator blows into the end having the smaller diameter, with a vibratory motion of the lips . . ."

PLATE LIII

Wordaman youth playing on the "drone pipe" or "bamboo trumpet."

“When using the ‘trumpet,’ the operator blows into the end having the smaller diameter, with a vibratory motion of the lips....”

In the course of conversation, an aboriginal aids his speech by gesticulation. When inclined to be cheerful, he smiles and laughs in a decidedly refreshing sort of way, and often smacks his lips out of sheer pleasure, but when his feelings have been ruffled and he is angry, he pouts out his lips and does not hesitate to let you know how he feels. He betrays his feelings by most apparent tone inflections in his voice. He frequently nods his head to indicate assent, but rarely shakes his head to betoken dissent. When in doubt over any matter, especially in answer to a question, he shrugs his shoulders. His hands are used a great deal during conversation. If he wishes to give one an idea of nearness, the finger is directed to the ground a short way off, and if distance is implied the finger points to the horizon.

In reply to a question concerned with the location of a thing, he does not point, as a European would, but turns his face in the direction he wants to indicate and thrusts out his lips. When beckoning with his hand, an aboriginal does not move his hand upwards towards his face as we do, but passes it downwards with a scooping action. The course a traveller has taken, or an animal decamped in, is indicated by extending an arm in the direction and snapping the fingers.

A native has a very good idea of time. The hour at which a past event has happened, or at which a coming event is to happen, is indicated by pointing at the assumed altitude of the sun. To fix a time definitely for a contemplated or proposed action, a stone is placed upon a cliff, or in the fork of a tree, at the hour decided upon, a day or two in advance, in such a position that a sunbeam just strikes it at the moment. When the correct day arrives, the stone is watched until it is again illumined in the same way; and the natives then know that it is time to act. This method is employed when, say, the men have left camp early, and they order the women to follow them at a later hour.

Days are reckoned by the number of sleeps they have had, and the biggest measurement of time goes by so many moons.

When he comes to computing numbers, his fingers are of the greatest service to him, and at times his toes as well. An aboriginal is not a

mathematician, and his vocabulary does not contain running series of numerals. The usual method of counting low figures is after the following pattern of the King Sound natives:

“*arra*” (one).

“*kwiarra*” (two).

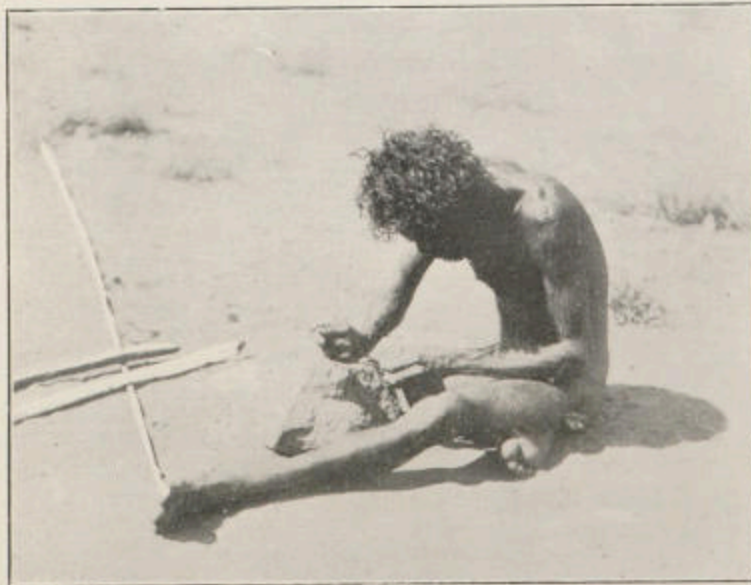
“*kwiarra arra*” (three).

“*kwiarra kwiarra*” (four).

Beyond four, counting either goes by “hands” or “feet,” or for ordinary purposes there are two comprehensive words in use which signify a “small-large number” and a “large-small number.” In some cases, such as the Aluridja, “one” (“*goitarada*”) appears as the diminutive form of “two” (“*goitara*”). In the same sense, a shorter distance is expressed by the Wongapitcha as the diminutive of a great distance by qualifying the word “*wurnma*,” meaning “far,” by affixing “*wimuggitta*,” which means “young.” Hence “*wimuggitta wurnma*” reads the “young (one) of far,” i.e. “close up.” This is really the same way of expressing a fraction of space as the same tribe has adopted for expressing minority in age; a youth or young man is known as “*wimuggitta wardi*,” the second word “*wardi*” standing for an adult man.



1. Making "vegetable down" by pounding grass between two stones.
Humbert River, Northern Territory.



2. Worora native making a stone spear-head, Northern Kimberleys,
Western Australia.

PLATE LIV

1. Making "vegetable down" by pounding grass between two stones. Humbert River, Northern Territory.
2. Worora native making a stone spear-head, Northern Kimberleys, Western Australia.

The Dieri of the Lake Eyre region have one of the most elaborate systems of numeration, which includes, at any rate, an expression for every number up to eleven.

“kulno,” one.

“mandru,” two.

“parkulu” or *“parkulintja,”* three.

“mandru mandru,” four.

“mandru ja parkulintja,” five.

or *“marra warra kulno,”* five, i.e. hand part one (one hand).

“marra pirri kulno,” six.

“marra pirri mandru,” seven.

“marra pirri parkulintja,” eight.

“marra pirri mandru mandru,” nine.

“marra warra mandru” ten, i.e. hand parts two (two hands).

“tjinna pirri kulno,” eleven.

“marrapu,” many.

“mörlla marrapu,” very many.

In the above table the following explanations will be found useful:

“ja” = and.

“marra” = hand.

“tjinna” = foot.

“warra” = part.

“pirri” = nail (toe or finger).

Thus the reading of the numerals runs: One, two, three, two-two (four), two and three (five), or one hand part (five), hand finger-nails one (six),

hand finger-nails two (seven), hand finger-nails three (eight), hand finger-nails four (nine), two hand parts (ten), foot toe-nails one (eleven).

To imply a repetition or continuance of action, the frequent use of the same verb is avoided by the Sunday Islanders, but the same effect is achieved by reiterating the word "*garra*" indefinitely. The expression of continued action is usually in the present tense and is mostly applied to words like "running," "walking," "jumping," "sleeping," "raining," "blowing," "bleeding," etc., etc. For example, a man from a tree or other look-out might be describing to his companions below, the doings of their hunting party out on the plain; while the latter keep moving, he conveys the fact to his hearers by ejaculating "*garra, garra, garra, ...*" The direction of the hunters' movements is indicated by the swing of his hand. The moment the observer in the tree stops saying "*garra*," the people below know that the hunters have ducked or have temporarily ceased the pursuit.

The terminal syllable of a word is never uttered very clearly by an aboriginal; and it is consequently difficult to distinguish between a short e, a, o, or u. This is particularly true of the language used by the old men, the defectiveness being often increased by the gaps between the incisors resulting from the initiation ceremonies.

There are occasions, however, when the last syllable receives especial emphasis. The Wongapitcha word for plenty is "*ura*"; when the idea of plenty is great the word becomes "*ura-ku*," with an intentional intonation upon the "*ku*"; and when the plenty is extraordinary, the word receives yet another syllable and becomes the superlative "*ura-ku-pu*."

The simple affirmative of the Aluridja, which has also been adopted by the Arundta, is "o," less commonly "u"; but when strong affirmation is intended the word is changed to "owa," or "owau."

The ordinary negative of the Arundta is "*itja*"; but forcibly expressed this may become "*itjama*" or "*itjingalai*."

Any sudden exclamation or accidental noise, like a cough or sneeze, might be exaggerated by the addition of a syllable at the end. When a Sunday Islander sneezes he makes a word out of the natural noise sounding like "*Tish-e!*"

An exclamation which is common practically to all Australian tribes, and may express surprise, fear, pain, or disgust, is a very liquid “*irr*.” The Arunndta have strengthened this monosyllabic cry by giving it the suffix “*ai*”; nowadays the word is, however, mostly pronounced “*Yerrai!*” A variety of the last-named, but not quite so forceful, is “*Yakai!*” A sudden fright or the anticipation of harm might produce a very short “*i*,” without any indication of the presence of the liquid consonant.

Appreciation is indicated by “*Aha!*” or “*Hm-hm!*” practically throughout the central and northern districts of Australia; in both cases special emphasis is applied to the second syllable.

A central Australian exclamation calling one to order or attention in a somewhat harsh manner is “*Wai!*” When one person is being irritated by another, such as a parent by a whining child, the offender is thus rebuked. This word may also become a suffix; it may be combined, for instance, with the radical “*irr*” and produce a word “*Yirrawai*,” which is perhaps the strongest in a sense of disgust and reproach available in the Arunndta tongue.

Standing at the end of an adjective, the expression “*’n-dora*” in the same dialect signifies a great or superlative degree; e.g. “*marra*” means “nice,” but “*marr’n-dora*” a conception more like “excellent.” In the opposite sense, “*kurrina*” (bad) becomes “*kurrina’n-dora*” (worst). The same suffix can be attached to an adverb. When, say, an emphatic denial is needed, “*itja*” (no) takes the form of “*itja’n-dora*.”

There are numerous other syllables, which, when fixed to the end of a word, convey a special significance. When, in the Arunndta, the syllable “*tu*” is added to a person’s name in address, it really stands in place of a sentence like “Is it not so?” For instance, “*Nani knulia atoa utnuriraka, Endola-tu?*” when literally translated reads: “This dog man bit Endola, is it not so?” Endola is the name of a woman who is being addressed. In place of “*tu*,” the longer form of “*ditjekwi*” might be used.

Again, the suffix “*lo*” is found in daily use in the same dialect; it stands for the phrase: “Where is?” Hence the completed sentence, “*Kwatche-lo?*” stands for “Where is the water?”

When one finds “*jara*” added to a word, plurality is indicated, the sense conveyed being that many of the kind specified by the noun are assumed. The word for girl in the Arunndta is “*kware*,” consequently “*kwarenjara*” means that a number of girls are being considered, the “*n*” being simply interposed for the sake of euphony.

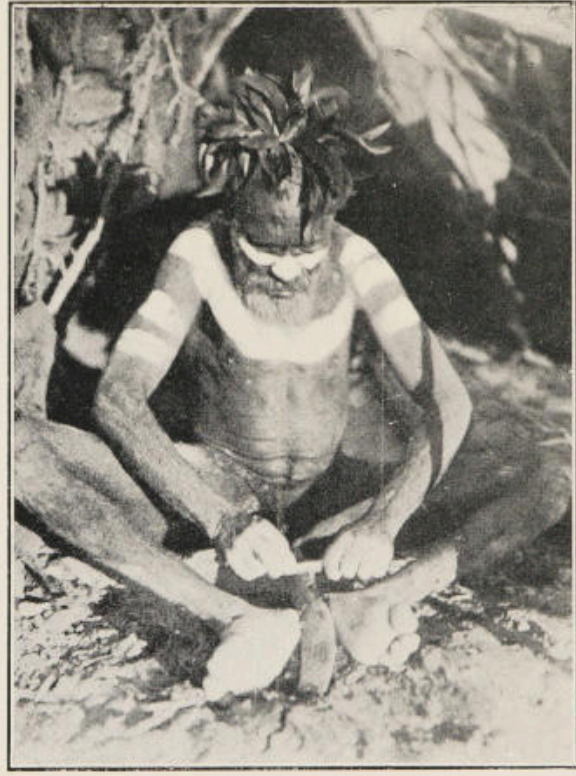
Any proper noun, like the name of a person, may be modified by adding “*ia*” to it, and, by so doing, one makes it a term of endearment in the vocative case. “*Ware*” ordinarily means “boy,” but by altering it to “*waria*” (i.e. “*ware-ia*”), the meaning becomes “dear boy” or “O boy.”

If the terminal “*a*” of a substantive is found to be changed to “*inna*,” the diminutive of the original is implied. A somewhat common name for a woman in the Arunndta is “*Unnruba*,” but during the years of childhood of a bearer of this name, the appellation is always in the form of “*Unnrubinna*”; in later years this changes to “*Unnruba*” more or less automatically.

A diminutive sense is also conveyed by duplicating an original word. In the eastern Aluridja dialect “*kaitji*” means an ordinary spear, whereas “*kaitji-kaitji*” is an expression applied to a toy spear. The Dieri recognize a Supreme Being whom they call “*Mura*”; any one of their numerous demi-gods, however, from whom they trace their descent is referred to as a “*Mura-Mura*.”



1. Wongapitcha man shaping a spear-thrower with an adze.



2. Aluridja man scraping a boomerang with a sharp stone flake.

PLATE LV

1. Wongapitcha man shaping a spear-thrower with an adze.

2. Aluridja man scraping a boomerang with a sharp stone flake.

Alterations in the ending of a verb indicate different moods and tenses. The verb “to come” in the Arundnta is “*pitchima*”; its inflections and their meanings will become apparent from the following short sentences:

“*Pitchai arrekutcha!*” Come here, old woman!

“*Pitchikarrerai arrekutch’njara!*” Come here, old women!

“*Arrekutcha pitchama.*” The old woman is coming.

“*Arrekutcha pitchika.*” The old woman came (or has come).

“*Arrekutcha pitchichinna.*” The old woman will come.

If we replace the verb “*pitchima*” by “*lama*” (“to go”), the inflections, taken in the same order as above, become: “*lai*,” “*larrirai*,” “*lakama*,”

“laka,” “litchinna.”

An adverb which finds considerable application in the Arunndta is “kalla,” indicating the completion of any deed or action. Most frequently the nearest translation would be supplied by the English word, “already.” If, therefore, we again consider one of the above sentences and interpose the word “kalla,” the meaning is strengthened considerably: “*Arrekutcha kalla pitchama*” then means “The old woman is already coming.” But “kalla” might further convey the sense of repletion. “*Einga knullia kalla kwatche n’ dai*” would mean, literally translated, “I dog enough water gave.” And finally “kalla” might express the completion of an action. A native, after losing the track of an animal, or having eaten as much as he wants, might be heard to say “kalla,” meaning “finished.” Finally “kalla” might even stand for “dead.” We have already noted a similar word in the vocabulary of the Sunday Islanders, viz. “Kaleya,” meaning “finish” or “good-bye.”

The personal pronouns are either used as separate words in a sentence, or they appear as prefixes to the principal verbs.

In the Arunndta, the following are used:

“einga,” I; “nuka,” mine; “eingana,” me.

“unda,” thou; “unguranga,” thine; “ngana,” thee.

“era,” he; “ekurra,” his; “erinna,” him.

“nuna,” we; “nunaka,” ours; “nungana,” us.

“rankara,” you; “rakankara,” yours; “rankarana,” you.

“etna,” they; “etnaka,” theirs; “etnana,” them.

As an illustration let us translate: “I like the boy”; we should have to transpose the words into the following order, “I boy like,” and the Arunndta would be: “*Einga ware unjinum.*” On the other hand, we might select the Sunday Island dialect as an example in which prefixes are used for representing the personal pronouns. The first personal prefix is “nun,” the second “min,” and the third “il” or “n’.” If, therefore, we take a simple verb like “*jakuli*” to “break,” we have:

“*nunjaluki,*” I break.

“*minjaluki*,” you break.

“*iljaluki*” or “*n’jaluki*,” he breaks.

The third person plural is represented by “*punjanga n*”; hence “*punjanga n’jaluki*” stands for “they (altogether) break.”

Whereas we have seen that an “*n*” might be interposed between two words for the sake of euphony, it no less frequently happens that two or more words are contracted into one for similar reasons. If, to quote a simple case, we wish to translate into Arundta “You give (it) me,” we have three separate words, “*unda*” (you), “*nuka*” (me), and “*dai*” (give), which in the above expression are fused into one word, “*nukundai*.”

Apart from the full and pure vowel sounds, the Australian aboriginal dialects include the modified sounds expressed by the German diphthongs, *œ*, *äu*, and *ue* (i.e. the French *u*). The pairs of consonants, *p* and *b*, *k* and *g*, and *t* and *d*, are often interchanged during speech. Sibilant sounds are extremely rare. My own name was very difficult, if not impossible, for the Aluridja to pronounce; the best they could make of it was “*Battedu*.” So, also, the imported pet name of the domestic cat, “pussy-pussy,” as it is used by the settlers, cannot be managed. The Aluridja call it “*putte-putte*,” and the Arundta “*put-pudtha*.”

Let us look into the construction of the Aluridja personal pronouns a little more carefully. Every pronoun consists of a stem, which is different for every person, and an ending, which varies with the case.

The stems are:

First person, singular, “*naiu*”; plural, “*nganan*.”

Second person, singular, “*nuntu*”; plural, “*nurangari*.”

Third person, singular, “*balu*”; plural, “*tana*.”

And the endings are as follows:

Genitive, “*-ba*.”

Dative, “*-lukuru*.”

Accusative, “*-na*.”

Ablative, “-languru” (from), or “-lawana” (with).

A simple sentence embodying three pronouns would be: “*Naiulu tanalawana nuntulukura ngalierra*,” which, literally translated, reads: “I with them to you have come.”

A few of the Arunndta interrogatives are:

“*Ewunna?*” What? or What is it?

“*Unndana?*” Where?

“*Nguna?*” Which?

“*Ngula?*” Who?

“*Ntakinya?*” How many?

The aspirate is very occasionally met with. The local group of the Arunndta, residing at Arltunga, which styles itself “*Herrinda*,” is one of the few exceptions. Even in the settled districts, where the natives have adopted European names, one christened “Harry” is usually spoken of as “*Yarri*.” The sounds which are most commonly aspirated are the exclamations, e.g. “*Hai!*” “*Aha*,” and “*Hm-hm!*”

Used in conjunction with a vowel, usually an “a,” the consonantal combination “*ng*” is common to all Australian dialects. Although to the modern tongue a little difficult to master, the sound is strikingly fluent with the aboriginal; indeed, its frequent inclusion in words of fundamentally simple origin suggests a primitive, natural derivation. Along the north-western coast, the words for “water” are “*ngawa*” (Wave Hill), “*ngauwa*” (Humbert River), “*kornga*” (Sunday Island).

In the Arunndta, “*nga*” stands for either “Here you are!” “There!” or “Take it!” all phrases being used in the sense of offering something to a person, such as mother to infant. “*Ngaboni*” is a modified form of the last-named, having any of the following meanings: “Here!” “Look!” or “Behold!” And “*ngarai*” is yet another modification of similar significance, usually placed immediately behind the object it is desired to draw attention to, e.g. “*Arre ngarai!*” (Look at the kangaroo).

Verbal greetings, akin to our “Good-day!” are rarely, if ever, made use of by natives when they meet; but some of the tribes recognize an orthodox term of salutation at partings, as indicating a friendly feeling which the speaker bears towards the person he is on the point of leaving. The Mulluk-Mulluk and some of their neighbours on the Daly River have adopted the word “*mummuk*” to express their farewell, whilst the Sunday Islanders’ vocabulary includes a similar expression in the word “*tchorrogu*.”

An alarm, such as we would raise in the form of “Look out!” or “Get out of the way!” is rendered by the Arunndta “*Par-le!*”

There are numerous examples which could be mentioned of words appearing in an aboriginal dialect which have a striking resemblance to European words of modern and ancient languages, which at this stage of our knowledge of primitive tongues must be recorded as curiosities rather than linguistic analogies. One of the most remarkable, perhaps, is the Arunndta for head, viz. “*kaputa*”; but to connect this in any way with the Latin word would be as unreasonable as calling the “patriarchal” type of aboriginal a semite.

It is, therefore, not my intention to suggest anything beyond mere coincidence, and it lies far from my mind to attempt theorizing upon little evidence, yet it must be admitted that similarities in points of evolution and culture, no matter how trivial and accidental they may be, are not without interest. Viewed entirely in this light, there is no harm in mentioning one or two similarities which exist between certain words of the Australian dialects and those of other tongues, articulate or otherwise.

Affirmation is expressed in the Cambridge Gulf district by “*yau*,” “*yo-au*,” or “*yowai*.” During latter years, the same expression has found its way south, even as far as the MacDonnell Ranges, where it has largely replaced the original form of “*owa*.” In the adjoining coastal districts of the Northern Territory, a similar word is in use as well as one which sounds more like the German “*ja*.” It is of more than passing interest to note that throughout the Malay Archipelago “*yo-au*” or “*yowai*” is the principal form of affirmation in use among the local primitive peoples. It is not unlikely, therefore, that the early Macassan trepang fishers, who used to visit the north coast of Australia long before the white man arrived, brought this word along with them.

We find, also, that the consonantal expression, “*ng*,” combined with a vowel sound, is common among other primitive languages. Professor Garner, too, has found it included in the inarticulate “speech” of apes in the form of a note indicating satisfaction, which can be represented by “*ngkw-a*.” The speculative mind naturally wonders whether this simian cry of satisfaction is perhaps fundamentally of similar, though independent, phonetic origin to the northern Australian word for water, the essential of life, which we noted above is “*ngawa*.”

The di-syllabic muttering of a European child-in-arms, moreover, which the happy parents flatter themselves sounds like “*pa-pa*” or “*ma-ma*,” is equally characteristic of the aboriginal babe, but the parents in the latter case interpret the note to be “*nga-nga*.” And, indeed, the Arunndta mother responds by handing the infant the fluid it needs, while she caressingly soothes it with the simple little word “*Nga*,” which in our own language would be equivalent to the sentence, “Here you are!”

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The Hassell Press, 104 Currie Street, Adelaide

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